

STABILITY-PRESERVING SEMIGROUPS, CAPACITY DYNAMICS, AND CLASSICAL GROUND-STATE ALGORITHMS

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ABSTRACT. We classify the linear generators that preserve all complex half-plane-stable polynomials on a finite box of degrees. Their canonical differential order is at most two, and their coefficients admit certificates of size independent of the capacities. In disk coordinates, strictly positive coordinate damping makes the actual semigroup kernel strictly zero-free and gives a sharp, capacity-independent real-part spectral gap, without self-adjointness or coefficient positivity. For positive degree-preserving evolutions, the homogeneous-sector spectral bounds form a concave sequence. This yields sharp token-graph spectral concavity and resolves the adjacency and signless-Laplacian monotonicity conjectures of Apte, Parekh, and Sud. For bounded-degree two-body generators on binary variables, this structure yields deterministic classical algorithms for the principal eigenvalue and relative approximations to stable product tests of the principal eigenvectors. The algorithms combine local perturbation coefficients with a nonvanishing reciprocal-field domain for the scalar tests. At fixed local norm bound and positive damping, their bit complexity is polynomial in the system size, inverse error, and input length. Applied to the full Suzuki–Fisher family, they compute ground-state energies, relative amplitudes against product states in a hemisphere, and the corresponding product-event probabilities, and give classical sampling of adaptive equatorial measurement records. We also prove an all-temperature Hamiltonian converse and classify quantum Markov semigroups that completely preserve the Lee–Yang property at arbitrary finite local spins: interacting generators are excluded, and every spin above one-half has isotropic noise and axial drift. A separate positive branch classifies finite, infinite, and mixed-capacity dynamics, with the generation and domain hypotheses stated explicitly.

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1. INTRODUCTION

A zero-free polynomial can encode a probability distribution, a quantum state, or a linear operator. In each setting the location of its zeros imposes a global constraint on the encoded object. For probability generating functions, real stability is a source of negative dependence. For quantum systems, the Lee–Yang property controls the coherent tensor of a state or a Gibbs operator. The same algebraic question appears in both settings: which continuous evolutions preserve the relevant zero-free region?

The classification of individual stability-preserving operators by their symbols gives a starting point for this question [10, 13, 11]. A semigroup imposes an additional condition: every operator e^{tA} , including those at arbitrarily small positive times, must be a preserver. This infinitesimal requirement is especially restrictive at collisions of roots. It is also useful. A generator description can identify all admissible interactions, distinguish physical positivity from complex stability, and supply a spectral structure that is not apparent from the entries of the operator.

Infinitesimal stability preservers have also been studied directly. Purbhoo constructs a family on multiaffine polynomials indexed by matrices with real diagonal and nonnegative off-diagonal entries [47, Proposition 4.5]. His result gives sufficient generators. The structural theorem below supplies necessary and sufficient conditions for arbitrary endomorphisms on every finite coordinate box.

This paper develops these consequences in the following chain:

$$\begin{aligned} &\text{root-collision constraints} \implies \text{generator structure} \\ &\implies \text{strict complex spectral control} \implies \text{classical computation.} \end{aligned}$$

There are two preservation classes. The first tests all complex stable polynomials. Its finite-dimensional generator cone is particularly rigid, and underlies our spectral and quantum results. The second tests nonnegative real stable polynomials,

as required for probability generating functions. It allows additional dynamics, including coalescence, and requires a separate capacity and analytic theory. Neither preservation class is silently substituted for the other.

1.1. The finite-dimensional structure theorem. Let V_κ be the space of polynomials with coordinatewise degree bounds $\kappa = (\kappa_1, \dots, \kappa_n)$. Every linear operator on this space has a canonical normally ordered differential representation. The first main theorem describes which such operators generate stability-preserving semigroups, without assuming positivity, locality, or a differential order in advance.

In real half-plane coordinates, the answer is second order. If

$$A = \sum_{\alpha \leq \kappa} P_\alpha(z) \partial^\alpha$$

is the canonical expression, preservation forces $P_\alpha = 0$ for $|\alpha| > 2$. Each diagonal second-order coefficient is a polynomial of its own variable and is nonpositive on the real line. Each mixed coefficient is a polynomial of the corresponding pair of variables and is nonpositive on the real plane. Together with the condition that A acts on V_κ , these conditions are also sufficient. This is Theorem 3.1. The finite-box constraint is substantial: it couples the highest terms of the lower-order coefficients to the second-order part.

For complex generators there is one further component. Every generator is the sum of such a real generator and

$$i \left(cI + \sum_i \left(q_i(z_i) \partial_i - \frac{\kappa_i}{2} q'_i(z_i) \right) \right), \quad c \in \mathbb{R}, \quad q_i \geq 0 \text{ on } \mathbb{R}, \quad (1)$$

where each q_i is a real polynomial of degree at most two. Conversely every such sum generates a complex stability-preserving semigroup (Theorem 3.2). The operators in (1) are the infinitesimal one-variable Möbius motions pointing into the half-plane. Thus arbitrary interactions in the imaginary part are excluded.

The proof has two different ingredients. Root-cluster asymptotics force the order bound and the signs. The local multiple-root mechanism has an antecedent in the variational analysis of Burke and Overton [21, Theorem 1.2 and Lemma 1.4]. A finite-dimensional algebraic decomposition then reduces sufficiency to explicit binary primitives. The required certificates use Gram matrices of orders at most four, independently of the numerical capacities. Polarization gives an exact binary-slot realization, with the binomial factors retained. These are generator statements; they are not obtained by differentiating a smooth boundary of the stable set. That boundary is singular, and its higher-multiplicity strata matter.

The fixed-degree one-variable geometry can nevertheless be expressed spectrally. If $H(p)$ is the Hermite matrix of a monic real-rooted polynomial p , then

$$T_{\mathcal{R}_n}(p) = \{h : DH_p[h]|_{\ker H(p)} \succeq 0\}.$$

Appendix A proves this formula for arbitrary root multiplicities and identifies exactly when the associated multiplier image is closed. This supplies a local geometric result, with its fixed-degree convention stated explicitly; it does not replace the degree-changing arguments used for the full generator theorem.

1.2. Strict damping and a complex spectral gap. Pass to the unit disk by a Cayley transform. Suppose A_0 generates a semigroup preserving all complex disk-stable polynomials in V_κ , and set

$$A = A_0 - 2 \sum_i \mu_i E_i, \quad E_i = z_i \partial_i, \quad \mu_i > 0. \quad (2)$$

The damping has a simple interpretation: it contracts each polynomial variable towards the center of the disk. The conclusion is stronger than preservation of the open disk. For every $t > 0$ the actual binomial-weighted kernel of e^{tA} is nonzero on the closed unit polydisk, and it is stable in a polydisk of some radius $r_t > 1$. For any fixed positive time, a common strict radius works at all later times (Theorem 4.3).

Consequently A has an algebraically simple principal eigenvalue a_* and

$$\Re a_* - \Re a \geq 2 \min_i \mu_i \quad \text{for every other eigenvalue } a. \quad (3)$$

The constant is sharp, and the bound is independent of the capacities and number of variables (Theorem 4.8). Neither self-adjointness nor coefficient positivity is assumed. The principal right and left objects are strictly stable.

The proof uses the finite-dimensional complex-cone theory of Rugh [50], together with a damping argument and contraction of stable kernels. The sharp radius-to-spectrum mechanism has a Hermitian Lee–Yang antecedent in [18]. Our extension concerns arbitrary complex finite-box stable generators and their actual kernels. It must not be read as a bound on all transients in a fixed ordinary operator norm. The appropriate quantity is instead the oscillation of a quotient h/p over the polydisk. Its contraction, including time-dependent flows, is developed in Section 4. The projective-metric and oscillation constructions themselves have antecedents in [24, 27].

1.3. Comparing homogeneous sectors. Coefficient positivity gives another spectral consequence. Suppose a finite-box generator preserves real stability, has nonnegative off-diagonal entries, and preserves every homogeneous degree. The spectral bounds of its degree blocks form a concave sequence (Theorem 4.16). This also holds for nonself-adjoint, reducible blocks. The proof evolves a single polynomial with positive coefficients, applies Newton’s inequalities to its diagonal restriction, and takes exponential growth rates. The underlying growth-exponent principle is classical [15].

For a nonnegatively weighted graph G , let A_k, D_k be the adjacency and degree matrices of its k -token graph. The generator classification applies to the whole family

$$B_k(a) = A_k + aD_k, \quad -1 \leq a \leq 1.$$

Its largest eigenvalues satisfy

$$2\lambda_{\max}(B_k(a)) \geq \lambda_{\max}(B_{k-1}(a)) + \lambda_{\max}(B_{k+1}(a)),$$

and are symmetric under $k \leftrightarrow n - k$. They are therefore nondecreasing up to half filling. This proves the adjacency and signless-Laplacian monotonicity assertions of [4, Conjectures 5–6], with a stronger weighted concavity statement and a sharp universal interval for a (Theorem 4.17). The $a = 0$ semigroup is already contained in Purbhoo’s construction [47]; the new conclusion compares its particle-number sectors. The corresponding EPR interpretation uses the established token-graph correspondence of [4].

1.4. A classical algorithm beyond self-adjoint operators. Here is the algorithmic consequence in a form that specifies its input model. On n binary variables let A_0 be given as a sum of one-site and two-site matrices with Gaussian-rational entries. Suppose the interaction graph has degree at most d , each local matrix has norm at most J , and e^{tA_0} preserves all complex disk-stable multiaffine polynomials. The last condition is a promise about the whole generator; individual displayed summands need not be preservers. For fixed d, J , and a fixed rational $\mu > 0$, the principal eigenvalue of

$$A_0 - 2\mu \sum_i E_i$$

can be approximated to absolute complex error δ by a deterministic classical algorithm in time

$$\text{poly}_{d,J,\mu}(n, \delta^{-1}, \text{input bit length}).$$

This is Theorem 5.1. The matrix has exponentially many entries, but the input is its sparse local description. There is no claim of a running time polynomial uniformly in μ^{-1} .

The same local input also gives relative access to the principal eigenvectors. Normalize the right and transpose-left vectors v, w to have vacuum coefficient one. For any tensor product $B = \bigotimes_i B_i$ of Gaussian-rational 2-by-2 matrices with $(B_i)_{00} = 1$ and disk-stable kernels, Theorem 5.11 computes

$$w^\top B v \quad \text{and} \quad \text{Tr}(B\Pi) = \frac{w^\top B v}{w^\top v}, \quad \Pi = \frac{v w^\top}{w^\top v},$$

to relative complex error ϵ in $\text{poly}_{d,J,\mu}(n, \epsilon^{-1}, L_{\text{in}})$ time. Both quantities are nonzero. This is scalar query access; no ordinary-norm conditioning bound for Π is required.

The algorithm uses two distinct analytic regions. A local expansion at a product vacuum has a size-independent convergence radius. The perturbation-coefficient algorithm of Bravyi, DiVincenzo, and Loss [16] supplies these coefficients. A complexification argument extends the coefficient bounds from Hermitian perturbations to arbitrary complex ones. The stability gap then continues the same principal branch through a right half-plane. An explicit disk map joining these regions turns logarithmically many local coefficients into a value estimate at the desired fixed damping. We give the branch-identification argument, the coefficient construction, and the rational bit bounds in Section 5.

For the state queries, the extra step is to obtain a nonvanishing scalar test on that same type of domain. The creation-coefficient bounds in [16] give a uniform vacuum disk for both eigenvectors. The half-plane continuation and strict disk contraction then show that $w^\top B v$ never vanishes. Its logarithm has a one-sided real-part bound of order n throughout the domain (Lemma 5.13). Connected coefficients of this logarithm can be computed from subsystems of logarithmic size, without requiring the induced subsystems to satisfy the global preservation promise.

This method does not introduce a new perturbation expansion or a new general interpolation principle; connected-cluster computation combined with analytic continuation is an established strategy [45, 62, 58]. Its point is that complex stability provides a global analytic domain for a class of sparse operators that need not be self-adjoint, positive, or Markovian. The computational statement uses that domain to go beyond the perturbative neighborhood where the coefficient expansion is initially justified.

1.5. Quantum consequences and their scope. For qubits, consider

$$H = - \sum_{ij} \left(J_{ij}^z Z_i Z_j + \sum_{a,b \in \{x,y\}} J_{ij}^{ab} \sigma_i^a \sigma_j^b \right) - \sum_i (h_i^x X_i + h_i^y Y_i + h_i^z Z_i), \quad (4)$$

$$J_{ij}^z \geq \|J_{ij}^\perp\|_{\text{op}}, \quad h_i^z \geq 0,$$

with real coefficients and $J_{ij}^\perp = (J_{ij}^{ab})_{a,b \in \{x,y\}}$. This is the phase-rotated Suzuki–Fisher family. Its Lee–Yang sufficiency, including the full transverse operator-norm condition, is classical: see [53, equation (2.47)] and [25, Theorem 2]. Asano’s contemporaneous account already uses a full matrix generating function and its composition law for the spin-one-half XXZ subfamily [6, equation (1.15), Theorem 1]. The tensor language of [61] is a modern formulation of this tradition. We do not claim the sufficient family or the use of full matrix generating functions as new. The generator theorem gives the converse: up to an additive scalar, these are precisely the finite-qubit Hamiltonians whose full Gibbs tensor is Lee–Yang at every temperature (Theorem 6.2). In particular no genuine higher-body term is possible under that full-tensor hypothesis. Ruelle’s all-temperature converse [49, Theorem 9] already forces pair interactions for classical coefficientwise Boltzmann Lee–Yang polynomials. The object here is the full matrix tensor of an arbitrary Hermitian Hamiltonian, including noncommuting interactions.

For rational Pauli coefficients, bounded degree and bounded local strength, and longitudinal fields bounded below by a fixed $\mu > 0$, Theorem 5.7 gives deterministic classical ground-energy approximation to arbitrary absolute accuracy. Theorem 5.16 additionally computes nonzero relative amplitudes against product states $(|0\rangle + s_i|1\rangle)/\sqrt{1+|s_i|^2}$, $|s_i| \leq 1$, and relative probabilities of the corresponding product events. It gives a classical randomized sampler for adaptive equatorial measurements, with any prescribed total variation error. Each site is measured at most once; the strategy’s own running time and the bit lengths of its phases are included explicitly. At nonnegative fields, adding a fixed field of size proportional to ε gives an additive εn energy-value approximation scheme (Theorem 5.9). This covers bounded-degree EPR models on arbitrary graphs and, after a local unitary transformation, bounded-degree bipartite Quantum MaxCut. In the unweighted latter setting the estimate also gives a relative approximation scheme for the optimum value.

Earlier classical algorithms treat important subclasses of quantum ferromagnets [17, 30]. The broader classical computation question is stated in [18]; a related interpolation strategy was announced in [59], with further background in [60]. Our comparison is between precise hypotheses and error guarantees, not between names for the methods. Relative X -basis amplitudes, phases, and measurement marginals already appear in [61, Section 4], for general Lee–Yang states with a fixed spatial zero-free radius greater than one and state-dependent quasipolynomial circuits. Here the local Hamiltonian is the input, the field is any fixed positive constant, and the complexity is polynomial; a size-independent spatial radius margin is not assumed. The new analytic input is the nonvanishing reciprocal-field domain for the principal-state tests. It is distinct from the thermal partition-function domains used for product measurements in [62, Theorem 15]. The zero-field energy-value scheme does not prepare an approximately optimal quantum state. It does not

solve the unrestricted inverse-polynomial-accuracy EPR problem [41], and it is not a replacement for approximation algorithms that also construct states [3].

Positive damping also proves strict zero-freeness of the actual Gibbs kernel and of its unique ground-state polynomial, addressing the qualitative radius question in [61]. The stronger proposed radius $s^{-1/2}$ for a deformed EPR ground state is false. Appendix H gives an unweighted six-vertex tree at $s = 1/2$, together with an exact rational certificate of a zero strictly inside that radius. It leaves the qualitative radius-greater-than-one conclusion intact.

Physical quantum Markov evolution exhibits a different rigidity. For at least two qubits, ordinary Lee–Yang preservation by a continuous completely positive trace-preserving semigroup is equivalent to full-Choi preservation. Its generator is a sum of independent one-qubit generators satisfying two fixed-size matrix inequalities (Theorem 7.1). Locality is already forced by testing finitely many fixed trine product states at times tending to zero (Theorem 7.3). This is a statement about physical positive inputs, for which the full complex preserver theorem cannot simply be assumed. It contrasts interacting equilibrium with independent Lee–Yang-preserving Markovian evolution.

Theorem 7.22 extends the dynamical classification to arbitrary finite local spins, without assuming a local or linear Lindblad representation. If at least one site has spin above one-half, it is enough to test pure Lee–Yang inputs near time zero. All intersite terms vanish. On every site of spin $j \geq 1$, the generator is

$$\mathcal{L}_i(\rho) = -i[h_i S_i^z, \rho] - \frac{\gamma_i}{2} \sum_{\alpha=x,y,z} [S_i^\alpha, [S_i^\alpha, \rho]], \quad h_i \in \mathbb{R}, \quad \gamma_i \geq 0.$$

Qubit sites retain the two matrix inequalities above. The isotropic spin semigroup itself is classical; the conclusion here is that Lee–Yang preservation forces this form out of an arbitrary finite GKLS generator. A higher-spin site also supplies the pure-state test that rules out the extra qubit-only boundary possibilities. As a short application, the spin-one channel identifies axial isotropic Brownian convolution on $\text{SO}(3)$ (Corollary 7.25). The finite-representation recognition mechanism is an instance of Voit’s Gaussian characterization [56], not a new general characterization principle for Lie-group processes.

The appendices give the corresponding static invertible classification, the isometric and pure-basis-output cases, one-qubit singular strata, and the independently covariant higher-capacity case. They retain their hypotheses: arbitrary singular noisy channels and arbitrary noncovariant higher-spin static channels are not classified here.

1.6. Positive dynamics and capacities. For probability generating functions the positivity restriction changes the answer. The positive branch includes exclusion, immigration, transfer, and coalescence mechanisms. Its infinitesimal tests arise from finite root collisions and from Poisson or Laguerre–Pólya limits. We prove arbitrary-jump Markov rigidity, a canonical logarithmic-symbol theorem, the finite quadratic-contact criterion, and the finite/infinite/mixed-capacity classification in Section 8. The connection with earlier stability-preserving particle systems is made explicitly [14, 39].

Infinite capacity requires more than an algebraic generator condition. The weighted spaces, polynomial-domain result, killed finite-capacity approximations, and generation criteria are proved separately in Appendix B. In particular, dilation

by a factor greater than one is not treated as a bounded similarity on unweighted ℓ^1 . We also distinguish polynomial-valued semigroups from entire-valued extensions; the latter need not preserve every entire zero-free function.

This organization separates three kinds of input: inherited symbol, complex-cone, and perturbation theory; the polynomial-specific generator and strictification arguments; and the analytic or physical conditions needed for each application. It permits the positive capacity theory and the complex spectral theory to share their root-collision mechanism without asserting that their preservation cones coincide.

1.7. Organization. Section 2 fixes all normalizations. Section 3 proves the finite-box generator theorem, Section 4 develops strictification, spectral contraction, and sector concavity; Section 5 proves the eigenvalue, principal-state query, and measurement-sampling algorithms. Sections 6 and 7 treat equilibrium and quantum Markov dynamics. Section 8 gives the positive capacity theory. The appendices contain the local Hermite geometry, analytic generation, static and higher-capacity classifications, quantitative diagonal rigidity, and the exact radius counterexample.

2. CONVENTIONS AND ZERO-FREE KERNELS

2.1. Regions, degrees, and preservation. Write $\mathbb{H} = \{z \in \mathbb{C} : \Im z > 0\}$ and $\mathbb{D}_r = \{z \in \mathbb{C} : |z| < r\}$, with $\mathbb{D} = \mathbb{D}_1$.

Definition 2.1 (Stability and preservation). A nonzero polynomial is Ω -stable if it has no zero when all its variables lie in Ω . For a product of different regions the corresponding meaning is used. A polynomial is *real stable* if its coefficients are real and it is $\mathbb{H}$ -stable. In one variable this is equivalent to real-rootedness. A *positive stable polynomial* has nonnegative real coefficients and is real stable. Positivity throughout refers to coefficients, unless an operator on a Hilbert space is explicitly said to be positive semidefinite. An operator *preserves* one of these classes when it sends every member to another member or to zero.

Let $\mathbb{N} = \{0, 1, 2, \dots\}$. For $\kappa \in \mathbb{N}^n$ put

$$V_\kappa = \{f \in \mathbb{C}[z_1, \dots, z_n] : \deg_{z_i} f \leq \kappa_i\}, \quad \binom{\kappa}{\alpha} = \prod_i \binom{\kappa_i}{\alpha_i}.$$

Variables of capacity zero can be deleted. A degree bound is not an assertion that the polynomial has that degree. All finite-box statements include degree drops and constants. For a real operator, its action on complex polynomials always means its complex-linear extension. Every operator on V_κ has a unique canonical expression

$$A = \sum_{\alpha \leq \kappa} P_\alpha(z) \partial^\alpha, \quad (5)$$

as an operator on V_κ . Uniqueness follows by applying the expression successively to monomials, starting with 1: the equation for z^α determines $\alpha! P_\alpha$ after all lower terms are known. The phrase *differential order* refers to this expression.

A finite-dimensional generator A preserves a class when e^{tA} preserves it for all $t \geq 0$. For an infinite-dimensional space, the existence and topology of a strongly continuous semigroup are additional requirements, treated in Appendix B. A formal differential expression alone does not assert generation.

Definition 2.2 (Complete stability preservation). An operator is *completely stable* if its tensor product with the identity in arbitrarily many extra polynomial variables preserves the same complex stable class, for every finite choice of the extra degree bounds.

This is a statement about polynomial stability, distinct from complete positivity of a quantum channel. Positive stability with nonnegative coefficients and complete complex stability must also be distinguished.

2.2. Polarization and the disk kernel. The normalized polarization of z_i^a into κ_i binary slots is

$$z_i^a \mapsto \binom{\kappa_i}{a}^{-1} e_a(z_{i1}, \dots, z_{i\kappa_i}).$$

It is symmetric in the slots and specializes to z_i^a on the diagonal. Polarization preserves stability in the convex circular regions used here, including open disks and half-planes; diagonalization preserves it directly. We use the Grace–Walsh–Szegő coincidence theorem in this form; see [10, 11, 12]. The binomial normalization is part of the definition, not an optional choice of basis.

For $T : V_\kappa \rightarrow V_\lambda$ define its disk kernel by

$$K_T(z, w) = T_z \prod_i (1 + z_i w_i)^{\kappa_i} = \sum_{\alpha \leq \kappa} \binom{\kappa}{\alpha} (Tz^\alpha)(z) w^\alpha. \quad (6)$$

The notation z_i in the first expression denotes the input variables on which T acts; its result is a polynomial in the output variables. The kernel has output degree bound λ and input degree bound κ . In particular $K_{\text{Id}}(z, w) = \prod_i (1 + z_i w_i)^{\kappa_i}$.

For two polynomials of degree at most κ in paired variables define their bilinear contraction by

$$[f(u), g(v)]_\kappa = \sum_{\alpha \leq \kappa} \binom{\kappa}{\alpha}^{-1} f_\alpha g_\alpha, \quad f(u) = \sum_\alpha f_\alpha u^\alpha, \quad g(v) = \sum_\alpha g_\alpha v^\alpha. \quad (7)$$

There is no complex conjugation in this formula. Matrix composition is exactly contraction of the corresponding kernels:

$$K_{ST}(z, w) = [K_S(z, u), K_T(v, w)]_\kappa,$$

where κ is the intermediate capacity.

Lemma 2.3 (Disk contraction). *Suppose $f(z, u)$ and $g(v, w)$ are stable in products of disks and $\deg_{u_i} f, \deg_{v_i} g \leq \kappa_i$. If their paired radii r_i, s_i satisfy $r_i s_i \geq 1$, then their contraction (7) is stable in the remaining disks or is identically zero. If all $r_i s_i > 1$, contraction at any point in the remaining disks is nonzero.*

Proof. First consider a multiaffine polynomial $F(u, v) = a + bu + cv + duv$ that is nonzero on $\mathbb{D}_r \times \mathbb{D}_s$. Then $a \neq 0$ and $F(rt, st)$ is nonzero for $|t| < 1$. If $d \neq 0$, the product of its two roots has modulus at least one, so $|d|rs \leq |a|$; if $d = 0$, this inequality is immediate. Thus $a + d \neq 0$ when $rs > 1$. The same argument applies after fixing any uncontracted variables in their respective disks. Consequently the operation $a + bu + cv + duv \mapsto a + d$ preserves stability under strict separation of the paired radii. It may be iterated on a general multiaffine polynomial, without a factorization assumption.

Polarize the paired variables of f and g into κ_i slots for coordinate i , and apply this operation to each pair of corresponding slots in their product. For a fixed multi-index α , the matching slot subsets have cardinality $\binom{\kappa}{\alpha}$, while the two normalized polarizations contribute the factor $\binom{\kappa}{\alpha}^{-2}$. The result is therefore exactly (7). This proves strict nonvanishing, including degree deficiencies. It is the weighted form of circular-domain contraction; see [6, 10, 11]. For equality of paired radii, replace every paired variable in each input polynomial by $(1 - \varepsilon)$ times that variable, apply the strict statement, and let $\varepsilon \downarrow 0$. Hurwitz's theorem gives the stable-or-zero conclusion. The argument is valid after fixing any values of the unpaired variables in their respective disks. $\square$

The finite-box symbol theorem [10, 11] identifies stability preservers of rank at least two by the appropriate stable kernel; the low-rank alternatives are retained whenever they arise. In particular an invertible disk-stability preserver has a stable kernel (6). We do not infer a kernel criterion for arbitrary positive-coefficient preservers from this theorem.

The Cayley isomorphism is

$$(\mathcal{C}_\kappa f)(x) = \prod_i (x_i + i)^{\kappa_i} f\left(\frac{x_1 - i}{x_1 + i}, \dots, \frac{x_n - i}{x_n + i}\right). \quad (8)$$

It is a linear isomorphism between the degree boxes, and f is $\mathbb{D}$ -stable if and only if $\mathcal{C}_\kappa f$ is $\mathbb{H}$ -stable. Thus the generator classifications in half-plane coordinates and the spectral results in disk coordinates concern conjugate finite-dimensional semigroups.

2.3. Coherent coordinates and physical tensors. Equip V_κ with the inner product for which $\binom{\kappa}{\alpha}^{1/2} z^\alpha$ is an orthonormal basis. Equivalently, identify the standard basis $|\alpha\rangle$ of $\bigotimes_i \mathbb{C}^{\kappa_i+1}$ with $\binom{\kappa}{\alpha}^{1/2} z^\alpha$. The polynomial of a vector ψ is

$$f_\psi(z) = \sum_{\alpha \leq \kappa} \binom{\kappa}{\alpha}^{1/2} \psi_\alpha z^\alpha.$$

An operator M on this Hilbert space has coherent tensor

$$F_M(z, w) = \sum_{\alpha, \beta \leq \kappa} \binom{\kappa}{\alpha}^{1/2} \binom{\kappa}{\beta}^{1/2} M_{\alpha\beta} z^\alpha w^\beta. \quad (9)$$

This is precisely (6) for its action in coherent coordinates. Bra variables are independent complex variables; they are not conjugated when testing zero-freeness. For a positive semidefinite density matrix ρ , the assertion that F_ρ is $\mathbb{D}$ -stable is called its *Lee–Yang property*.

For a linear map Φ between matrix algebras the full coherent Choi tensor is defined by replacing $M_{\alpha\beta}$ in (9) with the coefficients $\Phi(|\gamma\rangle\langle\delta|)_{\alpha\beta}$ and adding input monomials $u^\gamma v^\delta$ with the corresponding square-root binomial weights. Its stability is called *full-Choi Lee–Yang preservation*. For qubits all capacities are one, so these weights are one. They cannot be omitted for higher spin. Ordinary physical preservation tests only positive semidefinite Lee–Yang inputs; it is a weaker condition for isolated maps.

We use the Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus $E_i = z_i \partial_i$ represents $|1\rangle\langle 1|_i$ and $Z_i = I - 2E_i$ on a qubit. For general capacity $E_i = z_i \partial_i$ has eigenvalues $0, \dots, \kappa_i$. A *principal eigenvalue* below means an algebraically simple eigenvalue whose real part is strictly greater than the real parts of all other eigenvalues; for a discrete operator it means the corresponding strict modulus dominance.

3. GENERATORS ON A FINITE POLYNOMIAL BOX

We first classify semigroups acting on the whole stable class, with no assumption on the signs of coefficients of the polynomials or of the operators. The distinction from preservation of the coefficient-positive stable class is essential: the two problems have different generator cones.

Let $\kappa = (\kappa_1, \dots, \kappa_d)$ have finite positive integer coordinates, and write

$$V_\kappa^\mathbb{F} = \mathbb{F}[z_1, \dots, z_d]_{\deg_{z_i} \leq \kappa_i}, \quad \mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}.$$

Coordinates of capacity zero are omitted. If no coordinate remains, the space consists of constants and every scalar generator has the requisite preservation property. Stability and complete preservation have the meanings of Definitions 2.1 and 2.2; in this section the domain of stability is $\mathbb{H}^d$. A real operator acts on $V_\kappa^\mathbb{C}$ by complexification.

For a polynomial q in one variable of degree at most two, define

$$J_{i, \kappa_i}(q) = q(z_i) \partial_i - \frac{\kappa_i}{2} q'(z_i). \quad (10)$$

These are endomorphisms of the box. The three choices $q = 1, z_i, z_i^2$ give

$$J_i^- = \partial_i, \quad J_i^0 = z_i \partial_i - \frac{\kappa_i}{2}, \quad J_i^+ = z_i^2 \partial_i - \kappa_i z_i.$$

Their span is the sitewise $\mathfrak{sl}_2$ algebra.

Every endomorphism has a unique canonical normal representation

$$A = \sum_{\alpha \leq \kappa} P_\alpha(z) \partial^\alpha. \quad (11)$$

Here the coefficient polynomials are not assumed to belong to the original box. The order of A means the largest $|\alpha|$ for which its canonical coefficient P_α is nonzero. The uniqueness of this representation, proved below, is important at small capacities: in particular ∂_i^2 is absent when $\kappa_i = 1$.

Theorem 3.1 (Real generators). *Let $A \in \text{End}_\mathbb{R}(V_\kappa^\mathbb{R})$. The following conditions are equivalent.*

- (1) *For every $t \geq 0$, e^{tA} preserves real stability.*
- (2) *For every $t \geq 0$, its complexification preserves complex stability, also after adjoining arbitrary inert polynomial variables.*
- (3) *The canonical representation of A has order at most two, and its second-order coefficients satisfy*

$$P_{2e_i}(s) \leq 0 \quad \text{for all } s \in \mathbb{R}, \quad \kappa_i \geq 2, \quad (12)$$

$$P_{e_i+e_j}(s, t) \leq 0 \quad \text{for all } (s, t) \in \mathbb{R}^2, \quad i < j. \quad (13)$$

In the third condition, box preservation forces P_{2e_i} to depend only on z_i , with degree at most four, and $P_{e_i+e_j}$ to depend only on z_i, z_j , with bidegree at most $(2, 2)$. There are no further inequalities on the zeroth- and first-order coefficients beyond the identities already forced by $A(V_\kappa^\mathbb{R}) \subseteq V_\kappa^\mathbb{R}$.

The coefficient $P_{e_i+e_j}$ in this statement is the full coefficient of $\partial_i \partial_j$, with $i < j$. Thus a convention writing the second-order part as $\sum_{i,j} Q_{ij} \partial_i \partial_j$, with Q symmetric, has $P_{e_i+e_j} = 2Q_{ij}$. We use the full-coefficient convention throughout this section.

Theorem 3.2 (Complex generators). *Let $A \in \text{End}_\mathbb{C}(V_\kappa^\mathbb{C})$. The following conditions are equivalent.*

- (1) For every $t \geq 0$, e^{tA} preserves complex stability.
- (2) For every $t \geq 0$, e^{tA} preserves complex stability after adjoining arbitrary inert polynomial variables.
- (3) There are a real generator R satisfying Theorem 3.1, a real scalar c , and real polynomials q_i of degree at most two, nonnegative on $\mathbb{R}$, such that

$$A = R + i \left(c \text{Id} + \sum_{i=1}^d J_{i,\kappa_i}(q_i) \right). \quad (14)$$

Equivalently, the canonical representation has order at most two, all its second-order coefficients are real and satisfy (12)–(13), and every $\Im P_{e_i}$ is nonnegative on $\mathbb{R}^d$. In this formulation box preservation forces $\Im P_{e_i}(z) = q_i(z_i)$ with $\deg q_i \leq 2$, and forces the corresponding imaginary zeroth-order term in (14).

The two theorems identify ordinary and complete preservation for *semigroups starting at the identity*. We will prove this by constructing the generators. It is not an assertion that an arbitrary isolated real stability preserver preserves complex stability or admits every tensor extension.

3.1. Canonical differential operators and box preservation. We record the algebraic restrictions separately from the stability inequalities. All statements in this subsection hold over either $\mathbb{R}$ or $\mathbb{C}$.

Lemma 3.3 (Canonical normal representation). *Every linear endomorphism of $V_\kappa^\mathbb{F}$ has a unique representation (11).*

Proof. Apply the proposed representation successively to monomials in increasing total degree. The constant monomial determines $P_0 = A1$. Once P_β is known for all $|\beta| < |\alpha|$, the equation on z^α determines

$$P_\alpha = \frac{1}{\alpha!} \left(Az^\alpha - \sum_{\substack{\beta \leq \alpha \\ \beta \neq \alpha}} P_\beta(z) (\alpha)_\beta z^{\alpha-\beta} \right).$$

Here $(\alpha)_\beta = \prod_i \alpha_i! / (\alpha_i - \beta_i)!$. No derivative whose index is not bounded by α acts nontrivially on z^α . The recursion therefore proves existence and uniqueness on the monomial basis. $\square$

Descriptions of differential operators on finite polynomial modules by $\mathfrak{sl}_2$ enveloping algebras are classical in quasi-exact solvability [54, 55]; we include the following decomposition to make the box conventions and small-capacity degeneracies explicit.

Proposition 3.4 (Operators of order at most two on a box). *Every box endomorphism of canonical order at most two is a linear combination of Id, the operators J_i^-, J_i^0, J_i^+ , and their products of length two, including products at the same site. Every endomorphism of order at most one has a unique expression*

$$c \text{Id} + \sum_{i=1}^d J_{i, \kappa_i}(q_i), \quad c \in \mathbb{F}, \quad q_i \in \mathbb{F}[z_i]_{\leq 2}. \quad (15)$$

Consequently the second-order coefficients have the variable dependence and degree bounds stated in Theorem 3.1. The vector space of endomorphisms of order at most two has dimension

$$1 + 3d + 5\#\{i : \kappa_i \geq 2\} + 9\binom{d}{2}. \quad (16)$$

Proof. Write $p_{\alpha, \beta}$ for the coefficient of z^β in P_α , and group terms according to the integer monomial shift $\nu = \beta - \alpha$. For input exponent $x \in \prod_i \{0, \dots, \kappa_i\}$, the coefficient of $z^{x+\nu}$ is the restriction to this grid of

$$F_\nu(x) = \sum_{\substack{|\alpha| \leq 2, \\ \alpha + \nu \geq 0}} p_{\alpha, \alpha + \nu}(x)_\alpha. \quad (17)$$

This polynomial has total degree at most two and degree at most κ_i in each x_i . We use the polynomial (17) also when $x + \nu$ is outside the exponent grid.

Suppose first that $\nu_i < 0$. Every summand has $\alpha_i \geq -\nu_i$, so F_ν is divisible by $(x_i)_{-\nu_i}$. If $-\nu_i > \kappa_i$, there are no summands and $F_\nu = 0$. In particular, whenever a grid point would give a negative output exponent, this falling factorial makes F_ν vanish there.

If $\nu_i > 0$, box preservation gives vanishing on every overflow face

$$x_i \in \{\kappa_i - \nu_i + 1, \dots, \kappa_i\} \cap \{0, \dots, \kappa_i\}.$$

To justify the divisibility as a polynomial identity, fix one such value of x_i . On the remaining tensor grid the value of F_ν is zero: either the output has a negative coordinate, in which case the preceding falling-factorial argument applies, or it is a genuine monomial outside the box, whose coefficient must vanish. Tensor-product interpolation, using the coordinate degree bounds in (17), shows that the restriction to this face is identically zero as a polynomial in all remaining variables. Thus the corresponding linear factor in x_i divides F_ν . If $\nu_i > \kappa_i$, the same argument gives vanishing on all $\kappa_i + 1$ coordinate values and hence $F_\nu = 0$.

For every remaining nonzero shift polynomial, these factors give

$$F_\nu = D_\nu G_\nu, \quad D_\nu(x) = \prod_i (x_i)_{\nu_i^-} (\kappa_i - x_i)_{\nu_i^+}, \quad \deg G_\nu \leq 2 - |\nu|_1. \quad (18)$$

The factors belonging to distinct coordinate values are coprime. Thus $F_\nu \neq 0$ implies $|\nu|_1 \leq 2$.

Put $E_i = z_i \partial_i$. For $\nu = 0$, every possible F_ν is realized by a combination of Id, $E_i, E_i^2, E_i E_j$. For $\nu = \pm e_i$, the required constant or affine factor G_ν is realized by $J_i^\pm$ and $J_i^\pm E_j$. For $\nu = \pm 2e_i$ use $(J_i^\pm)^2$; these shifts can be nonzero only if $\kappa_i \geq 2$. For $\nu = \pm e_i \pm e_j$, with $i \neq j$, use the corresponding products $J_i^\pm J_j^\pm$. The sign of the raising coefficient $x_i - \kappa_i = -(\kappa_i - x_i)$ is absorbed in the scalar multiplier. This accounts for every possibility in (18) on the input grid, and hence for every matrix entry of A .

This argument also covers binary sites. For example, if $\kappa_i = 1$, the canonical coordinate degree of F_ν in x_i is at most one; any inadmissible quadratic dependence has already been excluded before the realization step. Some displayed products may then be linearly dependent or zero, which does not affect the spanning assertion.

Repeating the same argument with total degree at most one leaves only a scalar and the three sitewise operators at each coordinate. They are linearly independent: the off-diagonal shifts separate the raising and lowering terms, while an affine polynomial in the diagonal exponents vanishing on the full tensor grid has every coefficient zero. This proves (15), including uniqueness.

The first-order coefficients of J_i^-, J_i^0, J_i^+ span $1, z_i, z_i^2$. Products at one site therefore have principal coefficients spanning all polynomials of degree at most four. Products at two different sites span all biforms of bidegree at most $(2, 2)$. At a site with $\kappa_i \geq 2$, these are canonical second-order coefficients; at a binary site that derivative is absent. The kernel of this principal-coefficient map is exactly the order-at-most-one space. Its dimension is $1 + 3d$, and the respective principal-coefficient spaces have dimensions five and nine. Surjectivity gives (16). $\square$

The proposition says that the operators of canonical order at most two form the image of the degree-at-most-two part of the enveloping algebra of the sitewise $\mathfrak{sl}_2$'s. This is an algebraic statement about the whole box. Stability will impose precisely the signs of its principal coefficients and, in the complex case, an inward sign on the imaginary first-order fields.

3.2. Root clusters and infinitesimal necessity. The relevant localization uses an isolated root cluster and is valid even when the ambient polynomial degree changes. No full-degree tangent formula is applied to a lower-degree polynomial. The underlying multiple-root contact principle is classical: after rotation of the half-plane, it is contained in the polynomial abscissa tangent theorem of Burke and Overton [21, Theorem 1.2 and Lemma 1.4]. We give a direct moment proof to record the signs and the local factorization needed for the operator argument.

Lemma 3.5 (Lower-half-plane root clusters). *Let $p_t(s)$, $t \geq 0$, be a curve of complex polynomials of uniformly bounded degree whose coefficients are right differentiable at zero. Suppose*

$$p_0(s) = s^m g(s), \quad m \geq 1, \quad g(0) \neq 0,$$

and that p_t has no zero in $\mathbb{H}$ for all sufficiently small t . Let C_t be the monic factor consisting of the m roots converging to zero. Its coefficients are right differentiable, and

$$\left. \frac{d}{dt} C_t(s) \right|_{t=0} = c_1 s^{m-1} + c_2 s^{m-2}, \quad \Im c_1 \geq 0, \quad c_2 \in \mathbb{R}, \quad c_2 \leq 0. \quad (19)$$

When $m = 1$, the c_2 term is absent. If every p_t is real rooted, then c_1 is real as well.

Proof. Choose a small circle about zero containing no other zero of p_0 . Coefficient continuity and Rouché's theorem isolate exactly m roots inside it for small t . If these roots are $\lambda_1(t), \dots, \lambda_m(t)$, their power sums are

$$M_k(t) = \sum_{j=1}^m \lambda_j(t)^k = \frac{1}{2\pi i} \int s^k \frac{p'_t(s)}{p_t(s)} ds, \quad k \geq 1,$$

where the contour is the chosen circle. The integrands are right differentiable with a denominator bounded away from zero. Thus each M_k is right differentiable and $M_k(t) = O(t)$. Newton's identities give the same differentiability for the coefficients of C_t .

Write $\lambda_j = a_j + ib_j$. The roots lie in the closed lower half-plane, so $b_j \leq 0$, and

$$\sum_j |b_j| = -\Im M_1(t) = O(t), \quad \sum_j b_j^2 = O(t^2).$$

Moreover,

$$\sum_j a_j^2 = \Re M_2(t) + \sum_j b_j^2 = O(t).$$

Consequently

$$\Im M_2(t) = 2 \sum_j a_j b_j = O(t^{3/2}) = o(t),$$

and, for $k \geq 3$,

$$|M_k(t)| \leq \sum_j |\lambda_j(t)|^k \leq \left(\sum_j |\lambda_j(t)|^2 \right)^{k/2} = O(t^{k/2}) = o(t).$$

The second inequality uses the ordinary finite counting sum over the roots.

Write $C_t = s^m + \gamma_1(t)s^{m-1} + \dots + \gamma_m(t)$. At $t = 0$, all γ_j and M_j vanish. Differentiating Newton's identities gives $\gamma_j'(0) = -M_j'(0)/j$. Hence every derivative below the coefficient of s^{m-2} vanishes. The first coefficient has nonnegative imaginary part because $\Im M_1(t) \leq 0$. For the second coefficient, $M_2'(0)$ is real, and

$$\Re M_2'(0) = \lim_{t \downarrow 0} \frac{\sum_j a_j(t)^2 - \sum_j b_j(t)^2}{t} \geq 0.$$

This proves (19). Real roots make M_1 real. $\square$

Proposition 3.6 (Order and sign from product contacts). *Suppose $A \in \text{End}_{\mathbb{C}}(V_{\kappa}^{\mathbb{C}})$ generates a complex stability-preserving semigroup. Then A has canonical order at most two, every second-order canonical coefficient is real and nonpositive on $\mathbb{R}^d$, and $\Im P_{e_i} \geq 0$ on $\mathbb{R}^d$. If A is real and its semigroup is only assumed to preserve real stability, the order bound and the nonpositivity of its second-order coefficients still follow.*

Proof. For $0 \neq \alpha \leq \kappa$ and $y \in \mathbb{R}^d$, use the stable product

$$f_{\alpha,y}(z) = \prod_i (z_i - y_i)^{\alpha_i}.$$

In the canonical representation, every derivative at $z = y$ vanishes except for ∂^α , and therefore

$$(Af_{\alpha,y})(y) = \alpha! P_\alpha(y). \quad (20)$$

Choose $v \in \mathbb{R}_{>0}^d$ and restrict the polynomial curve to $z = y + sv$. Its initial value is

$$\left(\prod_i v_i^{\alpha_i} \right) s^{|\alpha|}.$$

Its coefficients are differentiable and its degree is bounded by $|\kappa|$. It has no zero for $s \in \mathbb{H}$, because such an s puts every active coordinate in $\mathbb{H}$. The restricted

polynomial cannot be identically zero, since that would contradict multivariate stability. Invertibility of e^{tA} also excludes the zero multivariate output.

Apply Lemma 3.5. Factoring the restriction as $C_t(s)G_t(s)$, its constant first variation is

$$G_0(0) \left. \frac{d}{dt} C_t(0) \right|_{t=0}.$$

The term involving the derivative of G_t vanishes because $C_0(0) = 0$. Here $G_0(0) = \prod_i v_i^{\alpha_i} > 0$. For $|\alpha| \geq 3$, the constant variation is zero, so (20) gives $P_\alpha(y) = 0$. For $|\alpha| = 2$, it is real and nonpositive, giving the corresponding assertion for $P_\alpha(y)$. For $\alpha = e_i$, its imaginary part is nonnegative.

Since y is arbitrary, polynomial identity proves the vanishing of all coefficients of order greater than two. A complex polynomial real-valued on all of $\mathbb{R}^d$ has real coefficients; this proves reality of the principal coefficients. For a real semigroup, the same inputs and restrictions are real rooted, so the real case follows from the same lemma. Proposition 3.4 then supplies the asserted variable dependence. $\square$

The moment calculation identifies the obstruction to higher differential order: one-sided control of the imaginary root depths, together with the second root moment, eliminates every higher first-order cluster moment. Positivity of coefficients of the input polynomial is not used; every real contact point is available.

3.3. Binary apolar generators and sums of squares. The sufficiency proof starts in two binary variables. For $f = f_{00} + f_{10}s + f_{01}t + f_{11}st \in V_{(1,1)}^{\mathbb{R}}$, put

$$\Delta(f) = f_{10}f_{01} - f_{00}f_{11}.$$

A nonzero real f is stable if and only if $\Delta(f) \geq 0$. Indeed, if $f = a + bs + ct + dst$ and $b + dt \neq 0$, its zero in s is $-(a + ct)/(b + dt)$, whose imaginary part equals $-\Delta(f)\Im t/|b + dt|^2$. The cases in which f is independent of s follow directly.

Lemma 3.7 (Low-degree nonnegative polynomials). *Every nonnegative real univariate polynomial of degree at most four is a sum of squares of real polynomials of degree at most two. Every nonnegative real polynomial of bidegree at most (2, 2) is a sum of squares of real bilinear polynomials.*

Proof. For a nonzero nonnegative univariate polynomial, every real root has even multiplicity and its nonreal roots occur in conjugate pairs. A positive real quadratic is a sum of two squares of linear polynomials. Repeated use of

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$$

gives the first assertion, with square degrees at most two. The zero polynomial presents no exception.

For the second assertion let $R(s, t) \geq 0$ and homogenize it to

$$\tilde{R}(x, y, u) = u^4 R(x/u, y/u).$$

The bidegree bound makes this a ternary quartic. It is nonnegative when $u \neq 0$, and hence everywhere by continuity. Hilbert's theorem on nonnegative ternary quartics writes it as a sum of squares of real quadratic forms; we use the version including all degenerate nonnegative forms [46]. The coefficients of x^4 and y^4 in $\tilde{R}$ are zero. They are sums of the squares of the x^2 and y^2 coefficients of the quadratic forms, respectively. Thus every form is a linear combination of xy, xu, yu, u^2 . Setting $u = 1$ yields bilinear squares. $\square$

Lemma 3.8 (Apolar rank-one primitive). *For a real bilinear polynomial $F = a + bs + ct + dst$, define*

$$\ell_F(f) = df_{00} - cf_{10} - bf_{01} + af_{11}, \quad B_F f = -F \ell_F(f).$$

The full mixed canonical coefficient of B_F is $-F^2$. Moreover

$$B_F^2 = 2\Delta(F)B_F, \quad (21)$$

$$\Delta(f + kB_F f) = \Delta(f) + k(1 + k\Delta(F))\ell_F(f)^2 \quad (k \in \mathbb{R}). \quad (22)$$

For all $t \geq 0$, e^{tB_F} preserves real stability.

Proof. For any endomorphism B of the binary box, its mixed coefficient is

$$P_B(s, t) = B(st) - sB(t) - tB(s) + stB(1). \quad (23)$$

The four evaluations of B_F give $P_{B_F} = -F^2$. Also $\ell_F(F) = 2ad - 2bc = -2\Delta(F)$, which gives (21). The polarization of the quadratic form Δ satisfies $D\Delta_f[F] = -\ell_F(f)$. Expanding $\Delta(f - kF\ell_F(f))$ proves (22).

If $D_F = \Delta(F)$, equation (21) gives

$$e^{tB_F} = \text{Id} + k_t B_F, \quad k_t = \begin{cases} (e^{2tD_F} - 1)/(2D_F), & D_F \neq 0, \\ t, & D_F = 0. \end{cases}$$

For $t \geq 0$, $k_t \geq 0$ and

$$1 + k_t D_F = \frac{1 + e^{2tD_F}}{2} > 0.$$

Thus (22) preserves the inequality $\Delta(f) \geq 0$. The exponential is invertible, so a nonzero input does not become the zero polynomial. These formulas also include $F = 0$ and $D_F = 0$. $\square$

For use below, we make explicit the passage from real to complex preservation in this binary semigroup.

Lemma 3.9 (Orientation at the identity). *Let B be a real endomorphism of $V_{(1,1)}^{\mathbb{R}}$. Then e^{tB} preserves real stability for all $t \geq 0$ if and only if its mixed coefficient P_B is nonpositive on $\mathbb{R}^2$. In this case every e^{tB} is a complete complex stability preserver.*

Proof. For necessity, take $f = (s - x)(t - y)$. It has $\Delta(f) = 0$. Differentiating $\Delta(e^{\tau B} f)$ at $\tau = 0$, and using (23), gives $-P_B(x, y)$. Real preservation implies that this derivative is nonnegative.

Conversely, Lemma 3.7 gives $-P_B = \sum_{\ell} F_{\ell}^2$. The operator $B - \sum_{\ell} B_{F_{\ell}}$ has zero mixed coefficient and hence has canonical order at most one. Proposition 3.4 writes it as a real scalar plus two sitewise operators $J_{i,1}(q_i)$, with real quadratic q_i , without a sign restriction on these two quadratics.

For clarity, the real sitewise flows preserve complex stability in both time directions. If $q(z) = q_0 + q_1 z + q_2 z^2$, let

$$M_{\tau} = \exp \left[\tau \begin{pmatrix} q_1/2 & q_0 \\ -q_2 & -q_1/2 \end{pmatrix} \right] = \begin{pmatrix} a_{\tau} & b_{\tau} \\ c_{\tau} & d_{\tau} \end{pmatrix}.$$

This is a real matrix of determinant one. The polynomial action

$$U_{\tau} f(z) = (c_{\tau} z + d_{\tau})^{\kappa} f \left(\frac{a_{\tau} z + b_{\tau}}{c_{\tau} z + d_{\tau}} \right) \quad (24)$$

has infinitesimal generator $J_\kappa(q)$. The quotient maps $\mathbb{H}$ to itself, since its imaginary part is $\Im z/|c_\tau z + d_\tau|^2$, and the denominator has no zero there. Formula (24) is polynomial after cancellation term by term and holds for every real τ . It also works with inert variables.

Lemma 3.8 and the finite-dimensional Lie product formula now show that $e^{\tau B}$ preserves real stability. Products of preserving factors converge coefficientwise; Hurwitz's theorem and invertibility of the limiting exponential exclude a zero output.

It remains to justify complex preservation of the apolar factors and of the resulting binary semigroup. We use the finite-box classification of Borcea and Brändén [10, Theorems 1.1 and 1.2]. A real stability preserver of rank greater than two has a stable plus symbol or a stable reflected-minus symbol. For the binary box these are

$$G_T^+(s, t; u, v) = T((s + u)(t + v)), \quad G_T^-(s, t; u, v) = T((s - u)(t - v)),$$

where T acts only in s, t . Every $e^{\tau B}$ has rank four. At $\tau = 0$ the minus symbol is $(s - u)(t - v)$, which is not stable: it vanishes when $s = u \in \mathbb{H}$. The cone consisting of stable polynomials and zero is closed in this fixed coefficient space. Continuity in τ therefore excludes the minus alternative for all sufficiently small nonnegative τ . The plus symbol is then stable, and the complex classification gives complex preservation.

More explicitly, adjoining an identity operator on an auxiliary box of capacity λ multiplies the plus symbol by $\prod_j (\xi_j + \eta_j)^{\lambda_j}$. This remains stable. The same complex classification proves preservation on every enlarged box. Finally, divide an arbitrary time into sufficiently small equal pieces and compose. This proves complete preservation for every time and completes the lemma. $\square$

This use of the symbol theorem is limited and precise. The real rank-two exceptional case is not invoked: the binary pair operator has rank four at every time. A single binary site is handled directly by (24).

3.4. Polarization and the proof of the real classification. Let $\mathcal{P}_\kappa$ be normalized polarization into $N = |\kappa|$ binary variables, grouped into κ_i slots at site i :

$$\mathcal{P}_\kappa(z_i^r) = \frac{e_r(z_{i,1}, \dots, z_{i,\kappa_i})}{\binom{\kappa_i}{r}}.$$

Let $\mathcal{D}_\kappa$ identify all slots belonging to each site. The two maps are inverse between $V_\kappa^\mathbb{C}$ and the subspace symmetric within each group of slots. Normalized polarization preserves upper-half-plane stability, and diagonalization preserves it in the reverse direction [10, Proposition 2.4], with the normalization fixed in the preliminaries.

We will need the exact derivative factors

$$\mathcal{D}_\kappa \partial_{i,a} \mathcal{P}_\kappa f = \frac{\partial_i f}{\kappa_i}, \tag{25}$$

$$\mathcal{D}_\kappa \partial_{i,a} \partial_{j,b} \mathcal{P}_\kappa f = \frac{\partial_i \partial_j f}{\kappa_i \kappa_j}, \quad i \neq j, \tag{26}$$

$$\mathcal{D}_\kappa \partial_{i,a} \partial_{i,b} \mathcal{P}_\kappa f = \frac{\partial_i^2 f}{\kappa_i(\kappa_i - 1)}, \quad a \neq b. \tag{27}$$

The last formula is used only for $\kappa_i \geq 2$. They follow by differentiating the elementary symmetric polynomial on a monomial and then diagonalizing.

Proof of Theorem 3.1. The implication from complete complex preservation to real preservation is immediate, since the operator is real. Necessity of the order and sign conditions is Proposition 3.6. We prove sufficiency constructively.

Write $a_i = P_{2e_i}$ and $b_{ij} = P_{e_i+e_j}$. For each $i < j$, apply Lemma 3.7 to obtain

$$-b_{ij}(s, t) = \sum_{\ell} F_{ij,\ell}(s, t)^2$$

with real bilinear $F_{ij,\ell}$. The binary operator $B_{ij} = \sum_{\ell} B_{F_{ij,\ell}}$ has mixed coefficient b_{ij} and generates a complete preserving semigroup. On the full slot space, place a copy of B_{ij} on each pair consisting of one slot at site i and one at site j , and sum these $\kappa_i \kappa_j$ copies. Every summand is a complete generator, so the sum is one by the Lie product formula. It commutes with permutations of slots at either site and therefore preserves the symmetric subspace.

Descend this operator by $\mathcal{D}_{\kappa}$ and $\mathcal{P}_{\kappa}$. Equation (26) shows that its full mixed coefficient is exactly $b_{ij}(z_i, z_j)$. No other second-order coefficient is created. Indeed, after writing each binary summand in normal order, zeroth- and first-order terms remain of those respective orders after diagonalization, and the only second-order derivative involves its two selected slots.

For a site with $\kappa_i \geq 2$, write

$$-a_i(z) = \sum_{\ell} p_{i,\ell}(z)^2, \quad p_{i,\ell}(z) = a_{\ell} z^2 + b_{\ell} z + c_{\ell},$$

using the univariate part of Lemma 3.7. Set

$$F_{i,\ell}(s, t) = a_{\ell} st + \frac{b_{\ell}}{2}(s + t) + c_{\ell}.$$

These forms are symmetric in their two variables and satisfy $F_{i,\ell}(z, z) = p_{i,\ell}(z)$. The symmetric binary operator $2 \sum_{\ell} B_{F_{i,\ell}}$ has mixed coefficient

$$P_i(s, t) = -2 \sum_{\ell} F_{i,\ell}(s, t)^2, \quad P_i(z, z) = 2a_i(z).$$

Symmetry of the operator follows directly from symmetry of both $F_{i,\ell}$ and its apolar functional. Sum copies of this operator over all unordered pairs of distinct slots at site i . It is again a complete generator and commutes with all slot permutations at that site. Its descended same-site principal coefficient, by (27), is

$$\binom{\kappa_i}{2} \frac{P_i(z_i, z_i)}{\kappa_i(\kappa_i - 1)} = a_i(z_i).$$

This is the reason for the factor two in the binary operator. There is no same-site construction when $\kappa_i = 1$.

Let A_{pr} be the sum of all descended generators. For each summand, permutation invariance implies that its exponential on the symmetric subspace is intertwined by polarization with the exponential of the descended operator. Polarization, complete preservation on the full slot space, and diagonalization therefore show that A_{pr} generates a complete preserving semigroup on the original box. It has exactly the same second-order canonical coefficients as A .

The difference $A - A_{\text{pr}}$ has order at most one. By Proposition 3.4, it is a real scalar plus a sum of real sitewise $J_{i,\kappa_i}(q_i)$. Formula (24) proves complete preservation for each of their flows, with no sign restriction on q_i . A final Lie product formula proves complete preservation for e^{tA} . The argument takes place on any fixed enlarged finite

box, and the limiting exponential is invertible there, so Hurwitz's zero alternative never occurs for a nonzero stable input. This proves the theorem. $\square$

3.5. Imaginary inward fields and the complex classification. The remaining complex generators have a particularly explicit one-site realization.

Lemma 3.10 (Imaginary parabolic primitive). *Let $a, b \in \mathbb{R}$ and $\kappa \geq 1$. The generator $iJ_\kappa((az + b)^2)$ has the semigroup*

$$(U_t f)(z) = d_t(z)^\kappa f(\phi_t(z)), \quad d_t(z) = 1 - iat(az + b), \quad \phi_t(z) = \frac{z + ibt(az + b)}{1 - iat(az + b)}. \quad (28)$$

For every $t \geq 0$, this is a complete complex stability preserver.

Proof. The corresponding fractional-linear matrix is

$$M_t = \text{Id} + it \begin{pmatrix} ab & b^2 \\ -a^2 & -ab \end{pmatrix}.$$

The displayed nonidentity matrix has square zero, so $M_{t+s} = M_t M_s$ and $\det M_t = 1$. Denominators in (28) cancel on each monomial of degree at most κ , giving a polynomial operator and an exact semigroup. Differentiating at zero gives

$$\left. \frac{d}{dt} U_t f \right|_{t=0} = i((az + b)^2 f' - \kappa a(az + b)f) = iJ_\kappa((az + b)^2)f.$$

For $z \in \mathbb{H}$ and $t \geq 0$, direct multiplication by the conjugate denominator gives

$$\Im \phi_t(z) = \frac{\Im z + t|az + b|^2}{|1 - iat(az + b)|^2} > 0. \quad (29)$$

When $a \neq 0$ and $t > 0$, the denominator has its only zero at $-b/a - i/(a^2 t)$, below the real axis. When $a = 0$, it is identically one and $\phi_t(z) = z + ib^2 t$. Thus the substitution maps $\mathbb{H}$ to itself and the prefactor does not vanish there. The same argument with all other variables held inert proves complete preservation. The cases $t = 0$ and $a = b = 0$ give the identity. $\square$

Proof of Theorem 3.2. Assume ordinary complex preservation and decompose $A = R + iS$ by entrywise real and imaginary parts in the ordinary monomial basis. Both R and S are real box endomorphisms. The canonical recursion in Lemma 3.3 is real linear, so it commutes with this decomposition.

By Proposition 3.6, A has order at most two and its second-order coefficients are real and nonpositive. The same coefficients are therefore the principal coefficients of R , which satisfies Theorem 3.1. Meanwhile S has order at most one. Proposition 3.4 gives

$$S = c \text{Id} + \sum_i J_{i, \kappa_i}(q_i), \quad c \in \mathbb{R}, \quad q_i \in \mathbb{R}[z_i]_{\leq 2}.$$

Its first-order coefficients are exactly $q_i = \Im P_{e_i}$; the product-contact test makes them nonnegative on $\mathbb{R}$. This proves the necessity of (14).

Conversely, every real quadratic nonnegative on $\mathbb{R}$ is a sum of squares of real linear polynomials. For $q(z) = az^2 + bz + c$ with $a > 0$, complete the square:

$$q(z) = \left(\sqrt{a}z + \frac{b}{2\sqrt{a}} \right)^2 + c - \frac{b^2}{4a}.$$

The last constant is nonnegative. If $a = 0$, nonnegativity forces $b = 0$ and $c \geq 0$. Lemma 3.10 and the Lie product formula therefore give complete preservation for each $iJ(q_i)$. The generator R has the same property by the real theorem, and $ic\text{Id}$ contributes a nonzero scalar phase. One further product formula proves complete preservation for their sum. Invertibility of the finite-dimensional exponential again excludes the zero limit. Complete preservation implies ordinary preservation, finishing the equivalence. $\square$

In particular, every complex second-order coefficient in an admissible generator is forced to be real. The imaginary part has no pair term: it consists entirely of sitewise inward quadratic fields and a scalar. The sign in (28) is fixed by the upper-half-plane convention: $i\partial$ shifts the argument upward and moves polynomial zeros downward.

3.6. Finite certificates, quotient cones, and exact lifts. The construction records both a finite membership certificate and a representation by local binary generators.

Corollary 3.11 (Gram certificates and lineality). *In Theorem 3.1, the principal inequalities are equivalent to the existence of real positive semidefinite matrices G_i of size three and G_{ij} of size four such that*

$$-P_{2e_i}(s) = (1, s, s^2)G_i(1, s, s^2)^\top, \quad \kappa_i \geq 2, \quad (30)$$

$$-P_{e_i+e_j}(s, t) = (1, s, t, st)G_{ij}(1, s, t, st)^\top, \quad i < j. \quad (31)$$

The extra inequalities in Theorem 3.2 have size-two certificates

$$q_i(s) = (1, s)K_i(1, s)^\top, \quad K_i \succeq 0. \quad (32)$$

These matrix sizes do not depend on the numerical capacities.

Let $\mathcal{N}_4, \mathcal{N}_{2,2}, \mathcal{N}_2$ denote, respectively, the cones of nonnegative real univariate quartics of degree at most four, nonnegative real biforms of bidegree at most $(2, 2)$, and nonnegative real univariate quadratics. The real generator cone has lineality

$$\mathfrak{g}_\kappa = \mathbb{R}\text{Id} + \sum_i J_{i, \kappa_i}(\mathbb{R}[z_i]_{\leq 2}),$$

and its quotient by this lineality is naturally isomorphic to

$$\prod_{i: \kappa_i \geq 2} \mathcal{N}_4 \times \prod_{i < j} \mathcal{N}_{2,2}. \quad (33)$$

The complex generator cone, regarded as a real cone, has lineality

$$\mathbb{C}\text{Id} + \sum_i J_{i, \kappa_i}(\mathbb{R}[z_i]_{\leq 2}),$$

and its quotient is the product in (33) times $\prod_i \mathcal{N}_2$.

Proof. The square decompositions in Lemma 3.7 give (30) and (31) by taking sums of outer products of the coefficient vectors. Conversely a real positive semidefinite matrix is a sum of rank-one positive semidefinite matrices, yielding the required square decomposition. The quadratic case is the same argument with the vector $(1, s)$.

The negative-principal-coefficient map has kernel precisely the order-at-most-one space, by Proposition 3.4. The polarization construction realizes every tuple of nonnegative principal polynomials, so it is onto the product in (33). A real A

and its negative are both admissible if and only if all their second-order coefficients vanish. This gives exactly the stated real lineality and quotient.

For complex generators also record the imaginary first-order polynomials q_i . Their nonnegativity is independent of the real principal coefficients, as (14) shows. Both signs of a generator can be admissible only if all these q_i and all second-order coefficients vanish. The remaining imaginary scalar is unrestricted. This proves the complex assertion. $\square$

Corollary 3.12 (Exact binary-slot lift). *For every generator A in Theorem 3.2, there is a complete stability-preserving generator $\tilde{A}$ on the full multiaffine space with $|\kappa|$ variables such that*

$$\tilde{A}\mathcal{P}_\kappa = \mathcal{P}_\kappa A, \quad e^{t\tilde{A}}\mathcal{P}_\kappa = \mathcal{P}_\kappa e^{tA} \quad (t \geq 0). \quad (34)$$

It commutes with every permutation of slots within a site. It can be chosen as a sum of a scalar, one-slot real Möbius generators, one-slot imaginary parabolic generators, and nonnegative multiples of the rank-one apolar generators B_F on pairs of slots.

Proof. For the real second-order part use exactly the slot sums constructed in the proof of Theorem 3.1. They commute with within-site permutations and descend to the chosen principal-coefficient realization. Lift each remaining sitewise drift by

$$J_{i,\kappa_i}(q) \mapsto \sum_{a=1}^{\kappa_i} \left(q(z_{i,a}) \partial_{i,a} - \frac{1}{2} q'(z_{i,a}) \right).$$

Equation (25) shows that its diagonalization on a polarized input is precisely $J_{i,\kappa_i}(q)$. The lifted sum is permutation invariant, so this equality upgrades to intertwining with $\mathcal{P}_\kappa$. The same lift applies to the imaginary quadratics, which are sums of the parabolic primitives of Lemma 3.10. The scalar lifts as the same scalar identity.

All summands generate complete preserving flows on the full slot space, hence so does their sum. This proves the first relation in (34); exponentiation gives the second. $\square$

The lift is a generator intertwining, not merely a static polarization followed by an unrelated preserving map. It is not asserted to have a nonnegative coefficient matrix. Consequently it imposes no coefficient-positive lifting principle on a different preservation problem. Likewise, the signed generation statements proved above are finite dimensional; they contain no infinite-capacity domain assertion.

3.7. A degree-two ancillary test for positive semigroups. The preceding classifications concern preservation of all stable inputs. For positive semigroups, preservation of stable inputs with nonnegative coefficients is a weaker condition. The exact gap can be detected with only two additional polynomial degrees. This assertion also has a mixed-capacity version, using the positive classification proved in Section 8.

Let $\kappa \in (\mathbb{N}_{>0} \cup \{\infty\})^d$, and let $T(t)$ be an existing coefficientwise-positive C_0 -semigroup on $X_\kappa = \ell^1(\Omega_\kappa)$, with generator A . For a finite list of ancillary capacities λ , the inert extension $T(t) \otimes \text{Id}_\lambda$ acts coefficientwise in the original variables. In the next theorem preservation means preservation of the corresponding positive Laguerre-Pólya class; equivalently, it suffices to test the allowed stable polynomial

inputs with nonnegative coefficients. No assertion of semigroup generation is inferred from a formal operator.

Theorem 3.13 (The sharp ancillary test). *The following conditions are equivalent.*

- (1) $T(t) \otimes \text{Id}_{(1,1)}$ preserves positive stability for every $t \geq 0$.
- (2) $T(t) \otimes \text{Id}_{(2)}$ preserves positive stability for every $t \geq 0$.
- (3) Every inert extension by finitely many finite- or infinite-capacity coordinates preserves positive stability for every $t \geq 0$.
- (4) All allowed polynomials belong to $D(A)$, the generator has the mixed-capacity rate form of Theorem 8.18, and its symmetric second-order matrix in (162) satisfies

$$Q_{ij}(y) \leq 0 \quad (y \in \mathbb{R}^d, 1 \leq i, j \leq d). \quad (35)$$

- (5) The same domain and rate form hold, and the rate parameters satisfy

$$\alpha_i = \chi_i = 0, \quad \tau_i \leq 0, \quad a_{ij} = g_{ij} = 0, \quad \theta_{ij}^2 \leq 4c_{ij}c_{ji}. \quad (36)$$

The capacity conventions of Theorem 8.18 remain in force; in particular $\tau_i = 0$ for binary sites and $c_{ij} = 0$ when the destination j has infinite capacity.

The two ancillary degrees are necessary uniformly over capacity vectors: one inert binary coordinate cannot replace the two in condition (1). For finite κ , these conditions are also equivalent to complete preservation of arbitrary complex stable polynomials by $T(t)$.

Proof. First note that $X_{\kappa \sqcup \lambda} = \ell^1(\Omega_\lambda; X_\kappa)$. The inert extension is therefore a positive C_0 -semigroup of norm $\|T(t)\|$. If polynomials belong to $D(A)$, each finite polynomial in the enlarged space belongs to its generator domain, and its generator is $A \otimes \text{Id}$ there. This verifies the analytic hypotheses whenever Theorem 8.18 is applied below. On a preserving side that theorem itself supplies the polynomial domain. Clearly (3) implies (1) and (2).

Suppose (1) holds. Inputs independent of the ancillary variables give ordinary positive preservation. Theorem 8.18 supplies the rate form, and its application on the enlarged space gives the contact test (163) with $\tilde{Q} = Q \oplus 0$. Let u, v denote the ancillary coordinates. For distinct original indices i, j , fix any $y \in \mathbb{R}^d$, and set

$$a = e_i + e_u, \quad b = e_j + e_v, \quad H = \frac{ab^\top + ba^\top}{8}, \quad \tilde{y} = (y, -y_i, -y_j).$$

Here e_i are coordinate vectors and e below is the all-ones vector. Direct calculation gives

$$\begin{aligned} H_{ab} &\geq 0, & e^\top H e &= 1, & H \tilde{y} &= 0, \\ (He)(He)^\top - H &= \frac{(a-b)(a-b)^\top}{16} \succeq 0. \end{aligned} \quad (37)$$

All diagonals required to vanish at binary sites do vanish. Thus this is an admissible contact in (159), and

$$0 \geq \text{tr}(\tilde{Q}(\tilde{y})H) = Q_{ij}(y)/4.$$

For $i = j$ use $a = e_i + e_u$, $b = e_i + e_v$. The same identities hold; the only nonzero original diagonal of H is at site i , so the construction is allowed when $\kappa_i \geq 2$. It again gives $Q_{ii}(y) \leq 0$. For binary sites that coefficient is zero by canonical convention. Hence (1) implies (4).

For (2), again obtain the original rate form first. Fix y and the indices i, j , permitting $i = j$ only at a nonbinary site. Choose $R \geq \max(y_i, y_j)$, and put $c = R - y_i$, $d = R - y_j$. With the single capacity-two ancillary coordinate w , set

$$a = e_i + e_w, \quad b = e_j + e_w, \quad H = (ab^\top + ba^\top)/8, \quad \tilde{y} = (y, -R).$$

The normalization and semidefinite identity in (37) still hold, while

$$H\tilde{y} = -(da + cb)/8 \leq 0, \quad \tilde{y}^\top H\tilde{y} = cd/4 \geq 0.$$

The diagonal in coordinate w is now allowed. The positive stable quadratic is a positive scalar multiple of $(z_i + w + c)(z_j + w + d)$. The contact inequality once more isolates $Q_{ij}(y)/4$, proving (2) $\Rightarrow$ (4).

A univariate polynomial $\alpha s + \tau s^2 + \chi s^3$ is nonpositive on $\mathbb{R}$ exactly when $\alpha = \chi = 0$ and $\tau \leq 0$. For the bivariate entry in (162), simultaneous scaling of its two variables to both signs of infinity kills the cubic homogeneous part; scaling towards zero kills the linear part. The remaining quadratic is nonpositive exactly when

$$\begin{pmatrix} c_{ji} & -\theta_{ij}/2 \\ -\theta_{ij}/2 & c_{ij} \end{pmatrix} \succeq 0.$$

The nonnegativity of the off-diagonal rate parameters makes this condition precisely (36). This proves (4) $\Leftrightarrow$ (5).

Assume (4) and adjoin any finite list of ancillary capacities. The extended rate form is unchanged, with all new rates zero. Every contact matrix H in (159) is entrywise nonnegative, so

$$\text{tr}((Q(y) \oplus 0)H) \leq 0.$$

Theorem 8.18, with the tensor-domain observation above, gives preservation. This proves (4) $\Rightarrow$ (3). In finite dimensions, (4) is exactly the second-order sign condition of Theorem 3.1: the full mixed differential coefficient is $2Q_{ij}$, and the diagonal coefficient is Q_{ii} . That theorem proves the final complete-preservation assertion and its converse.

For sharpness take an active capacity-two coordinate z and $A = (z - z^2)\partial_z^2$. With one inert binary coordinate its contact matrix has the form $H = \begin{pmatrix} h & a \\ a & 0 \end{pmatrix}$, where $h, a \geq 0$. If $h > 0$, the condition $H\tilde{y} \leq 0$ forces $y_z \leq 0$, using the second row when $a > 0$, and the first when $a = 0$. Consequently $\text{tr}(Q(y)H) = h(y_z - y_z^2) \leq 0$; for $h = 0$ it is zero. The finite positive criterion proves one-binary preservation. With two inert binary variables the stable input $(z + u)(z + v)$ evolves to

$$e^{-2t}z^2 + z(u + v) + uv + (1 - e^{-2t})z.$$

Identifying $u = v = w$ and specializing $z = r \in (0, 1)$ gives a real quadratic in w of discriminant $-4(1 - e^{-2t})r(1 - r) < 0$ for $t > 0$. Stability is preserved by both operations, so the evolved polynomial is not stable. This proves the uniform sharpness, without asserting that every fixed capacity vector requires two ancillary degrees. $\square$

The complete positive cone has a particularly small pairwise representation. Apart from scalar and linear Euler terms, its one-site and directed first-order terms are deaths, finite births, infinite immigration, transfers to infinite sites, and the damping operators $-\eta_i E_i(E_i - 1)$, where $E_i = z_i \partial_i$ and $\eta_i \geq 0$. A pair with one finite and one infinite site additionally permits the saturated transfer into the finite

site, $c_{ij}z_j\partial_i(\kappa_j - E_j)$ for $i \in U$, $j \in F$, with $c_{ij} \geq 0$; its pair diagonal parameter is necessarily zero. For two finite sites the remaining terms are exactly

$$c_{ij}z_j\partial_i(\kappa_j - E_j) + c_{ji}z_i\partial_j(\kappa_i - E_i) + \theta_{ij}E_iE_j, \quad \theta_{ij}^2 \leq 4c_{ij}c_{ji}. \quad (38)$$

Indeed expansion recovers (162) and the prescribed first-order rates; all remaining diagonal coefficients are scalar and linear Euler terms. Each pair cone is linearly isomorphic to $\mathbb{S}_+^2$. Spectral decomposition therefore expresses every pair as the sum of at most two rank-one terms, whose parameters have the form $c_{ij} = u^2$, $c_{ji} = v^2$, $\theta_{ij} = \pm 2uv$. This representation concerns positive stability and is distinct from the signed apolar slot lift in Corollary 3.12.

4. SPECTRAL CONTRACTION AND SECTOR GROWTH

We first study arbitrary complex stability-preserving semigroups on a finite polynomial space. No self-adjointness, positivity of coefficients, or differential representation of the generator is assumed. Positive damping of polynomial degree has two consequences: it improves the zero-free domain of the actual evolution, and it gives a sharp contraction estimate in a norm determined by stable polynomials. The last subsection returns to real half-plane stability and assumes coefficient positivity and preservation of homogeneous degree. It compares spectral bounds between sectors without damping.

4.1. Weighted kernels and the stable cone. Zero-capacity coordinates may be omitted. If no coordinate remains, the kernel and eigenvector assertions reduce to nonzero constants. For the gap statements assume that at least one coordinate remains. Fix $d \geq 1$ and $\kappa \in \mathbb{N}_{>0}^d$, write

$$V_\kappa = \{p \in \mathbb{C}[z_1, \dots, z_d] : \deg_{z_i} p \leq \kappa_i\}, \quad b_\alpha = \binom{\kappa}{\alpha},$$

and let $\mathbb{D}_r^d = \{z : |z_i| < r \text{ for all } i\}$. Let $\mathcal{C}_\kappa$ consist of zero and all polynomials in V_κ that do not vanish on $\mathbb{D}_1^d$. We call a polynomial *strictly stable* if it does not vanish on $\overline{\mathbb{D}_1^d}$. For $T \in \text{End}_{\mathbb{C}}(V_\kappa)$, use the kernel of (6), namely

$$K_T(z, w) = T_z \prod_{i=1}^d (1 + z_i w_i)^{\kappa_i} = \sum_{\alpha \leq \kappa} b_\alpha (T z^\alpha)(z) w^\alpha. \quad (39)$$

The corresponding coefficient pairing is bilinear:

$$[f, g]_\kappa = \sum_{\alpha \leq \kappa} b_\alpha^{-1} f_\alpha g_\alpha. \quad (40)$$

In particular, $[K_T(z, \cdot), p]_\kappa = Tp(z)$. There is no complex conjugation in this identity. In the orthonormal coherent basis $\sqrt{b_\alpha} z^\alpha$, the matrix kernel has one square-root binomial weight on each of its two legs.

We shall use the classical disk contraction theorem, Lemma 2.3, in the following form. If two nonzero stable coefficient tensors have matched variables of the same capacities, their contraction by (40) is stable or zero whenever each pair of contracted radii has product at least one. If every such product is greater than one, the contraction is nonzero. The radii of all uncontracted variables are retained. This is the Grace–Asano contraction principle, with the finite-capacity formulation obtained by polarization; see [10, 11, 61]. Indeed, a coefficient of multidegree α is divided by b_α under normalized polarization, and summation over its b_α equal

subsets leaves exactly the reciprocal weight in (40). Applied to two kernels, this pairing gives the kernel of their operator product.

Lemma 4.1. *An invertible operator T preserves $\mathcal{C}_\kappa$ if and only if K_T is zero-free on $\mathbb{D}_1^d \times \mathbb{D}_1^d$. If K_T is zero-free on $\mathbb{D}_r^d \times \mathbb{D}_r^d$, where $r > 1$, then TP is nonzero and zero-free on $\mathbb{D}_r^d$ for every $p \in \mathcal{C}_\kappa \setminus \{0\}$, without an invertibility assumption on T in this last assertion.*

Proof. For fixed $w \in \mathbb{D}_1^d$, the polynomial $\prod_i (1 + z_i w_i)^{\kappa_i}$ is stable and nonzero. Its image under an invertible stability preserver is stable and nonzero. Evaluating at any $z \in \mathbb{D}_1^d$ proves the forward implication. Conversely, contract the input variables of K_T against a stable polynomial. The contraction theorem gives a stable polynomial or zero; invertibility excludes zero on a nonzero input. For the final assertion the contracted radii have product $r > 1$, so the strict contraction theorem itself excludes zero and retains output radius r . $\square$

Lemma 4.2. *The set $\mathcal{C}_\kappa$ is a closed, complex-scalar-invariant cone containing no complex two-dimensional subspace. Its interior consists exactly of the strictly stable polynomials. Every invertible operator preserving this cone maps its interior into its interior.*

Proof. Closedness follows from Hurwitz's theorem on the connected polydisk. Every complex two-dimensional subspace contains a nonzero polynomial annihilated by evaluation at zero, and such a polynomial is not stable. A polynomial nonzero on the closed polydisk remains so under sufficiently small coefficient perturbations, by compactness. Conversely, a boundary zero ζ of a stable polynomial moves into the open polydisk for the arbitrarily small perturbation $p((1 + \varepsilon)z)$, at $z = \zeta/(1 + \varepsilon)$. Zero is not an interior point, since arbitrarily small nonzero multiples of z_1 do not belong to the cone. Finally, an invertible linear operator is open, so its image of the cone interior is an open subset of the cone. $\square$

4.2. Strictification by degree damping. Put $E_i = z_i \partial_i$, $E = \sum_i E_i$, and $P_0 p = p(0)$. The following theorem improves the zero-free domain of the exponential itself, rather than that of an auxiliary splitting operator.

Theorem 4.3 (Degree-damping strictification). *Suppose $A_0 \in \text{End}_{\mathbb{C}}(V_\kappa)$ satisfies $e^{tA_0} \mathcal{C}_\kappa \subseteq \mathcal{C}_\kappa$ for $t \geq 0$. Let $\mu_i > 0$ and*

$$A = A_0 - 2 \sum_{i=1}^d \mu_i E_i.$$

For every $t > 0$, the kernel $K_{e^{tA}}$ is nonzero on the closed unit polydisk in all $2d$ variables. Consequently it has a zero-free radius $r_t > 1$. The same radius works for the image of every nonzero unit-disk-stable input polynomial.

Moreover, for each $t_0 > 0$, one radius $r > 1$ works simultaneously for the kernels $K_{e^{tA}}$, $t \geq t_0$, and for their action on all nonzero unit-disk-stable inputs.

Proof. First consider uniform damping $A = A_0 - 2\mu E$. Fix $t > 0$ and allow μ to vary in the complex half-plane $\Re \mu > 0$. For $s \geq 0$,

$$e^{-2s\mu E} p(z) = p(e^{-2s\mu} z)$$

preserves disk stability. Thus every Lie product

$$\left(e^{tA_0/m} e^{-2t\mu E/m} \right)^m$$

preserves stability and is invertible. The finite-dimensional product formula, followed by Hurwitz and invertibility of the limiting exponential, proves preservation by $T_t(\mu) = e^{t(A_0 - 2\mu E)}$. Lemma 4.1 gives

$$K_{T_t(\mu)}(z, w) \neq 0 \quad \text{if } z, w \in \mathbb{D}_1^d, \quad \Re \mu > 0. \quad (41)$$

Each kernel coefficient is an entire function of μ .

We next establish a real large-damping limit, valid without normality. Let $a_{00} = (A_0 1)(0)$. In the Euclidean norm on monomial coefficients, E is diagonal, nonnegative, and has the constants as its zero eigenspace. Hence, for real $\mu > 0$,

$$\|e^{-2s\mu E}\| \leq 1, \quad e^{-2s\mu E} \longrightarrow P_0 \quad (s > 0).$$

Expand $T_t(\mu)$ in its Dyson series about $-2\mu E$. The term of order $k \geq 1$ is the integral over $0 < s_1 < \dots < s_k < t$ of

$$e^{-2\mu(t-s_k)E} A_0 e^{-2\mu(s_k-s_{k-1})E} \dots A_0 e^{-2\mu s_1 E}.$$

Every time interval is positive almost everywhere. The integrand therefore converges to $a_{00}^k P_0$, and its norm is bounded by $\|A_0\|^k$, independently of μ . Dominated convergence on the simplex gives the limit $t^k a_{00}^k P_0 / k!$. These integrated terms are uniformly bounded by the summable sequence $t^k \|A_0\|^k / k!$. Including the order-zero term and passing the limit through the sum proves

$$T_t(\mu) \longrightarrow e^{ta_{00}} P_0 \quad \text{in operator norm as real } \mu \longrightarrow +\infty. \quad (42)$$

This is a finite-dimensional instance of strong-coupling limits such as [19, Theorem 1]; the displayed argument fixes its normalization directly. Since $K_{P_0} = 1$, the limiting kernel is the nonzero constant $e^{ta_{00}}$.

Fix any $(z, w) \in \overline{\mathbb{D}}_1^d \times \overline{\mathbb{D}}_1^d$. For $0 < \rho < 1$, the functions

$$f_\rho(\mu) = K_{T_t(\mu)}(\rho z, \rho w)$$

are holomorphic and nonvanishing on $\Re \mu > 0$. Their limit as $\rho \uparrow 1$ is locally uniform in that half-plane, because the kernel has finitely many entire coefficient functions. Hurwitz therefore says that $K_{T_t(\mu)}(z, w)$, as a function of μ , is identically zero or nowhere zero. The limit (42) excludes the first alternative. This proves closed-polydisk nonvanishing at the desired real $\mu > 0$. Compactness then supplies a radius $r_t > 1$. Lemma 4.1 gives the common input-output assertion.

For nonuniform rates, let $\mu = \min_i \mu_i > 0$. The operator $A_0 - 2 \sum_i (\mu_i - \mu) E_i$ generates a stable semigroup by the same product-formula argument. Apply the uniform result to this operator and the remaining damping $-2\mu E$.

Finally fix $t_0 > 0$, and choose $r > 1$ for $B = e^{t_0 A/2}$. For every $t \geq t_0$,

$$e^{tA} = B e^{(t-t_0)A} B.$$

The middle kernel has radius one, including at $t = t_0$, and the outer kernels have radius r . The two kernel contractions pair radii whose products are $r > 1$, retaining radius r on both external legs. A further contraction with any stable input proves the uniform input assertion. $\square$

Remark 4.4. The radius can depend on the capacity, generator, and initial positive time. The theorem gives neither a dimension-independent lower bound for $r - 1$ nor an improvement on inert ancillary variables. Complex scalar shifts of the generator do not affect its conclusions. No normalization by a trace is used.

4.3. Dominant eigenvectors and their dual certificates. We use the following finite-dimensional complex-cone theorem of Rugh [50, Theorem 8.4]: if a nontrivial closed complex-scalar-invariant cone contains no complex plane and a linear map sends its nonzero part into its interior, then the map has an algebraically simple nonzero eigenvalue of strictly largest modulus. Convexity of the cone and self-adjointness of the map are not hypotheses.

Proposition 4.5. *Under the hypotheses of Theorem 4.3, A has an algebraically simple eigenvalue a_* uniquely largest in real part. Choose its right eigenvector v and left eigenfunctional ℓ so that $\ell(v) = 1$. Both v and*

$$Q_\ell(w) = \sum_{\alpha \leq \kappa} b_\alpha \ell(z^\alpha) w^\alpha \quad (43)$$

are zero-free on a common polydisk of radius greater than one. Furthermore $\ell(p) \neq 0$ for every nonzero stable p , and $e^{-ta_} e^{tA} p \rightarrow \ell(p)v$.*

Proof. For each fixed $t > 0$, Theorem 4.3 and Lemma 4.2 verify all the hypotheses of Rugh's theorem for $T = e^{tA}$; the cone contains all constant polynomials and is therefore nontrivial. Its simple dominant eigenspace is invariant under A , because A commutes with T . Thus it is an A -eigenline, say for a_* . Spectral mapping gives strict real-part dominance: another generator eigenvalue with equal real part would give the same modulus for the exponential. If two distinct generator eigenvalues exponentiated to the same value, that value would have algebraic multiplicity at least two. A nontrivial Jordan block at a_* would also give a nontrivial exponential Jordan block, since $te^{ta_*} \neq 0$. Both alternatives contradict algebraic simplicity for T .

Fix a kernel radius $r > 1$ for T , and put $\Lambda = e^{ta_*}$. Finite-dimensional spectral theory gives

$$\Lambda^{-m} T^m \longrightarrow P = v \otimes \ell, \quad \ell(v) = 1.$$

Every power has kernel radius r , because its internal contraction radii have product $r^2 > 1$. The kernel transform is injective, so K_P is not the zero polynomial. Hurwitz yields

$$K_P(z, w) = v(z) Q_\ell(w) \neq 0 \quad (z, w \in \mathbb{D}_r^d).$$

Both factors consequently have radius r . The strict weighted contraction of Q_ℓ with a nonzero unit-stable polynomial is $\ell(p)$, so it is nonzero. The strict real-part separation and the finite-dimensional spectral expansion then give the asserted convergence, including when the remaining spectrum has Jordan blocks. $\square$

4.4. Sharp spectral bounds. We first record the scalar estimate used in both the spectral and dynamical arguments. It is the diameter form of the Schwarz lemma used in [18, Theorem 2].

Lemma 4.6. *If g is holomorphic on $\mathbb{D}_1^d$, continuous on its closure, and $0 \leq q \leq 1$, then*

$$\text{diam } g(q\overline{\mathbb{D}_1^d}) \leq q \text{diam } g(\overline{\mathbb{D}_1^d}).$$

The same conclusion holds for a coordinatewise dilation all of whose moduli are at most q .

Proof. The cases $q = 1$, zero diameter, and $q = 0$ are immediate. For $x, y \in q\overline{\mathbb{D}}_1^d$, put $\rho = \max(\|x\|_\infty, \|y\|_\infty)$. If $\rho > 0$,

$$f(\zeta) = \frac{g(\zeta x/\rho) - g(\zeta y/\rho)}{\text{diam } g(\overline{\mathbb{D}}_1^d)}$$

is holomorphic on the unit disk, bounded in modulus by one, and vanishes at zero. Schwarz at $\zeta = \rho < 1$ bounds $|g(x) - g(y)|$ by ρ times the denominator. Take the supremum over x, y . A coordinatewise dilation has its image in $q\overline{\mathbb{D}}_1^d$. $\square$

Theorem 4.7 (Complex radius bound). *Suppose $r > 1$ and K_T is zero-free on $\mathbb{D}_r^d \times \mathbb{D}_r^d$. Then T has an algebraically simple nonzero eigenvalue $\Lambda_\star$ of strictly largest modulus, with an eigenvector of radius r . Every other eigenvalue satisfies*

$$|\Lambda/\Lambda_\star| \leq r^{-2}. \quad (44)$$

No invertibility, Hermiticity, or diagonalizability is assumed.

Proof. The kernel maps the nonzero stable cone into its interior by Lemma 4.1. Rugh's theorem therefore supplies the simple dominant eigenvalue. Its rank-one spectral projection is nonzero on some interior cone vector, since a proper linear hyperplane cannot contain an open set. Normalized iterates of that vector converge to a nonzero eigenvector ψ in the closed cone. Applying T once improves ψ to radius r .

Let ϕ be an eigenvector of a different nonzero eigenvalue, and put $g = \phi/\psi$, $\alpha = \Lambda/\Lambda_\star$. The function g is nonconstant and holomorphic on $\mathbb{D}_r^d$. For $1/r < s < r$, we claim

$$\alpha g(\mathbb{D}_r^d) \subseteq g(\mathbb{D}_s^d).$$

Otherwise, for some $v \in \mathbb{D}_r^d$, the nonzero polynomial $f = \phi - \alpha g(v)\psi$ would be stable on $\mathbb{D}_s^d$, but $Tf = \Lambda(\phi - g(v)\psi)$ would vanish at v . This contradicts strict contraction, since $rs > 1$.

For $1/r < s < t < r$, let $\Delta_u = \text{diam } g(\overline{\mathbb{D}}_u^d)$. These diameters are finite and positive. The inclusion gives $|\alpha|\Delta_t \leq \Delta_s$. Applying Lemma 4.6 after rescaling radius t gives $\Delta_s \leq (s/t)\Delta_t$. Letting $s \downarrow 1/r$ and $t \uparrow r$ proves (44). Zero eigenvalues satisfy the inequality trivially. $\square$

The quantitative mechanism in this proof is the ratio-squeezing argument of [18]; the complex-cone theorem supplies the eigenvector without a positive-semidefinite hypothesis.

Theorem 4.8 (Sharp degree-damping gap). *Under the hypotheses of Theorem 4.3, set $\mu = \min_i \mu_i$. The dominant generator eigenvalue satisfies*

$$\Re a_\star - \Re a \geq 2\mu \quad (a \in \text{spec}(A), a \neq a_\star). \quad (45)$$

The constant is sharp for every positive finite capacity vector.

Proof. For small $h > 0$, put

$$D_h = e^{-h \sum_i \mu_i E_i}, \quad T_h = D_h e^{h A_0} D_h.$$

The kernel of $e^{h A_0}$ has radius one by Lemma 4.1. If $q_i = e^{-h \mu_i}$, direct coefficient calculation gives

$$K_{T_h}(z, w) = K_{e^{h A_0}}(q_1 z_1, \dots, q_d z_d, q_1 w_1, \dots, q_d w_d).$$

It therefore has radius $e^{h\mu}$. Theorem 4.7 gives a simple dominant eigenvalue $\Lambda_\star(h)$ and bounds every other eigenvalue by $e^{-2h\mu}|\Lambda_\star(h)|$.

Now $B_h = (T_h - I)/h \rightarrow A$ in norm. Its eigenvalues are exactly $b(h) = (\Lambda(h) - 1)/h$, with algebraic multiplicities. These multisets are bounded and converge to the spectrum of A . Uniformly for bounded complex b ,

$$h^{-1} \log |1 + hb| = \Re b + O(h).$$

The modulus inequality consequently implies $\Re b_\star(h) - \Re b(h) \geq 2\mu + O(h)$. Choose a sequence along which $b_\star(h)$ converges. After removing this one eigenvalue from each multiset, the remaining eigenvalues converge, with multiplicity, to the remaining copies of the spectrum of A . The limiting inequality separates all those copies by 2μ . In particular, the distinguished limit is algebraically simple and is the unique maximal-real-part eigenvalue, hence is $a_\star$. This proves (45) without differentiable eigenvalue labels or a normal-matrix perturbation bound. For $A_0 = 0$, the eigenvalues are $-2 \sum_i \mu_i \alpha_i$; a degree-one monomial at a site of minimum rate attains equality. $\square$

4.5. Contraction along the actual evolution. For strictly stable p and arbitrary $h \in V_\kappa$, define

$$\text{Osc}_p(h) = \text{diam}\{h(z)/p(z) : z \in \overline{\mathbb{D}_1^d}\}. \quad (46)$$

It is a complex seminorm with kernel $\mathbb{C}p$: homogeneity and the triangle inequality follow from the diameter definition, and zero diameter means that the rational function is constant.

Lemma 4.9. *If T is an invertible stability preserver, p is strictly stable, and $h \in V_\kappa$, then Th is strictly stable and*

$$(Th/Thp)(\mathbb{D}_1^d) \subseteq (h/p)(\mathbb{D}_1^d).$$

In particular $\text{Osc}_{Thp}(Th) \leq \text{Osc}_p(h)$.

Proof. Strictness follows from Lemma 4.2. If w does not belong to the input ratio image, then $h - wp$ is a nonzero stable polynomial. Its image $Th - wThp$ is nonzero and stable, excluding w from the output ratio image. Taking diameters proves the assertion; the open and closed polydisks give the same diameter by continuity and density. $\square$

Theorem 4.10 (Moving-denominator contraction). *Let A_0 generate a stability-preserving semigroup, and let $A = A_0 - 2 \sum_i \mu_i E_i$, with $\mu_i \geq \mu \geq 0$. For every strictly stable p , every $h \in V_\kappa$, and $t \geq 0$,*

$$\text{Osc}_{e^{tA}p}(e^{tA}h) \leq e^{-2\mu t} \text{Osc}_p(h). \quad (47)$$

The constant is sharp.

Proof. The damping operator $D_\tau = e^{-2\tau \sum_i \mu_i E_i}$ acts by a coordinatewise dilation of maximum modulus $e^{-2\mu\tau}$. Lemma 4.6, applied to h/p , gives the corresponding oscillation contraction. Each $e^{\tau A_0}$ is nonexpansive by Lemma 4.9. Iterating along $(e^{tA_0/m} D_{t/m})^m$ therefore gives exactly the factor $e^{-2\mu t}$, with all intermediate denominators strictly stable.

The products converge to e^{tA} , which is invertible and preserves stability by the product formula and Hurwitz. Its image of p is strictly stable. The minimum modulus of that limiting polynomial on the closed polydisk is positive, so coefficient

convergence gives uniform convergence of the ratios there. Diameters are continuous under uniform convergence, proving (47). Equality holds for pure damping with $p = 1$ and $h = z_j$ at a minimum-rate site. $\square$

Corollary 4.11 (Nonautonomous evolution). *On each compact time interval, suppose $A_0(t)$ and $\mu_i(t)$ are bounded and piecewise continuous, with finitely many continuity pieces, each $A_0(t)$ a stability generator and $\mu_i(t) \geq 0$. Let $U(t, s)$ solve the linear evolution equation with generator $A_0(t) - 2 \sum_i \mu_i(t) E_i$. Then, for strictly stable p ,*

$$\text{Osc}_{U(t,s)p}(U(t, s)h) \leq \exp\left(-2 \int_s^t \min_i \mu_i(u) du\right) \text{Osc}_p(h).$$

Proof. Approximate on each continuity piece by step functions taking actual values of the coefficients. They converge in L^1 . Their frozen evolutions satisfy Theorem 4.10, including when the minimum rate is zero. Multiplying the bounds gives the exponential of the integral of the stepwise minimum rates. These integrals converge to the stated integral, since the minimum of finitely many real coordinates is Lipschitz. The integral-equation estimate for finite matrix ODEs gives uniform convergence of the propagators. The limiting fundamental matrix is invertible and preserves stability by Hurwitz; Lemma 4.2 gives a strictly stable limiting denominator. Uniform convergence of the ratios passes the estimate to the limit, as in the autonomous proof. $\square$

Corollary 4.12 (Adapted norm and resolvent). *Assume positive damping, and normalize v, ℓ as in Proposition 4.5. On $W = \ker \ell$, set $\|h\|_v = \text{Osc}_v(h)$ and $B = (A - a_\star I)|_W$. Then $\|\cdot\|_v$ is a norm and*

$$\|e^{tB}\|_{v \rightarrow v} \leq e^{-2\mu t}, \quad \|(\zeta I - B)^{-1}\|_{v \rightarrow v} \leq \frac{1}{\Re \zeta + 2\mu} \quad (\Re \zeta > -2\mu).$$

Every eigenvalue of B with real part -2μ is semisimple.

Proof. The only possible null vectors of the seminorm are multiples of v , and $\ell(v) = 1$. Thus it is a norm on W . The left eigenfunctional identity makes W invariant. Since $e^{tA}v = e^{ta_\star}v$, Theorem 4.10 gives the first estimate, including the complex phase of the scalar. For $\Re \zeta > -2\mu$, the absolutely convergent integral $\int_0^\infty e^{-t\zeta} e^{tB} dt$ is the inverse of $\zeta I - B$, by integration of its derivative. Integrating the norm estimate gives the stated bound. A nontrivial Jordan block at a boundary eigenvalue would produce unbounded polynomial growth after multiplication by $e^{2\mu t}$, contradicting the first estimate and norm equivalence on its finite-dimensional generalized eigenspace. $\square$

Example 4.13 (No uniform strict radius). Fix $\mu, t > 0$. On V_1 , in the coefficient basis $(1, z)$, let X and Z be the usual Pauli matrices and put

$$A_{0,R} = RX, \quad A_R = A_{0,R} - 2\mu E = \begin{pmatrix} 0 & R \\ R & -2\mu \end{pmatrix}, \quad R > 0.$$

The undamped flow preserves disk stability, also with arbitrary inert variables. Indeed, writing $c = \cosh(Rs)$, $b = \sinh(Rs)$, its action is the explicit weighted substitution

$$(e^{sA_{0,R}p})(z) = (c + bz)p\left(\frac{b + cz}{c + bz}\right).$$

The prefactor is nonzero on the disk and the fraction is a disk automorphism, since $c^2 - b^2 = 1$. Put $\omega_R = \sqrt{R^2 + \mu^2}$. The identity $(A_R + \mu I)^2 = \omega_R^2 I$ gives

$$e^{tA_R} = e^{-\mu t} \left[\cosh(\omega_R t) I + \frac{\sinh(\omega_R t)}{\omega_R} (A_R + \mu I) \right].$$

Thus the image of the fixed input 1 has its unique zero at

$$z_R = -\frac{\omega_R \coth(\omega_R t) + \mu}{R}.$$

For every finite R , $|z_R| > 1$, but $|z_R| \rightarrow 1$ as $R \rightarrow \infty$. Since $K_{e^{tA_R}}(z, 0) = e^{tA_R} 1(z)$, the same zero obstructs a common kernel radius depending only on μ, t . This failure already occurs for Hermitian generators on one binary variable. The dominant eigenvector has its zero at modulus $(\omega_R + \mu)/R$, which also tends to one.

Example 4.14 (Unbounded mixing remainder at a fixed gap). Fix $\mu > 0$, and on V_1 put

$$G_R = R(X + iZ), \quad B_R = G_R - 2\mu E = \begin{pmatrix} iR & R \\ R & -iR - 2\mu \end{pmatrix}, \quad R > 0.$$

Here $G_R^2 = 0$ and $G_R^* Z + Z G_R = 0$. Its stability-preserving action is explicit: with $a = 1 + iRs$ and $b = Rs$,

$$(e^{sG_R} p)(z) = (a + bz)p\left(\frac{b + \bar{a}z}{a + bz}\right), \quad |a + bz|^2 - |b + \bar{a}z|^2 = 1 - |z|^2.$$

The denominator is nonzero in the disk, and the fraction maps the disk to itself. The formula also proves preservation with inert variables.

Choose the square root $\omega_R = \sqrt{\mu^2 + 2i\mu R}$ with $x_R = \Re \omega_R > 0$. The two eigenvalues of B_R are $a_{\pm} = -\mu \pm \omega_R$, and

$$x_R = \sqrt{\frac{\mu \sqrt{\mu^2 + 4R^2} + \mu^2}{2}} \geq \mu.$$

The dominant spectral projection and normalized exponential are

$$P_R = \frac{1}{2} \left(I + \frac{B_R + \mu I}{\omega_R} \right), \quad S_R(t) = e^{-ta_+} e^{tB_R} = P_R + e^{-2\omega_R t} (I - P_R).$$

Their off-diagonal entries give

$$\|P_R\|_2 \geq \frac{R}{2|\omega_R|}, \quad |[S_R(t)]_{10}| = \frac{R}{2|\omega_R|} |1 - e^{-2\omega_R t}| \geq \frac{R}{2|\omega_R|} (1 - e^{-2x_R t}).$$

Since $|\omega_R| = (\mu^4 + 4\mu^2 R^2)^{1/4}$, both quantities diverge as $R \rightarrow \infty$, the second for every fixed $t > 0$.

To keep the actual spectral gap fixed, pass to $V_{(1,1)}$ and set

$$\mathcal{A}_R = G_R \otimes I - 2\mu(E_1 + E_2) = B_R \otimes I + I \otimes (-2\mu E).$$

The undamped part is still a joint stability generator by the displayed weighted substitution. Its four eigenvalues are $a_+, a_-, a_+ - 2\mu, a_- - 2\mu$. Hence its dominant eigenvalue is a_+ , its real-part gap is exactly 2μ , and its dominant projector is $\mathcal{P}_R = P_R \otimes P_0$. For the fixed strictly stable input $p(z_1, z_2) = 1 + z_2/2$, define $\mathcal{R}_R(t) = e^{-ta_+} e^{t\mathcal{A}_R} - \mathcal{P}_R$. The coefficient of $z_1 z_2$ is exactly

$$[z_1 z_2] \mathcal{R}_R(t) p = \frac{1}{2} e^{-2\mu t} \frac{R}{2\omega_R} (1 - e^{-2\omega_R t}).$$

The projector term contains no z_2 , which proves this identity directly from the tensor exponential. Its modulus tends to infinity for every fixed $\mu, t > 0$. Thus even after subtraction of the dominant projector there is no coefficient-norm mixing bound depending only on the damping, time, and exact spectral gap, in a fixed two-variable box and on one fixed strictly stable input. Equivalence of fixed finite-dimensional norms gives the same obstruction for every fixed coefficient norm. This is consistent with Corollary 4.12, whose norm depends on the dominant polynomial. The example uses linear normalization by e^{ta+} ; it makes no claim about nonlinear normalization by an evolving coefficient.

Remark 4.15. For the stable cone, the forbidden-pencil projective distance of [24] is $\text{osc}_{\mathbb{D}_1^d} \log |q/p|$. Its real infinitesimal form is $\text{osc } \Re(h/p)$; maximizing over a scalar phase of h gives (46). Thus the metric and the general complex-cone contraction principle are inherited. The estimates above specify the exact degree-damping rate for arbitrary tangent vectors along the actual, possibly nonautonomous evolution.

4.6. Sector growth and token-graph spectral concavity. We now return to real upper-half-plane stability. There is a second spectral consequence when coefficient positivity and total degree are preserved. It does not require degree damping or self-adjointness. A single stable evolution compares the exponential growth rates of all homogeneous sectors.

Theorem 4.16 (Concavity of sector growth). *Let V_κ be a finite real coordinate box and $N = \sum_i \kappa_i$. Suppose that A has nonnegative off-diagonal entries in the monomial basis, preserves every homogeneous subspace $V_{\kappa,k}$, and that e^{tA} preserves nonnegative real stability for every $t \geq 0$. Write*

$$s_k = \max\{\Re \lambda : \lambda \in \text{spec}(A|_{V_{\kappa,k}})\}, \quad 0 \leq k \leq N.$$

Then

$$2s_k \geq s_{k-1} + s_{k+1} \quad (1 \leq k < N).$$

Neither irreducibility nor symmetry of A is assumed.

Proof. Start at the stable polynomial $f_0(z) = \prod_i (1 + z_i)^{\kappa_i}$. Every allowed monomial has a strictly positive coefficient. Choose α such that $A + \alpha I$ is entrywise nonnegative. Then $e^{tA} = e^{-\alpha t} e^{t(A+\alpha I)}$ is entrywise nonnegative and has a strictly positive diagonal. Consequently $f_t = e^{tA} f_0$ still has every allowed coefficient strictly positive. Its diagonal restriction is a degree- N polynomial

$$P_t(x) = f_t(x, \dots, x) = \sum_{k=0}^N c_k(t) x^k, \quad c_k(t) > 0,$$

whose roots are all strictly negative.

Newton's inequalities give

$$\left(\frac{c_k(t)}{\binom{N}{k}} \right)^2 \geq \frac{c_{k-1}(t)}{\binom{N}{k-1}} \frac{c_{k+1}(t)}{\binom{N}{k+1}}. \quad (48)$$

Here is the particular form needed. The derivative $R = P_t^{(k-1)}$ has $m = N - k + 1$ negative roots, so $R(x) = R(0) \prod_{j=1}^m (1 + u_j x)$ with $u_j > 0$. Cauchy's inequality gives

$$\sum_{i < j} u_i u_j \leq \frac{m-1}{2m} \left(\sum_i u_i \right)^2.$$

The first three coefficients of R are $(k-1)!c_{k-1}$, $k!c_k$, and $(k+1)!c_{k+1}/2$. Substitution gives (48), including its binomial normalization.

Let $A_k = A|_{V_{\kappa,k}}$, and let v_k be the coefficient vector of the degree- k part of f_0 . Degree preservation gives

$$c_k(t) = \mathbf{1}^\top e^{tA_k} v_k.$$

Since every entry of v_k is positive, this quantity is bounded above and below by fixed positive multiples of the sum of all entries of e^{tA_k} . For an entrywise nonnegative m_k -by- m_k matrix M , that sum lies between $\|M\|_1$ and $m_k\|M\|_1$, where $\|\cdot\|_1$ is the induced maximum column-sum norm. Spectral mapping gives $\|e^{tA_k}\|_1 \geq e^{ts_k}$, and Jordan normal form gives

$$\|e^{tA_k}\|_1 \leq C_k(1 + t^{m_k-1})e^{ts_k} \quad (t \geq 0)$$

for a fixed constant C_k . Therefore

$$\lim_{t \rightarrow \infty} t^{-1} \log c_k(t) = s_k.$$

Take logarithms in (48), divide by t , and let $t \rightarrow \infty$. The fixed binomial factors disappear, proving the assertion. For $N \leq 1$ there is no inequality to check. $\square$

The proof uses coefficient sums, not $\text{Tr}(e^{tA_k})$. No real-rooted generating function for the latter traces is asserted. The passage from coefficient log-concavity to concavity of growth exponents is the one-variable tropical-stability principle [15, Theorem 5 and the discussion following it]. The proof above establishes the needed exponential version directly, including nonsymmetric blocks and their Jordan factors. The preservation theorem supplies the input needed to apply it.

Let G be a finite simple graph on $[n]$, with nonnegative edge weights w_{ij} . Its k -token graph has the k -subsets as vertices. The edge from S to $S - \{i\} + \{j\}$ has weight w_{ij} whenever $i \in S$, $j \notin S$, and ij is an edge of G . Denote its adjacency and weighted degree matrices by $A_k(G)$ and $D_k(G)$. The sectors $k = 0, n$ are singletons with zero matrices.

Theorem 4.17 (Token-graph spectral concavity). *For every $a \in [-1, 1]$, put*

$$\Lambda_k(a) = \lambda_{\max}(A_k(G) + aD_k(G)).$$

For every nonnegatively weighted G ,

$$2\Lambda_k(a) \geq \Lambda_{k-1}(a) + \Lambda_{k+1}(a) \quad (1 \leq k < n), \quad \Lambda_k(a) = \Lambda_{n-k}(a).$$

In particular the sequence is nondecreasing up to half filling. The interval $[-1, 1]$ is the exact interval of real parameters for which this concavity holds for every such graph.

Proof. On the binary polynomial box define, for every unordered edge,

$$\mathcal{B}_{ij} = w_{ij}[(z_j + az_i)\partial_i + (z_i + az_j)\partial_j - (z_i^2 + 2az_iz_j + z_j^2)\partial_i\partial_j].$$

It annihilates 1, z_iz_j and sends z_i to $w_{ij}(az_i + z_j)$ and z_j to $w_{ij}(z_i + az_j)$. Thus $\mathcal{B} = \sum_{ij} \mathcal{B}_{ij}$ preserves total degree, is a real symmetric Metzler matrix, and restricts on the degree- k sector to $A_k(G) + aD_k(G)$. Its mixed second-order coefficient is

$$-w_{ij}[(s + at)^2 + (1 - a^2)t^2] \leq 0.$$

Theorem 3.1 therefore gives stability preservation. Theorem 4.16 yields concavity, since for a real symmetric matrix its spectral bound is its largest eigenvalue.

Complementation of subsets gives a weight-preserving isomorphism between the k - and $(n-k)$ -token graphs, proving symmetry. Set $\delta_k = \Lambda_k - \Lambda_{k-1}$. Concavity says that the δ_k are nonincreasing, while symmetry gives $\delta_{n-k+1} = -\delta_k$. If $k \leq (n+1)/2$, it follows that $\delta_k \geq -\delta_k$, proving monotonicity.

For sharpness, take the unweighted star $K_{1,3}$. Its two-token graph is a six-cycle. On the equal-leaf subspace, the one-token matrix is $\begin{pmatrix} 3a & \sqrt{3} \\ \sqrt{3} & a \end{pmatrix}$; its orthogonal complement has eigenvalue a . Thus

$$\Lambda_1(a) = 2a + \sqrt{a^2 + 3}, \quad \Lambda_2(a) = 2a + 2.$$

For $a > 1$, $\Lambda_1(a) > \Lambda_2(a)$; symmetry then contradicts concavity at $k = 2$. For $a < -1$, a single edge has $\Lambda_0 = \Lambda_2 = 0$ and $\Lambda_1 = a + 1 < 0$, again violating concavity. At $a = 1$, the star has $\Lambda_1 = \Lambda_2 = 4$, so connectedness does not imply strict monotonicity. $\square$

For completeness, the local preservation used here has a short verification independent of the generator classification. For $\tau \geq 0$, set $b = e^{a\tau} \sinh \tau$ and $c = e^{a\tau} \cosh \tau$. The unweighted edge exponential fixes $1, xy$ and sends x, y to $cx + by, bx + cy$. Its algebraic symbol is

$$xy + uv + b(xu + yv) + c(xv + yu).$$

Writing this symbol as $\zeta^T M \zeta / 2$ in the order $\zeta = (x, y, u, v)^T$, one has

$$M = \begin{pmatrix} 0 & 1 & b & c \\ 1 & 0 & c & b \\ b & c & 0 & 1 \\ c & b & 1 & 0 \end{pmatrix}.$$

Its eigenvalues are

$$1 + e^{(a+1)\tau}, \quad 1 - e^{(a+1)\tau}, \quad -1 - e^{(a-1)\tau}, \quad -1 + e^{(a-1)\tau}.$$

Exactly one is positive when $-1 \leq a \leq 1$. The quadratic form is positive on every strictly positive real vector Y . A zero at $X + iY$ would imply $X^T M Y = 0$ and $X^T M X = Y^T M Y > 0$, contradicting the one-positive-eigenvalue property. Hence the symbol is stable. The classical symbol theorem [10] gives complete preservation for each edge. The finite-dimensional product formula and coefficientwise closure give preservation for their sum; invertibility of the limiting exponential excludes zero outputs.

Taking $a = 1$ and $a = 0$ gives, respectively, monotonicity of the largest eigenvalues of the signless Laplacian and the adjacency matrix, as conjectured in [4, Conjectures 5–6]. Theorem 4.17 proves the stronger weighted concavity throughout the displayed interpolation interval. The adjacency-preserving generator at $a = 0$ is already contained in Purbhoo's construction [47, Proposition 4.5]. The contribution here is the cross-sector spectral conclusion. The bounds in Conjectures 1–4 of [4] were settled by [8] and are not claimed here.

In the occupation basis, the same block matrices arise from

$$\mathcal{B}_a(G) = \frac{1}{2} \sum_{ij} w_{ij} [X_i X_j + Y_i Y_j + a(I - Z_i Z_j)].$$

Thus its maximum energy over fixed particle number is concave in that number. The negative Hamiltonian has the corresponding convex sector ground-state energies. At $a = 1$, the common one-qubit rotation $R = e^{-i\pi X/4}$, with $RYR^* = Z$ and

$RZR^* = -Y$, identifies this operator with the EPR Hamiltonian in [4]. There is a maximum-energy state in a half-filled sector in the rotated basis. This locates a maximizing sector; it does not give an efficient algorithm within that exponentially large sector. For $a < 0$ the quantity in Theorem 4.17 is the largest eigenvalue, not in general the spectral radius.

5. CLASSICAL COMPUTATION OF PRINCIPAL EIGENVALUES AND STATES

The spectral theorem of the preceding section has an algorithmic consequence for operators with a bounded-degree local description. A positive damping rate selects a holomorphic eigenvalue throughout a complex half-plane. This domain can be joined to a disk on which perturbation coefficients are efficiently computable. An explicit change of variable then reduces evaluation at any fixed positive damping rate to logarithmically many coefficients.

The coefficient algorithm is due to Bravyi, DiVincenzo, and Loss [16]. Our use of it includes a complexification argument, so that every invocation of their algorithm remains within its Hermitian hypotheses. Neither an extension of Hermitian eigenvalue ordering to nonnormal matrices nor numerical continuation of eigenvectors is needed. We first estimate the principal eigenvalue. Starting in Section 5.7, we then compute stable product tests of its eigenvectors, including relative ground-state amplitudes, product-event probabilities, and adaptive equatorial measurement records. These scalar queries do not require a bound on the condition number of the spectral projection.

5.1. Local input and the approximation theorem. Identify the coefficient space

$$\mathcal{P}_n = \{p \in \mathbb{C}[z_1, \dots, z_n] : \deg_{z_i} p \leq 1\} \cong (\mathbb{C}^2)^{\otimes n}, \quad z^\alpha \longleftrightarrow |\alpha\rangle.$$

We use the Euclidean norm in this coefficient basis and the induced operator norm. Let $\mathcal{S}_n$ be the set of nonzero polynomials in $\mathcal{P}_n$ that do not vanish in $\mathbb{D}^n$, where $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$. The degree operator is

$$E = \sum_{i=1}^n |1\rangle\langle 1|_i = \sum_{i=1}^n z_i \partial_{z_i}. \quad (49)$$

All local matrices below are supplied in the displayed tensor-product basis. A Gaussian rational means an element of $\mathbb{Q}(i)$.

Theorem 5.1 (Principal-eigenvalue approximation). *Fix $d \in \mathbb{N} \cup \{0\}$ and positive rational numbers J, μ . For $n \geq 1$, suppose that the input specifies*

$$A_0 = \sum_{i=1}^n K_i + \sum_{\{i,j\} \in \mathcal{E}} K_{ij}, \quad (50)$$

where the graph $(\{1, \dots, n\}, \mathcal{E})$ has maximum degree at most d , every local matrix has Gaussian-rational entries and operator norm at most J , and

$$e^{tA_0} \mathcal{S}_n \subseteq \mathcal{S}_n \quad (t \geq 0). \quad (51)$$

Let $a_\star$ be the eigenvalue of $A = A_0 - 2\mu E$ with largest real part. This eigenvalue is unique and algebraically simple. Given a positive rational δ , a deterministic classical algorithm returns $\hat{a} \in \mathbb{Q}(i)$ such that

$$|\hat{a} - a_\star| \leq \delta.$$

For fixed d, J, μ , its bit complexity is polynomial in n , the input bit length, and δ^{-1} .

Only the sum A_0 is assumed to satisfy (51); the displayed local terms need not individually generate preserving semigroups. The theorem uses the supplied local representation and does not include an algorithm for finding such a representation or verifying the preservation promise. No positivity, Hermiticity, stochastic normalization, or diagonalizability is assumed.

We will give explicit choices of all continuation parameters. In particular, put

$$N = 2n, \quad D = d + 1, \quad \rho = \frac{2^{-18}}{DJ}, \quad c = \frac{1}{\mu}. \quad (52)$$

The number p of perturbation coefficients required by the proof satisfies

$$p = O\left(\left(1 + \frac{c}{\rho}\right)^2 \log\left(2 + \frac{C_{d,J,\mu}n}{\delta}\right)\right), \quad (53)$$

for a constant $C_{d,J,\mu}$ independent of n, δ . The arithmetic operation count is $N \exp(O_D(p))$, with a polynomial bit overhead. The large constants in (52) are inherited from the perturbation estimate. They are sufficient for a complexity statement, not a claim of practical efficiency. In particular, Theorem 5.1 does not assert polynomial dependence on μ^{-1} .

5.2. Perturbation coefficients and complexification. We first state precisely the published perturbation input. For a graph on N qubits with maximum degree at most $D \geq 1$, set

$$H_0 = 2 \sum_{u=1}^N |1\rangle\langle 1|_u, \quad \Omega = |0\rangle^{\otimes N}. \quad (54)$$

For any matrix V , the eigenvalue of $H_0 + xV$ issuing from the simple eigenvalue 0 has a convergent germ at $x = 0$, which we write

$$F_V(x) = \sum_{k \geq 1} e_k(V) x^k. \quad (55)$$

Proposition 5.2 (Bravyi–DiVincenzo–Loss). *Let $W = \sum_{\{u,v\}} W_{uv}$ be Hermitian, with each W_{uv} Hermitian and of norm at most $L > 0$. For H_0 as in (54), put*

$$R_L = \frac{2^{-16}}{DL}.$$

Then

$$|e_k(W)| \leq 2^{-15} N R_L^{-k} \quad (k \geq 1). \quad (56)$$

For every $p \geq 1$, the coefficients $e_1(W), \dots, e_p(W)$ can be computed in $N \exp(O_D(p))$ arithmetic operations.

These are Lemma 6 and Theorem 3 of [16], specialized to unperturbed gap 2. Their Theorem 2 also gives convergence of the energy expansion throughout the corresponding high-field disk. The coefficient algorithm is formal: computing the first p coefficients does not require the eventual evaluation point to lie in that disk. We use the coefficient bound and algorithm exactly in this Hermitian form. The required rational-input bit implementation is specified below.

Lemma 5.3 (Complex perturbation coefficients). *Let $V = \sum_{\{u,v\}} V_{uv}$, where the local matrices are arbitrary complex matrices with $\|V_{uv}\| \leq J$. The germ (55) extends to a holomorphic eigenvalue function on*

$$|x| < \rho, \quad \rho = \frac{2^{-18}}{DJ},$$

and its coefficients satisfy

$$|e_k(V)| \leq 2^{-15} N \rho^{-k} \quad (k \geq 1). \quad (57)$$

If the entries of the local matrices are Gaussian rational, for each integer $p \geq 1$ its first p coefficients can be computed exactly using $N \exp(O_D(p))$ arithmetic operations on Gaussian rationals.

Proof. Let $Q = \text{Id} - |\Omega\rangle\langle\Omega|$, and let S be the inverse of H_0 on $Q(\mathbb{C}^2)^{\otimes N}$, extended by zero on Ω . Normalize the eigenvector germ by

$$v(x) = \Omega + \sum_{k \geq 1} v_k x^k, \quad \langle \Omega, v_k \rangle = 0.$$

Writing $v_0 = \Omega$, coefficient comparison in $(H_0 + xV)v(x) = F_V(x)v(x)$ gives

$$e_k(V) = \langle \Omega, V v_{k-1} \rangle, \quad (58)$$

$$v_k = -SQ \left(V v_{k-1} - \sum_{j=1}^{k-1} e_j(V) v_{k-j} \right). \quad (59)$$

The vacuum coordinate functional in these formulas is fixed; it does not conjugate any variable matrix entry of V . Induction shows that v_k and $e_k(V)$ are homogeneous polynomials of degree k in the entries of V , with no conjugate variables. The same recursion also proves uniqueness of the formal normalized eigenpair. Thus the coefficients returned by Proposition 5.2 are the restrictions of these polynomials to Hermitian inputs.

Decompose the local terms as

$$B_{uv} = \frac{V_{uv} + V_{uv}^*}{2}, \quad C_{uv} = \frac{V_{uv} - V_{uv}^*}{2i},$$

and write $V = B + iC$ for the resulting sums. Both B_{uv} and C_{uv} are Hermitian of norm at most J . For real θ , the local perturbation

$$W_\theta = B \cos \theta + C \sin \theta = \frac{1}{2} e^{-i\theta} V + \frac{1}{2} e^{i\theta} V^*$$

is Hermitian, with each local norm at most $2J$. Since e_k is homogeneous of degree k , the Fourier coefficient of frequency $-k$ in $e_k(W_\theta)$ is $2^{-k} e_k(V)$. Indeed, a monomial of degree k contributes that frequency only when its V -term is selected in all k factors. Consequently

$$e_k(V) = \frac{2^k}{2\pi} \int_0^{2\pi} e^{ik\theta} e_k(W_\theta) d\theta. \quad (60)$$

Apply (56) with $L = 2J$. The corresponding radius is $2^{-17}/(DJ)$, so (60) gives

$$|e_k(V)| \leq 2^k 2^{-15} N \left(\frac{2^{-17}}{DJ} \right)^{-k} = 2^{-15} N \rho^{-k}.$$

This proves normal convergence on compact subsets of $|x| < \rho$. Near zero the sum is the actual eigenvalue germ, by (58)–(59) and the simple eigenvalue of H_0 . The identity theorem applied to

$$\det(H_0 + xV - F_V(x) \text{Id})$$

therefore proves the eigenvalue identity on the entire disk. This step does not assert simplicity there or make any ordering claim about the other eigenvalues.

For the coefficient algorithm, take the distinct rational nodes

$$s_\ell = -1 + \frac{2\ell}{p}, \quad 0 \leq \ell \leq p.$$

At each node, $B + s_\ell C$ is a Hermitian local perturbation with norm bound $2J$. Apply Proposition 5.2 at each of these $p + 1$ inputs. For $k \leq p$, the function

$$s \mapsto e_k(B + sC)$$

is a polynomial of degree at most k . Its values at the nodes determine its value at $s = i$, namely $e_k(V)$, by exact rational interpolation. The factor $p + 1$ in the number of calls and the polynomial interpolation cost are absorbed in $N \exp(O_D(p))$. Thus the integral in (60) is used only for a bound; the algorithm uses no numerical quadrature. $\square$

Remark 5.4. The factor four between R_J and ρ has two sources: the Hermitian family has local norm bound $2J$, and extracting the extremal Fourier coefficient contributes 2^k . The argument transfers the coefficient estimate, not a Hermitian spectral-ordering or spectral-simplicity theorem. This distinction is necessary for arbitrary nonnormal inputs.

One-site perturbations can be incorporated without altering this published input. Attach a private auxiliary qubit i' to each physical site i , and represent a one-site term V_i by $V_i \otimes \text{Id}_{i'}$ on the edge $\{i, i'\}$. Give every auxiliary qubit the unperturbed term $2|1\rangle\langle 1|_{i'}$. The resulting operator is exactly

$$\tilde{H}(x) = (2E + xV) \otimes \text{Id}_{\text{aux}} + \text{Id}_{\text{phys}} \otimes 2 \sum_{i'} |1\rangle\langle 1|_{i'}. \quad (61)$$

Its vacuum eigenvalue germ and all its coefficients agree with those of the physical operator. The number of qubits is $N = 2n$, the maximum degree is at most $D = d + 1$, and the local norm bound is unchanged. The identity on the auxiliary factor is important: the auxiliary qubit is not coupled to the physical perturbation. This construction is only a device for using the coefficient algorithm; it imposes no stability assumption on the auxiliary description.

5.3. A global eigenvalue branch.

Lemma 5.5 (Half-plane continuation). *Under (51), the matrix*

$$A(h) = A_0 - 2hE$$

has an algebraically simple eigenvalue $a(h)$, uniquely maximizing real part, for every h with $\Re h > 0$. The function a is holomorphic on that half-plane. For $V = -A_0$, the function

$$F(x) = -x a(1/x), \quad \Re x > 0, \quad (62)$$

is a holomorphic eigenvalue of $2E + xV$. It agrees with the vacuum germ of Lemma 5.3 near the positive real origin.

Proof. Write $h = u + iv$, where $u > 0$. The imaginary Euler flow acts on polynomials by

$$e^{-2itvE}p(z_1, \dots, z_n) = p(e^{-2itv}z_1, \dots, e^{-2itv}z_n),$$

and therefore preserves $\mathcal{S}_n$. The finite-dimensional product formula gives

$$e^{t(A_0 - 2ivE)} = \lim_{m \rightarrow \infty} (e^{tA_0/m} e^{-2itvE/m})^m.$$

For each stable input, every approximating output is stable. The limit is stable or zero by Hurwitz's theorem, and it cannot be zero on a nonzero input because the limiting exponential is invertible. Hence $A_0 - 2ivE$ also generates a preserving semigroup.

Theorem 4.8, applied with damping u , supplies an algebraically simple eigenvalue of $(A_0 - 2ivE) - 2uE = A(h)$ whose real part exceeds those of all the others by at least $2u$. At each point, the implicit function theorem for the characteristic polynomial makes this eigenvalue locally holomorphic. Spectral continuity preserves its strict real-part inequality in a sufficiently small neighborhood. Thus the locally defined functions agree wherever their neighborhoods overlap: they select the same uniquely distinguished eigenvalue. This proves the global holomorphic assertion.

The reciprocal map takes the right half-plane into itself, so (62) is holomorphic there. The matrix identity

$$-xA(1/x) = 2E + xV$$

proves the eigenvalue assertion. For positive real x , the scalar $-x$ reverses real-part order, so $F(x)$ is the unique eigenvalue of $2E + xV$ with smallest real part. We do not make this ordering assertion for nonreal x .

It remains to identify the high-field germ without assuming Hermiticity of V . Choose positive x so small that $x\|V\| < 1/4$. Normality of $2E$ implies that every eigenvalue of $2E + xV$ is within distance $1/4$ of $\{0, 2, \dots, 2n\}$. To see the inclusion, outside these disks the resolvent of $2E$ has norm at most 4, so its product with xV has norm less than 1. The Neumann series therefore makes the perturbed resolvent invertible, including when $V = 0$. The Riesz projection inside $|z| = 1$ has rank one: the contour stays in the resolvent set throughout $2E + \tau xV$, $0 \leq \tau \leq 1$, and its rank at $\tau = 0$ is one. Hence the eigenvalue near zero is precisely the vacuum branch. All other eigenvalues have real part greater than one. The vacuum branch is therefore the uniquely minimal real-part eigenvalue, proving its agreement with $F(x)$ on a positive real interval. $\square$

Apply the identity theorem on the connected overlap $\{\Re x > 0\} \cap \{|x| < \rho\}$. Lemma 5.5 and Lemma 5.3, with the auxiliary construction when needed, yield one holomorphic function on

$$\Omega_\rho = \{x \in \mathbb{C} : \Re x > 0\} \cup \{x \in \mathbb{C} : |x| < \rho\}. \quad (63)$$

It remains an eigenvalue of $\tilde{H}(x)$ throughout this domain, by (61) and the characteristic-polynomial identity. In particular,

$$|F(x)| \leq \|\tilde{H}(x)\| \leq 2N + |x|W, \quad W = NDJ. \quad (64)$$

The displayed W is a convenient rational upper bound on the sum of all local norms. The scalar estimate $|\lambda| \leq \|T\|$ for an eigenvalue of a matrix T does not require normality. Finally,

$$a_\star = -\mu F(1/\mu). \quad (65)$$

5.4. Effective continuation and exact arithmetic. The next lemma gives the continuation step independently of its operator-theoretic application.

Lemma 5.6 (An explicit continuation map). *Let $\rho, c > 0$, and suppose that F is holomorphic on Ω_ρ . Define*

$$\eta = \frac{\rho^2}{4c + 2\rho}, \quad R = c + \eta, \quad q = \frac{c}{R}, \quad b = \frac{R^2 - c^2}{R}, \quad \phi(w) = \frac{bw}{1 - qw}. \quad (66)$$

Then $0 < q < 1$, $\phi(\mathbb{D}) \subset \Omega_\rho$, $\phi(0) = 0$, and $\phi(q) = c$. If $|F(x)| \leq L_0 + L_1|x|$ on Ω_ρ , the function $G = F \circ \phi$ satisfies

$$|G(w)| \leq M := L_0 + L_1(c + R) \quad (w \in \mathbb{D}).$$

Writing $F(x) = \sum_{j \geq 0} f_j x^j$ at zero and $G(w) = \sum_{k \geq 0} g_k w^k$, one has $g_0 = f_0$ and

$$g_k = \sum_{j=1}^k f_j b^j q^{k-j} \binom{k-1}{j-1} \quad (k \geq 1). \quad (67)$$

For every $p \geq 0$,

$$\left| F(c) - \sum_{k=0}^p g_k q^k \right| \leq \frac{M q^{p+1}}{1 - q}. \quad (68)$$

Proof. The definitions give $q \in (0, 1)$, $\eta \leq \rho/2$, and

$$0 < R^2 - c^2 = 2c\eta + \eta^2 < \rho^2.$$

If $|x - c| < R$ and $\Re x \leq 0$, then

$$|x|^2 = |x - c|^2 + 2c\Re x - c^2 < R^2 - c^2 < \rho^2.$$

Thus the disk of center c and radius R lies in Ω_ρ . The identity

$$\phi(w) = c + R \frac{w - q}{1 - qw}$$

shows that ϕ maps $\mathbb{D}$ biholomorphically onto this disk. Direct substitution gives its values at 0 and q . Since $|\phi(w)| < c + R$, the bound on G follows.

For $j \geq 1$, expansion at zero gives

$$\phi(w)^j = b^j w^j (1 - qw)^{-j} = b^j \sum_{k \geq j} \binom{k-1}{j-1} q^{k-j} w^k.$$

Only $j \leq k$ contributes to the coefficient of w^k , which proves (67). Applying Cauchy's estimate on circles with radii increasing to one gives $|g_k| \leq M$. The geometric tail at $w = q$ now proves (68). No holomorphic extension to the boundary of $\mathbb{D}$ is used. $\square$

Proof of Theorem 5.1. The existence, uniqueness, and simplicity of a_* follow from Theorem 4.8. Use $V = -A_0$, the auxiliary construction (61), and the parameters (52). Lemmas 5.3 and 5.5 give the function F on Ω_ρ , with (64). Apply Lemma 5.6 with

$$L_0 = 2N, \quad L_1 = W, \quad M = 2N + (c + R)W.$$

All the parameters in (66) are rational.

Choose the smallest integer $p \geq 0$ for which

$$\mu M q^{p+1} \leq \delta(1 - q). \quad (69)$$

The algorithm finds this integer by rational multiplication and comparison; evaluation of logarithms is unnecessary. If $p > 0$, compute $e_1(V), \dots, e_p(V)$ by Lemma 5.3, and then compute $g_0, \dots, g_p$ by (67). Here $g_0 = F(0) = 0$. Return the Gaussian rational

$$\hat{a} = -\mu \sum_{k=0}^p g_k q^k. \quad (70)$$

For $p = 0$, no coefficient call is made and the same formula returns zero. Equations (65), (68), and (69) give $|\hat{a} - a_\star| \leq \delta$.

We first bound the number of arithmetic operations. A useful exact identity is

$$\frac{1}{1-q} = \frac{R}{\eta} = 1 + \frac{4c^2}{\rho^2} + \frac{2c}{\rho}. \quad (71)$$

Since $-\log q \geq 1 - q$, the stopping rule implies

$$p = O\left(\frac{1}{1-q} \log\left(2 + \frac{\mu M}{\delta(1-q)}\right)\right).$$

The bound (53) follows, because $M = O_{d,J,\mu}(n)$. For fixed d, J, μ , this is $p = O_{d,J,\mu}(\log(2 + n/\delta))$. The coefficient algorithm uses $N \exp(O_D(p))$ arithmetic operations. Interpolation, the triangular transform, and evaluation in (70) add polynomially many operations in p .

We finish by verifying that exact arithmetic has only polynomial bit overhead. This also specifies a finite-input algorithm rather than a real-arithmetic oracle. Let B_0 bound the bit length of each numerator and denominator in the original local matrices. For $p > 0$, the interpolation nodes $s_\ell = -1 + 2\ell/p$ have $O(\log(p+1))$ bits. Taking local Hermitian parts and forming $B + s_\ell C$ therefore produces Gaussian-rational inputs with bit lengths polynomial in B_0 and $\log(p+1)$.

Consider one Hermitian coefficient computation at these nodes. Let Q_0 be the product of the positive denominators of the real and imaginary parts of its local entries. There are $O_D(N)$ such entries, so $\log Q_0$ is polynomial in the input length. Set

$$K_p = 48(p+1)!, \quad D_j = (Q_0 K_p)^j \quad (0 \leq j \leq p). \quad (72)$$

We recall the properties of the coefficient recursion used in Sections 3–5 of [16]. An order- j creation coefficient $C_j(M)$ is zero unless M is contained in a connected set of at most $j+1$ vertices. Each nonzero term in its recursion contains one local matrix entry, previous creation coefficients of total order $j-1$, and a divisor

$$k! E_0(M), \quad k \leq 4, \quad E_0(M) = 2|M|.$$

The basis creation matrices $\prod_{u \in M} |1\rangle\langle 0|_u$ have entries zero or one; expanding their nested commutators therefore produces signed sums of local perturbation entries. The corresponding energy recursion has the same form without the outer divisor $E_0(M)$. These are the explicit recursions used to implement the published coefficient algorithm, rather than an assumption about arbitrary symbolic perturbation expressions.

For $j \leq p$, every product $k! E_0(M)$ just displayed divides K_p . Induction in j consequently shows that

$$D_j C_j(M) \in \mathbb{Z}[i], \quad D_j e_j(B + s_\ell C) \in \mathbb{Z}[i]. \quad (73)$$

Indeed, the previous coefficients contribute denominator dividing $(Q_0 K_p)^{j-1}$, the new local entry contributes a factor dividing Q_0 , and the remaining divisor divides

K_p . All terms at a fixed order may be accumulated over this common denominator. Its bit length is

$$O(p \log Q_0 + p^2 \log(p+1)).$$

Intermediate numerators have polynomial bit length as well. Here a deliberately loose bound suffices. Fully unfold a term of perturbation order at most p . It has at most p local insertions and bounded-arity recursion nodes. Choose its local edges in at most $(ND)^{O(p)}$ ways. Their union has $O(p)$ vertices, and all intermediate creation sets lie in this union. Choosing these sets at $O(p)$ nodes, together with the tree shape, order partitions, and bounded-depth commutator terms, gives at most

$$\exp(O_D(p^2 + p \log(N+1)))$$

fully unfolded summands. Each summand contains at most p bounded local entries; the positive integer energy and factorial divisors cannot increase its absolute value. Its magnitude is therefore at most $\max(1, 2J)^{O(p)}$ times a fixed exponential factor. After introducing (72), this bounds the logarithmic magnitude of every intermediate numerator by a polynomial in p, N, B_0 . This argument concerns bit sizes only. The algorithm does not unfold these trees: its sharper operation count is the inherited $N \exp(O_D(p))$ count.

Exact interpolation also has polynomial bit cost. For the nodes $s_\ell = -1 + 2\ell/p$, its weights at i can be written

$$L_\ell(i) = \frac{\prod_{\substack{0 \leq r \leq p \\ r \neq \ell}} (pi + p - 2r)}{2^p (-1)^{p-\ell} \ell! (p-\ell)!}. \quad (74)$$

Both numerator and denominator have $O(p \log(p+1))$ bits. The coefficients $e_k(V)$ are obtained by multiplying these weights by the computed node values and adding $p+1$ terms. The rational powers and binomial coefficients in (67), and the powers in (69)–(70), increase bit length only polynomially. Thus a safe total bit bound is

$$N \exp(O_D(p)) \text{poly}(N, B_0, p, L_{\text{in}}),$$

where L_{in} is the total input bit length, including the requested accuracy. Together with (53), this proves the theorem. $\square$

5.5. Suzuki–Fisher ground energies. The preceding result applies to the Suzuki–Fisher class of Theorem 6.2. We keep its sign convention: the longitudinal term is $-\sum_i \mu_i^z Z_i$, with $\mu_i^z \geq 0$. The Lee–Yang theorem gives disk stability preservation by the Gibbs semigroup e^{-tH} ; see [53, 61, 18].

Theorem 5.7 (Fixed positive longitudinal field). *Fix a maximum degree d , a positive rational bound J on the absolute values of the Pauli coefficients, and a positive rational μ . Let H be a Suzuki–Fisher Hamiltonian with rational Pauli coefficients on an n -vertex graph of maximum degree at most d , and suppose*

$$\mu_i^z \geq \mu \quad (1 \leq i \leq n).$$

For every positive rational δ , its ground energy $E_0(H)$ can be approximated to absolute error at most δ by a deterministic classical algorithm polynomial in n , the input bit length, and δ^{-1} . The polynomial may depend on d, J, μ .

Proof. Define

$$V = H + \mu \sum_i Z_i.$$

The longitudinal fields of V are $\mu_i^z - \mu \geq 0$; all its interaction inequalities are unchanged. Thus V is again Suzuki–Fisher, and $A_0 = -V$ satisfies (51). Its local matrices have Gaussian-rational entries. Combining the finitely many Pauli terms on each edge or site gives a local norm bound depending only on J, μ , and leaves the degree bound unchanged. For example, sums of the absolute Pauli coefficients give explicit rational norm upper bounds, so the algorithm need not compute exact matrix norms.

Since $2E = n \text{Id} - \sum_i Z_i$,

$$A_0 - 2\mu E = -H - \mu n \text{Id}. \quad (75)$$

The matrix on the right is Hermitian. Its eigenvalue with largest real part is therefore $-E_0(H) - \mu n$. Apply Theorem 5.1 with error δ . If it returns $\hat{a}$, return the real rational

$$\hat{E} = -\Re \hat{a} - \mu n.$$

Equation (75) gives the required error bound and the asserted complexity. $\square$

Remark 5.8 (Hermitian coefficient bound). For this application, the complexification step can be omitted: the perturbation V is Hermitian. Applying Proposition 5.2 directly gives the larger guaranteed disk $2^{-16}/(DJ')$, where J' is the local norm bound after the auxiliary construction. The same continuation proof then applies to the ground-energy branch of $2E + xV$. The factor-four improvement in this disk changes constants, not the fixed-parameter complexity conclusion.

Theorem 5.7 includes nonstoquastic interactions and does not require conservation of total degree. Its computational output and parameter dependence differ from several earlier results. Bravyi and Gosset [17] give randomized algorithms for a two-axis stoquastic family on arbitrary graphs. Harrow, Mehraban, and Soleimanifar [30, Section 7 of the full version] give an all-temperature quasipolynomial partition-function algorithm for a number-conserving XXZ subclass. Their fugacity-polynomial reduction uses the commutation of the interaction with total magnetization. Bravyi, Gosset, Liu, and Wong [18] give a polynomial-time quantum ground-energy algorithm for the full Suzuki–Fisher family, including zero fields and inverse-polynomial absolute accuracy. Their sharp Hermitian gap and quantum algorithm are not consequences newly claimed here. Theorem 5.7 gives a deterministic classical conclusion on bounded-degree inputs at every fixed positive field. It does not replace the unrestricted quantum theorem.

5.6. Energy-value approximation at zero field. A fixed positive field can also be used as a controlled perturbation when the requested error is proportional to the number of sites.

Theorem 5.9 (Additive energy-value approximation). *Fix a maximum degree d , a bound J on the absolute Pauli coefficients, and a positive rational ε . For every rational Suzuki–Fisher Hamiltonian on an n -vertex graph satisfying these bounds, a deterministic classical algorithm polynomial in n and the input bit length returns $\hat{E} \in \mathbb{Q}$ such that*

$$|\hat{E} - E_0(H)| \leq \varepsilon n.$$

The longitudinal fields may vanish. The polynomial may depend on d, J, ε . The output is an energy value; no approximating state is asserted.

Proof. It suffices to consider $0 < \varepsilon \leq 1$, since a smaller accuracy parameter also meets a larger requested tolerance. Put $\nu = \varepsilon/4$ and

$$H_\nu = H - \nu \sum_i Z_i.$$

This is a Suzuki–Fisher Hamiltonian whose longitudinal fields are all at least the fixed positive number ν . Its local coefficients are bounded in terms of J, ε . The variational characterization of the lowest eigenvalue gives

$$|E_0(H_\nu) - E_0(H)| \leq \|H_\nu - H\| \leq \nu n.$$

Use Theorem 5.7 to estimate $E_0(H_\nu)$ with absolute error at most $\varepsilon n/2$. The returned value differs from $E_0(H)$ by at most

$$\frac{\varepsilon n}{4} + \frac{\varepsilon n}{2} = \frac{3\varepsilon n}{4} \leq \varepsilon n.$$

For fixed d, J, ε , the field ν and all local bounds are fixed. The runtime in Theorem 5.7 is therefore polynomial in the input size and n . $\square$

We spell out the consequence for Quantum MaxCut and EPR Hamiltonians, keeping value approximation separate from state construction. For a graph $G = (V, \mathcal{E})$ with nonnegative rational weights, let

$$Q_G = \sum_{\{i,j\} \in \mathcal{E}} w_{ij} |\Psi^-\rangle \langle \Psi^-|_{ij}, \quad |\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}},$$

and write $\text{OPT}(G) = \lambda_{\max}(Q_G)$. Similarly, define the EPR objective

$$P_G = \sum_{\{i,j\} \in \mathcal{E}} w_{ij} |\Phi^+\rangle \langle \Phi^+|_{ij}, \quad |\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Corollary 5.10 (EPR and bipartite Quantum MaxCut values). *For fixed degree and upper weight bounds and fixed rational $\varepsilon > 0$, the optimal values of P_G on arbitrary graphs and of Q_G on bipartite graphs admit deterministic classical additive- $\varepsilon|V|$ approximation in polynomial time. For unweighted bipartite Quantum MaxCut, this gives a relative energy-value approximation scheme after isolated vertices are deleted. In each case the algorithm estimates the optimum value and does not produce a state attaining the estimate.*

Proof. The negative EPR objective is Suzuki–Fisher up to a scalar:

$$-P_G = -\frac{1}{4} \sum_{\{i,j\} \in \mathcal{E}} w_{ij} (\text{Id} + X_i X_j - Y_i Y_j + Z_i Z_j). \quad (76)$$

On each edge the transverse matrix has operator norm $w_{ij}/4$, equal to its longitudinal interaction coefficient. Subtract the known scalar $-\frac{1}{4} \sum_{\{i,j\}} w_{ij}$ when applying Theorem 5.9, and restore it afterward. This proves the assertion for P_G .

If G is bipartite with parts A, B , let $U = \prod_{i \in A} Y_i$. On every edge, U maps the singlet vector to the EPR vector up to a scalar of modulus one. Hence

$$U Q_G U^* = P_G.$$

The spectra agree, proving the additive assertion for Q_G . This is the same local change of basis used in [18, Corollary 2].

For the unweighted bipartite Quantum MaxCut statement, remove isolated vertices and let N and m be the remaining vertex and edge counts. If $m = 0$, the value is zero and is returned exactly. Otherwise $N \leq 2m$. The product vector which is 0 on one part and 1 on the other has expectation $m/2$. Thus $\text{OPT}(G) \geq m/2$. Apply the additive algorithm with error $\varepsilon N/4$. Its error is bounded by

$$\frac{\varepsilon N}{4} \leq \frac{\varepsilon m}{2} \leq \varepsilon \text{OPT}(G).$$

For each fixed ε , the runtime is polynomial, which proves the relative value approximation assertion. $\square$

The additive statement for bounded weights does not automatically give a relative guarantee when positive weights may be arbitrarily small. Nor is the value scheme a state-producing approximation algorithm of the type studied, for example, in [3]. The inverse-polynomial absolute-accuracy conjectures for general EPR problems in [41] remain outside its scope. Indeed, introducing a field of order δ/n would make the factor c/ρ in (53) grow with n . The resulting coefficient count need no longer be logarithmic, and the algorithm proved here need no longer run in polynomial time.

5.7. Stable tests of principal eigenvectors. We now pass from one eigenvalue to queries about its principal eigenvectors. The input remains (50)–(51). For the binary coefficient space, the transpose exchanges the two groups of variables in a matrix kernel. Proposition 4.5 therefore gives strictly disk-stable right eigenvectors of both $A = A_0 - 2\mu E$ and $A^\top$. Normalize them by

$$v_\emptyset = w_\emptyset = 1.$$

The transpose, rather than the adjoint, makes the dependence on a complex parameter holomorphic. A *stable product test* is $B = \bigotimes_{i=1}^n B_i$, where $B_i \in M_2(\mathbb{C})$, $(B_i)_{00} = 1$, and

$$K_{B_i}(z, w) = (B_i)_{00} + (B_i)_{10}z + (B_i)_{01}w + (B_i)_{11}zw$$

does not vanish in $\mathbb{D}^2$. The matrices B_i need not be Hermitian, positive, or invertible.

Theorem 5.11 (Classical product tests of principal states). *Fix d, J, μ as in Theorem 5.1, and suppose that A_0 satisfies that theorem's input assumptions. Given a stable product test with Gaussian-rational entries, both*

$$G_B = w^\top B v, \quad \text{Tr}(B\Pi) = \frac{w^\top B v}{w^\top v}, \quad \Pi = \frac{v w^\top}{w^\top v},$$

are nonzero. For every rational $0 < \epsilon < 1$, a deterministic classical algorithm returns Gaussian-rational estimates of these two quantities with relative complex error at most ϵ . Its bit complexity is

$$\text{poly}_{d,J,\mu}(n, \epsilon^{-1}, L_{\text{in}}),$$

where L_{in} includes the local generator, the test matrices, and the accuracy. Relative complex error means $|\hat{z}/z - 1| \leq \epsilon$.

This is query access to specified scalar tests, not a listing of an exponentially long vector. No polynomial bound on $\|\Pi\|$ is assumed or obtained. The proof instead constructs a nonvanishing scalar function with a controlled logarithm in the reciprocal-field parameter.

5.8. A uniform domain for normalized eigenvectors. Write $V = -A_0$, $H(x) = 2E + xV$, and set

$$r_0 = \frac{2^{-18}}{(d+1)J}, \quad \rho_s = \frac{r_0}{8}. \quad (77)$$

The following use of the Kirkwood–Thomas expansion concerns its creation coefficients, not just the energy estimate of Proposition 5.2. For a nonempty set $M \subseteq [n]$, put

$$a_M^\dagger = \prod_{i \in M} |1\rangle\langle 0|_i.$$

These operators commute, and products with overlapping supports vanish. The vacuum-normalized eigenvector germ has a unique expression

$$v(x) = \exp \left(- \sum_{\emptyset \neq M \subseteq [n]} C(M; x) a_M^\dagger \right) \Omega, \quad C(M; x) = \sum_{k \geq 1} C_k(M) x^k.$$

Uniqueness follows by taking the finite logarithm in the nilpotent commutative algebra generated by the $a_i^\dagger$.

Lemma 5.12 (Creation coefficients and a zero-free vacuum disk). *For every local complex V with the bounds above,*

$$\max_i \sum_{M \ni i} |C_k(M)| \leq 2^{-15} r_0^{-k}, \quad C_k(M) = 0 \quad \text{if } |M| > k + 1. \quad (78)$$

The normalized right eigenvector germs of $H(x)$ and $H(x)^\top$ extend holomorphically to $|x| < \rho_s$. Their coefficient polynomials are nonzero on the closed polydisk $\{|z_i| \leq 2\}$ throughout this parameter disk.

Proof. For Hermitian perturbations with local norm bound L on a graph of maximum degree D , Lemma 5, equation (32), and Lemma 7, equation (47), of [16], with unperturbed gap 2, give

$$\max_i \sum_{M \ni i} |C_k(M)| \leq 2^{-15} R_L^{-k}, \quad R_L = 2^{-16}/(DL),$$

and say that a nonzero $C_k(M)$ is supported inside a connected set of at most $k + 1$ vertices. In particular $|M| \leq k + 1$. Use the private auxiliary qubits in (61) for one-site terms, with $D = d + 1$. The normalized eigenvector germ of the augmented operator is exactly the physical germ tensored with the auxiliary vacuum. Its creation coefficients involving an auxiliary vertex vanish; all other coefficients agree with the physical ones.

The recursion (58)–(59) and the finite logarithm show that $C_k(M)$ is a homogeneous polynomial of degree k in the local entries of V , without their conjugates. With W_θ from the proof of Lemma 5.3,

$$C_k(M; V) = \frac{2^k}{2\pi} \int_0^{2\pi} e^{ik\theta} C_k(M; W_\theta) d\theta.$$

Taking absolute values, summing over $M \ni i$, and applying the Hermitian estimate with $L = 2J$ proves (78). The support conclusion is inherited by the same coefficient identity. Transposition gives the identical estimates for the left germ. For $|x| < r_0/8$, it follows that

$$\max_i \sum_{M \ni i} 4^{|M|} |C(M; x)| \leq 2^{-15} \sum_{k \geq 1} 4^{k+1} (|x|/r_0)^k < 2^{-13}. \quad (79)$$

All these series converge locally uniformly.

We give the elementary polymer argument that turns this estimate into zero-freeness. For weights u_M on nonempty subsets of a finite set, define

$$Z(S) = \sum_{\substack{\mathcal{A} \text{ a collection of} \\ \text{pairwise disjoint nonempty subsets of } S}} \prod_{M \in \mathcal{A}} u_M.$$

Suppose that $\sup_i \sum_{M \ni i} |u_M| 2^{|M|-1} \leq 1/2$. Induction on $|S|$ proves

$$Z(S) \neq 0, \quad \left| \frac{Z(S)}{Z(S \setminus \{i\})} - 1 \right| \leq \frac{1}{2} \quad (i \in S). \quad (80)$$

Indeed,

$$Z(S) = Z(S \setminus \{i\}) + \sum_{\substack{M \subseteq S \\ i \in M}} u_M Z(S \setminus M).$$

By the induction hypothesis, deleting the $|M| - 1$ additional vertices gives $|Z(S \setminus M)/Z(S \setminus \{i\})| \leq 2^{|M|-1}$. The claimed ratio estimate follows, and its disk does not contain zero. The empty set starts the induction.

The polynomial of $v(x)$ is this polymer sum with $u_M = -C(M; x)z^M$. The factorials in the exponential cancel the orderings of pairwise disjoint sets. For $|z_i| \leq 2$, the left side of the polymer condition is at most half the left side of (79), hence is less than $1/2$. This proves the asserted zero-freeness.

The convergent creation series and its finite exponential give holomorphic vectors on $|x| < \rho_s$. Their formal eigenvalue is the convergent energy series of Lemma 5.3. The eigenvector equations hold near zero and then throughout the disk by the identity theorem. No assertion of simplicity on the entire disk is needed. $\square$

Lemma 5.13 (Nonvanishing principal tests). *The vectors in Lemma 5.12 and the normalized principal vectors of $A(1/x)$ glue to holomorphic vectors $v(x), w(x)$ on*

$$\Omega_{\rho_s} = \{\Re x > 0\} \cup \{|x| < \rho_s\}.$$

For every stable product test,

$$G_B(x) = w(x)^T B v(x) \quad (81)$$

is nonvanishing on this domain, $G_B(0) = 1$, and

$$|G_B(x)| \leq 4^n, \quad \Re F_B(x) \leq n \log 4, \quad F_B = \log G_B, \quad F_B(0) = 0. \quad (82)$$

Proof. The half-plane argument in Lemma 5.5 applies to the principal eigenvectors as well. Local analytic eigenvectors exist for a simple eigenvalue; their vacuum coordinates are nonzero because their coefficient polynomials are strictly stable. Vacuum normalization makes these local choices agree on overlaps. Use $A(h)^T$ for the left vectors. On a sufficiently small positive real interval they coincide with the vacuum vectors, by the same Riesz-projection argument as in Lemma 5.5. That

interval may depend on n ; it is used only to identify the branches, not to choose ρ_s . The identity theorem on the connected overlap gives the required gluing.

In the half-plane, $v(x)$ and $w(x)$ each have a strict polydisk radius greater than one. In the small disk they have radius at least two by Lemma 5.12. Contract their legs against the two legs of every K_{B_i} . The strict version of the disk contraction rule, equivalently polarized apolarity, applies because each contracted pair has product of radii greater than one; see Lemma 2.3. The resulting scalar is (81), hence is nonzero. This argument allows singular tests and needs no uniform lower bound for the strict radius.

For completeness, a multiaffine stable polynomial $f(z) = \sum_S f_S z^S$ with $f_\emptyset = 1$ satisfies $|f_S| \leq 1$. Set variables outside S to zero and all remaining variables to t . If $f_S \neq 0$, the resulting polynomial has degree $|S|$, constant term one, leading coefficient f_S , and all roots of modulus at least one. Their product proves the bound. The case $f_S = 0$ is immediate. Apply this observation to v, w and each K_{B_i} . The expansion of $w^\top B v$ has 4^n terms of modulus at most one. Also $G_B(0) = \prod_i (B_i)_{00} = 1$. The domain is star-shaped about zero, so its nonvanishing holomorphic function has a unique logarithm vanishing there. Taking its real part gives (82). $\square$

Lemma 5.14 (Continuation with a one-sided logarithmic bound). *Use the parameters (66) with $\rho = \rho_s$ and $c = 1/\mu$. Write $F_B(x) = \sum_{j \geq 1} f_j x^j$ and $F_B(\phi(u)) = \sum_{k \geq 1} g_k u^k$. The transform (67) remains valid, and*

$$|g_k| \leq 4n, \quad \left| F_B(c) - \sum_{k=1}^p g_k q^k \right| \leq \frac{4nq^{p+1}}{1-q}. \quad (83)$$

In particular $|F_B(c)| \leq Kn$, where $K = 4q/(1-q)$.

Proof. The map lemma puts $\phi(\mathbb{D})$ in Ω_{ρ_s} . The holomorphic function $U(u) = 2n - F_B(\phi(u))$ has nonnegative real part and $U(0) = 2n$. On a circle of radius $r < 1$, the positive function $\Re U(re^{it})$ has average $2n$. Its k -th Fourier coefficient, for $k \geq 1$, equals $-g_k r^k/2$, so $|g_k| r^k \leq 4n$. Let $r \uparrow 1$ and sum the geometric tail. The coefficient transform is a formal identity already proved in Lemma 5.6. $\square$

5.9. Connected coefficients and finite bit complexity. We compute the coefficients of F_B without expanding the full eigenvectors. This is an instance of the connected-logarithm method for multiplicative analytic graph functions; compare [45, 62]. The needed multiplicativity is formal at the vacuum and does not require a preservation promise on induced subgraphs.

For $U \subseteq [n]$, keep just the local terms supported inside U , obtaining V_U , and let $B_U = \bigotimes_{i \in U} B_i$. Define the vacuum-normalized formal vectors v_U, w_U for $2E_U + xV_U$ and its transpose. Put

$$F_{B,U}(x) = \log(w_U(x)^\top B_U v_U(x)), \quad \Phi_{B,U}(x) = \sum_{W \subseteq U} (-1)^{|U|-|W|} F_{B,W}(x).$$

For the empty subsystem all logarithms are zero.

Lemma 5.15 (Local computation of test logarithms). *For $k \geq 1$,*

$$[x^k] F_B(x) = \sum_{\substack{U \subseteq [n] \text{ connected} \\ |U| \leq k+1}} [x^k] \Phi_{B,U}(x).$$

For Gaussian-rational input, the first p coefficients can be computed exactly with $n \exp(O_d(p)) \text{poly}(L_{\text{in}}, n, p)$ bit operations.

Proof. If the induced graph on U splits into components, the normalized formal vectors tensor across components and the test does likewise. Thus $F_{B,U}$ is the sum of the component logarithms. Subset inversion shows that $\Phi_{B,U} = 0$ when U is disconnected.

To obtain the order bound as well, give each local perturbation term its own parameter. An order- k coefficient is a homogeneous polynomial of degree k in these parameters, by the unique formal eigenvector recursion. If a parameter monomial has disconnected union of supports, set all unused parameters to zero. Tensor factorization makes the logarithm additive across those components, so that monomial has coefficient zero. A connected union of at most k one- or two-site supports has at most $k + 1$ vertices. Finally the coefficient of a monomial with support union T is unchanged in every induced subsystem containing T . The alternating sum over $W \subseteq U$ retains it only if $T = U$. This proves the formula, including both vanishings.

Here is a direct finite-input implementation. In each subsystem of size $m \leq p + 1$, use the exact recursions (58)–(59), and their transposes, through order p . If

$$a_k = \sum_{i+j=k} w_{U,i}^T B_U v_{U,j}, \quad \log \left(1 + \sum_{k \geq 1} a_k x^k \right) = \sum_{k \geq 1} f_k x^k,$$

then coefficient comparison after logarithmic differentiation gives

$$f_k = a_k - \frac{1}{k} \sum_{j=1}^{k-1} j f_j a_{k-j}. \quad (84)$$

Every vector has dimension at most 2^{p+1} . Even dense matrix arithmetic and enumeration of every $W \subseteq U$ cost only $\exp(O(p)) \text{poly}(p)$ operations per U . Connected subsets of size at most $p + 1$ can be enumerated by traversals of their spanning trees, with deduplication, in $n \exp(O_d(p))$ time; a traversal has length at most $2p$. The case $d = 0$ consists just of singletons.

We record denominator and magnitude bounds to justify the bit claim. Let Q_0 be a common positive denominator for all local entries of V , let Q_B be the product of the denominators of the entries of the B_i , and put

$$D_p = 2(p+1)!, \quad T = Q_0 D_p.$$

In a subsystem of size at most $p + 1$, each nonzero diagonal entry of the reduced inverse S_U is $1/(2r)$, $1 \leq r \leq p + 1$. Induction in the vector recursion gives denominators dividing T^{2k} for $v_{U,k}, w_{U,k}$ and T^{2k-1} for the energy coefficient. The denominator of a_k divides $Q_B T^{2k}$. Equation (84) then bounds the denominator of f_k by $k! Q_B^k T^{2k}$. At each order, accumulate the sums over these common denominators; their logarithms are polynomial in p and the total input length.

Intermediate numerators also have polynomial bit length. For a subsystem U , $\|V_U\| \leq m(d+1)J$. On $|x| \leq r_U = [16(1+m(d+1)J)]^{-1}$, the resolvent Neumann series on the circle $|z| = 1$ bounds the spectral projection at the vacuum and its difference from the unperturbed projection. In particular, the projection has norm at most $16/15$, its vacuum entry differs from one by at most $1/15$, and both

vacuum-normalized eigenvectors have norm less than two. Cauchy's estimate gives $\|v_{U,k}\|, \|w_{U,k}\| \leq 2r_U^{-k}$. As $\|B_U\| \leq 2^m$,

$$|a_k| \leq 4(k+1)2^m r_U^{-k}.$$

For $k \leq p$, choose $A_p = 4(p+1)2^{p+1}$. The finite formula $\log(1+a) = \sum_{\ell=1}^p (-1)^{\ell+1} a^\ell / \ell \pmod{x^{p+1}}$ bounds $|f_k|$ by $(2A_p)^k r_U^{-k}$. Matrix products, vector sums and the partial sums in (84) have the same type of bound, up to factors $2^{O(p)} \text{poly}(p)$. Subset sums add at most 2^{p+1} terms, and the final connected sum has at most $n \exp(O_d(p))$ terms. Combining these magnitude bounds with the common denominators controls every stored numerator, including before cancellation. This proves the bit complexity. $\square$

Proof of Theorem 5.11. Use $c = 1/\mu$, ρ_s from (77), and the rational map parameters of Lemma 5.6. Given $0 < \tau < 1$, choose the least $p \geq 0$ for which

$$4nq^{p+1} \leq \tau(1-q).$$

Rational multiplication and comparison find p ; (71) gives $p = O_{d,J,\mu}(\log(2+n/\tau))$. Use Lemma 5.15, the triangular transform, and (83) to compute a Gaussian rational S_B with $|S_B - F_B(c)| \leq \tau$. Also $|S_B| \leq Kn + 1$. The preceding lemma and the rational transform give polynomial bit cost.

We make explicit that exponentiation does not introduce an unbounded-precision oracle. Let T_0 be an integer upper bound on $Kn + 1$, and choose an integer ℓ with $2^{-\ell} \leq \tau$. Truncate the Taylor series of e^{S_B} at degree

$$M = \lceil 10(2T_0 + \ell + 1) \rceil$$

by taking $\sum_{j=0}^M S_B^j / j!$. The remainder is at most $e^{T_0} T_0^{M+1} / (M+1)!$. Using $r! \geq (r/3)^r$, it is less than τe^{-T_0} . Since $|e^{S_B}| \geq e^{-T_0}$, this is relative error less than τ . All terms are Gaussian rational, $M = O_{d,J,\mu}(n + \log \tau^{-1})$, and the numerators and denominators have polynomial bit length. Furthermore $|e^{S_B} / G_B(c) - 1| \leq e^\tau - 1$. Choosing $\tau = \epsilon/32$ supplies the required margin. Repeat for $B = I$, and divide the two nonzero rational estimates to obtain the estimate of G_B/G_I . For errors at most $\epsilon/8$ in each factor, the ratio error is at most $(\epsilon/4)/(1 - \epsilon/8) < \epsilon$.

The nonvanishing assertions follow from Lemma 5.13; I is a stable product test because its local kernel is $1 + zw$. For a simple eigenvalue the spectral projection is $vw^\top / (w^\top v)$, so this quotient is indeed $\text{Tr}(B\Pi)$. $\square$

5.10. Product-state queries and equatorial measurement sampling. For $s \in \mathbb{C}$ with $|s| \leq 1$, define

$$|s\rangle = \frac{|0\rangle + s|1\rangle}{\sqrt{1 + |s|^2}}.$$

These are the pure states in the closed northern hemisphere of the Bloch sphere in our fixed basis. The amplitude and probability queries extend to this whole hemisphere. Complete two-outcome projective measurement sampling is stated on the equator, where the two states $|s\rangle, |-s\rangle$ are orthogonal and both satisfy the condition.

Theorem 5.16 (Classical access to Suzuki–Fisher ground states). *Fix d, J, μ as in Theorem 5.7. Let H be a rational Suzuki–Fisher Hamiltonian on $n \geq 1$ qubits satisfying that theorem's hypotheses, and let $0 < \epsilon < 1$ be rational. Its unique unit*

ground vector ψ can be normalized by $\langle 0^n | \psi \rangle > 0$. The query algorithms in (1)–(2) have bit complexity $\text{poly}_{d,J,\mu}(n, \epsilon^{-1}, L_{\text{in}})$.

- (1) Given $s_1, \dots, s_n \in \mathbb{Q}(i)$ with $|s_i| \leq 1$, approximate the nonzero complex amplitude $\langle s_1 \cdots s_n | \psi \rangle$ with relative complex error at most ϵ .
- (2) Given $S \subseteq [n]$ and $s_i \in \mathbb{Q}(i)$, $|s_i| \leq 1$ for $i \in S$, approximate the positive probability

$$\left\langle \psi \left| \bigotimes_{i \in S} |s_i\rangle \langle s_i| \otimes I_{S^c} \right| \psi \right\rangle$$

by a positive rational with relative error at most ϵ .

- (3) Given a polynomial-time adaptive strategy that measures each site at most once, in a basis $\{|s\rangle, |-s\rangle\}$ with $s \in \mathbb{Q}(i)$, $|s| = 1$, sample its outcome record by a classical randomized algorithm with total variation error at most ϵ .

The strategy in (3) has a finite description, takes previous outcomes as input, and outputs the next unmeasured site and a phase. More precisely, if its total evaluation time along any record is at most T_{strategy} , and all generated phases have bit length at most L_{phase} , the sampler uses at most

$$T_{\text{strategy}} + \text{poly}_{d,J,\mu}(n, \epsilon^{-1}, L_{\text{in}} + L_{\text{phase}})$$

bit operations. In particular the sampler is polynomial-time for each fixed polynomial-time strategy.

Proof. Use $A_0 = -(H + \mu \sum_i Z_i)$ as in (75). Its local norm bound is a fixed rational J' depending on the Pauli bound J and on μ ; all radii in the preceding lemmas are evaluated with this bound, not with an unadjusted J . At positive real x , $H(x)$ is Hermitian, so its vacuum-normalized transpose eigenvector is $w = \bar{v}$. Consequently $G_I = \|v\|^2 > 0$ and $\psi = v/\sqrt{G_I}$. Along the positive real path the normalized logarithm F_I is the real logarithm, because it starts at zero and G_I stays positive.

For amplitudes take

$$B_i^{\text{amp}} = \begin{pmatrix} 1 & \bar{s}_i \\ 0 & 0 \end{pmatrix}.$$

Its kernel $1 + \bar{s}_i w$ is disk-stable, and

$$\langle s_1 \cdots s_n | \psi \rangle = \left(\prod_i (1 + |s_i|^2)^{-1/2} \right) \exp(F_{B^{\text{amp}}}(c) - F_I(c)/2).$$

For the event in (2), take $B_i = I$ off S and

$$B_i^{\text{prob}} = \begin{pmatrix} 1 & \bar{s}_i \\ s_i & |s_i|^2 \end{pmatrix} = (1 + |s_i|^2) |s_i\rangle \langle s_i| \quad (i \in S).$$

Its kernel factors as $(1 + s_i z)(1 + \bar{s}_i w)$. Thus

$$P_S = \left(\prod_{i \in S} (1 + |s_i|^2)^{-1} \right) \exp(F_B(c) - F_I(c)).$$

Nonvanishing follows from Lemma 5.13; positivity of the projector expectation then gives $P_S > 0$.

Compute the logarithms with a fixed fraction of the requested error, using the constructive proof of Theorem 5.11. For (2), take the real part of the logarithmic difference before exponentiating; its error does not increase because the exact difference is real on the positive path. For either logarithmic difference use

$T_0 = \lceil 2Kn + 2 \rceil$ in that finite Taylor procedure. In case (2), the resulting rational estimate is real and positive when its relative error is less than one. The probability normalization is rational and exact. For (1), the product $R_s = \prod_i (1 + |s_i|^2)$ is a positive rational between 1 and 2^n . Rational bisection for $\sqrt{R_s}$, followed by inversion, computes its reciprocal square root to any required relative error with $O(n + \log \epsilon^{-1} + L_{\text{in}})$ bits and polynomial arithmetic cost. Splitting the error budget between the two logarithms, exponentiation and this factor proves (1) and (2), including the phase assertion.

For (3), fix an already generated history. Its two extensions have true probabilities $p_+, p_- > 0$. Part (2) computes positive estimates $\hat{p}_\pm$ with relative error at most $\eta = \epsilon/(16n)$. Normalizing their sum gives a conditional probability with error at most $2\eta/(1 - \eta) < 4\eta$, regardless of the probability of the parent history. Round it to a dyadic probability within $\epsilon/(4n)$ and draw the corresponding Bernoulli variable using $O(\log(n/\epsilon))$ unbiased random bits. On a fixed history, projectors on distinct sites commute, so its probability is exactly the product event in (2), even though the bases were chosen adaptively. Coupling the two processes at each step bounds their total variation distance by the sum of the at most n conditional errors, which is less than ϵ . At most $2n$ queries are made. Their accuracy and phase bit lengths give the displayed polynomial overhead; evaluating the strategy contributes T_{strategy} . $\square$

The novelty of this conclusion is its uniform local input and its fixed-parameter polynomial complexity on the full positive-field Suzuki–Fisher family. For general Lee–Yang states with a fixed spatial radius $r > 1$, [61, Section 4, Theorems 5–6] already provides state-dependent quasipolynomial circuits for relative X -basis amplitudes, with marginal probabilities and phases in the state-preparation proof; see also [60]. Our proof does not assume a size-independent spatial radius margin. It uses the common reciprocal-field domain instead. Connected coefficient computation and zero-free interpolation are established methods [45, 62, 58]; the new input to those methods is Lemma 5.13 for principal-state tests. In particular [62, Theorem 15] already treats tensorized measurements of thermal partition functions. Its thermal zero-free hypothesis does not itself supply the ground-state parameter domain used here.

The queries above do not include arbitrary product bases: the projector onto $|1\rangle$ has no nonzero vacuum entry and is not an allowed normalized test. Nor does the fixed-field theorem extend the zero-field energy-value scheme to state approximation or sampling. The broad quantum algorithms in [18, 48] have different parameter ranges and outputs; the present classical result does not subsume them.

6. ALL-TEMPERATURE LEE–YANG HAMILTONIANS

We now specialize the complex generator classification to Hermitian operators on qubits. The resulting inequalities are precisely the Suzuki–Fisher conditions, including their full transverse operator-norm form. Their sufficiency is classical; the generator theorem supplies necessity and excludes additional interactions of arbitrary support. Positive longitudinal fields then improve the zero-free domain of the actual Gibbs tensor, at every positive temperature.

6.1. From matrix tensors to polynomial generators. Fix the computational basis $\{|a\rangle : a \in \{0, 1\}^n\}$, with $Z|0\rangle = |0\rangle$. Associate to a matrix M the polynomial

$$F_M(z, w) = \sum_{a, b \in \{0, 1\}^n} M_{ab} z^a w^b. \quad (85)$$

We say that M has the tensor Lee–Yang property at radius r if F_M is nonzero on $\mathbb{D}_r^n \times \mathbb{D}_r^n$. The two groups of variables in this definition are independent complex variables. In particular, (85) is not the sesquilinear form $\langle z | M | z \rangle$.

For a Hermitian Hamiltonian H , the Gibbs operator is $M_\beta = e^{-\beta H}$, and its normalized density matrix is $\rho_\beta = M_\beta / \text{Tr } M_\beta$. Since $\text{Tr } e^{-\beta H} > 0$ for real β , the two matrices have exactly the same polynomial zeros.

The following bridge fixes both the tensor and operator conventions. It is the binary instance, with a unitary scalar normalization, of the Cayley correspondence (8).

Lemma 6.1. *Let*

$$C_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} i & -i \\ 1 & 1 \end{pmatrix}, \quad C = C_1^{\otimes n}, \quad D_1 = \text{diag}(-1, 1).$$

For an invertible matrix M , its tensor has the unit-disk Lee–Yang property if and only if CMC^{-1} , acting on multiaffine coefficient vectors, preserves upper-half-plane stability. Consequently, the Gibbs family corresponds to the generator

$$A = -CHC^*. \quad (86)$$

Here operator conjugation is a similarity, whereas transforming both groups of tensor variables gives the congruence CMC^T .

Proof. The coefficient action of C is

$$(Cf)(s) = 2^{-n/2} \prod_i (s_i + i) f\left(\frac{s_1 - i}{s_1 + i}, \dots, \frac{s_n - i}{s_n + i}\right).$$

The fractions map the upper half-plane onto the unit disk, and the prefactors are nonzero there. Thus C identifies the two stability classes. By the binary form of the contraction theorem, or Lemma 4.1, the disk-stable tensor of an invertible M is equivalent to preservation of disk-stable coefficient vectors. This proves the equivalence by similarity.

For completeness, the reverse tensor conversion can also be checked directly. The unitary matrix C_1 satisfies $C_1 C_1^T = D_1$, so the coefficient matrix of

$$\prod_i (s_i t_i - 1)$$

is $D_1^{\otimes n} = CC^T$. Each factor is stable in the two upper-half-plane variables: if $s_i t_i = 1$, then $t_i = 1/s_i$ has negative imaginary part. Acting on its first group of variables by $N = CMC^{-1}$ gives coefficient matrix $NCC^T = CMC^T$. For fixed $t \in \mathbb{H}^n$, invertibility excludes a zero output, so stability preservation gives nonvanishing jointly in $(s, t) \in \mathbb{H}^{2n}$. Finally,

$$F_{CMC^T}(s, t) = 2^{-n} \prod_i (s_i + i)(t_i + i) F_M\left(\frac{s - i}{s + i}, \frac{t - i}{t + i}\right).$$

This verifies the congruence and the sign of the half-plane kernel. As $C^{-1} = C^*$, exponential similarity yields $Ce^{-\beta H}C^{-1} = e^{\beta A}$. $\square$

6.2. The exact Hamiltonian classification. Write X, Y, Z for the Pauli matrices, with $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$. For a real 2×2 matrix, $\|\cdot\|_{\text{op}}$ denotes its largest singular value.

Theorem 6.2 (All-temperature qubit classification). *For a Hermitian operator H on $(\mathbb{C}^2)^{\otimes n}$, the following conditions are equivalent:*

- (1) $e^{-\beta H}$ has the tensor Lee–Yang property at radius one for every $\beta \geq 0$.
- (2) The same property holds for every β in some interval $[0, \varepsilon)$, where $\varepsilon > 0$.
- (3) There are real coefficients such that

$$H = h_0 I + \sum_i (h_i^x X_i + h_i^y Y_i + h_i^z Z_i) + \sum_{i < j} (a_{ij} X_i X_j + b_{ij} X_i Y_j + c_{ij} Y_i X_j + d_{ij} Y_i Y_j + e_{ij} Z_i Z_j), \quad (87)$$

where

$$h_i^z \leq 0, \quad e_{ij} \leq - \left\| \begin{pmatrix} a_{ij} & b_{ij} \\ c_{ij} & d_{ij} \end{pmatrix} \right\|_{\text{op}}. \quad (88)$$

There is no restriction on h_0, h_i^x, h_i^y .

Equivalently, in the ferromagnetic convention

$$H = h_0 I - \sum_i \sum_{a=x,y,z} \mu_i^a \sigma_i^a - \sum_{i < j} \left(J_{ij}^{zz} Z_i Z_j + \sum_{a,b=x,y} J_{ij}^{ab} \sigma_i^a \sigma_j^b \right), \quad (89)$$

the conditions are $\mu_i^z \geq 0$ and $J_{ij}^{zz} \geq \|J_{ij}^\perp\|_{\text{op}}$, where the rows and columns of $J_{ij}^\perp$ are ordered x, y .

Proof. The first condition implies the second. By Lemma 6.1, the second says that e^{tA} preserves upper-half-plane stability for small $t \geq 0$. The semigroup identity extends preservation to all $t \geq 0$: for any t , choose an integer m with $t/m < \varepsilon$ and compose $e^{(t/m)A}$ m times. We may therefore apply Theorem 3.2 on the binary capacity box.

We first spell out how differential order becomes physical support. On one binary site a basis of endomorphisms is

$$I, \quad J^- = \partial_s, \quad J^0 = s\partial_s - \frac{1}{2}, \quad J^+ = s^2\partial_s - s.$$

The last three operators are traceless; their first-order coefficients are the linearly independent polynomials $1, s, s^2$. Expand A in tensor products of this basis and choose a support set S of maximal cardinality occurring with a nonzero coefficient. The canonical coefficient of $\partial_S = \prod_{i \in S} \partial_i$ is then the sum of its exact-support terms, because no strictly larger support occurs. Those terms have distinct principal monomials $\prod_{i \in S} s_i^{r_i}$, with $r_i \in \{0, 1, 2\}$. The order-two conclusion of Theorem 3.2 forces this coefficient to vanish if $|S| \geq 3$. Linear independence then eliminates every term of that support. Descending in support eliminates all interactions on three or more sites. Local similarity preserves the identity and traceless subspaces at each site, so the same support conclusion holds for H .

Next write $A = R + iS$ with entrywise real matrices R, S . The imaginary-part conclusion of Theorem 3.2 has the form

$$S = c_0 I + \sum_i J_i(q_i), \quad q_i(s) = \alpha_i + \eta_i s + \gamma_i s^2 \geq 0 \quad (s \in \mathbb{R}), \quad (90)$$

where $J(q) = q\partial - q'/2$. On coefficients $(1, s)$,

$$J(q) = \begin{pmatrix} -\eta/2 & \alpha \\ -\gamma & \eta/2 \end{pmatrix}.$$

Since A is Hermitian, $R^T = R$ and $S^T = -S$. Taking the trace in (90) first gives $c_0 = 0$. Independence of the remaining one-site traceless components shows that each $J(q_i)$ is skew-symmetric. Thus $\eta_i = 0$ and $\gamma_i = \alpha_i$; nonnegativity gives $\alpha_i \geq 0$. In particular,

$$q_i(s) = \alpha_i(1 + s^2), \quad iS = -\sum_i \alpha_i Y_i. \quad (91)$$

The local Pauli conjugations are

$$C_1 X C_1^* = -Z, \quad C_1 Y C_1^* = X, \quad C_1 Z C_1^* = -Y. \quad (92)$$

Therefore the physical field $h_i^x X_i + h_i^y Y_i + h_i^z Z_i$ contributes $h_i^x Z_i - h_i^y X_i + h_i^z Y_i$ to A . Comparison with (91) gives $h_i^z = -\alpha_i \leq 0$. A two-site physical Pauli term containing exactly one Z becomes a term containing exactly one Y , and is entrywise imaginary. The four such terms on each pair are linearly independent. Since S has no two-site component, all physical XZ, YZ, ZX, ZY coefficients vanish. This leaves exactly the form (87).

It remains to determine the pair inequality. Suppress the pair indices. A physical term $aXX + bXY + cYX + dYY + eZZ$ contributes

$$-aZZ + bZX + cXZ - dXX - eYY$$

to A . The first-order coefficients of the coefficient operators X, Z, Y , respectively, are

$$p_X(s) = 1 - s^2, \quad p_Z(s) = -2s, \quad p_Y(s) = -i(1 + s^2).$$

Hence the full coefficient of $\partial_i \partial_j$ is

$$q_{ij}(s, t) = e(1 + s^2)(1 + t^2) - 4ast - 2bs(1 - t^2) - 2ct(1 - s^2) - d(1 - s^2)(1 - t^2). \quad (93)$$

There is no factor of two in this expression: it is the full mixed coefficient in Theorem 3.2. Define

$$u(s) = \left(\frac{2s}{1 + s^2}, \frac{1 - s^2}{1 + s^2} \right), \quad J = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Then

$$\frac{q_{ij}(s, t)}{(1 + s^2)(1 + t^2)} = e - u(s)^T J u(t). \quad (94)$$

The set $u(\mathbb{R})$ is dense in the unit circle. Consequently the required inequality $q_{ij}(s, t) \leq 0$ for all real s, t is equivalent to

$$e \leq \min_{\|u\|_2 = \|v\|_2 = 1} u^T J v = -\|J\|_{\text{op}}.$$

This proves necessity of (88).

Conversely, suppose (87)–(88) hold. The displayed conjugations show that A has canonical differential order at most two. Its mixed coefficients are real and nonpositive by (94), and its imaginary part is (90) with $q_i(s) = -h_i^z(1 + s^2) \geq 0$. Its real part therefore satisfies the real criterion in Theorem 3.2, and its imaginary part satisfies that theorem's one-site condition. The theorem proves preservation

for all $\beta \geq 0$; the bridge then gives the tensor Lee–Yang property of $e^{-\beta H}$. Finally, changing the signs of the coefficients converts (88) into (89). $\square$

Remark 6.3. The sufficient Suzuki–Fisher family originates in [53, 6, 5]. The full transverse operator-norm condition is already contained, in equivalent algebraic form, in [53, equation (2.47)] and [25, Theorem 2, equation (8)]; see also the modern notation in [18, Definition 1 and Fact 3]. Indeed, for a real matrix $J = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$2\|J\|_{\text{op}} = \sqrt{(a-d)^2 + (b+c)^2} + \sqrt{(a+d)^2 + (b-c)^2}.$$

To verify the identity, the squared radicals are $\|J\|_F^2 - 2\det J$ and $\|J\|_F^2 + 2\det J$. If the singular values are $s_1 \geq s_2 \geq 0$, the two radicals are therefore $s_1 + s_2$ and $s_1 - s_2$, in some order. Full matrix generating functions and their composition are also explicit in Asano’s contemporaneous account for the XXZ subfamily [6, equations (1.15), (3.7)–(3.10)]. Thus a distinction between classical traces and modern full tensors alone would not identify the contribution. Theorem 6.2 adds the converse among all finite Hermitian qubit Hamiltonians, including the exclusion of higher-support interactions. The proof also shows that testing only a sufficiently small interval of inverse temperatures is enough.

Corollary 6.4. *Let H be an arbitrary complex qubit matrix. If $e^{-\beta H}$ has the unit-disk tensor Lee–Yang property for all sufficiently small real $\beta \geq 0$, then its Pauli expansion contains no term supported on three or more sites.*

Proof. The Cayley bridge and the semigroup identity still apply. The complex generator theorem bounds the canonical order by two. The maximal-support basis argument in the proof of Theorem 6.2 does not use Hermiticity and therefore gives the assertion. $\square$

Remark 6.5. For a complex Hamiltonian, tensor stability does not justify trace normalization. For example, $H = \text{diag}(0, i\pi)$ has Gibbs polynomial $1 + e^{-i\pi\beta}zw$, which is nonzero on the open unit bidisk for every real β , but $\text{Tr } e^{-H} = 0$. The support conclusion in Corollary 6.4 applies to this complex setting; the normalized Gibbs-state statements use Hermiticity.

6.3. Strict positive fields and the actual Gibbs kernel.

Theorem 6.6 (Strict-field Gibbs radius). *Let H have the form (89), with $J_{ij}^{zz} \geq \|J_{ij}^\perp\|_{\text{op}}$ and $\mu_i^z > 0$ at every site. Then, for every $\beta > 0$, $F_{e^{-\beta H}}$ is nonzero on the closed unit polydisk in its full $2n$ variables. In particular, it has some radius $r_\beta > 1$.*

The ground eigenspace of H is one-dimensional. Its nonzero coefficient vector ψ defines a polynomial $f_\psi(z) = \sum_a \psi_a z^a$ of radius greater than one. For every $\beta_0 > 0$, one radius $r > 1$ works for all Gibbs kernels at $\beta \geq \beta_0$, and also for the image under each $e^{-\beta H}$ of every nonzero unit-disk-stable coefficient vector.

Proof. Let $\mu = \min_i \mu_i^z > 0$, and write $H = H_0 - \mu \sum_i Z_i$. The Hamiltonian H_0 satisfies the nonstrict Suzuki–Fisher conditions. By Theorem 6.2 and the disk version of Lemma 6.1, its coefficient operator $A_0 = -H_0$ generates a disk-stability preserver. In the multiaffine coefficient basis, $Z_i = I - 2E_i$. Thus

$$-H = A_0 - 2\mu E + \mu nI.$$

The harmless scalar factor $e^{\beta\mu n}$ reduces every kernel and evolution assertion to Theorem 4.3.

Proposition 4.5 supplies an algebraically simple dominant eigenvalue of $-H$. Since H is Hermitian, this is exactly the negative of its ground energy, so the ground space is one-dimensional. The same proposition gives the strict radius of its right eigenvector. Equivalently, one may fix a Gibbs kernel radius $r > 1$ and normalize its powers by their largest eigenvalue. These powers retain radius r by contraction, and converge to $|\psi\rangle\langle\psi|$. Their nonzero limiting polynomial is

$$f_\psi(z)\overline{f_\psi(\bar{w})}.$$

Hurwitz gives nonvanishing on $\mathbb{D}_r^{2n}$, and therefore nonvanishing of each factor on $\mathbb{D}_r^n$. $\square$

Remark 6.7. The spectral conclusion

$$E_1(H) - E_0(H) \geq 2 \min_i \mu_i^z$$

is the Hermitian specialization of Theorem 4.8 and was already proved for this family in [18, Theorem 1]. Here the additional conclusion is strict nonvanishing of the actual Gibbs tensor at every positive inverse temperature, and its transfer to the ground-state polynomial. No numerical lower bound for $r_\beta - 1$, uniform in system size, is asserted. This distinction also separates the actual kernel from the strict kernels used in a product-formula approximation.

6.4. A qualitative radius for deformed EPR ground states. Consider a finite graph without isolated vertices, with positive edge weights w_e . On an edge $e = \{i, j\}$, fix $0 \leq s_e < 1$ and $\theta_e \in \mathbb{R}$, and define

$$|\eta_e\rangle = |00\rangle + s_e e^{i\theta_e} |11\rangle, \quad H_{\text{EPR}} = - \sum_e w_e |\eta_e\rangle\langle\eta_e|_{ij}.$$

The edge vectors here are unnormalized; replacing them by normalized vectors simply rescales the positive weights.

Corollary 6.8. *The ground space of H_{EPR} is one-dimensional, and its ground-state polynomial is zero-free on a polydisk of some radius $r > 1$.*

Proof. On one edge, direct expansion in Pauli matrices gives

$$\begin{aligned} |\eta_e\rangle\langle\eta_e| &= \frac{1+s_e^2}{4}(I + Z_i Z_j) + \frac{1-s_e^2}{4}(Z_i + Z_j) \\ &\quad + \frac{s_e}{2} [\cos \theta_e (X_i X_j - Y_i Y_j) + \sin \theta_e (X_i Y_j + Y_i X_j)]. \end{aligned}$$

Its ferromagnetic coefficients are therefore

$$J_e^{zz} = \frac{w_e(1+s_e^2)}{4}, \quad J_e^\perp = \frac{w_e s_e}{2} \begin{pmatrix} \cos \theta_e & \sin \theta_e \\ \sin \theta_e & -\cos \theta_e \end{pmatrix}.$$

The displayed real matrix is orthogonal, so $\|J_e^\perp\|_{\text{op}} = w_e s_e/2$ and $J_e^{zz} - \|J_e^\perp\|_{\text{op}} = w_e(1-s_e)^2/4 \geq 0$. The longitudinal field at site i is

$$\mu_i^z = \sum_{e \ni i} \frac{w_e(1-s_e^2)}{4} > 0.$$

All hypotheses of Theorem 6.6 follow, including strictness because every vertex has an incident edge. Apply that theorem to obtain the stated uniqueness and radius. $\square$

For a common deformation parameter $s < 1$, the corollary gives the qualitative strict-radius conclusion asked for in [61, Conjecture 1]. It does not assert a prescribed sharp radius as a function of s . Edge phases may be chosen independently; no simultaneous gauge removal is used in the proof.

7. PHYSICAL QUANTUM MARKOV DYNAMICS

The zero-freeness of a Gibbs operator and the preservation of zero-freeness by a trace-preserving physical evolution impose different restrictions. We now classify the latter. The physical tensor decomposition and the computational basis are fixed throughout this section.

For a matrix M on n qubits, put

$$F_M(z, w) = \sum_{a, b \in \{0, 1\}^n} M_{ab} z^a w^b. \quad (95)$$

A density matrix is called *Lee–Yang* if this polynomial does not vanish on $\mathbb{D}^{2n}$. Thus the row and column variables are independent. For a pure state, $F_{|\psi\rangle\langle\psi|}(z, w) = f_\psi(z) \overline{f_\psi(w)}$, where $f_\psi(z) = \sum_a \psi_a z^a$. A channel is *ordinarily preserving* if it preserves these density matrices on its given system, and is *completely preserving* if it does so after adjoining any number of inert qubits. Complete positivity of a channel and complete preservation in this sense are distinct conditions.

Write $\sigma_x = X$, $\sigma_y = Y$, and $\sigma_z = Z$, with $Z = \text{diag}(1, -1)$. The Pauli matrices in all generator formulas below are unnormalized. We use the standard finite-dimensional GKLS representation [29, 40]. If $\mathcal{P}_n$ denotes the nonidentity Hermitian Pauli strings, every generator of a continuous CPTP semigroup has a representation

$$\mathcal{L}(\rho) = -i[H, \rho] + \sum_{P, Q \in \mathcal{P}_n} K_{PQ} (P\rho Q - \tfrac{1}{2}\{QP, \rho\}), \quad H = H^*, \quad K \succeq 0. \quad (96)$$

Identity components in the noise have been absorbed into the Hamiltonian.

Theorem 7.1 (Ordinary and complete preservation). *Let $n \geq 2$, and let $\Phi_t = e^{t\mathcal{L}}$ be a continuous CPTP semigroup on n qubits. The following conditions are equivalent.*

- (1) Φ_t ordinarily preserves Lee–Yang density matrices for every $t \geq 0$.
- (2) The same preservation holds for all t in some interval $[0, \varepsilon)$.
- (3) Φ_t completely preserves Lee–Yang density matrices for every $t \geq 0$.
- (4) The full Choi tensor of Φ_t is Lee–Yang for every $t \geq 0$.
- (5) The generator is a sum of one-qubit generators,

$$\mathcal{L} = \sum_{j=1}^n \mathcal{L}_j \otimes \text{Id}_j, \quad \mathcal{L}_j(\rho) = -i[h_j Z, \rho] + \sum_{\alpha, \beta = x, y, z} (K_j)_{\alpha\beta} (\sigma_\alpha \rho \sigma_\beta - \tfrac{1}{2}\{\sigma_\beta \sigma_\alpha, \rho\}), \quad (97)$$

where $h_j \in \mathbb{R}$ and, for a real symmetric 2×2 matrix B_j , a vector $d_j \in \mathbb{R}^2$, and $c_j \in \mathbb{R}$,

$$K_j = \begin{pmatrix} B_j & id_j \\ -id_j^\top & c_j \end{pmatrix} \succeq 0, \quad \begin{pmatrix} c_j I_2 - \text{adj } B_j & d_j \\ d_j^\top & \text{Tr } B_j \end{pmatrix} \succeq 0. \quad (98)$$

In this case $\Phi_t = \bigotimes_j e^{t\mathcal{L}_j}$. The equivalence between ordinary and complete preservation is false on one qubit.

We prove the theorem in four steps. A fixed finite family of product inputs first excludes every interaction. Two-site pure states then fix the reality of the remaining local coefficients. Finally, positive definite separable states recover the individual local inequalities even though both sites evolve. The resulting inequalities are reduced to (98) without dividing by a possibly vanishing trace.

7.1. A finite family detecting every interaction.

Lemma 7.2 (A one-sided root collision). *Let g_t be a differentiable curve in a finite-dimensional polynomial space, with $g_0 \neq 0$. Suppose 0 is a root of g_0 of multiplicity m , and g_t is upper-half-plane stable along a sequence $t \downarrow 0$ of positive times. If $m \geq 3$, then $\dot{g}_0(0) = 0$. If $m = 2$ and $g_0(x) = cx^2 + O(x^3)$, then $\dot{g}_0(0)/c$ is real and nonpositive.*

Proof. Choose a small circle about zero containing no other root of g_0 . The nearby cluster has a monic factor $b_t(x) = x^m + a_1(t)x^{m-1} + \cdots + a_m(t)$ with $b_0 = x^m$. Its coefficients are differentiable: the contour integrals of $x^k g'_t(x)/g_t(x)$ give the cluster power sums, and Newton identities give its elementary symmetric functions. This construction also applies when the ambient degree changes. The complementary factor is nonzero at zero.

At each of the specified stable times, write the cluster roots as $\lambda_j = x_j + iy_j$, where $y_j \leq 0$. Differentiability of the first power sum gives $\sum_j |y_j| = O(t)$. The second power sum is also $O(t)$, so its real part gives $\sum_j x_j^2 = O(t)$, since $\sum_j y_j^2 = O(t^2)$. Thus every $\lambda_j = O(\sqrt{t})$ and $\sum_j \lambda_j^k = o(t)$ for $k \geq 3$. Newton identities imply

$$a_k(t) = o(t) \quad (k \geq 3), \quad a_2(t) = -\frac{1}{2} \sum_j x_j^2 + o(t).$$

For the second equality, the imaginary part of the second power sum is $O(t^{3/2})$ and $a_1(t)^2 = O(t^2)$. Consequently $a_2'(0)$ is real and nonpositive, while $a_k'(0) = 0$ for $k \geq 3$. Multiplication by the complementary factor proves the assertions. All derivatives are fixed by the differentiable curve, so estimates along the given sequence suffice. $\square$

Theorem 7.3 (Finite equatorial tests). *Let $\Phi_t = e^{t\mathcal{L}}$ be a continuous CPTP semigroup on n qubits. Set $\mathcal{E} = \{0, \sqrt{3}, -\sqrt{3}\}$ and*

$$\psi_a = \frac{(1+ia)|0\rangle + (1-ia)|1\rangle}{\sqrt{2(1+a^2)}}, \quad \rho_a = \bigotimes_{j=1}^n |\psi_{a_j}\rangle\langle\psi_{a_j}|, \quad \mathbf{a} \in \mathcal{E}^n.$$

Suppose that, for each of these 3^n densities, there is a sequence of positive times tending to zero along which its image is Lee–Yang. Then $\mathcal{L}$ is a sum of one-qubit GKLS generators. The sequences may depend on the input.

Proof. The coefficient Cayley transform is

$$C = \frac{1}{\sqrt{2}} \begin{pmatrix} i & -i \\ 1 & 1 \end{pmatrix}, \quad f(z) \mapsto \frac{s+i}{\sqrt{2}} f\left(\frac{s-i}{s+i}\right). \quad (99)$$

It takes disk stability to upper-half-plane stability. Applied to all row and column legs of a density tensor, it is a coefficient transformation, not a unitary conjugation of that density matrix. Direct multiplication gives $C\psi_a = (-a, 1)^T/\sqrt{1+a^2}$

and $\overline{C\psi_a} = (a, 1)^T / \sqrt{1 + a^2}$. The transformed input is therefore a nonzero scalar multiple of

$$p_{\mathbf{a}}(z, w) = \prod_j (z_j - a_j)(w_j + a_j). \quad (100)$$

For a binary coefficient matrix $M = (M_{ab})$, its polynomial action has the first-order form

$$M_{00} + M_{10}s + [M_{01} + (M_{11} - M_{00})s - M_{10}s^2]\partial_s.$$

The identities $CXC^* = -Z$, $CYC^* = X$, $CZC^* = -Y$ consequently give the derivative coefficients

$$v(s) = (2s, 1 - s^2, i(1 + s^2)) \quad (101)$$

for left multiplication. Right multiplication uses transposed Paulis; its derivative vector at a general column coordinate t is $\tilde{v}(t) = (2t, -(1 - t^2), i(1 + t^2))$. In particular $\tilde{v}(-a) = -\overline{v(a)}$.

Use (96). A zero diagonal entry of a positive semidefinite matrix has a zero row and column. Suppose that $k \geq 2$ is the greatest weight of a Pauli string with $K_{PP} > 0$, and choose such a string with support S , $|S| = k$. All noise rows of larger weight vanish. Put both coordinates of each site in S at their roots in (100), keeping every other coordinate away from its root, and restrict to a line with every coordinate slope positive. The initial polynomial has a root of multiplicity $2k \geq 4$. Lemma 7.2 makes its first variation zero at this point.

Hamiltonian and anticommutator terms act only on row legs or only on column legs, so leave selected roots untouched. A sandwich $P\rho Q$ can survive only if both supports contain S . Maximality makes both supports exactly S . Dividing out the nonzero complementary product and the input scalar therefore gives

$$0 = (-1)^k \sum_{\text{supp } P = \text{supp } Q = S} K_{PQ} V_P(\mathbf{a}) \overline{V_Q(\mathbf{a})}, \quad V_P(\mathbf{a}) = \prod_{j \in S} v_{P_j}(a_j). \quad (102)$$

The sum, without its sign, is a positive semidefinite quadratic form. Thus the exact-support block K_{SS} kills $\overline{V(\mathbf{a})}$ for every $\mathbf{a} \in \mathcal{E}^S$. The three components in (101) form a basis of the quadratic polynomials. Evaluation at the three distinct points of $\mathcal{E}$ is invertible. The resulting tensor products span $\mathbb{C}^{3^k}$, forcing $K_{SS} = 0$, a contradiction. Every surviving noise index consequently has weight one. This still permits cross-site noise blocks; we eliminate them below.

The anticommutator operator $\sum K_{PQ}QP$ now has support at most two. If a Hamiltonian term has maximal support S with $|S| = k \geq 3$, put just the row coordinates of S at their roots. All column coordinates and remaining row coordinates stay away from their roots. The k -fold contact again has zero first variation. No noise term, anticommutator, or column Hamiltonian removes all these row roots. The surviving row Hamiltonian equals a nonzero common factor times

$$-i \sum_{\text{supp } P = S} h_P \prod_{j \in S} v_{P_j}(a_j).$$

The same finite tensor evaluation basis forces every displayed h_P to vanish. Descending in the maximal support leaves only Hamiltonian terms of weight at most two. The transformed superoperator now acts on at most two individual row or column legs at a time.

Fix distinct physical sites i, j . A row-row double contact with coordinates $s, t \in \mathcal{E}$ uses input parameters $a_i = s, a_j = t$. A row-column contact uses

$a_i = s, a_j = -t$, which is available because $\mathcal{E} = -\mathcal{E}$. Keep all other coordinates off their roots. Only the corresponding mixed principal coefficient survives. Dividing by the initial quadratic coefficient divides that coefficient by the positive product of the two selected slopes. Lemma 7.2 therefore makes both mixed coefficients real and nonpositive on this nine-point grid.

Let K^{ij} be the 3×3 cross-site noise block, and write the Hamiltonian pair as $\sum_{\alpha,\beta} h_{\alpha\beta} \sigma_{i,\alpha} \sigma_{j,\beta}$ with h real. The row-column coefficient matrix is K^{ij} . Since distinct-site Paulis commute, the two conjugate noise blocks in the anticommutator combine to give the row-row coefficient matrix

$$-\Re K^{ij} - ih. \quad (103)$$

Here the real part is taken entrywise; there is no transpose in this formula.

Put

$$u(s) = \frac{(2s, 1 - s^2)}{1 + s^2}, \quad R = \text{diag}(1, -1).$$

The normalized row and column derivative vectors are $(u(s), i)$ and $(Ru(t), i)$. Uniform averaging over $\mathcal{E}$ gives

$$\mathbb{E}u = 0, \quad \mathbb{E}uu^\top = \frac{1}{2}I_2. \quad (104)$$

Initially write $K^{ij} = \begin{pmatrix} A & B \\ D & e \end{pmatrix}$ with complex blocks. The imaginary part of its normalized row-column coefficient is the imaginary part of $u^\top ARv + iu^\top B + iDRv - e$. Its nine values vanish. Averaging these values, and then their products with u , Rv , and $u(Rv)^\top$, proves $\Im e = 0$, $\Re B = 0$, $\Re D = 0$, and $\Im A = 0$. Hence $B = ib$, $D = id^\top$ with A, b, d, e real, and

$$u^\top ARv - b^\top u - d^\top Rv - e \leq 0. \quad (105)$$

The imaginary part obtained from (103) is $-u^\top h_{TT}v + h_{zz}$, where $T = \{x, y\}$. The same moments give $h_{TT} = 0 = h_{zz}$. Write $k = h_{Tz}$ and $\ell^\top = h_{zT}$. The real row-row inequality is

$$e - u^\top Av + k^\top u + \ell^\top v \leq 0. \quad (106)$$

Both inequalities have already been divided by the positive factor $(1 + s^2)(1 + t^2)$. Their unweighted averages are respectively $-e$ and e , so $e = 0$. Each of the nine nonpositive values in each inequality has zero average, and thus each value is zero. Applying (104) once more gives $A = b = d = k = \ell = 0$. Consequently $K^{ij} = 0$ and the pair Hamiltonian vanishes.

Every remaining block K^{ii} is positive semidefinite. The remaining terms of (96) are therefore genuine one-qubit GKLS generators. They commute on distinct tensor factors. The proof compared only derivatives at zero, so the different stable time sequences present no compatibility issue. $\square$

Corollary 7.4. *If a CPTP generator is not a sum of one-qubit generators, one of the fixed inputs in Theorem 7.3 has a non-Lee-Yang output at every sufficiently small positive time.*

Proof. Otherwise each input would have Lee-Yang output at positive times arbitrarily close to zero, providing the sequences in that theorem. $\square$

The theorem makes no assertion that its number of tests is minimal or that testing the output polynomial is computationally efficient. Nor can a single isolated positive time replace the sequences: the interaction $H = Z_1 Z_2$ generates the identity channel at $t = \pi$.

7.2. The local reality forced by a second physical site. We first record the elementary one-qubit state geometry.

Lemma 7.5. *A qubit density matrix $\rho = (I + r \cdot \sigma)/2$ is Lee–Yang if and only if $r_z \geq 0$.*

Proof. Write $\rho = \begin{pmatrix} a & b \\ \bar{b} & d \end{pmatrix}$, so $a + d = 1$, $|b|^2 \leq ad$, and $F_\rho(z, w) = g(w) + zh(w)$ with $g = a + bw$, $h = \bar{b} + dw$. On $|w| = 1$,

$$|g(w)|^2 - |h(w)|^2 = (a - d)[a + d + 2\Re(bw)]. \quad (107)$$

If $a > d$, then $a > |b|$, so g is zero-free on the closed disk. The right side is nonnegative. The maximum principle applied to h/g proves $|h/g| \leq 1$ on the disk and hence the desired zero-freeness. If $a = d$, add a positive multiple of $|0\rangle\langle 0|$ and take its nonzero coefficient limit; Hurwitz’s theorem gives the assertion. If $a < d$, positivity gives $d > |b|$, and (107) is strictly negative on the circle. Thus a nearby interior point has $|h| > |g|$ and $h \neq 0$; choosing $z = -g/h$ gives an interior zero. This proves both directions. $\square$

Proposition 7.6. *Suppose $n \geq 2$ and a CPTP semigroup ordinarily preserves Lee–Yang densities on a neighborhood of time zero. In the local representation given by Theorem 7.3, each noise matrix and Hamiltonian have the form*

$$K = \begin{pmatrix} B & id \\ -id^\top & c \end{pmatrix}, \quad B = B^\top \in \mathbb{R}^{2 \times 2}, \quad d \in \mathbb{R}^2, \quad H = hZ. \quad (108)$$

Writing $T = \text{Tr } B$ and $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, the local Bloch equation is

$$\dot{r}_\perp = Mr_\perp + k, \quad \dot{r}_z = -2Tr_z, \quad M = 2[B - (T + c)I_2] + 2hJ, \quad k = 4Jd. \quad (109)$$

In particular the finite-time map has the form

$$(r_\perp, r_z) \mapsto (D_t r_\perp + \tau_t, \lambda_t r_z), \quad D_t \in GL_2(\mathbb{R}), \quad \lambda_t = e^{-2Tt} > 0. \quad (110)$$

Proof. After locality is known, let $q_j(s, w)$ denote the coefficient of $\partial_s \partial_w$ in the Cayley-transformed generator at site j . For a general Hermitian noise matrix write

$$K = \begin{pmatrix} \alpha & \delta + i\eta & r + i\mu \\ \delta - i\eta & \beta & t + i\nu \\ r - i\mu & t - i\nu & \gamma \end{pmatrix}.$$

Multiplying the row and column derivative vectors in (101) gives

$$\Im q(s, w) = 2(s + w)[- \eta(1 - sw) + r(1 + sw) + t(w - s)]. \quad (111)$$

The Hamiltonian and anticommutator act on single legs and do not enter this coefficient.

Fix two sites. For real a, b and $c_0 > 0$, the polynomial $f(z_1, z_2) = (z_1 - a)(z_2 - b) - c_0$ is upper-half-plane stable: solving for z_2 when $z_1 \in \mathbb{H}$ gives a point in the lower half-plane. Its inverse Cayley transform defines a pure physical Lee–Yang input. The transformed conjugate factor is $g(w_1, w_2) = (w_1 + a)(w_2 + b) - c_0$, since $\bar{C} = \text{diag}(-1, 1)C$ and there are two sites. For arbitrary nonzero real u, v , choose the real contact

$$z_1 = a + u, \quad z_2 = b + c_0/u, \quad w_1 = -a + v, \quad w_2 = -b + c_0/v.$$

Here $f = g = 0$, $\nabla f = (c_0/u, u)$ and $\nabla g = (c_0/v, v)$. Their directional derivatives along any strictly positive direction are nonzero. The initial product has a double

root with nonzero real leading coefficient. Lower-order generator terms vanish at the contact, and its first variation is

$$h = \frac{c_0^2}{uv} q_1(a+u, -a+v) + uv q_2(b+c_0/u, -b+c_0/v). \quad (112)$$

Lemma 7.2 makes h real. Additional sites may be padded by equatorial factors kept away from their roots; their generators contribute zero at this contact.

Formula (111) gives $\Im q_2(b, -b) = 0$. The coefficient of c_0 in the imaginary part of (112) is therefore

$$v \partial_s \Im q_2(b, -b) + u \partial_w \Im q_2(b, -b) = 0.$$

Independent variation of u, v makes both derivatives vanish. By (111), this says $-\eta(1+b^2) + r(1-b^2) - 2tb = 0$ for every real b . Its three coefficients give $\eta = r = t = 0$. Interchanging the sites proves the same conclusion at the other site, and every site has a partner. This proves the stated shape of K .

At an equatorial Bloch vector the dissipator with this shape has zero longitudinal velocity. A Hamiltonian $h_x X + h_y Y + h_z Z$ contributes $\dot{r}_z = 2(h_x r_y - h_y r_x)$. Product inputs and Lemma 7.5 require this quantity to be nonnegative at every equatorial point. Replacing that point by its negative gives $h_x = h_y = 0$. Finally, multiplication of the Pauli matrices in the generator gives (109); in particular $\mathcal{L}(I) = -4d_y X + 4d_x Y$, fixing the displacement sign. The linear differential equation gives $D_t = e^{tM}$ and $\lambda_t = e^{-2Tt}$, as claimed. $\square$

7.3. Separable conditional blocks prevent compensation.

Lemma 7.7 (Conditional-block cancellation). *Let a qubit channel Ψ have the affine form $(r_\perp, r_z) \mapsto (Dr_\perp + \tau, \lambda r_z)$, with $\lambda > 0$. If $\alpha, \beta \in \mathbb{C}$ satisfy $|\beta| > |\alpha|$, the polynomial of $\alpha\Psi(|0\rangle\langle 0|) + \beta\Psi(|1\rangle\langle 1|)$ has a zero in $\mathbb{D}^2$.*

Proof. The scalar $\alpha + \beta$ is nonzero. Dividing the matrix by $(\alpha + \beta)/2$ gives $I + \tau \cdot \sigma_\perp + qZ$, where

$$q = \lambda \frac{\alpha - \beta}{\alpha + \beta}, \quad \Re q = \lambda \frac{|\alpha|^2 - |\beta|^2}{|\alpha + \beta|^2} < 0.$$

Its polynomial is $(1+q) + (\tau_x + i\tau_y)z + (\tau_x - i\tau_y)w + (1-q)zw$. Choose a phase θ making $(\tau_x + i\tau_y)e^{i\theta} = |\tau|$ and substitute $z = e^{i\theta}u$, $w = e^{-i\theta}u$; if $\tau = 0$ any phase works. The quadratic has nonzero leading coefficient $1-q$ and root product $(1+q)/(1-q)$ of modulus less than one. At least one root has modulus less than one, giving the asserted pair of disk coordinates. $\square$

The lemma applies to the actual evolving channel at a second site. It does not require that this channel fix the reference state. In particular, if positive matrices A, B give a Lee–Yang input proportional to $A \otimes |0\rangle\langle 0| + B \otimes |1\rangle\langle 1|$, ordinary preservation by $\Phi \otimes \Psi$ forces

$$|F_{\Phi(B)}(z, w)| \leq |F_{\Phi(A)}(z, w)| \quad (z, w \in \mathbb{D}). \quad (113)$$

Indeed a strict reverse inequality supplies α, β for Lemma 7.7. This also covers $\alpha = 0$.

Proposition 7.8 (Recovery of each local inequality). *Under the assumptions of Proposition 7.6, each local matrix in (108) satisfies*

$$r^\top Bv + d \cdot (v - r) \leq c \quad (r, v \in \mathbb{R}^2, |r| = |v| = 1). \quad (114)$$

Necessity follows from positive definite separable inputs classical on one of two selected sites.

Proof. Fix an active site and a distinct reference site. For $0 < \kappa < 1/2$ and $e \in \mathbb{R}^2$ with $|e| = 1$, put

$$U = I + \frac{1}{2}Z, \quad V = \frac{1}{8}I + \frac{\sqrt{\kappa(1-\kappa)}}{4}\sigma_e + \frac{\kappa}{4}Z, \quad \sigma_e = e_x X + e_y Y,$$

and define

$$A = \frac{1}{2}(U + V), \quad B_0 = \frac{1}{2}(U - V), \quad \rho_{\kappa,e} = \frac{1}{2}[A \otimes |0\rangle\langle 0| + B_0 \otimes |1\rangle\langle 1|]. \quad (115)$$

Both blocks are positive definite, because $\lambda_{\min}(U) = 1/2$ and $\|V\| = 1/8 + \sqrt{\kappa}/4 < 1/2$. Their traces sum to 2, so this is a normalized positive definite separable density matrix.

For a Hermitian matrix $Q = q_0 I + q_{\perp} \cdot \sigma_{\perp} + q_z Z$, parameterize the distinguished boundary by $z = e^{i(\gamma+\delta)}$, $w = e^{i(\gamma-\delta)}$, and set $x = \cos \gamma$, $y = \sin \gamma$, $n = (\cos \delta, -\sin \delta)$. Expansion of (95) gives

$$\frac{1}{2}e^{-i\gamma}F_Q(z, w) = q_0 x + q_{\perp} \cdot n - iq_z y. \quad (116)$$

Write $u = 1/2$, $v_0 = 1/8$, $w_0 = 2v_0\sqrt{\kappa(1-\kappa)}$ and $\zeta = \kappa v_0/u$, so $U = I + uZ$ and $V = v_0 I + w_0 \sigma_e + \zeta Z$. Formula (116) yields

$$\begin{aligned} \mathcal{D}_0(x, n) &:= \frac{1}{4}(|F_A|^2 - |F_{B_0}|^2) \\ &= v_0[(1-\kappa)x^2 + 2\sqrt{\kappa(1-\kappa)}x(e \cdot n) + \kappa] \\ &\geq v_0(\sqrt{1-\kappa}|x| - \sqrt{\kappa})^2 \geq 0. \end{aligned} \quad (117)$$

The matrix A is positive definite with strictly positive Z coefficient. Consequently its diagonal entries obey $a > d > 0$, and (107) is strictly positive on the circle. The proof of Lemma 7.5 then gives zero-freeness of F_A on the closed bidisk. Applying the maximum principle separately in z, w to F_{B_0}/F_A extends (117) to that bidisk. Hence $F_A + z_2 w_2 F_{B_0}$ is nonzero on $\mathbb{D}^4$, proving that (115) is Lee–Yang. Other sites may be padded by maximally mixed states.

For the active map write D_t, τ_t, λ_t as in (110). At time zero their derivatives are $M, k, -2T$ in (109). The reference map has $\lambda_t > 0$ at every finite time. By (113), ordinary preservation forces the active output block difference to be nonnegative on the disk and, by continuity, on its boundary. Its exact normalized expression is

$$\mathcal{D}_t(x, n) = (x + \tau_t \cdot n)[v_0(x + \tau_t \cdot n) + w_0 n^{\top} D_t e] + \lambda_t^2 u \zeta (1 - x^2). \quad (118)$$

Choose $n = -e$ and $x = \rho := \sqrt{\kappa/(1-\kappa)} \in (0, 1)$. The input difference is zero. Its derivative must therefore be nonnegative. Differentiating (118) gives

$$\begin{aligned} \frac{\mathcal{D}'_0}{2v_0\kappa} &= \rho k \cdot n - n^{\top} M n - 2T \frac{1-2\kappa}{1-\kappa} \\ &= 2(c - n^{\top} B n) + 2T \rho^2 + \rho k \cdot n \geq 0. \end{aligned} \quad (119)$$

For explicit verification, the three derivative contributions before division are $(k \cdot n)(2v_0\rho - w_0)$, $-\rho w_0 n^{\top} M n$, and $-4T u \zeta (1 - \rho^2)$. The Hamiltonian term drops out because $n^{\top} J n = 0$.

Set $n = Jm$, $|m| = 1$. Since $k = 4Jd$ and $(Jm)^{\top} B (Jm) = T - m^{\top} B m$, we obtain

$$c - T + m^{\top} B m + T \rho^2 + 2\rho m \cdot d \geq 0 \quad (|m| = 1, 0 \leq \rho \leq 1). \quad (120)$$

The endpoints follow by continuity. For arbitrary unit r, v , set $(v - r)/2 = \rho m$. Their mean is perpendicular to m and has squared length $1 - \rho^2$; when $\rho = 0$, choose either perpendicular direction for m . In two dimensions this gives

$$r^\top Bv + d \cdot (v - r) = T - m^\top Bm - T\rho^2 + 2\rho m \cdot d.$$

Apply (120) with $-m$ to prove (114). $\square$

7.4. The exact local semidefinite cone.

Lemma 7.9 (Reduction of the circle inequality). *Suppose $B = B^\top \in \mathbb{R}^{2 \times 2}$ and $\begin{pmatrix} B & id \\ -id^\top & c \end{pmatrix} \succeq 0$. Then (114) is equivalent to*

$$\begin{pmatrix} cI_2 - \text{adj } B & d \\ d^\top & \text{Tr } B \end{pmatrix} \succeq 0. \quad (121)$$

If $T = \text{Tr } B > 0$, the equivalent scalar condition is $c \geq \lambda_{\max}(\text{adj } B + dd^\top/T)$. If $T = 0$, positivity forces $B = d = 0$, and every $c \geq 0$ is allowed.

Proof. Unitary congruence, followed if necessary by changing the sign of the last coordinate, gives $\begin{pmatrix} B & d \\ d^\top & c \end{pmatrix} \succeq 0$. In particular $B \succeq 0$ and $dd^\top \preceq cB$. The assertion for $T = 0$ follows immediately. Suppose $T > 0$. Writing $(v - r)/2 = \rho m$ as above, the expression to maximize is

$$T - m^\top Bm - T\rho^2 + 2\rho d \cdot m, \quad |m| = 1, \quad 0 \leq \rho \leq 1.$$

If $c < T$, then $|d|^2 \leq c\lambda_{\max}(B) < T^2$. Choose the sign of m so $d \cdot m \geq 0$. The unconstrained maximum in ρ is attained at $\rho = |d \cdot m|/T < 1$. The maximum is consequently

$$\lambda_{\max} \left(TI_2 - B + \frac{dd^\top}{T} \right) = \lambda_{\max} \left(\text{adj } B + \frac{dd^\top}{T} \right).$$

This proves equivalence in this case.

If $c \geq T$, put $b_m = m^\top Bm$ and use $|d \cdot m| \leq \sqrt{cb_m}$. The difference between c and the expression above is bounded below by

$$c - T + b_m + T\rho^2 - 2\sqrt{cb_m}\rho = (\sqrt{b_m} - \sqrt{c}\rho)^2 + (c - T)(1 - \rho^2) \geq 0.$$

The scalar matrix condition also holds automatically, because

$$dd^\top \preceq cB \preceq T[B + (c - T)I_2],$$

the second inequality being equivalent to $(c - T)(TI_2 - B) \succeq 0$. Finally, the Schur complement with positive last entry T proves equivalence of the scalar condition and (121). The separate $c \geq T$ argument avoids an invalid unconstrained radial maximization when $|d| > T$. $\square$

Lemma 7.10 (Choi tensors and complete preservation). *For a CPTP map Φ on n qubits, the following are equivalent: its full Choi tensor is Lee–Yang; it completely preserves physical Lee–Yang states; and its extension by n inert qubits takes the maximally entangled state to a Lee–Yang state. For any invertible complex-linear map, without a positivity or trace assumption, its full Choi tensor is Lee–Yang if and only if its coefficient action preserves all complex disk-stable polynomials in the $2n$ row and column variables.*

Proof. With $|\Omega\rangle = \sum_a |a\rangle|a\rangle$, the Choi entries are $J_{ai,bj} = [\Phi(|i\rangle\langle j|)]_{ab}$. The normalized maximally entangled input is Lee–Yang, since its polynomial is proportional to $\prod_k (1 + z_k z'_k)(1 + w_k w'_k)$. Its output is the normalized Choi matrix. Conversely, for a joint input ρ , the output coefficients are

$$[(\Phi \otimes \text{Id})(\rho)]_{ar,bs} = \sum_{i,j} J_{ai,bj} \rho_{ir,js}.$$

This is bilinear tensor contraction with no conjugation of entries. Lemma 2.3 gives a Lee–Yang polynomial or zero; trace preservation excludes zero. This proves the physical equivalences.

Let M be the superoperator matrix in coefficient coordinates. Its full tensor differs from the Choi tensor only by a permutation of indices. Contraction with an arbitrary stable coefficient vector gives stability or zero, and invertibility excludes zero. In the other direction, if M preserves all stable polynomials, apply it for each fixed $w \in \mathbb{D}^{2n}$ to $\prod_j (1 + z_j w_j)$. The resulting joint polynomial is exactly the full tensor of M and is nonzero for every $(z, w) \in \mathbb{D}^{4n}$. $\square$

Proposition 7.11 (One-qubit complete generators). *A one-qubit CPTP semigroup is completely preserving if and only if its generator has the form (108) and satisfies the two positive semidefinite conditions in (98).*

Proof. For necessity, extend the semigroup by one inert qubit. Propositions 7.6 and 7.8 give the real local shape and (114) for the active site. Lemma 7.9 gives the two matrix conditions.

For sufficiency, transform both coefficient legs by (99). Write $B = \begin{pmatrix} a & u \\ u & b \end{pmatrix}$ and $d = (x, y)^\top$. The transformed generator is real. This follows directly from the Pauli identities: the mixed terms pair the purely imaginary transverse–longitudinal noise entries with the purely imaginary Z derivative; the anticommutator is $-(a + b + c)I + y(Z_1 + Z_2) + x(X_1 - X_2)$; and the longitudinal Hamiltonian is real after the transform. Its full coefficient of $\partial_s \partial_t$ is

$$\begin{aligned} q(s, t) = & 4ast - b(1 - s^2)(1 - t^2) - c(1 + s^2)(1 + t^2) \\ & + 2u[t(1 - s^2) - s(1 - t^2)] \\ & + 2x[t(1 + s^2) - s(1 + t^2)] - y[(1 - s^2)(1 + t^2) + (1 + s^2)(1 - t^2)]. \end{aligned} \quad (122)$$

There is no extra factor of two in this convention. With

$$r(s) = \frac{(2s, 1 - s^2)}{1 + s^2}, \quad v(t) = \frac{(2t, t^2 - 1)}{1 + t^2},$$

we have

$$\frac{q(s, t)}{(1 + s^2)(1 + t^2)} = r(s)^\top B v(t) + d \cdot (v(t) - r(s)) - c \leq 0. \quad (123)$$

The real two-variable specialization of the finite-box generator theorem (Theorem 3.2) says that a real endomorphism of the binary polynomial box generates upper-half-plane stability preservers precisely when this mixed coefficient is nonpositive on $\mathbb{R}^2$. It applies here because the transformed operator is an endomorphism of that box. Thus its exponential preserves complex stability. Undoing the Cayley similarity and applying Lemma 7.10, with invertibility of the exponential, proves complete preservation. $\square$

Proof of Theorem 7.1. Assume preservation on a neighborhood of zero. Theorem 7.3 gives locality. Propositions 7.6, 7.8, and Lemma 7.9 give (97)–(98). Conversely those conditions give complete preservation of every local semigroup by Proposition 7.11. Their tensor product completely preserves by successive application to the individual sites. Lemma 7.10 equates this with the full Choi condition. Complete preservation implies ordinary preservation, which implies its small-time version. The one-qubit exception is proved next. $\square$

7.5. The Hermiticity-preserving branch without positivity. For the full Choi condition, locality does not require complete positivity. The local semidefinite reduction does require it, and must be replaced by the circle inequality when positivity is absent.

Theorem 7.12 (Hermiticity- and trace-preserving semigroups). *Let $T_t = e^{t\mathcal{L}}$ be a continuous semigroup of complex-linear, Hermiticity-preserving, trace-preserving maps on n qubits. Its full Choi tensor is Lee–Yang for every $t \geq 0$ if and only if $\mathcal{L} = \sum_j \mathcal{L}_j \otimes \text{Id}_j$, where each $\mathcal{L}_j$ has the algebraic form (97), with*

$$H_j = h_j Z, \quad K_j = \begin{pmatrix} B_j & id_j \\ -id_j^\top & c_j \end{pmatrix}, \quad B_j = B_j^\top \in \mathbb{R}^{2 \times 2}, \quad d_j \in \mathbb{R}^2, \quad c_j, h_j \in \mathbb{R},$$

and satisfies

$$c_j \geq \max_{r, v \in S^1} [r^\top B_j v + d_j \cdot (v - r)]. \quad (124)$$

Here no positive semidefiniteness of K_j is imposed. Requiring the Choi property only on a neighborhood of zero is equivalent. If complete positivity is also imposed, the local conditions reduce exactly to (98).

Proof. We first justify the algebraic representation used in the proof. A Hermiticity-preserving, trace-annihilating generator has a representation (96) with $H = H^*$ and $K = K^*$, without the condition $K \succeq 0$. Indeed, its Choi matrix is Hermitian. In the orthogonal vectorized Pauli basis, its compression to the orthogonal complement of $|I\rangle\rangle$ uniquely specifies the Hermitian coefficients K_{PQ} of the sandwich terms. Subtract the corresponding dissipator, which is Hermiticity-preserving and trace-annihilating. The remaining Hermitian Choi matrix has zero compression to that complement, hence is of the form $|B\rangle\rangle\langle\langle I| + |I\rangle\rangle\langle\langle B|$ for some matrix B . It represents $\rho \mapsto B\rho + \rho B^*$. Trace annihilation gives $B + B^* = 0$, so $B = -iH$ with H Hermitian. This proves the claimed representation.

The last assertion of Lemma 7.10 and the Cayley transform show that the full Choi hypothesis places the coefficient generator in Theorem 3.2 on $2n$ binary legs. It consequently has differential order at most two. This implies tensor support at most two on those individual legs: expand in the local endomorphism basis $I, \partial_s, s\partial_s - 1/2, s^2\partial_s - s$. At a nonzero support of maximal size, its full derivative coefficient is a linear combination of independent products of $1, s, s^2$. If the support size exceeds two, the order bound forces all its coefficients to vanish. Descending eliminates every larger support. Local Cayley similarity preserves the identity and traceless subspaces, so this conclusion also holds before the transform.

For nonidentity Pauli strings P, Q , the tensor coefficient supported on their respective row and column legs is exactly the sandwich coefficient K_{PQ} , up to the harmless sign of a transposed Y . Neither a Hamiltonian nor an anticommutator

term can cancel a coefficient involving both row and column legs. Therefore

$$\text{wt}(P) + \text{wt}(Q) > 2 \implies K_{PQ} = 0.$$

In particular every row indexed by a string of weight at least two vanishes, directly and without a PSD argument. The anticommutator now has support at most two, so the same support bound excludes every Hamiltonian term of higher weight.

For distinct physical sites, the cross-site matrices and their normalized mixed coefficients are precisely (103)–(106). The general generator theorem makes these coefficients real and nonpositive at every real pair. Restricting to the trine grid, the discrete moment calculation (104) eliminates every cross-site block and pair Hamiltonian exactly as in Theorem 7.3. That calculation uses Hermiticity and trace annihilation, but never PSD of K . Thus the generator is a sum of local Hermiticity-preserving, trace-annihilating generators.

At one site, the nine products of the two Pauli derivative triples are independent. Reality of their mixed coefficient forces K_{xy} to be real and K_{xz}, K_{yz} to be purely imaginary, giving the displayed local shape. Its dissipator is then real after the Cayley transform. For a Hamiltonian $h_x X + h_y Y + h_z Z$, the remaining imaginary first-order coefficients are

$$h_y s^2 - 2h_x s - h_y, \quad h_y t^2 + 2h_x t - h_y.$$

The complex generator criterion requires each quadratic to be nonnegative on the real line. Comparing leading and constant terms forces $h_y = 0$, and then the linear term forces $h_x = 0$. The real mixed coefficient is (122), whose normalization (123) is nonpositive exactly under (124). This proves necessity without any positivity assumption.

Conversely, the displayed local algebraic generators are Hermiticity-preserving and trace-annihilating. Their Cayley transforms are real binary-box endomorphisms with nonpositive mixed coefficient. Theorem 3.2 and Lemma 7.10 prove full Choi stability of their exponentials and their tensor product. Small-time preservation extends to all times by the semigroup identity and Lemma 2.3, with invertibility excluding zero. Finally, complete positivity imposes $K_j \succeq 0$ in the canonical representation, and Lemma 7.9 gives exactly the second matrix condition in (98). $\square$

7.6. Cancellation at a fixed positive time. The mechanism used above has a finite-time version. It applies to different local channels and does not assume that they belong to a semigroup.

Theorem 7.13 (Static tensor cancellation). *Let $n \geq 2$. For each j , let Ψ_j be an invertible CPTP qubit map with Bloch action*

$$(r_\perp, r_z) \mapsto (D_j r_\perp + \tau_j, \lambda_j r_z), \quad D_j \in GL_2(\mathbb{R}), \quad \lambda_j > 0.$$

Then $\bigotimes_j \Psi_j$ ordinarily preserves all physical Lee–Yang densities if and only if every Ψ_j has a Lee–Yang Choi tensor. If one factor fails this condition, a positive definite separable input, classical on a second selected site and maximally mixed on the other sites, detects the failure, whatever the parameters of the second factor.

Proof. The outputs of the two computational basis states give $|\tau|^2 + \lambda^2 \leq 1$, so $|\tau| < 1$. The static one-qubit criterion in Theorem D.1 says that the Choi condition for the active factor Ψ is equivalent to

$$\|D^\top n\|^2 \leq (x - \tau \cdot n)^2 + \lambda^2(1 - x^2) \quad (n \in S^1, -1 \leq x \leq 1). \quad (125)$$

Changing x to $-x$ gives the alternative sign convention for its transverse displacement. Suppose the condition fails. By strictness and continuity choose $x \in (-1, 1)$ and $n \in S^1$ such that, with $v = D^\top n$, $V = |v|$ and $y = x - \tau \cdot n$,

$$V^2 > y^2 + \lambda^2(1 - x^2). \quad (126)$$

In particular $V > 0$. Set $u = yv/V^2$, so $|u| < 1$ and $u \cdot v = y$. The matrix $W = I_2 - uu^\top$ is positive definite and $v^\top W v = V^2 - y^2 > \lambda^2(1 - x^2)$. Choose $0 < \theta < 1$ sufficiently close to one and put

$$d_0 = \theta \frac{Wv}{\sqrt{v^\top W v}}.$$

Then $W - d_0 d_0^\top \succ 0$, as follows by congruence with $W^{-1/2}$, and

$$d_0 \cdot v > \lambda \sqrt{1 - x^2}. \quad (127)$$

Choose $\beta > 0$ with $\beta(1 - |u|) > |d_0|$ and define

$$A_R = R(I + u \cdot \sigma_\perp) + Z, \quad B_0 = \beta I + (\beta u + d_0) \cdot \sigma_\perp.$$

The second matrix is positive definite. The first is positive definite once $R^2(1 - |u|^2) > 1$.

We claim that, for all sufficiently large finite R , F_{A_R} is zero-free on the closed bidisk and

$$|F_{B_0}(s, w)| < |F_{A_R}(s, w)| \quad (|s|, |w| \leq 1). \quad (128)$$

On the distinguished boundary use (116) with $X = \cos \gamma$, $m \in S^1$, and $\ell = X + u \cdot m$. The squared-modulus difference divided by four is

$$G_R(X, m) = (R^2 - \beta^2)\ell^2 - 2\beta\ell(d_0 \cdot m) + 1 - X^2 - (d_0 \cdot m)^2.$$

On $\ell = 0$, its last two terms are at least $\delta := \lambda_{\min}(I_2 - uu^\top - d_0 d_0^\top) > 0$. If $|\ell| \leq \epsilon$, then $|X^2 - (u \cdot m)^2| \leq 2\epsilon$. Choose $\epsilon > 0$ so $2\epsilon + 2\beta\epsilon|d_0| < \delta/2$. For $R \geq \beta$, this gives $G_R > 0$ on that set. On $|\ell| \geq \epsilon$, using $|\ell| \leq 2$ gives

$$G_R \geq (R^2 - \beta^2)\epsilon^2 - 4\beta|d_0| - |d_0|^2 > 0$$

for sufficiently large R . This proves strict torus domination.

The diagonal entries of A_R are $R + 1, R - 1$ and its off-diagonal magnitude is $R|u|$. The constant coefficient in the representation $F_{A_R} = g(w) + sh(w)$ is nonzero on the closed disk because $R + 1 > R|u|$. On the circle,

$$|g(w)|^2 - |h(w)|^2 = 4R + 4\Re((A_R)_{01}w) > 0.$$

The maximum principle gives closed-bidisk zero-freeness of F_{A_R} . Apply it separately in both coordinates to F_{B_0}/F_{A_R} to obtain (128). Consequently the density proportional to $A_R \otimes |0\rangle\langle 0| + B_0 \otimes |1\rangle\langle 1|$ is positive definite, separable, and Lee–Yang.

At the output torus point in (116) with $X = -x$, $m = n$ and $Y = \sqrt{1 - x^2}$, the affine action gives

$$\frac{1}{2}e^{-i\gamma}F_{\Psi(A_R)} = R[-x + \tau \cdot n + u \cdot v] - i\lambda Y = -i\lambda Y,$$

and

$$\frac{1}{2}e^{-i\gamma}F_{\Psi(B_0)} = \beta[-x + \tau \cdot n + u \cdot v] + d_0 \cdot v = d_0 \cdot v.$$

The first value is nonzero, while (127) makes the second strictly larger in modulus. The strict inequality persists when the active coordinates move slightly into the disk. Lemma 7.7 then gives a zero of the two-site output after any allowed partner channel. Padding by maximally mixed states proves necessity for every $n \geq 2$. Conversely,

if all factors have Lee–Yang Choi tensors, Lemma 7.10 gives complete preservation at each site and therefore ordinary preservation of their tensor product. $\square$

The theorem assumes the displayed local affine form. It is not a factorization theorem for arbitrary isolated channels. In particular, the continuous-time locality theorem does not by itself imply such a static conclusion.

7.7. The one-qubit exception and dynamical consequences.

Proposition 7.14 (The ordinary one-qubit cone). *Let a one-qubit CPTP semigroup have Bloch equation $\dot{r} = Ar + b$, with real A, b . It ordinarily preserves Lee–Yang states if and only if*

$$b_z \geq \sqrt{A_{zx}^2 + A_{zy}^2}. \quad (129)$$

Proof. By Lemma 7.5, the additional condition is invariance of the northern Bloch hemisphere. At $r_z = 0$, its minimum outward-coordinate velocity is $b_z - \sqrt{A_{zx}^2 + A_{zy}^2}$, since the transverse vector ranges over the closed unit disk. A negative value gives immediate failure. Conversely, complete positivity keeps $|r(t)| \leq 1$. Under (129), $\dot{r}_z \geq A_{zz}r_z$ at every time. Multiplication by $e^{-A_{zz}t}$ and integration give $r_z(t) \geq e^{A_{zz}t}r_z(0) \geq 0$. $\square$

Example 7.15 (Amplitude damping). The amplitude-damping channel toward $|0\rangle$ has Kraus operators $K_0 = \text{diag}(1, \sqrt{\eta})$ and $K_1 = \sqrt{1-\eta}|0\rangle\langle 1|$, where $0 \leq \eta \leq 1$. The semigroup $\eta = e^{-\gamma t}$ satisfies $\dot{r}_z = \gamma(1 - r_z)$ and therefore ordinarily preserves one-qubit Lee–Yang states. It is not completely preserving for $0 < \eta < 1$. Indeed, its action on one half of a Bell state gives, up to a positive scalar, the polynomial

$$(1 + \sqrt{\eta}z_1z_2)(1 + \sqrt{\eta}w_1w_2) + (1 - \eta)z_2w_2.$$

At $z_1 = z_2 = w_1 = w_2 = ir$ its value is $1 - (2\sqrt{\eta} + 1 - \eta)r^2 + \eta r^4$. It is positive at $r = 0$ and equals $2\eta - 2\sqrt{\eta} < 0$ at $r = 1$. An interior root $0 < r < 1$ proves the claim.

Remark 7.16. For diagonal $K = \text{diag}(a, b, c)$ the complete criterion is $c \geq \max(a, b)$, recovering the Pauli-semigroup portion of the known sufficient channel family [61, Theorem 3]. The full cone also permits nonunital noise: $\sqrt{\gamma}(X + iZ)/2$ is an allowed jump and damps toward an equatorial pure state. Within the completely preserving cone, only rotations around the fixed Z axis are allowed as coherent motion. More generally, ordinary one-qubit dynamics with Hamiltonian $(\omega_x X + \omega_y Y + \omega_z Z)/2$, downward and upward jump rates $\gamma_\downarrow, \gamma_\uparrow$, and arbitrary Z dephasing satisfies (129) exactly when $\gamma_\downarrow - \gamma_\uparrow \geq \sqrt{\omega_x^2 + \omega_y^2}$.

Corollary 7.17. *An ordinarily preserving CPTP semigroup on $n \geq 2$ qubits cannot take a product input to a nonproduct state, either at a finite time or in a limit as $t \rightarrow \infty$. In particular, a semigroup converging from every input to a fixed nonproduct density matrix cannot ordinarily preserve the full Lee–Yang class.*

Proof. Theorem 7.1 keeps every product input a tensor product. The set of product densities is closed, being the continuous image of a compact product of one-site state spaces. $\square$

This conclusion concerns universal preservation and preparation from product inputs. It does not exclude a correlated stationary state of a nonergodic product semigroup, nor preservation along one specially chosen trajectory. The identity semigroup illustrates the first distinction.

7.8. Explicit boundaries of restricted tests. The following examples distinguish preservation of selected inputs or classical populations from preservation of the whole physical class. Their zeros provide direct certificates independent of the classification.

Lemma 7.18 (Pair defects of real hyperbola contacts). *Let $\mathcal{L}$ be a sum of local GKLS generators of the real form (108), on $n \geq 2$ sites, and let q_j be its local mixed coefficient (122). Define*

$$\Delta_j = \sup_{s, w \in \mathbb{R}, s+w \neq 0} \frac{q_j(s, w)}{(s+w)^2}.$$

All the real double-contact inequalities from the two-site pure hyperbola family (112) hold if and only if every Δ_j is finite and

$$\Delta_i + \Delta_j \leq 0 \quad (i \neq j). \quad (130)$$

In particular, these are necessary conditions for ordinary preservation. They are not asserted to be sufficient for ordinary preservation.

Proof. Fix a pair of sites and number them one and two. The double-root leading coefficient of the product of the two hyperbola factors in the proof of Proposition 7.6 has sign uv . Thus its contact inequality is

$$\frac{c_0^2}{u^2 v^2} q_1(a_0 + u, -a_0 + v) + q_2(b_0 + c_0/u, -b_0 + c_0/v) \leq 0.$$

Writing $S = u + v$ and $R = c_0 S/(uv)$, when $S \neq 0$ this is the sum of the two ratios defining Δ_1, Δ_2 . Every pair of real points with nonzero sums S, R is attained: choose a nonzero real P with the sign of S/R , taking $0 < P < S^2/4$ when it is positive; let u, v be the roots of $X^2 - SX + P$, put $c_0 = PR/S > 0$, and choose a_0, b_0 to fit the first coordinate in each pair. Thus the contact inequalities with nonzero sums say that every value of the first ratio plus every value of the second is nonpositive. Fixing a finite value of either ratio bounds the other ratio above. Each ratio also has at least one finite value, so both suprema are real and their sum is nonpositive. The converse follows directly from the definition of the suprema.

When $S = 0$, also $R = 0$. The paired Pauli derivative vectors obey $\tilde{v}(-s) = -\overline{v(s)}$ in (101), so positivity of the local noise matrix gives

$$q_j(s, -s) = -v(s)^T K_j \overline{v(s)} \leq 0.$$

These contacts therefore impose no additional condition. For $n > 2$, pad the input by pure local factors that are nonzero at the selected contact. Local terms on the other sites then vanish to second order at that contact, leaving exactly the displayed two-site inequality. This proves the claim for every pair and the necessity statement. $\square$

Proposition 7.19 (Real hyperbola contacts are insufficient). *For $a > 0$ and $c \geq 0$, the two-qubit generator*

$$\mathcal{L}(\rho) = a(X_1 \rho X_1 - \rho) + c(Z_2 \rho Z_2 - \rho)$$

does not ordinarily preserve Lee–Yang densities. Nevertheless, when $c \geq a$, it satisfies every real double-contact inequality obtained from the pure input family used in (112).

Proof. Here $q_1(s, w) = 4asw$ and $q_2(s, w) = -c(1 + s^2)(1 + w^2)$. The identities $4sw \leq (s + w)^2$ and $(1 + s^2)(1 + w^2) - (s + w)^2 = (sw - 1)^2$ give $\Delta_1 = a$, $\Delta_2 = -c$, with equality attained in each supremum. Lemma 7.18 proves the assertion about all real hyperbola contacts.

For actual failure, use the positive input

$$\rho = \frac{2}{7} \left[\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \otimes |0\rangle\langle 0| + \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \otimes |1\rangle\langle 1| \right].$$

It has trace one. Up to its positive scalar its polynomial is $2 + sw + \frac{1}{4}(1 + s)(1 + w)zv$, which is zero-free on $\mathbb{D}^4$ because $|2 + sw| > 1$ and $|\frac{1}{4}(1 + s)(1 + w)zv| < 1$ there. The second-site dephasing vanishes on this input and its entire first-site trajectory. At $t = (\log 2)/a$, the first diagonal block is $\text{diag}(13/8, 11/8)$, whereas the second block is fixed. At $s = w = 19i/20$ the output polynomial becomes

$$\frac{1229}{3200} + \frac{39 + 760i}{1600}zv.$$

Choose $zv = -1229/[2(39 + 760i)]$. Its modulus is $1229/1522 < 1$, since $39^2 + 760^2 = 761^2$. Taking both z, v to be the same square root puts all variables in the open disk and makes the output zero. $\square$

Example 7.20 (Random interchange and incoherent exclusion). Let S interchange two qubits. For $0 < p < 1$, the channel $\rho \mapsto p\rho + (1 - p)S\rho S$ fails ordinary preservation. The Lee–Yang input $|+0\rangle\langle +0|$ gives an output polynomial proportional to $p(1 + z_1)(1 + w_1) + (1 - p)(1 + z_2)(1 + w_2)$. It vanishes at

$$z_1 = w_1 = -1 + \sqrt{1 - p}e^{i\pi/4}, \quad z_2 = w_2 = -1 + \sqrt{p}e^{-i\pi/4}.$$

Every coordinate is in the open disk: for $0 < x < 1$, $|-1 + xe^{\pm i\pi/4}|^2 = 1 + x^2 - \sqrt{2}x < 1$. In particular the random-interchange semigroup fails at each positive time, since its mixture parameter is $(1 + e^{-2\gamma t})/2$.

The two jumps $\sqrt{\gamma}|01\rangle\langle 10|$ and $\sqrt{\gamma}|10\rangle\langle 01|$, with $\gamma > 0$, also fail at all sufficiently small positive times. On the same input, let $u = e^{-\gamma t/2}$. The 00–10 coherence decays by u , and the two excited populations exchange at rate 2γ , giving

$$2F_t = 1 + u(z_1 + w_1) + \frac{1+u^4}{2}z_1w_1 + \frac{1-u^4}{2}z_2w_2.$$

At $z_1 = w_1 = -r$, $z_2 = w_2 = ir$, this is $1 - 2ur + u^4r^2$. Its value at $r = 0$ is positive. At $r = 1$ it is $u^4 - 2u + 1$, negative for $u < 1$ sufficiently close to one, because this function vanishes at one and has derivative two there. The intermediate value theorem supplies an interior zero. Thus this natural quantum extension of population exchange does not preserve the entire physical class.

Example 7.21 (Equal local damping does not restore preservation). Apply the same amplitude-damping channel to both halves of a Bell state. Its output polynomial, up to the positive factor $1/2$, is

$$1 + (1 - \eta)^2 + \eta(1 - \eta)(z_1w_1 + z_2w_2) + \eta^2z_1z_2w_1w_2 + \eta(z_1z_2 + w_1w_2).$$

At all four variables ir , the polynomial is $1 + (1 - \eta)^2 - 2\eta(2 - \eta)r^2 + \eta^2r^4$. It is positive at zero and equals $2(1 - \eta)(1 - 2\eta) < 0$ at one for $1/2 < \eta < 1$. Hence equal

local damping fails ordinary joint preservation at every sufficiently small positive time. For instance, $\eta = 3/4$ has the exact interior root $r^2 = (5 - 2\sqrt{2})/3$. Vacuum padding extends the example to any larger system.

7.9. Physical dynamics at arbitrary finite spins. The physical classification becomes more restrictive when a site has spin at least one. We keep the physical basis fixed and use the coherent normalization of Appendix E. For positive integers $\kappa_i = 2j_i$, put

$$\mathcal{H} = \bigotimes_{i=1}^n \mathbb{C}^{\kappa_i+1}, \quad b_a = \prod_i \binom{\kappa_i}{a_i}, \quad f_\psi(z) = \sum_{a \leq \kappa} \sqrt{b_a} \psi_a z^a,$$

and define

$$F_\rho(z, w) = \sum_{a, b \leq \kappa} \sqrt{b_a b_b} \rho_{ab} z^a w^b. \quad (131)$$

A density matrix is coherently Lee–Yang when this polynomial is nonzero on $\mathbb{D}^{2n}$, with independent row and column variables. Ordinary preservation refers to this class on $\mathcal{H}$. Complete preservation means preservation after adjoining any finite collection of inert finite spins, with the same encoding. A pure input means a rank-one density matrix with disk-stable vector polynomial; the vector need not be a spin-coherent state. Adjoints are always taken in the physical orthonormal basis, equivalently in the polynomial inner product $\langle z^a, z^b \rangle = \delta_{ab}/b_a$.

Theorem 7.22 (Finite-spin physical generators). *Suppose that $\kappa_i \geq 2$ for at least one site, and let $\Phi_t = e^{t\mathcal{L}}$ be a continuous CPTP semigroup on $\mathcal{H}$. The following conditions are equivalent.*

- (1) Φ_t ordinarily preserves coherently Lee–Yang densities for every $t \geq 0$.
- (2) For some $\varepsilon > 0$, Φ_t preserves every pure coherently Lee–Yang input for $0 \leq t < \varepsilon$.
- (3) Φ_t completely preserves coherently Lee–Yang densities for every $t \geq 0$.
- (4) The generator is a sum of local generators,

$$\mathcal{L} = \sum_i \mathcal{L}_i \otimes \text{Id}_i. \quad (132)$$

At a site with $j_i \geq 1$, its local term is

$$\mathcal{L}_i(\rho) = -i[h_i S_i^z, \rho] - \frac{\gamma_i}{2} \sum_{\alpha=x,y,z} [S_i^\alpha, [S_i^\alpha, \rho]], \quad h_i \in \mathbb{R}, \quad \gamma_i \geq 0. \quad (133)$$

At a qubit site, its local term is

$$\mathcal{L}_i(\rho) = -i[h_i Z, \rho] + \sum_{\alpha, \beta=x,y,z} (K_i)_{\alpha\beta} \left(\sigma_\alpha \rho \sigma_\beta - \frac{1}{2} \{ \sigma_\beta \sigma_\alpha, \rho \} \right), \quad h_i \in \mathbb{R},$$

where $B_i = B_i^\top \in \mathbb{R}^{2 \times 2}$, $d_i \in \mathbb{R}^2$, and

$$K_i = \begin{pmatrix} B_i & id_i \\ -id_i^\top & c_i \end{pmatrix} \succeq 0, \quad \begin{pmatrix} c_i I_2 - \text{adj } B_i & d_i \\ d_i^\top & \text{Tr } B_i \end{pmatrix} \succeq 0. \quad (134)$$

In particular, $\Phi_t = \bigotimes_i e^{t\mathcal{L}_i}$.

The dissipator in (133) is the familiar isotropic spin diffusion, with Casimir multipole rates [23, 26]. The assertion here is its necessity from physical zero-freeness, without an assumption of covariance, linearity of the noise, locality, or

unitarity. The axial Hamiltonian commutes with the isotropic dissipator; a nonzero axial drift need not make the full semigroup rotation covariant.

We first establish the reduction from arbitrary GKLS operators using only physical pure inputs. The spin realization is

$$S_i^x = \frac{1}{2}[(1 - z_i^2)\partial_i + \kappa_i z_i], \quad S_i^y = \frac{1}{2i}[(1 + z_i^2)\partial_i - \kappa_i z_i], \quad S_i^z = \kappa_i/2 - z_i\partial_i. \quad (135)$$

The capacity-dependent Cayley transform is

$$(Cf)(s) = 2^{-|\kappa|/2} \prod_i (s_i + i)^{\kappa_i} f\left(\frac{s_1 - i}{s_1 + i}, \dots, \frac{s_n - i}{s_n + i}\right). \quad (136)$$

It is the symmetric-power realization of (99), and is unitary in the weighted inner product. Applied separately to both coefficient legs of a pure density, it gives

$$p(z)\mathcal{R}p(w), \quad \mathcal{R}p(w) = (-1)^{|\kappa|} \overline{p(-\bar{w})}, \quad p = Cf_\psi. \quad (137)$$

Indeed this follows in capacity one from $\bar{C} = \text{diag}(-1, 1)C$, and then by symmetric powers and tensor products. This two-leg transformation is not unitary conjugation of the density matrix. Every stable nonzero p in the capacity box is obtained from a physical pure input after normalization. Its degree may be strictly below the capacity; the inverse of (136) still belongs to the prescribed box.

Lemma 7.23 (Physical order reduction). *Under condition (2) of Theorem 7.22, every jump in any finite GKLS representation has, after removal of its scalar part, the form*

$$V_\nu \in \text{span}_{\mathbb{C}}\{S_i^x, S_i^y, S_i^z : 1 \leq i \leq n\}.$$

The Hamiltonian has spin degree at most two: it is a sum of a scalar, linear spin terms, on-site symmetric quadratics, and two-site bilinears. At qubit sites the on-site quadratics reduce to a scalar and linear terms.

Proof. Write a finite-dimensional GKLS representation

$$\mathcal{L}(\rho) = -i[H, \rho] + \sum_\nu (V_\nu \rho V_\nu^* - \frac{1}{2}\{V_\nu^* V_\nu, \rho\}).$$

In the vector-polynomial coordinates, let

$$Q_\nu = CV_\nu C^{-1} = \sum_{\alpha \leq \kappa} P_{\nu, \alpha}(z) \partial^\alpha$$

be its canonical representation. Let R be the largest order among all these operators. Suppose $R \geq 2$, and fix an index $|\alpha| = R$. For arbitrary $a \in \mathbb{R}^n$ take the stable pure input $p_a(z) = \prod_i (z_i - a_i)^{\kappa_i}$. In local coordinates $u = z - a$, $v = w + a$, its doubled polynomial is exactly $u^\kappa v^\kappa$, up to a positive normalization constant. Put $M = 2|\kappa|$.

Restrict the transformed output to $(u, v) = \tau\xi$ for any $\xi \in \mathbb{R}_{>0}^{2n}$. The initial polynomial is a nonzero multiple of τ^M , and the output remains upper-half-plane stable near time zero. Lemma 3.5 makes every coefficient of its first variation below degree $M - 2$ vanish. Varying ξ over the positive orthant shows, by polynomial identity, that each multivariate homogeneous part below that degree vanishes as well.

Consider the coefficient of $u^{\kappa-\alpha} v^{\kappa-\alpha}$. Hamiltonian and anticommutator terms act on one leg only, leaving the opposite full root factor untouched, and do not contribute. The sandwich term from V_ν is $(Q_\nu p_a)(z)\mathcal{R}(Q_\nu p_a)(w)$. To obtain $u^{\kappa-\alpha}$ from a canonical term with derivative index β , even allowing Taylor terms of its

coefficient polynomial at a , one must have $\beta \geq \alpha$. Since $|\beta| \leq R = |\alpha|$, this requires $\beta = \alpha$ and the constant Taylor coefficient. Reflection therefore gives the total coefficient

$$(-1)^R (\kappa)_\alpha^2 \sum_\nu |P_{\nu,\alpha}(a)|^2, \quad (\kappa)_\alpha = \prod_i \frac{\kappa_i!}{(\kappa_i - \alpha_i)!}. \quad (138)$$

Its degree is $M - 2R < M - 2$, so it is zero. Every square vanishes for every real a . Thus every coefficient of maximal order vanishes identically, contradicting the definition of R .

All jumps have order at most one. Proposition 3.4 identifies this space as a scalar plus the local $\mathfrak{sl}_2$ actions. Formula (135) and the Cayley rotation identify it with the stated spin span. Absorb scalar jump parts into the Hamiltonian. The dissipative superoperator now has differential order at most two on the doubled box.

If CHC^{-1} had maximal canonical order $R \geq 3$, use the same input and choose $|\alpha| = R$. The coefficient of $u^{\kappa-\alpha}v^\kappa$ in the first variation can come only from the row Hamiltonian and its constant maximal-order coefficient at a . The column Hamiltonian retains u^κ , and the dissipator cannot lower total degree by R . Its degree $M - R < M - 2$ forces that Hamiltonian coefficient to vanish for every a , again a contradiction. Proposition 3.4 identifies the resulting order-two space with products of at most two local spin operators. Hermiticity gives real coefficients in the Hermitian basis consisting of scalars, local spins, local symmetric quadrupoles, and products on distinct sites. The antisymmetric local products are spin commutators, and the scalar symmetric component is the Casimir. At capacity one the remaining quadrupoles vanish. This proves the assertion. $\square$

Lemma 7.24 (Locality at finite spins). *Under the same hypothesis, the generator is a sum of local GKLS generators as in (132).*

Proof. Put $T_i^\alpha = 2S_i^\alpha$. Lemma 7.23 gives a PSD Gram matrix K in the basis of these local spin operators. Its (i, j) block is denoted K^{ij} . The transformed left and right actions have respective derivative vectors

$$\mathbf{v}(s) = (2s, 1 - s^2, i(1 + s^2)), \quad \tilde{\mathbf{v}}(t) = (2t, t^2 - 1, i(1 + t^2)). \quad (139)$$

These vectors are independent of the capacity. Only their zeroth-order coefficients depend on κ_i .

Fix distinct sites i, j . The physical pure product with transformed vector $\prod_k (z_k - a_k)$ allows independent row-row contacts at $z_i = s, z_j = t$ and row-column contacts at $z_i = s, w_j = t$, by choosing $a_i = s$ and respectively $a_j = t$ or $a_j = -t$. Keep all other legs off their roots and restrict to a line with all slopes positive. The initial root has multiplicity two. Only the chosen mixed principal coefficient survives at the contact; after division by the initial leading coefficient its multiplier is the reciprocal of a positive product of slopes. Lemma 3.5 makes it real and nonpositive for every real s, t . On-site quadrupoles and other terms leave at least one selected root untouched.

Write the pair Hamiltonian as $\sum_{\alpha,\beta} h_{\alpha\beta} T_i^\alpha T_j^\beta$, with h real. The row-column coefficient is $\mathbf{v}(s)^\top K^{ij} \tilde{\mathbf{v}}(t)$. The row-row coefficient is $\mathbf{v}(s)^\top (-\Re K^{ij} - ih) \mathbf{v}(t)$: the conjugate Gram blocks combine in the anticommutator, since the two sites commute. Set

$$u(s) = \frac{(2s, 1 - s^2)}{1 + s^2}, \quad R_0 = \text{diag}(1, -1).$$

Its range is dense in the unit circle. The normalized derivative vectors are $(u(s), i)$ and $(R_0 u(t), i)$.

Reality of the row–column coefficient, by comparison of its constant, linear, and bilinear circle components, gives

$$K^{ij} = \begin{pmatrix} A & ib \\ id^\top & e \end{pmatrix}, \quad A, b, d, e \text{ real.}$$

Its sign condition is

$$u^\top A R_0 v - b \cdot u - d \cdot R_0 v - e \leq 0 \quad (u, v \in S^1). \quad (140)$$

For completeness, the comparison follows by averaging the imaginary part first alone, then against u , $R_0 v$, and $u(R_0 v)^\top$; the circle moments are $\mathbb{E}u = 0$ and $\mathbb{E}uu^\top = I_2/2$. Reality of the row–row coefficient similarly gives $h_{\perp\perp} = 0$ and $h_{zz} = 0$. With $k = h_{\perp z}$ and $\ell^\top = h_{z\perp}$, its remaining sign condition is

$$e - u^\top A v + k \cdot u + \ell \cdot v \leq 0. \quad (141)$$

Independent circle averages of (140) and (141) are $-e$ and e , so $e = 0$. Each continuous nonpositive function then has zero average and vanishes everywhere. The same circle moments give $A = b = d = k = \ell = 0$. Thus all cross Gram blocks and pair Hamiltonians vanish. The diagonal Gram blocks are PSD and define local GKLS generators, proving locality. $\square$

Proof of Theorem 7.22. We prove (2) $\Rightarrow$ (4) first. The preceding lemmas give locality. Each local semigroup preserves its pure Lee–Yang inputs near zero: pad such an input with arbitrary pure Lee–Yang states at the other sites. Its evolved global polynomial is a product of the local output polynomials, each nonzero since its density has trace one. Stability of that product implies stability of each factor.

Fix a site of capacity $\kappa \geq 2$, retaining the normalization $T = 2S$. Let $q(s, t)$ denote its row–column principal coefficient. Use the pure vector polynomial $p(z) = (z - s)(z + t)$, with $s + t \neq 0$. At row coordinate s and column coordinate t both factors in (137) have simple roots. A positive-slope line has a double root, and only $q(s, t)$ contributes at the contact. The common reflection sign cancels on dividing by the initial leading coefficient. Thus $q(s, t)$ is real and nonpositive; continuity adds the omitted line $s + t = 0$.

For a general Hermitian Gram matrix write

$$K = \begin{pmatrix} a & u + i\eta & r + i\mu \\ u - i\eta & b & v + i\nu \\ r - i\mu & v - i\nu & c \end{pmatrix}.$$

Multiplication of (139) gives

$$\Im q(s, t) = 2(s + t)[- \eta(1 - st) + r(1 + st) + v(t - s)].$$

Hence $\eta = r = v = 0$, and

$$K = \begin{pmatrix} B & id \\ -id^\top & c \end{pmatrix}, \quad B = B^\top \in \mathbb{R}^{2 \times 2}, \quad d \in \mathbb{R}^2. \quad (142)$$

With

$$r(s) = \frac{(2s, 1 - s^2)}{1 + s^2}, \quad v(t) = \frac{(2t, t^2 - 1)}{1 + t^2},$$

the mixed coefficient satisfies

$$\frac{q(s, t)}{(1 + s^2)(1 + t^2)} = r(s)^\top Bv(t) + d \cdot (v(t) - r(s)) - c \leq 0. \quad (143)$$

Both circle variables range densely and independently. Taking them equal gives $B \preceq cI_2$.

There is a second, independent contact available at this capacity. Take $p(z) = (z - s)^2$ and a column coordinate away from its reflected root. Its pure-row principal coefficient must be real and nonpositive. This follows either by real specialization of the column, or by a positive-slope line through the contact; the nonzero complementary factor cancels from the root-cluster coefficient. Write the quadratic Hamiltonian as $H_2 = T^\top QT$, with Q real symmetric. Put $u = r(s)$. The pure-row principal coefficient, divided by $(1 + s^2)^2$, is

$$\frac{1}{2}(c - u^\top Bu) + 2Q_{\perp z} \cdot u - i(u^\top Q_{\perp\perp} u - Q_{zz}). \quad (144)$$

Indeed its dissipative part is $-\mathbf{v}(s)^\top (\Re K) \mathbf{v}(s)/2$, and its Hamiltonian part is $-i\mathbf{v}(s)^\top Q \mathbf{v}(s)$, using the three-component vector in (139). Reality gives $Q_{\perp\perp} = Q_{zz}I_2$. Remove the corresponding Casimir scalar and set $k = Q_{\perp z}$. Nonpositivity at u and $-u$ now gives $c - u^\top Bu \leq 0$. Together with $B \preceq cI_2$, this yields $B = cI_2$; the remaining inequality $2k \cdot u \leq 0$ gives $k = 0$. The quadratic Hamiltonian is consequently scalar.

Equation (143) reduces to

$$c(r \cdot v - 1) + d \cdot (v - r) \leq 0 \quad (r, v \in S^1).$$

If $d \neq 0$, take $r \perp d$ and $v = (\cos \theta)r + (\sin \theta)d/|d|$ for small positive θ . The left side is $c(\cos \theta - 1) + |d| \sin \theta > 0$, a contradiction. Thus $d = 0$, $K = cI_3$, and positivity gives $c \geq 0$.

The transformed isotropic dissipator has the explicit form

$$\begin{aligned} \tilde{\mathcal{D}} = & -2c\kappa(\kappa zw + 1) + 2c\kappa z(zw - 1)\partial_z + 2c\kappa w(zw - 1)\partial_w \\ & - 2c(zw - 1)^2\partial_z\partial_w. \end{aligned} \quad (145)$$

To verify it directly, the left actions of (T_x, T_y, T_z) are

$$2z\partial_z - \kappa, \quad (1 - z^2)\partial_z + \kappa z, \quad i[(1 + z^2)\partial_z - \kappa z],$$

and the right actions are the same formulas in w except that the middle action has its sign reversed. Substitute them into $c \sum_\alpha (L_\alpha R_\alpha - (L_\alpha^2 + R_\alpha^2)/2)$. In particular this is a real operator and its pure-row and pure-column second-order coefficients vanish. For the remaining Hamiltonian $H = \hat{h}_x T_x + \hat{h}_y T_y + \hat{h}_z T_z$, use $p(z) = z - s$ and keep the reflected column off its root. The imaginary simple-root velocity comes only from the row Hamiltonian and is

$$\hat{h}_y s^2 - 2\hat{h}_x s - \hat{h}_y \geq 0 \quad (s \in \mathbb{R}),$$

by Lemma 3.5. Hence $\hat{h}_x = \hat{h}_y = 0$. Returning to $S = T/2$ gives (133), with $h = 2\hat{h}_z$ and $\gamma = 4c$.

It remains to recover the full qubit condition in a mixed-spin system. Fix a qubit and a high-spin partner whose local form has just been proved. Their mixed

principal coefficients will be denoted by $q_q(s, t)$ and $q_h(u, v) = -2c(uv - 1)^2$. For arbitrary real s, t with $s + t \neq 0$, choose

$$\begin{aligned} x &= (s + t)/2, & a &= (s - t)/2, & c_0 &> |x|, \\ b &= \sqrt{(c_0/x)^2 - 1}, & u_* &= b + c_0/x, & v_* &= -b + c_0/x. \end{aligned} \quad (146)$$

Then $u_*v_* = 1$. The polynomial

$$p(z, u) = (z - a)(u - b) - c_0 \quad (147)$$

is upper-half-plane stable, since for $z \in \mathbb{H}$ its root in u is $b + c_0/(z - a)$, which lies in the lower half-plane. It is a valid physical vector in capacities $(1, \kappa)$, including when its degree is below κ . Apart from the common parity sign, its reflected factor is $(w + a)(v + b) - c_0$.

At $(z, u, w, v) = (s, u_*, t, v_*)$ both factors vanish. Along the line with all four slopes one, the initial quadratic coefficient, apart from the same common sign, is $(c_0/x + x)^2 > 0$. Locality makes its first variation at the contact equal, with that sign, to

$$\frac{c_0^2}{x^2}q_q(s, t) + x^2q_h(u_*, v_*) = \frac{c_0^2}{x^2}q_q(s, t). \quad (148)$$

All lower-order terms vanish because one input factor remains zero; the high-spin pure-row and pure-column coefficients are zero by (145). Other sites can be padded by pure stable factors kept nonzero at the contact. The root-cluster condition thus makes $q_q(s, t)$ real and nonpositive. Continuity again adds $s + t = 0$. Consequently (142) and (143) hold at every qubit as well. For this Gram shape the dissipator is real after Cayley transformation, as in the proof of Proposition 7.11. The same simple-root test excludes its transverse Hamiltonian. Lemma 7.9 gives precisely (134), including $\text{Tr } B = 0$. This proves (2) $\Rightarrow$ (4), without treating the evolving partner as inert.

Conversely assume (4). Formula (145) is a real endomorphism of the doubled capacity box whose only second-order coefficient is $-2c(zw - 1)^2 \leq 0$. The axial Hamiltonian contributes only a real first-order endomorphism. At qubit sites, the proof of Proposition 7.11 gives a real box endomorphism with its only mixed coefficient nonpositive by Lemma 7.9. The sum therefore satisfies Theorem 3.1 on the full doubled capacity box. Its exponential preserves all complex stable polynomials after arbitrary inert-variable extensions. Undoing the coefficient Cayley transforms gives the asserted physical complete preservation, in particular for arbitrary inert finite spins. No convexity of the stable class is used. This proves (4) $\Rightarrow$ (3). Finally (3) $\Rightarrow$ (1) and (1) $\Rightarrow$ (2) are immediate. $\square$

The all-qubit ordinary classification is Theorem 7.1; it is not asserted there or here to follow from pure inputs alone. On a single qubit the ordinary cone is instead Proposition 7.14, and ordinary preservation need not be complete. Theorem 7.22 also makes no classification claim for isolated noncovariant high-spin channels: its semigroup hypothesis is essential to the argument.

7.10. An application to rotation convolution semigroups. Gaussian processes on compact groups can already be recognized through a faithful finite-dimensional representation and its tensor products; see [56, Theorem 2.8], which restates the earlier compact-group characterization. For the real defining representation of $SO(3)$, the spin-one conjugation action is its tensor square and contains the defining representation as its antisymmetric part. The following application

combines that recognition mechanism with the physical classification above. We give the short character proof in this case using Hunt's formula [32] and its twirling realization [2, Theorem 5.1].

Corollary 7.25 (Lee–Yang rotation dynamics). *Let $(\mu_t)_{t \geq 0}$ be a weakly continuous probability convolution semigroup on $SO(3)$ with $\mu_0 = \delta_e$. Let U be its spin-one representation and set*

$$\Phi_t(\rho) = \int_{SO(3)} U(g)\rho U(g)^* \mu_t(dg).$$

Then Φ_t preserves every pure coherently Lee–Yang input in a neighborhood of zero if and only if the group generator is

$$\mathcal{A} = hX_z + \frac{\gamma}{2}(X_x^2 + X_y^2 + X_z^2), \quad h \in \mathbb{R}, \quad \gamma \geq 0. \quad (149)$$

Here X_α are the left-invariant fields normalized by $U(\exp(tE_\alpha)) = e^{-itS_\alpha}$ and $X_\alpha f(g) = \frac{d}{dt}f(g \exp(tE_\alpha))|_{t=0}$. In particular the Lévy measure is zero, without a prior assumption of continuous sample paths or rotation invariance of μ_t .

Proof. The twirling maps form a continuous CPTP semigroup. Under the stated preservation hypothesis Theorem 7.22 gives (133). The spin-one operator space decomposes as the angular-momentum representations $0 \oplus 1 \oplus 2$. On its ℓ component the double-commutator Casimir is $\ell(\ell+1)I$; the axial commutator has trace zero. Thus the generator restrictions satisfy

$$\text{Tr } \mathcal{L}_1 = -3\gamma, \quad \text{Tr } \mathcal{L}_2 = -15\gamma.$$

Let χ_1, χ_2 be the characters of the three- and five-dimensional representations. For a rotation of angle θ , put

$$F(g) = 2 - \chi_1(g) + \frac{1}{5}\chi_2(g) = \frac{4}{5}(1 - \cos \theta)^2. \quad (150)$$

This smooth function is strictly positive off the identity and has zero jet through order two at the identity. Since $\int \chi_\ell d\mu_t = \text{Tr}(e^{t\mathcal{L}_\ell})$, the displayed traces give $\mathcal{A}F(e) = 3\gamma - 15\gamma/5 = 0$.

Write Hunt's generator with drift, diffusion matrix $a \succeq 0$, and Lévy measure ν , taking its second-order part to be $\frac{1}{2} \sum_{\alpha, \beta} a_{\alpha\beta} X_\alpha X_\beta$. At the identity all local terms and the linear jump compensation vanish on F . Therefore

$$0 = \mathcal{A}F(e) = \int_{SO(3) \setminus \{e\}} F(g) \nu(dg).$$

The integral is finite: near the identity $F = O(\text{dist}(g, e)^4)$, while a Lévy measure has the requisite local second-moment integrability; away from the identity compactness applies. Positivity of F off e implies $\nu = 0$, including for an initially infinite-activity Lévy measure.

On the dipole space the symmetric part of the remaining generator is $\frac{1}{2}(a - (\text{Tr } a)I_3)$. Comparing with $-\gamma I_3$, first taking traces and then substituting back, gives $a = \gamma I_3$. Its antisymmetric part identifies the drift as hX_z . This proves (149). Conversely, that group generator induces (133) under twirling, so Theorem 7.22 gives complete preservation. $\square$

The group and the tested spin matter. Conjugation cannot detect jumps in the center $\{\pm I\}$ of $SU(2)$, so the zero-Lévy-measure conclusion is stated on $SO(3)$. Moreover, Poisson jumps of rate λ by angle π about the z axis induce on a qubit

$$\Phi_t(\rho) = \frac{1}{2}(1 + e^{-2\lambda t})\rho + \frac{1}{2}(1 - e^{-2\lambda t})Z\rho Z.$$

This is an allowed dephasing semigroup, but its rotation process has nonzero jumps. Thus spin one cannot be replaced uniformly by spin one-half in Corollary 7.25.

8. POSITIVE COEFFICIENTS AND PARTICLE CAPACITIES

The restriction to nonnegative coefficients changes the generator problem substantially. In particular, collective particle loss is compatible with stability, whereas its extension by an arbitrary unchanged coordinate need not be. We first classify the unrestricted positive problem and then give a capacity-sensitive criterion. Throughout this section, coefficients are ordinary monomial coefficients, and matrix entries use the convention $Az^x = \sum_y A_{y,x} z^y$.

Write $\mathcal{S}_+$ for the nonzero real stable polynomials with nonnegative coefficients. Write $\mathcal{LP}_+$ for their nonzero locally uniform entire limits. When taking closed cones we adjoin zero explicitly. On a countable state space we use the coefficient space ℓ^1 . Thus every member of $\mathcal{LP}_+$ belongs to ℓ^1 , although a general member of ℓ^1 need not be entire.

Lemma 8.1 (Positive Laguerre–Pólya coefficient closure, [39]). *Coefficientwise limits of members of $\mathcal{LP}_+$, when the limiting coefficient sum is finite, belong to $\mathcal{LP}_+ \cup \{0\}$. Every member of $\mathcal{LP}_+$ has stable Jensen-polynomial approximants converging in ℓ^1 , whose coefficients are bounded by those of the limiting series.*

This is the positive Laguerre–Pólya closure and Jensen characterization. We use this precise standard theorem when passing from coefficient norm convergence to an entire stable limit.

8.1. Root contacts and Poisson tangency.

Lemma 8.2 (A repeated-root first variation). *Suppose p_t is holomorphic near zero, has real Taylor coefficients, and*

$$p_t(u) = cu^m + th(u) + o(t), \quad c > 0,$$

locally uniformly as $t \downarrow 0$. Suppose that all zeros sufficiently near zero are real. Then $h(0) = 0$ if $m \geq 3$, and $h(0) \leq 0$ if $m = 2$.

Proof. Fix a small circle isolating the zero of p_0 . The argument principle and p'_t/p_t express the power sums of the enclosed m roots as contour integrals. Consequently their monic root polynomial has a right first variation, and its first two nonleading coefficients are $O(t)$. If its roots are $r_1(t), \dots, r_m(t)$, Newton's identity gives $\sum_j r_j(t)^2 = O(t)$. Since these roots are real, each is $O(\sqrt{t})$. Their product is $o(t)$ when $m \geq 3$. The complementary analytic factor tends to c , so this proves the first assertion. For $m = 2$, the discriminant of the root polynomial is nonnegative. Its linear coefficient is $O(t)$, hence its constant coefficient has upper first derivative at most zero. Multiplication by the positive limiting factor c proves the second assertion. $\square$

For a capacity vector $\kappa \in (\mathbb{N} \cup \{\infty\})^d$, let $\Omega_\kappa = \{x \in \mathbb{N}^d : x_i \leq \kappa_i\}$. Coordinates of capacity zero are removed. Every linear map on the allowed polynomials has a unique locally finite normal-ordered expression

$$A = \sum_{\alpha \leq \kappa} P_\alpha(z) \partial^\alpha, \quad P_\alpha = \frac{1}{\alpha!} \left(Az^\alpha - \sum_{\substack{\beta \leq \alpha \\ \beta \neq \alpha}} (\alpha)_\beta z^{\alpha-\beta} P_\beta \right). \quad (151)$$

Here the P_α may have degrees outside the original coordinate box. The expression is locally finite because only finitely many derivatives act on a fixed polynomial.

Lemma 8.3 (Forced differential order). *Suppose the polynomial action of a positive stability-preserving semigroup has a locally uniform generator first variation near a real open negative box. Then $P_\alpha = 0$ whenever $|\alpha| \geq 3$. For unrestricted coordinates it is enough to assume preservation of every $(c + a \cdot z)^m$, where $c \geq 0$, $a > 0$, and $m \geq 0$.*

Proof. For an allowed α , use $f(z) = \prod_i (z_i + s_i)^{\alpha_i}$, with $s_i > 0$ small. At $z = -s$, every normal-ordered term vanishes except the term with derivative α ; thus $Af(-s) = \alpha! P_\alpha(-s)$. Along a strictly positive direction through $-s$, the initial polynomial has a root of multiplicity $|\alpha|$, with positive leading coefficient. Lemma 8.2 annihilates P_α on an open box, hence identically by analytic continuation. For the last assertion choose $y < 0$, $c = -a \cdot y$, and evaluate $A(c + a \cdot z)^m$ at y . This gives $m! \sum_{|\alpha|=m} a^\alpha P_\alpha(y) = 0$ for every $a > 0$. Coefficient comparison in a gives the same conclusion. $\square$

The local first variation in this lemma is automatic in finite dimension. For an existing positive C_0 -semigroup, the required domain and local analytic statements are supplied by Theorem B.1; the subsequent extension to arbitrary real contacts is kept separate from this initial local argument.

Lemma 8.4 (Poisson tangency). *Let $G_t \in \mathcal{LP}_+$, $G_0(z) = c \exp(a \cdot z)$, where $c > 0$, $a \geq 0$. Suppose $G_t = G_0 + tH + o(t)$ locally uniformly near the origin. Then H/G_0 is a polynomial of total degree at most two. Its homogeneous quadratic part is nonpositive on every strictly positive vector.*

Proof. The one-variable positive Laguerre–Pólya factorization, including its exponential factor, is

$$g(s) = g(0) e^{\sigma s} \prod_j (1 + \rho_j s), \quad \sigma, \rho_j \geq 0, \quad \sum_j \rho_j < \infty.$$

It follows from the standard Laguerre–Pólya product representation [39]; positive coefficients put every finite zero on the negative real axis. If $\log(g(s)/g(0)) = \sum_{k \geq 1} \ell_k s^k$, then for $k \geq 2$

$$\ell_k = \frac{(-1)^{k-1}}{k} \sum_j \rho_j^k, \quad |\ell_k| \leq \frac{(-2\ell_2)^{k/2}}{k}.$$

The second inequality follows from $\sum \rho_j^k \leq (\sum \rho_j^2)^{k/2}$. Apply this to $G_t(sv)$ for fixed $v > 0$. Its logarithm has a first variation, $\ell_2(t) = O(t)$, and $\ell_2(t) \leq 0$. Therefore $\ell_k(t) = o(t)$ for every $k \geq 3$. The first variation of its logarithm is $H(sv)/G_0(sv)$. Each homogeneous Taylor coefficient of degree at least three vanishes on the open

positive orthant, and is therefore the zero polynomial. The degree-two coefficient is nonpositive there, as claimed. $\square$

8.2. Collective loss and an exact Euler step. Put $E = \sum_i z_i \partial_i$, $D = \sum_i \partial_i$, and

$$C = ED - E(E - 1) = \sum_{i,j} z_j (1 - z_i) \partial_i \partial_j. \quad (152)$$

For $|x| = n$, $Cz^x = (n-1) \sum_i x_i z^{x-e_i} - n(n-1)z^x$. Thus C is a Markov generator of single-particle losses, with the rate of losing a type i particle equal to $x_i(|x| - 1)$.

Proposition 8.5 (Sharp positive Euler step). *On polynomials of total degree at most $N \geq 2$, the operator $I + hC$ preserves $\mathcal{S}_+$ and evaluation at $\mathbf{1}$ whenever $0 \leq h \leq 1/[N(N-1)]$. This is the exact interval on which it is a Markov Euler step on the whole cutoff. Consequently e^{tC} preserves $\mathcal{S}_+$.*

Proof. Write $f = \sum_{n=0}^N f_n$ in homogeneous parts and homogenize positively: $F(w, z) = \sum_n w^{N-n} f_n(z)$. Positive homogenization preserves stability [14, Theorem 4.5]. Hence

$$G(\xi, \eta, z) = F(1 + \xi + \eta, z + \eta \mathbf{1})$$

is stable. Set $a = 1 - hN(N-1)$, $b = h(N-1)$, and $\mathcal{L} = a + b(\partial_\xi + \partial_\eta) - h\partial_\xi \partial_\eta$. For $h > 0$ its negative differential symbol is $a - b(u+v) - huv$. Solving for v gives

$$\Im \frac{a - bu}{b + hu} = -\frac{b^2 + ah}{|b + hu|^2} \Im u < 0, \quad b^2 + ah = h[1 - h(N-1)] > 0.$$

Thus this symbol is stable. The constant-coefficient differential-symbol theorem [10, 13] shows that $\mathcal{L}G$ is stable or zero, including in the unchanged z -variables. For $k = N - n$, direct differentiation of $(1 + \xi + \eta)^k f_n(z + \eta \mathbf{1})$ gives, at $\xi = \eta = 0$, $\mathcal{L}G_n = [a + 2bk - hk(k-1)]f_n + (b - hk)Df_n = [1 - hn(n-1)]f_n + h(n-1)Df_n$. Therefore $(\mathcal{L}G)|_{\xi=\eta=0} = (I + hC)f$. Real specialization preserves stability or zero. The displayed monomial formula for C proves nonnegativity of the coefficients and preservation of their sum, so the output is nonzero. For $h > 1/[N(N-1)]$, the coefficient of z_1^N in $(I + hC)z_1^N$ is negative. Finally take Euler limits on each fixed finite total-degree cutoff; coefficient closure and preservation of mass give the assertion for e^{tC} . $\square$

8.3. The unrestricted logarithmic symbol. For $a \in \mathbb{R}^d$, define the logarithmic symbol, when the indicated exponential belongs to the generator domain, by $\Psi_A(z, a) = e^{-a \cdot z} A(e^{a \cdot z})$.

Theorem 8.6 (Unrestricted positive classification). *Let $(T_t)_{t \geq 0}$ be a coefficientwise-positive C_0 -semigroup of bounded operators on $\ell^1(\mathbb{N}^d)$, with generator A . The following are equivalent:*

- (1) $T_t \mathcal{LP}_+ \subseteq \mathcal{LP}_+$ for every $t \geq 0$.
- (2) $T_t \mathcal{S}_+ \subseteq \mathcal{LP}_+$ for every $t \geq 0$.
- (3) All polynomials belong to $D(A)$, and on polynomials

$$\begin{aligned} A = cI + \lambda \cdot z + \sum_i \left(b_i + \sum_j m_{ij} z_j \right) \partial_i \\ + ED_\nu - \beta E(E - 1) - \sum_i \eta_i z_i^2 \partial_i^2, \quad D_\nu = \sum_i \nu_i \partial_i. \end{aligned} \quad (153)$$

where $\lambda_i, b_i, \nu_i, \beta, \eta_i \geq 0$, $m_{ij} \geq 0$ for $i \neq j$, and $c, m_{ii} \in \mathbb{R}$.

For $d \geq 2$ the parameters are unique. For $d = 1$, combine $\beta + \eta_1$ into one nonnegative parameter. Equivalently,

$$\Psi_A(z, a) = c + \lambda \cdot z + b \cdot a + a^\top M z + (\nu \cdot a)(z \cdot a) - \beta(z \cdot a)^2 - \sum_i \eta_i (z_i a_i)^2. \quad (154)$$

The theorem concerns an existing semigroup. The separate existence criterion for a formal expression is Theorem B.4.

Proof. The first implication is immediate. Assume the second statement. Jensen approximation and Lemma 8.1 give preservation of all $\mathcal{LP}_+$, initially allowing zero. Zero is impossible: for each basis vector, strong continuity gives $(T_{t/n})_{x,x} > 0$ for sufficiently large n , and positivity and the semigroup law give $(T_t)_{x,x} \geq ((T_{t/n})_{x,x})^n > 0$. Every nonzero positive input dominates a positive multiple of a basis vector. The domain conclusion follows from Corollary B.3. Lemma 8.3 gives $A = \sum_{|\alpha| \leq 2} P_\alpha \partial^\alpha$, with $P_\alpha \in \ell^1$. Consequently

$$\|Az^x\|_1 \leq \sum_{|\alpha| \leq 2} (x)_\alpha \|P_\alpha\|_1 \leq C(1 + |x|)^2. \quad (155)$$

The polynomial partial sums of $e^{a \cdot z}$ converge in graph norm by this bound, since Poisson factorial moments are finite. Closedness of A gives

$$e^{a \cdot z} \in D(A), \quad A(e^{a \cdot z}) = e^{a \cdot z} \sum_{|\alpha| \leq 2} a^\alpha P_\alpha(z).$$

The stable approximants $(1 + a \cdot z/m)^m$ converge in coefficient norm. Their images and positive coefficient closure imply $T_t e^{a \cdot z} \in \mathcal{LP}_+$. Lemma 8.4, followed by coefficient comparison in $a > 0$, now gives $\deg P_\alpha \leq 2$ and

$$P_0^{[2]}(v) + \sum_i a_i P_i^{[2]}(v) + \sum_{|\alpha|=2} a^\alpha P_\alpha^{[2]}(v) \leq 0 \quad (a, v > 0). \quad (156)$$

Positivity of the semigroup makes the off-diagonal entries of A nonnegative. The nonconstant coefficients of $P_0 = A1$ are therefore nonnegative. Let $a \rightarrow 0$ in (156); its quadratic part must vanish, giving $P_0 = c + \lambda \cdot z$, $\lambda \geq 0$. If $|\rho| = 2$, the coefficient of $z^{(n-1)e_i + \rho}$ in Az_i^n is $n[z^\rho]P_i + O(1)$. No second-order term has sufficiently high output degree to contribute. Positivity for arbitrarily large n gives $[z^\rho]P_i \geq 0$. Replacing a by εa in (156) and taking $\varepsilon \downarrow 0$ eliminates these quadratic coefficients. Thus $P_i = b_i + \sum_j m_{ij} z_j$, with the asserted signs obtained from Az_i .

At this point $A - M_\lambda$, where $M_\lambda f = (\lambda \cdot z)f$, preserves every finite total-degree cutoff, and M_λ is bounded on ℓ^1 . Theorem B.4 gives compatible weighted semigroups: if the cutoff-preserving semigroup has bound $Me^{\omega t}$, then $\|T_t f\|_R \leq MR^N e^{(\omega + M\Lambda R)t} \|f\|_1$ for $\deg f \leq N$, $R \geq 1$, where $\Lambda = \sum_i \lambda_i$. Since $A^2 f$ has degree at most $N + 2$, the Taylor integral gives

$$\|T_t f - f - tAf\|_R \leq \frac{t^2}{2} MR^{N+2} e^{(\omega + M\Lambda R)\tau} \|A^2 f\|_1 \quad (0 \leq t \leq \tau).$$

Thus generator first variations are legitimate at every real point, not only in the unit polydisc. We now determine the second-order part algebraically.

For $|\alpha| = 2$,

$$[z^\rho]P_\alpha \geq 0 \quad (\rho \neq \alpha). \quad (157)$$

For $\alpha = 2e_i$, divide the relevant off-diagonal coefficient in Az_i^n by $n(n-1)$ and let $n \rightarrow \infty$. For $\alpha = e_i + e_j$, at least one of ρ_i, ρ_j is zero if $|\rho| \leq 2$ and $\rho \neq \alpha$. If $\rho_i = 0$, use source $e_i + ne_j$ and target $(n-1)e_j + \rho$. Terms differentiating only z_j retain z_i , and ∂_i^2 vanishes. The coefficient is therefore $n[z^\rho]P_\alpha + O(1)$. This proves (157).

The square test $(c + a \cdot z)^2$ gives

$$H_a(z) := \sum_{|\alpha|=2} a^\alpha P_\alpha(z) \leq 0 \quad \text{whenever } a > 0, a \cdot z < 0.$$

Scaling z to zero first removes the nonnegative constant terms. The linear part $\ell(a) \cdot z$ is nonpositive on the half-space $a \cdot z < 0$, so $\ell(a) = k(a)a$ with $k(a) \geq 0$. The polynomial identities $a_j \ell_i = a_i \ell_j$ show that every ℓ_i is divisible by a_i , and the linear quotients agree. Thus $k(a) = \nu \cdot a$, with $\nu_i \geq 0$. Scaling z to infinity shows that the quadratic part of H_a is nonpositive on the same half-space. Being even in z , it is nonpositive everywhere.

Here is the full elimination of off-diagonal quadratic outputs. Set $a = e_i$. On $z_i = 0$, the nonnegative coefficients in (157) force every quadratic term not containing z_i to vanish. A surviving $z_i z_j$ term is also impossible, because z_j can have either sign and arbitrarily large magnitude. For mixed sources, the coefficient of z_k^2 in the nonpositive quadratic form is $-h_{kk}a_k^2 + \sum_{i < j} [z_k^2]P_{e_i+e_j}a_i a_j$. Put $a_k = \varepsilon$, all other entries equal to one, and let $\varepsilon \downarrow 0$. This first removes all coefficients whose source avoids k . Divide the remaining inequality by ε and take the limit again to remove sources containing k . Thus every displayed nonnegative mixed-source coefficient vanishes. Finally, a mixed output different from its mixed source has an index outside the source. Support a on the two source indices. The corresponding quadratic form has zero square coefficient at the outside index; a nonpositive quadratic form with such a zero has an identically zero row there. This removes that output as well. We have proved

$$P_{2e_i} = \nu_i z_i - h_{ii} z_i^2, \quad P_{e_i+e_j} = \nu_j z_i + \nu_i z_j - 2h_{ij} z_i z_j,$$

where $H = (h_{ij})$ is positive semidefinite.

Write the principal matrix as $Q(z) = (\nu z^\top + z \nu^\top)/2 - \text{diag}(z)H \text{diag}(z)$. Different affine factors supply the additional restriction. At a real common zero of $(c_1 + a \cdot z)(c_2 + b \cdot z)$, with nonnegative factors, Lemma 8.2 gives $a^\top Q(z)b \leq 0$. For distinct i, j, k , apply it to $(z_i + z_k)(1 + z_j)$ at $(z_i, z_k, z_j) = (s, -s, -1)$. The result is

$$-\frac{\nu_i + \nu_k}{2} + s(h_{ij} - h_{kj}) \leq 0 \quad (s \in \mathbb{R}).$$

Hence all off-diagonal entries of H are a common value β when $d \geq 3$; for $d = 2$, put $\beta = h_{12}$. The contact $(s + z_i)(s + z_j)$, at $z_i = z_j = -s$, gives $-s(\nu_i + \nu_j)/2 - \beta s^2 \leq 0$, whence $\beta \geq 0$. The contact $(z_i + z_j)(s + z_i)$, at $(z_i, z_j) = (-s, s)$, gives $-s(\nu_i + \nu_j)/2 - s^2(h_{ii} - \beta) \leq 0$, whence $h_{ii} \geq \beta$. Thus $H = \beta \mathbf{1}\mathbf{1}^\top + \text{diag}(\eta)$, $\beta, \eta \geq 0$. This proves necessity and uniqueness.

For sufficiency, remove the bounded multiplication M_λ and work first on a fixed total-degree cutoff. The affine first-order part has flow

$$f(z) \mapsto e^{ct} f \left(e^{tM} z + \int_0^t e^{sM} b ds \right).$$

The matrix M is Metzler, so e^{tM} is nonnegative with positive diagonal; this substitution preserves $\mathcal{S}_+$. For $\beta > 0$ and $\nu_i > 0$, conjugate βC by positive coordinate

scaling $S_s f(z) = f(s_1 z_1, \dots, s_d z_d)$, with $s_i = \nu_i / \beta$:

$$S_s^{-1}(\beta C)S_s = ED_\nu - \beta E(E - 1).$$

Proposition 8.5 proves preservation. Zero parameters follow by parameter limits on the same finite-dimensional cutoff. For individual damping $K_i = -z_i^2 \partial_i^2$, the active algebraic symbol of $I + hK_i$ at coordinate degree N factors as

$$(z + w)^{N-2} [w + (1 - \sqrt{\theta})z][w + (1 + \sqrt{\theta})z], \quad \theta = hN(N - 1) \in [0, 1].$$

It is stable, and the coefficients of the operator on monomials are nonnegative. The finite-degree symbol theorem and Euler limits give preservation by $e^{t\eta_i K_i}$. Finite-dimensional Lie products combine these elementary flows. Stability follows by closure, and invertibility of each finite exponential excludes zero outputs. Finally the bounded-perturbation product formula, justified in Theorem B.4, adds M_λ ; its own flow is multiplication by $e^{t\lambda \cdot z}$. Jensen approximation and coefficient closure extend the result to $\mathcal{LP}_+$. Nonzero outputs follow from $T_t z^x \geq e^{A_{x,x} t} z^x$, the variation-of-constants inequality for a positive semigroup and a basis vector in its domain. $\square$

Corollary 8.7 (Arbitrary-jump Markov rigidity). *Let T_t be the transition semigroup of a conservative nonexplosive continuous-time Markov chain on $\mathbb{N}^d$, with finite exit rate at every state. No restriction on jump sizes or rate dependence is imposed. The following are equivalent:*

- (1) *Every input in $\mathcal{S}_+$ is mapped into $\mathcal{LP}_+$ at every time.*
- (2) *The same assertion holds for all $(c + a \cdot z)^m$, where $c \geq 0$, $a > 0$, and $m \geq 0$.*
- (3) *The generator is*

$$A = \sum_i \lambda_i (z_i - 1) + \sum_i \left[\delta_i (1 - z_i) + \sum_{j \neq i} r_{ij} (z_j - z_i) \right] \partial_i + \gamma C, \quad (158)$$

with $\lambda_i, \delta_i, r_{ij}, \gamma \geq 0$.

In particular, the only transitions are constant immigration, independent migration, and the single deaths $x \rightarrow x - e_i$ at rate $x_i[\delta_i + \gamma(|x| - 1)]$. If all outputs from polynomial inputs must remain polynomials, then $\lambda = 0$, and this condition is sufficient.

Proof. Finite exit rates put every basis vector in the ℓ^1 generator domain: conditioning on the first jump gives $\|(T_t z^x - z^x)/t - Az^x\|_1 \rightarrow 0$ and $\|T_t z^x - z^x\|_1 \leq 2tq(x)$. Under the second assertion, the multinomial version of Lemma 8.3 applies. The diagonal coefficient gives

$$q(x) = - \sum_{|\alpha| \leq 2} (x)_\alpha [z^\alpha] P_\alpha \leq C(1 + |x|)^2.$$

Normalized multinomial approximants to a product-Poisson law converge in total variation. Markov contraction and positive coefficient closure therefore imply preservation of those Poisson inputs. The graph-norm and Poisson-tangency argument in Theorem 8.6 applies unchanged. It yields constant immigration, affine first-order coefficients, and a second-order part

$$P_{2e_i} = \nu_i z_i - h_{ii} z_i^2, \quad P_{e_i + e_j} = \nu_j z_i + \nu_i z_j - 2h_{ij} z_i z_j, \quad H \succeq 0.$$

Only square multinomial contacts were used to reach this point. Mass preservation gives $P_\alpha(\mathbf{1}) = 0$, hence $h_{ii} = \nu_i$ and $2h_{ij} = \nu_i + \nu_j$. The quadratic part of the

square-contact expression is consequently $-(a \cdot z)(\sum_i \nu_i a_i z_i)$, nonpositive for every real z . Two nonzero real linear forms whose product is everywhere nonnegative are nonnegative multiples of one another: the second must vanish on the kernel of the first. Taking $a = \mathbf{1}$ shows that every ν_i equals a common $\gamma \geq 0$. The constant and linear mass identities give exactly (158). This proves the difficult implication without assuming preservation of products of different affine factors. Sufficiency follows from Theorem 8.6, or from the same elementary flows and bounded immigration construction. The chain in (158) is nonexplosive: its population is bounded by the initial population plus a rate $\sum_i \lambda_i$ Poisson immigration process, and every bounded-population set is finite. If $\lambda_i > 0$, from the empty state a path with any prescribed number of type i immigrations and no other jumps has positive probability before any fixed positive time. The output therefore has infinite support. Conversely $\lambda = 0$ leaves every total-degree cutoff invariant. $\square$

Modulo the lineality generated by $I, z_1 \partial_1, \dots, z_d \partial_d$, Theorem 8.6 describes, for $d \geq 2$, the formal cone generated by

$$z_i, \quad \partial_i, \quad z_j \partial_i \ (i \neq j), \quad E \partial_i, \quad -E(E-1), \quad -z_i^2 \partial_i^2.$$

This is a statement about the formal cone; it does not assert that each ray individually generates bounded operators on every fixed coefficient space.

Corollary 8.8 (Diagonal selection). *Suppose $v : \mathbb{N}^d \rightarrow \mathbb{R}$ is bounded above. The positive semigroup $T_t z^x = e^{tv(x)} z^x$ preserves $\mathcal{LP}_+$ if and only if*

$$v(x) = c + \mu \cdot x - \beta |x|(|x| - 1) - \sum_i \eta_i x_i (x_i - 1), \quad \beta, \eta_i \geq 0,$$

with the same one-dimensional merging convention as above.

Proof. The upper bound gives a bounded-operator positive C_0 -semigroup on ℓ^1 . In (153), every nondiagonal parameter must vanish because its monomial matrix entry is nonnegative and cannot cancel another off-diagonal entry. The remaining diagonal terms are exactly the displayed expression. The converse is Theorem 8.6. $\square$

Example 8.9 (The square-symbol test is insufficient). The diagonal contraction generated by $A = -z_1^2 \partial_1^2 - 4z_1 z_2 \partial_1 \partial_2 - 4z_2^2 \partial_2^2$ has logarithmic symbol $-(a_1 z_1 + 2a_2 z_2)^2$. Nevertheless, at $t = (\log 2)/2$, it sends the stable polynomial $(z_1 + z_2)(1 + z_1)$ to $z_1 + z_2 + z_1^2/2 + z_1 z_2/4$, which vanishes at $(z_1, z_2) = (-3 + i, 2 + 6i)$. Thus positive semidefiniteness of the preliminary damping matrix is strictly weaker than the common-damping condition in Theorem 8.6. By contrast, $E(\partial_1 + 2\partial_2) - 2E(E-1)$ is a positive contraction preserving stability: its column sum at z^x is $-x_1(|x| - 1) \leq 0$, and it has the form (153). It represents unequal collective loss together with killing.

8.4. Finite capacities and quadratic contacts. Now let every κ_i be finite and positive, put $V_\kappa = \text{span}\{z^x : x \leq \kappa\}$, and let $J = \{i : \kappa_i = 1\}$. The numerical capacities enter the boundary factors of the rates. The intrinsic contact domain remembers exactly the binary coordinates. With $e = \mathbf{1}$, define $\mathfrak{Q}_J$ by

$$\begin{aligned} y \in \mathbb{R}^d, \quad H = H^\top, \quad H_{ij} \geq 0, \quad H_{ii} = 0 \ (i \in J), \quad e^\top H e = 1, \\ (He)(He)^\top - H \succeq 0, \quad Hy \leq 0, \quad y^\top Hy \geq 0. \end{aligned} \tag{159}$$

Here $H_{ij} \geq 0$ is entrywise nonnegativity, and is not a positive-semidefinite condition on H . Under the normalization, the displayed matrix inequality is equivalent to

H having exactly one positive eigenvalue. Indeed, decompose $u = (e^\top H u)e + v$, with $e^\top H v = 0$. A quadratic form with one positive direction is nonpositive on this orthogonal complement, which gives $u^\top H u \leq (e^\top H u)^2$; the converse follows by restriction.

Theorem 8.10 (Finite-capacity classification). *A real operator $A : V_\kappa \rightarrow V_\kappa$ generates a positive semigroup preserving $\mathcal{S}_+ \cap V_\kappa$ if and only if its only off-diagonal entries are*

$$\begin{aligned} A_{x-e_i, x} &= x_i \left[\delta_i + \alpha_i(x_i - 1) + \sum_{j \neq i} a_{ij} x_j \right], \\ A_{x+e_i, x} &= (\kappa_i - x_i) \left[b_i + \chi_i(\kappa_i - x_i - 1) + \sum_{j \neq i} g_{ij}(\kappa_j - x_j) \right], \\ A_{x-e_i+e_j, x} &= c_{ij} x_i (\kappa_j - x_j) \quad (i \neq j), \end{aligned} \quad (160)$$

all the displayed parameters are nonnegative, and the diagonal is

$$A_{x, x} = \omega + \sum_i \mu_i x_i + \sum_i \tau_i x_i (x_i - 1) + \sum_{i < j} \theta_{ij} x_i x_j, \quad (161)$$

with real coefficients. Set $\alpha_i = \chi_i = \tau_i = 0$ for $i \in J$. Writing the second-order part as $\sum_{i, j} Q_{ij}(z) \partial_i \partial_j$, where Q is symmetric, its entries are

$$\begin{aligned} Q_{ii}(z) &= \alpha_i z_i + \tau_i z_i^2 + \chi_i z_i^3, \\ 2Q_{ij}(z) &= a_{ij} z_j + a_{ji} z_i + \theta_{ij} z_i z_j - c_{ij} z_j^2 - c_{ji} z_i^2 + g_{ij} z_i^2 z_j + g_{ji} z_i z_j^2. \end{aligned} \quad (162)$$

The additional necessary and sufficient condition is

$$\text{tr}(Q(y)H) \leq 0 \quad ((y, H) \in \Omega_J). \quad (163)$$

Within the structural family (160)–(161), every failure of preservation has a quadratic stable input witnessing failure at first order in time.

We give the structural and geometric parts of the proof separately.

Lemma 8.11 (Boundary factors and finite rate reduction). *Every positive stability-preserving generator on V_κ has the rate form (160)–(161), with the stated signs.*

Proof. Lemma 8.3 gives order at most two. For a net shift σ , write $p_\sigma(x) = [z^{x+\sigma}] A z^x$. Normal ordering gives a polynomial of total degree at most two in x , represented in reduced degree on every finite coordinate grid. Put $r_i = (-\sigma_i)_+$, $s_i = (\sigma_i)_+$. The rate vanishes when a source has fewer than r_i particles or fewer than s_i holes. Reduced interpolation on each boundary slice makes these vanishings polynomial identities. Successive division gives

$$p_\sigma(x) = \prod_i (x_i)_{r_i} (\kappa_i - x_i)_{s_i} h_\sigma(x), \quad \deg h_\sigma \leq 2 - \sum_i (r_i + s_i). \quad (164)$$

A shift with no admissible source has zero reduced polynomial. Thus a nonzero shift has $\|\sigma\|_1 \leq 2$.

To exclude double births, set $g_t(s) = (e^{tA} 1)(s, \dots, s)$. It has nonnegative coefficients, negative real zeros, and $g_0 = 1$. The coefficient of s^2 in $\log g_t$ is nonpositive. Its derivative at zero is $[s^2](A1)(s, \dots, s)$, the sum of the nonnegative double-birth rates at the empty state. Each is zero. Formula (164) shows that a double-birth rate

is a constant times its two hole factors, so it vanishes everywhere. The involution $\mathcal{J}f(z) = z^\kappa f(1/z)$ preserves nonnegative coefficients and real stability: reciprocal upper-half-plane variables lie in the lower half-plane, where a real stable polynomial is also nonvanishing. Conjugating by $\mathcal{J}$ therefore excludes double deaths. Only single births, single deaths, transfers, and a quadratic diagonal remain.

The single-death affine factor has a unique expansion in $1, x_i - 1, x_j$ ($j \neq i$), and the birth factor has a unique expansion in the corresponding hole coordinates; ineffective binary terms are set to zero. The transfer boundary factors exhaust the degree budget, giving $c_{ij} \geq 0$. The pair test $(z_i + s)(z_j + t)$ at $(-s, -t)$ gives

$$c_{ji}s^2 + c_{ij}t^2 + a_{ji}s + a_{ij}t - \theta_{ij}st + g_{ij}s^2t + g_{ji}st^2 \geq 0 \quad (s, t > 0).$$

Let $s \downarrow 0$ and then $t \downarrow 0$ to obtain $a_{ij} \geq 0$; exchange indices for a_{ji} . Let $s \rightarrow \infty$ with t fixed, then $t \rightarrow \infty$, to obtain $g_{ij} \geq 0$; exchange indices again. For $\kappa_i \geq 2$, the square test gives $\alpha_i - \tau_i s + \chi_i s^2 \geq 0$ for $s > 0$, so $\alpha_i, \chi_i \geq 0$. A state with one particle at site i and no others gives $\delta_i \geq 0$. A state with exactly one hole at i and no other holes gives $b_i \geq 0$. These are all asserted signs. Direct normal ordering of the rates yields (162). $\square$

Lemma 8.12 (Quadratic witnesses). *For $(y, H) \in \mathfrak{Q}_J$, the polynomial $f_{y,H}(z) = (z - y)^\top H(z - y)$ belongs to $\mathcal{S}_+ \cap V_\kappa$, and $Af_{y,H}(y) = 2 \operatorname{tr}(Q(y)H)$.*

Proof. Nonnegative quadratic, linear, and constant coefficients are respectively the conditions $H \geq 0$ entrywise, $-2Hy \geq 0$, and $y^\top Hy \geq 0$. The binary restrictions are exactly $H_{ii} = 0$. For $v > 0$, $v^\top Hv > 0$. Since H has one positive eigenvalue, its form is nonpositive on the H -orthogonal complement of v . If $(u + iv)^\top H(u + iv) = 0$, the imaginary part makes u belong to this complement, whereas the real part requires $u^\top Hu = v^\top Hv > 0$, a contradiction. Thus $z^\top Hz$, and its real translate, are stable. At y both the value and the gradient vanish, leaving precisely the asserted second-order trace. Lemma 8.2 proves necessity of (163); a strict violation proves the witness assertion. $\square$

For sufficiency we need a precise finite-dimensional boundary argument. We first record explicitly the homogeneous phase property used in its localizations.

Lemma 8.13 (Homogeneous common phase). *A nonzero homogeneous complex stable polynomial is a nonzero complex scalar times a polynomial with nonnegative real coefficients.*

Proof. Let its degree be m , and fix $e > 0$. Homogeneity gives $P(ie) = i^m P(e) \neq 0$. For every real x , the degree- m polynomial $P(x + te)$ has no upper-half-plane zero by stability and no lower-half-plane zero, since $P(x + te) = (-1)^m P(-x - te)$. Dividing by its leading coefficient $P(e)$ gives a monic real-rooted polynomial. In particular $P(x)/P(e)$ is real for every real x , so $P_0 = P/P(e)$ has real coefficients. It does not vanish on the positive orthant, and equals one at e , hence is positive there. Fix positive real values of all coordinates except one. The resulting univariate polynomial is real-rooted, positive on the positive half-line, and has positive leading coefficient. Its roots are nonpositive, so all its coefficients are nonnegative. Each of these coefficient polynomials in the remaining variables is homogeneous and stable or zero, by differentiation and real specialization, and is nonnegative on the remaining positive orthant. Induction on the number of variables therefore makes every coefficient of P_0 nonnegative. $\square$

Lemma 8.14 (Projective boundary and interior density). *Let $K = (\mathcal{S}_+ \cap V_\kappa) \cup \{0\}$, and let O be the open set of polynomials with every allowed coefficient strictly positive. A stable section of multidegree κ on $(\mathbb{CP}^1)^d$ having no real projective singularity is interior to the real-stable set. Moreover $K = \overline{\text{int } K} \cap \overline{O}$.*

Proof. Use real Möbius charts of positive determinant at infinity; they preserve the upper half-plane. Suppose a stable section has a zero with a nonempty set I of real coordinates and a nonempty complementary set J strictly in the upper half-plane. Real specialization and Hurwitz's theorem imply $f(x_I, z_J) \equiv 0$. Let $h(u_I, z_J)$ be the first nonzero homogeneous term of $f(x_I + u_I, z_J)$ in u_I . Positive rescaling and coefficient limits show that h is jointly stable. For each $z_J \in \mathbb{H}^J$, its homogeneous coefficients in u_I have one common phase. Every nonzero such coefficient is itself stable, by derivative and specialization closure, and is nonvanishing on $\mathbb{H}^J$. Ratios of any two of them are therefore holomorphic and real-valued on this open connected set, hence constant. Polynomial continuation gives $h(u_I, z_J) = h_0(u_I)g(z_J)$, where h_0 is real homogeneous up to a phase and g is a real stable section up to the opposite phase. If the order in u_I is at least two, every point of this real slice is a singularity. If it is one, the first derivatives along the slice are multiples of g . This section has a real projective zero: if it depends genuinely on some coordinate, specialize the others generically to real values and use a nonconstant real-rooted polynomial; if a prescribed positive degree is missing, use infinity in that coordinate. At such a zero all first derivatives of f vanish. Thus a partially nonreal boundary zero forces a real singularity.

At an all-real regular zero, linear localization shows that the nonzero gradient components have the same sign. If the components in a nonempty set J vanish, the slice in those coordinates must vanish identically. To justify this last assertion, otherwise choose a positive direction v_J on which the slice is not identically zero and a coordinate i with nonzero partial derivative. The real specialization $q(s, t) = f(p + se_i + tv_J)$ is stable, $q_s(0, 0) \neq 0$, and $q(0, t) = ct^m + O(t^{m+1})$, with $m \geq 2$, $c \neq 0$. The analytic implicit function theorem gives a zero branch $s = \phi(t) = -(c/q_s(0, 0))t^m + O(t^{m+1})$. For an appropriate ray $t = \rho e^{i\vartheta}$, $0 < \vartheta < \pi$, its leading term has positive imaginary part. For small $\rho > 0$, both s and t lie in the upper half-plane, contradicting stability. The slice therefore vanishes identically and the preceding slice argument again supplies a real projective singularity.

In the absence of singularities, every real zero thus has all gradient components nonzero and of the same sign. Choose that sign σ in a small complex coordinate neighborhood, so that $\Re(\sigma \partial_i f) > 0$ there. The inequality persists for sufficiently small real coefficient perturbations g . For real u and positive v within this neighborhood,

$$\Im(\sigma g(u + iv)) = \sum_i v_i \int_0^1 \Re(\sigma \partial_i g(u + itv)) dt > 0.$$

Outside these neighborhoods the section is nonzero on the product of closed upper hemispheres, by the exclusion of partially nonreal zeros. Compactness gives a finite cover and a positive lower bound on the complementary compact set. Every sufficiently small real perturbation is therefore stable. This proves the first assertion.

To prove density, apply small independent birth–death flows, with strictly positive birth and death rates, to each finite site. Their two-state Möbius matrices have positive entries and positive determinant, so the induced degree- κ_i action preserves stability. Their transition matrices are strictly positive at positive times, since

every state communicates through nearest-neighbor moves. Thus every nonzero member of K is approximated by members of $K \cap O$. For such an f , apply $D_s = e^{-sE(E-1)}$, which preserves positive stability by Theorem 8.6's finite-cutoff proof and leaves the box invariant. For any real x , the radial polynomial $r \mapsto f(rx)$ is real-rooted. Indeed, positive homogenization $F(w, z) = w^N f(z/w)$ is stable, and $t \mapsto F((0, x) + t(1, \varepsilon, \dots, \varepsilon))$ is real-rooted for $\varepsilon > 0$. Its limit $F(t, x)$ has nonzero leading coefficient $F(1, 0) = f(0) > 0$; reciprocity gives the assertion for $f(rx)$.

Put $p_s(r) = (D_s f)(rx)$. It remains real-rooted and satisfies $\partial_s p_s = -r^2 p_s''$. Every nonzero root is simple when $s > 0$. Otherwise let $r_0 \neq 0$ have multiplicity $m \geq 2$ at an interior time $s_0 > 0$, and factor the nearby root cluster as $u^m + a_1(s)u^{m-1} + a_2(s)u^{m-2} + \dots$, $u = r - r_0$. Contour power sums make its coefficients differentiable. Dividing the evolution identity by the nonvanishing complementary factor shows that the derivative of this monic factor has no terms below u^{m-2} , and that $a_2'(s_0) = -r_0^2 m(m-1)$. The sum of squares of the real root displacements is $a_1^2 - 2a_2$, nonnegative and zero at s_0 , but with derivative $2r_0^2 m(m-1) > 0$, a contradiction. A finite critical zero of $D_s f$ would give a repeated radial root at $r = 1$, so none exists. At a projective face with coordinates I at infinity, let $m = \sum_{i \in I} \kappa_i$ and let g be the leading-coefficient face of f . The corresponding face of $D_s f$ is

$$e^{-sm(m-1)}(D_s g)(e^{-2sm} z_{I^c}).$$

The face g is stable, has all coefficients positive and has positive constant coefficient; the same radial argument excludes a critical zero there. A zero with all coordinates at infinity is excluded by the positive top coefficient. The first assertion makes $D_s f$ an interior point. Letting the two smoothing times tend to zero, and finally scaling a positive interior polynomial to zero, proves the density assertion. $\square$

Lemma 8.15 (Ordinary boundary contacts). *Let $M = \dim V_\kappa$. Except for a semialgebraic subset of dimension at most $M - 2$, every point of $\partial K \cap O$ has a unique real projective singularity. It is finite, has nonsingular Hessian, and admits a local defining function*

$$h(g) = \sigma g(y(g)), \quad K \text{ locally} = \{h \leq 0\},$$

where $y(g)$ is its continued critical point and $\sigma \in \{1, -1\}$. At the original point f , the matrix $H = \sigma \text{Hess } f(y)/2$, after positive normalization, belongs to $\mathfrak{Q}_J$.

Proof. The univariate capacity-one case is immediate: the strictly positive affine polynomials are interior. In all other cases, for fixed projective p , the conditions $f(p) = 0$, $df(p) = 0$ impose $d + 1$ independent linear conditions. First jets are surjective since every capacity is positive. The incidence space therefore has dimension $M - 1$. Hessian degeneracy imposes a further nontrivial polynomial equation in each fiber: the allowed local quadratic $\sum_{i < j} u_i u_j$ has invertible Hessian for $d \geq 2$; in one variable use u^2 . A projective coordinate at infinity likewise reduces the dimension of the point parameter by at least one. At a finite nondegenerate singularity, the implicit function theorem continues the critical point as $y(g)$; the differential of its critical value is evaluation at y . Evaluations at two distinct finite points are linearly independent, as is seen on $1, z_1, \dots, z_d$. Two distinct ordinary singularities thus lie on transverse critical-value hypersurfaces and impose codimension at least two. All these sets are projections of real algebraic incidence sets in finitely many projective charts, hence semialgebraic. Their dimension bounds give finite smooth

stratifications, each covered by countably many charts of dimension at most $M - 2$. This proves the exceptional-set assertion.

At a remaining boundary point, Lemma 8.14 supplies its singularity. The quadratic localization of the stable positive homogenization at $(1, y)$ is, with its common sign chosen, $(v - sy)^T H(v - sy)$. The homogeneous common-phase property and quadratic stability yield $H \geq 0$ entrywise, $Hy \leq 0$, $y^T Hy \geq 0$, and exactly one positive eigenvalue. The box imposes $H_{ii} = 0$ for binary coordinates. Its positive normalization is therefore in Ω_J .

The Hessian remains invertible nearby, and compactness excludes other projective singularities outside the chosen neighborhood. Lemma 8.14 implies that the local boundary is contained in $h(g) = \sigma g(y(g)) = 0$. Here $dh \neq 0$, since its differential is signed evaluation at y . If $h(g) > 0$, a positive-direction restriction through $y(g)$ has a positive critical value and a positive quadratic leading coefficient near the former double root; its two nearby zeros are nonreal. That side is unstable. Interior density from Lemma 8.14 gives stable interior points on the side $h < 0$. Shrink to an implicit-function chart in which this side is connected. Membership in K cannot change within the side without another boundary point, and every boundary point has $h = 0$. Thus the whole side $h < 0$ is stable. Closedness then includes $h = 0$, giving exactly the stated local half-space description. $\square$

Lemma 8.16 (Integration through ordinary boundary points). *Let $K \subset \mathbb{R}^M$ be closed, O open, and $K = \overline{\text{int } K} \cap \overline{O}$. Suppose a complete ambient flow generated by a C^1 vector field leaves O forward invariant. Suppose the boundary in O , except for countably many C^1 images of dimension at most $M - 2$, has C^2 local defining functions h with $dh \neq 0$ and $K = \{h \leq 0\}$. Then the flow preserves K if and only if $dh(V) \leq 0$ at every such ordinary boundary point, where V is its vector field.*

Proof. Necessity is differentiation from a boundary point. For sufficiency fix a time horizon T . If Σ is the exceptional set, the initial points whose trajectories meet Σ before time T lie in the union of the images $(t, u) \mapsto \Phi_{-t}(\psi(u))$, $0 \leq t \leq T$, where ψ ranges over its charts. Each parameter space has dimension at most $M - 1$, and countable compact covers make these images Lipschitz images of lower-dimensional parameter sets relative to M -dimensional volume. Their union has measure zero. Only the ambient inverse flow is used; O need not be backward invariant.

In a regular chart, the ordinary boundary points are dense in its smooth hypersurface, since Σ has zero $(M - 1)$ -dimensional measure. Continuity therefore extends $dh(V) \leq 0$ to that entire hypersurface. A local projection π to it satisfies $\|x - \pi(x)\| \leq C|h(x)|$. The function $a(x) = dh(x)V(x)$ is locally Lipschitz, so for $h(x) \geq 0$, $a(x) \leq LC h(x)$. Along a trajectory the positive part of h consequently satisfies the upper right derivative inequality $D^+ h_+ \leq LC h_+$. Gronwall's inequality forbids a crossing from $h \leq 0$ to $h > 0$. Every initial point of $\text{int } K \cap O$ outside the null hitting set therefore remains in K until T , because a first exit would occur in a regular chart. Such initial points are dense in K . Continuity of the flow and closedness of K finish the proof, since T was arbitrary. $\square$

Completion of the proof of Theorem 8.10. Necessity was proved in Lemmas 8.11 and 8.12. Conversely the rate conditions make A Metzler. Its ambient flow e^{tA} is complete, and the strictly positive coefficient orthant is forward invariant, since $(e^{tA} f)_x \geq e^{tA_{x,x}} f_x > 0$ for $f \in O$. Lemmas 8.14 and 8.15 verify the geometric hypotheses of Lemma 8.16. At an ordinary boundary point, $f(y) = 0$ and $\nabla f(y) = 0$,

so

$$dh_f(Af) = \sigma(Af)(y) = 2 \operatorname{tr}(Q(y)H) \leq 0.$$

The normalization of H changes only a positive factor. Lemma 8.16 now proves preservation of K . Invertibility of the finite matrix exponential excludes zero outputs from nonzero inputs. This proves sufficiency. $\square$

Remark 8.17 (A compact certificate). Write $Q = Q^{[1]} + Q^{[2]} + Q^{[3]}$ by homogeneous degree and put $\widehat{Q}(r, y) = r^2 Q^{[1]}(y) + r Q^{[2]}(y) + Q^{[3]}(y)$. Condition (163) is equivalent to $\operatorname{tr}(\widehat{Q}(r, y)H) \leq 0$ under (159), $r \geq 0$, and $r^2 + \|y\|^2 = 1$. For $r > 0$, this is the identity $\widehat{Q}(r, y) = r^3 Q(y/r)$; the face $r = 0$ follows by continuity. The set is compact, since $H \geq 0$ entrywise and $e^\top H e = 1$ bound every matrix entry. The number of auxiliary variables is $O(d^2)$, independently of the numerical capacities. This does not remove the capacities from the actual rates.

8.5. Mixed finite and infinite capacities. Let $F = \{i : \kappa_i < \infty\}$, $U = \{i : \kappa_i = \infty\}$, and $J = \{i : \kappa_i = 1\}$. Let $\mathcal{LP}_{+, \kappa}$ denote the positive Laguerre–Pólya class supported on Ω_κ .

Theorem 8.18 (Mixed-capacity classification). *Let T_t be a coefficientwise-positive C_0 -semigroup of bounded operators on $\ell^1(\Omega_\kappa)$ with generator A . Preservation of $\mathcal{LP}_{+, \kappa}$ at every time is equivalent to preservation for every allowed stable polynomial input, and to the following structural assertion together with (163): all allowed polynomials belong to $D(A)$, the only off-diagonal entries are*

$$\begin{aligned} A_{x-e_i, x} &= x_i \left[\delta_i + \alpha_i(x_i - 1) + \sum_{j \neq i} a_{ij} x_j \right], \\ A_{x+e_i, x} &= (\kappa_i - x_i) \left[b_i + \chi_i(\kappa_i - x_i - 1) \right. \\ &\quad \left. + \sum_{j \in F \setminus \{i\}} g_{ij}(\kappa_j - x_j) \right] \quad (i \in F), \\ A_{x+e_j, x} &= \lambda_j \quad (j \in U), \\ A_{x-e_i+e_j, x} &= c_{ij} x_i (\kappa_j - x_j) \quad (j \in F, i \neq j), \\ A_{x-e_i+e_j, x} &= r_{ij} x_i \quad (j \in U, i \neq j). \end{aligned} \tag{165}$$

Every displayed off-diagonal parameter is nonnegative. The diagonal is (161). In (162) and (159) use the conventions

$$\begin{aligned} \alpha_i &= \tau_i = 0 \quad (i \in J), & \chi_i &= 0 \quad (i \in U \cup J), \\ g_{ij} &= 0 \text{ unless } i, j \in F, & c_{ij} &= 0 \text{ unless } j \in F. \end{aligned}$$

Entries with targets outside Ω_κ are omitted. The existence criterion for the formal operator is separate and is given in Theorem B.4.

Proof. Assume polynomial-input preservation. Jensen approximation, positive coefficient closure, and the strictly positive diagonal entries of T_t , proved in Theorem 8.6, extend it to all $\mathcal{LP}_{+, \kappa}$. Corollary B.3 supplies the polynomial domain and the local first variation needed for Lemma 8.3. The order is at most two. The growth-reduction theorem, Theorem B.6, applies with Lemma 8.4: it identifies the

bounded part that strictly increases the total occupation of the unrestricted sites as

$$R = \sum_{j \in U} \lambda_j z_j + \sum_{i \in F, j \in U} r_{ij} z_j \partial_i, \quad \lambda_j, r_{ij} \geq 0,$$

and supplies the locally uniform first variation at arbitrary real points. In particular, $L = A - R$ cannot increase that unrestricted occupation. These analytic statements precede the signed contacts used below.

For completeness, we give the remaining algebraic reduction. For a net shift σ , the rate is a reduced polynomial of total degree at most two. Boundary interpolation, exactly as in (164), gives

$$p_\sigma(x) = \prod_i (x_i)_{(-\sigma_i)_+} \prod_{i \in F} (\kappa_i - x_i)_{(\sigma_i)_+} h_\sigma(x), \quad \deg h_\sigma \leq 2 - \sum_i (-\sigma_i)_+ - \sum_{i \in F} (\sigma_i)_+. \quad (166)$$

For unrestricted coordinates, vanishing on every integer point is a polynomial identity; for finite ones, use reduced interpolation. Thus at most two source or finite-target boundary factors can occur. Double finite births have constant times two hole factors and are excluded by A1: its simultaneous two-birth coefficient is non-negative but the first variation of the second logarithmic coefficient along a positive diagonal is nonpositive. Hence that coefficient is zero.

For every allowed multi-index α of total degree two, apply the product-root test to $\prod_i (z_i + s_i)^{\alpha_i}$. It gives

$$P_\alpha(-s_{\text{supp } \alpha}, z_{\text{spectators}}) \leq 0$$

for every real spectator vector. We classify all contributions to this complete coefficient polynomial, including catalytic one-particle shifts. Quadratic monomials in unrestricted destinations outside the source have nonnegative coefficients, by (166) and positivity. Set these spectators equal and let them tend to positive infinity. Their quadratic coefficients must vanish. A remaining free destination enters linearly. For two distinct source sites its coefficient has the form

$$d + t_i z_i + t_j z_j - b_i z_i^2 - b_j z_j^2;$$

for a repeated source it has the form $d + t_i z_i$. These are all possibilities because (166) leaves at most two total factors: a two-to-one replacement gives the constant d , a source-dependent transfer gives a term $t_i z_i$, and a simultaneous finite birth gives its negative hole term $-b_i z_i^2$. The destination can have either real sign, so its coefficient vanishes identically on the negative source orthant. Coefficient comparison removes every such replacement, every source-self or third-site catalytic transfer, and every simultaneous finite birth and unrestricted transfer. A pure double death is the remaining nonnegative constant in P_α ; sending the negative source roots to zero removes it as well.

A transfer to a finite site already has its two boundary factors and is $c_{ij} x_i (\kappa_j - x_j)$, $c_{ij} \geq 0$. After the preceding eliminations a transfer to an unrestricted site can only be $x_i (r_{ij} + h_{ij} x_j)$, with $r_{ij}, h_{ij} \geq 0$. The pair contact at $(z_i, z_j) = (-s, -t)$ contains the term $h_{ij} t^2$ in its second-order coefficient, while the other remaining terms are at most linear in t for fixed s , apart from a possible reverse finite-target contribution $-c_{ji} s^2$. Letting $t \rightarrow \infty$ forces $h_{ij} = 0$. Finally a finite birth with unrestricted catalyst has rate $h(\kappa_i - x_i) x_\ell$, $h \geq 0$, and contributes $-h z_i^2 z_\ell$ to the mixed principal coefficient. At $(-s, -t)$, divide the contact inequality by t and let $t \rightarrow \infty$. Its leading dependence on s is $h s^2$; the remaining terms have degree

at most one. Letting $s \rightarrow \infty$ forces $h = 0$. The surviving shifts are precisely (165). The small-root, large-root, and boundary-state arguments in Lemma 8.11 give all their asserted signs, also when the other occupancies are unrestricted. The diagonal is the reduced polynomial of degree at most two given by normal ordering. Lemma 8.12 proves necessity of the contact inequality.

Conversely, impose artificial capacities $N_j \geq 2$ at the unrestricted sites. Keep the original diagonal, and replace immigration and transfer into those sites by $\lambda_j(1 - x_j/N_j)$ and $r_{ij}x_i(1 - x_j/N_j)$, respectively. The principal trace changes by

$$\mathrm{tr}(Q_N(y)H) - \mathrm{tr}(Q(y)H) = - \sum_{j \in U, i \neq j} \frac{r_{ij}}{N_j} y_j^2 H_{ij} \leq 0.$$

The immigration correction is first order and the binary set is unchanged. Every approximant therefore satisfies Theorem 8.10. Theorem B.8 proves convergence of these finite flows, started from their coordinate projections, to the given semigroup; its proof uses the bounded increasing part and weighted estimates, not an unbounded similarity on ordinary ℓ^1 . Theorem B.9 and Lemma 8.1 pass preservation to all $\mathcal{LP}_{+, \kappa}$, and exclude zero outputs. This proves all implications. $\square$

8.6. One-site cones, conservation, and exact obstructions.

Corollary 8.19 (One finite site). *For capacity $K \geq 2$, the exact positive stability-preserving cone is*

$$Az^n = d_n z^{n-1} + u_n z^{n+1} + v_n z^n,$$

where

$d_n = n[\delta + \alpha(n-1)]$, $u_n = (K-n)[b + \chi(K-n-1)]$, $v_n = c + \mu n + \tau n(n-1)$, $\delta, b, \alpha, \chi \geq 0$, $c, \mu \in \mathbb{R}$, and $\tau \leq 2\sqrt{\alpha\chi}$. For $K = 1$, every Metzler 2×2 matrix qualifies. For every $K \geq 2$, the cone modulo $I, z\partial_z$ is nonpolyhedral.

Proof. In one variable the normalized contact has $H = 1$, $y \leq 0$. The condition $\alpha y + \tau y^2 + \chi y^3 \leq 0$ is therefore $\alpha - \tau s + \chi s^2 \geq 0$ for all $s > 0$. With $\alpha, \chi \geq 0$, this is equivalent to $\tau \leq 2\sqrt{\alpha\chi}$, including either zero parameter by limits. For $K = 1$, every nonzero affine polynomial with nonnegative coefficients is stable, and every positive invertible matrix exponential preserves this class. For nonpolyhedrality, for each $u > 0$ the functional

$$\alpha/u + u\chi - \tau + \delta + b$$

is nonnegative on the quotient cone. Equality forces $\delta = b = 0$, $\alpha = u^2\chi$, and $\tau = 2u\chi$, by the arithmetic–geometric mean inequality and the cone inequality. Its zero set is exactly the ray generated by $(\alpha, \chi, \tau) = (u^2, 1, 2u)$. These distinct exposed rays, for every $u > 0$, exclude polyhedrality. $\square$

Corollary 8.20 (Conserving mixed systems). *A finite-exit, particle-number-conserving Markov generator on Ω_κ preserves all allowed positive stable polynomials if and only if its only rates are*

$$q(x, x - e_i + e_j) = \begin{cases} c_{ij}x_i(\kappa_j - x_j), & i, j \in F, \quad c_{ij} = c_{ji} \geq 0, \\ r_{ij}x_i, & j \in U, \quad r_{ij} \geq 0, \end{cases}$$

and there is no transfer from an unrestricted to a finite site. The preservation quantifier ranges over the entire allowed polynomial space, including mixtures of different total particle numbers.

Proof. Every fixed-total state space is finite, so the chain is nonexplosive. Conservation and nonnegative off-diagonal entries remove all births and deaths from Theorem 8.18. For two finite sites the pair part of the principal coefficient is

$$(c_{ij} + c_{ji})z_i z_j - c_{ij}z_j^2 - c_{ji}z_i^2.$$

The contact at $(-s, -t)$ requires $(t - s)(c_{ij}t - c_{ji}s) \geq 0$ for every $s, t > 0$. Taking t on either side of s forces $c_{ij} = c_{ji}$. For $i \in U, j \in F$, a rate $c_{ij}x_i(\kappa_j - x_j)$ gives the pair coefficient $c_{ij}(z_i z_j - z_j^2)$. Taking $z_j = -s, z_i = -t$ and $t \rightarrow \infty$ forces $c_{ij} = 0$. No second-order term restricts independent transfer into an unrestricted site.

For sufficiency, a finite pair contributes

$$G_{ij}^\kappa = (z_j - z_i)(\kappa_j \partial_i - \kappa_i \partial_j) - (z_j - z_i)^2 \partial_i \partial_j. \quad (167)$$

Normalized polarization replaces each site by κ_i binary slots. Summing the swap generators over every cross-block pair of slots intertwines with G_{ij}^κ : on a state with occupancies x_i, x_j , exactly $x_i(\kappa_j - x_j)$ swaps move a particle from i to j . A binary swap τ has flow

$$e^{t(\tau - I)} = \frac{1 + e^{-2t}}{2} I + \frac{1 - e^{-2t}}{2} \tau,$$

which preserves stability by partial symmetrization [14]. Polarization, specialization, and finite-dimensional Lie products prove preservation for finite exclusion. Independent transfers into unrestricted sites have the affine-substitution flow from Theorem 8.6. On each fixed total-degree cutoff all these operators act in finite dimension, so a final Lie product combines them. $\square$

Corollary 8.21 (Reversibility and conserving time-one embeddings). *On a fully finite box put $w(x) = \prod_i \binom{\kappa_i}{x_i}$ and $D = \text{diag}(\sqrt{w(x)})$. Every generator in Corollary 8.20 is reversible with respect to w . If its site graph is connected, its unique stationary law at total population n is $w(x)/(\sum_n \kappa_i)$. A stochastic, number-conserving operator T is the time-one map of a positive stability-preserving Markov semigroup if and only if $S = D^{-1}TD$ is symmetric positive definite and*

$$D(\log S)D^{-1} = \sum_{\{i,j\}} c_{ij} G_{ij}^\kappa, \quad c_{ij} \geq 0.$$

The embedding generator in this class is unique.

Proof. For $y = x - e_i + e_j$, the binomial identities give

$$\frac{w(y)}{w(x)} = \frac{x_i}{\kappa_i - x_i + 1} \frac{\kappa_j - x_j}{x_j + 1}.$$

This is exactly detailed balance for the symmetric rates. Connectivity permits moving particles along site paths until any prescribed occupancy of the same total is reached, so the fixed-total chain is irreducible. The normalization is the coefficient of s^n in $\prod_i (1 + s)^{\kappa_i}$.

If $T = e^A$ with A Markov and positive, choose a so that $B = A + aI$ is entrywise nonnegative. If an entry of e^A between different total layers is zero, every term in $e^{-a} \sum_{m \geq 0} B^m / m!$ at that entry is zero, in particular the corresponding off-diagonal entry of A . Thus time-one conservation forces generator conservation. The previous corollary applies, and $D^{-1}AD$ is symmetric. Its exponential is symmetric positive definite, and its unique symmetric logarithm is $\log S$. This proves necessity and uniqueness. Conversely the displayed logarithm is a nonnegative combination of

the conserving generators (167); its exponential is T and the previous corollary proves preservation. $\square$

For two binary sites, $T_p = (1 - p)I + p\tau$ preserves stability for all $0 \leq p \leq 1$, but its conserving Markov embedding exists exactly for $0 \leq p < 1/2$. Indeed the antisymmetric eigenvalue is $1 - 2p$, and the embedding generator is $-\frac{1}{2}\log(1 - 2p)(\tau - I)$. Furthermore the one-particle rates of any finite conserving system obey every cycle identity: the product of forward rates along a labeled cycle equals the product along its reverse, since $a_{ij} = c_{ij}\kappa_j$. Pointwise limits preserve these identities. The independent directed three-cycle on unrestricted sites violates them. Thus unrestricted conserving systems need not be dilute limits of finite conserving systems on the same labeled sites. This does not apply to the killed approximants used in Theorem 8.18.

Corollary 8.22 (Support restriction for universal conserving mixing). *On a full binary box, suppose a universally positive stability-preserving conserving Markov flow leaves a fixed-size family $\mathcal{B}$ invariant and is irreducible on it. Then, for a partition into site-graph components $C_1, \dots, C_r$ and integers $k_1, \dots, k_r$,*

$$\mathcal{B} = \{S : |S \cap C_\ell| = k_\ell \text{ for every } \ell\}.$$

Proof. Corollary 8.20 makes the flow symmetric exclusion on a site graph. Every edge transposition must preserve $\mathcal{B}$, since a positive rate cannot leave an invariant family. Edge transpositions generate the full symmetric group of each connected component. Their orbits are exactly the sets with prescribed component occupancies. Irreducibility makes $\mathcal{B}$ one such orbit. $\square$

Proposition 8.23 (Exact compensation for a capped transfer). *Let z have finite capacity $K \geq 1$, and let w be unrestricted. Add to common coalescence of rate $\gamma \geq 0$ the transition $(m, n) \rightarrow (m+1, n-1)$ at rate $cn(K-m)$, where $c \geq 0$. The resulting Markov flow preserves positive stability if and only if $c \leq 2\gamma$.*

Proof. The additional differential operator is $cK(z-w)\partial_w + cz(w-z)\partial_z\partial_w$. For a contact matrix $H = \begin{pmatrix} h & a \\ a & k \end{pmatrix}$ and $y = (z, w)$, common coalescence has principal trace $\gamma(e^\top Hy - y^\top Hy) \leq 0$. At $c = 2\gamma$, the total trace simplifies exactly to

$$\gamma e^\top Hy - \gamma(h+2a)z^2 - \gamma kw^2 \leq 0.$$

The endpoints $c = 0$ and $c = 2\gamma$ therefore pass every contact, and linearity gives the entire interval. Theorem 8.18 proves sufficiency. For necessity take $h = k = 0, a = 1/2, z = -s, w = -t, s, t > 0$. Twice the trace is $-\gamma(s+t) - (2\gamma - c)st - cs^2$. If $c > 2\gamma$, choose $s > \gamma/(c - 2\gamma)$ and then sufficiently large t ; the trace is positive. This is an admissible contact, including when $K = 1$, and proves necessity. For example $K = 1, \gamma = 1, c = 3, f = (z+2)(w+15)$, and $t = \log 2$ give

$$T_t f = 30 + \frac{139}{8}z + \frac{3}{8}w + \frac{1}{4}zw, \quad \frac{139}{8}\frac{3}{8} - 30\frac{1}{4} = -\frac{63}{64} < 0.$$

The last inequality is the bivariate multiaffine stability discriminant. $\square$

Proposition 8.24 (Capacity and unchanged coordinates). *The operator $A = (z - z^2)\partial_z^2$ preserves positive stability on $V_{(2,1)}$, with the binary coordinate w unchanged. It fails on $V_{(2,1,1)}$ with two unchanged binary coordinates. Moreover there is no*

positive stability-preserving generator $\tilde{A}$ on three binary variables (u, v, w) satisfying the exact compression $D\tilde{A}P = A$, where

$$Pz = (u + v)/2, \quad Pz^2 = uv, \quad Pw = w, \quad Dg(z, w) = g(z, z, w).$$

Proof. For one unchanged coordinate, the contact matrix is $H = \begin{pmatrix} h & a \\ a & 0 \end{pmatrix}$, $h, a \geq 0$. If $h > 0$, the condition $Hy \leq 0$ forces $y_z \leq 0$: use the second row when $a > 0$, and the first row when $a = 0$. Thus $\text{tr}(Q(y)H) = h(y_z - y_z^2) \leq 0$; if $h = 0$ it is zero. Theorem 8.10 proves preservation. With two unchanged binary variables, the stable input $(z + w_1)(z + w_2)$ evolves into

$$\theta z^2 + z(w_1 + w_2) + w_1 w_2 + (1 - \theta)z, \quad \theta = e^{-2t}.$$

Identify $w_1 = w_2 = w$ and specialize $z = r \in (0, 1)$. Its discriminant in w is $-4(1 - \theta)r(1 - r) < 0$ for $t > 0$, proving failure.

Suppose the asserted lift exists. Average it with the swap of u, v . The positive preserving generator cone is convex, by finite-dimensional Lie products, and this averaging preserves the compression identity. The averaged lift commutes with the swap. Since PD is the projection onto its symmetric subspace and D kills the antisymmetric subspace, we have $D\tilde{A} = D\tilde{A}PD = AD$. Evaluation at all ones makes $\tilde{A}$ Markov, because A is Markov. Positivity and this lumping identity forbid any change of w , any jump from zero u, v -occupancy, or any occupancy change from a singleton. From the double occupancy uv , symmetry forces deaths to u and v at rate one each. The only remaining transitions are swaps of the singleton states, at rates $r_0, r_1 \geq 0$ according to the value of w .

For $F_s = (u + w)(v + s)$, direct application gives at $(u, v, w) = (2s, -s, -2s)$

$$(\tilde{A}F_s)(2s, -s, -2s) = s + (4 - 3r_0 - 6r_1)s^2.$$

Put $c = 3r_0 + 6r_1$ and $s = 1/[2(1 + c)]$. This value is $(c + 6)/[4(c + 1)^2] > 0$, although both nonnegative affine factors vanish at the chosen real point. Lemma 8.2 gives a contradiction. $\square$

For unrestricted z -coordinates, the Markov flow (158) with $\lambda = 0$ preserves all joint positive stable inputs after adjoining one unrestricted unchanged coordinate if and only if $\gamma = 0$. Sufficiency is affine substitution. For necessity, apply Corollary 8.7 in the enlarged system: the common quadratic rate must vanish because the unchanged coordinate has zero second-degree action. It is enough to test $(w + a \cdot z)^m$, since positive scaling and translation of w supply all enlarged multinomial tests. This quantifier is different from the one-binary-spectator assertion above.

Example 8.25 (The semigroup assumption in multinomial testing). Multinomial tests do not characterize arbitrary positive operators. Define a stochastic linear map from two variables to one by

$$K(z_1 z_2) = (1 + w^2)/2, \quad K(z^\alpha) = w \quad (\alpha \neq (1, 1)).$$

For $F = (c + az_1 + bz_2)^m$, put $M = (c + a + b)^m$, $U = [z_1 z_2]F$. Then $KF = U(1 + w^2)/2 + (M - U)w$. For $m \geq 2$,

$$M \geq \binom{m}{2} c^{m-2} (a + b)^2 \geq 2m(m - 1) c^{m-2} ab = 2U.$$

For $m < 2$, $U = 0$. Thus the quadratic discriminant is $M(M - 2U) \geq 0$, and every such output is a nonzero positive stable polynomial. Nevertheless the stable

monomial $z_1 z_2$ is sent to a polynomial vanishing at $w = i$. Even adjoining all monomial tests does not suffice if zero outputs are allowed. The projection retaining only $z_1 z_2, z_1 z_4, z_2 z_3, z_3 z_4$ sends every multinomial, for $m \geq 2$, to

$$m(m-1)c^{m-2}(a_1 z_1 + a_3 z_3)(a_2 z_2 + a_4 z_4),$$

and sends every monomial to a stable monomial or zero. It sends $(z_1 + z_2)(z_3 + z_4)$ to $z_1 z_4 + z_2 z_3$, which vanishes at $z_1 = z_4 = 1+i, z_2 = z_3 = -1+i$. These examples concern isolated maps and do not contradict Corollary 8.7.

Corollary 8.26 (A finite-degree witness under an order bound). *Suppose a particle-number-conserving Markov generator on unrestricted sites is known to have differential order at most $R < \infty$. If it is not an independent-particle migration generator, then some multinomial of degree at most $\max(2, R)$ fails stability preservation at arbitrarily small positive times.*

Proof. If every first variation at a multinomial root contact of degree $3, \dots, R$ passed Lemma 8.2, coefficient comparison as in Lemma 8.3 would remove all derivatives of order greater than two. Degree conservation already supplies the output-degree bound. If all degree-two square contacts also passed, the positivity and mass arguments in Corollary 8.7 would give common coalescence and independent migration. Conservation removes coalescence. This contradicts the hypothesis. Thus one of the stated root-contact conditions fails; its local root argument produces nonreal zeros for arbitrarily small positive times. $\square$

For example, the positive conserving binary-consensus generator $A = (z_1 - z_2)^2 \partial_1 \partial_2$ sends $(z_1 + z_2)^2$ at time t to $(2 - \theta)(z_1^2 + z_2^2) + 2\theta z_1 z_2$, where $\theta = e^{-2t}$. Specializing $z_2 = 1$ gives discriminant $16(\theta - 1) < 0$ at every positive time.

APPENDIX A. HERMITE COMPRESSION AND BOUNDARY LOCALIZATION

The Hermite–Sylvester criterion represents real-rootedness by a positive semidefinite matrix of Newton sums; see, for example, [43]. At a polynomial with repeated roots, the differential of this matrix must be tested only on its kernel. We show that this necessary condition is also sufficient for coefficient tangency, at every root multiplicity. The proof identifies the compressed matrix in independent root-jet coordinates and constructs a real-rooted curve realizing each admissible direction. Root-cluster tangent descriptions belong to the variational theory developed in [21, 20]; the matrix formulation also records exactly when semidefinite normal multipliers have a closed image.

A.1. Newton sums and the compressed differential. Fix an integer $n \geq 1$. Let

$$\mathcal{R}_n = \{z^n + a_1 z^{n-1} + \dots + a_n : \text{all roots are real}\}.$$

We regard $\mathcal{R}_n$ as a closed subset of the affine space of real monic degree- n polynomials. Its ambient tangent space is $V_{n-1} = \mathbb{R}[z]_{\leq n-1}$, with the Euclidean coefficient inner product in the basis $1, z, \dots, z^{n-1}$. The tangent cone throughout this section is the contingent cone

$$T_{\mathcal{R}_n}(p) = \left\{ h \in V_{n-1} : \frac{p_j - p}{t_j} \longrightarrow h \text{ for some } p_j \in \mathcal{R}_n, t_j > 0, t_j \downarrow 0 \right\}. \quad (168)$$

For any monic real polynomial p of degree n , let $s_k(p)$ be the sum of the k th powers of its roots, counted with multiplicity, and set $s_0(p) = n$. Newton's identities make every s_k a polynomial function of the coefficients of p . Define

$$H(p) = (s_{i+j}(p))_{0 \leq i, j \leq n-1}. \quad (169)$$

Thus $H(p)$ has size n , whereas the vector space of polynomials of degree at most n has dimension $n + 1$. The derivative $DH_p[h]$ is taken in the monic affine space, so that $\deg h < n$ and $Ds_0[h] = 0$.

Lemma A.1 (Hermite representation). *A monic real degree- n polynomial p belongs to $\mathcal{R}_n$ if and only if $H(p) \succeq 0$. If*

$$p(z) = \prod_{a \in \mathcal{A}} (z - a)^{q_a}, \quad \mathcal{A} \subset \mathbb{R}, \quad q_a \geq 1, \quad \sum_{a \in \mathcal{A}} q_a = n, \quad (170)$$

then, upon identifying a coefficient vector with its polynomial v ,

$$v^\top H(p) v = \sum_{a \in \mathcal{A}} q_a v(a)^2, \quad \ker H(p) = \{v \in V_{n-1} : v(a) = 0 \text{ for every } a \in \mathcal{A}\}. \quad (171)$$

Proof. Expanding the quadratic form in (169) gives the sum of $v(\lambda)^2$ over all roots λ of p , counted with their multiplicities. This proves (171) and the forward implication. Conversely, suppose that p has a nonreal root λ . Its conjugate has the same multiplicity. Interpolate the values i and $-i$ at λ and $\bar{\lambda}$, respectively, and the value zero at every other distinct root. The interpolating polynomial has degree less than the number of distinct roots, hence less than n . Conjugating its coefficients preserves the prescribed data, so uniqueness of interpolation shows that this polynomial has real coefficients. Its Hermite quadratic form equals minus twice the multiplicity of λ , contradicting positive semidefiniteness. $\square$

Theorem A.2 (Exact Hermite tangent criterion). *For every $p \in \mathcal{R}_n$, put $K_p = \ker H(p)$. Then*

$$T_{\mathcal{R}_n}(p) = \left\{ h \in V_{n-1} : DH_p[h]|_{K_p} \succeq 0 \right\}. \quad (172)$$

The restriction denotes compression of the quadratic form to K_p ; positive semidefiniteness of the entire matrix $DH_p[h]$ is not required. Every direction on the right of (172) is realized by a curve $p_t \in \mathcal{R}_n$, $t \geq 0$, whose coefficients are polynomials in t and which satisfies $p_0 = p$ and $p'_0 = h$.

We first compute the compressed differential. For a fixed polynomial F , write $L_F(p)$ for the sum of F over the roots of p , with multiplicities. This is again a polynomial coefficient functional.

Lemma A.3 (Residue formula and independent jet blocks). *Let p have the factorization (170), and let $h \in V_{n-1}$. Write its unique partial-fraction expansion as*

$$\frac{h(z)}{p(z)} = \sum_{a \in \mathcal{A}} \sum_{\ell=1}^{q_a} \frac{b_{a,\ell}}{(z-a)^\ell}. \quad (173)$$

The map $h \mapsto (b_{a,\ell})$ is a linear isomorphism onto $\mathbb{R}^n$. For every fixed polynomial F ,

$$DL_F(p)[h] = - \sum_{a \in \mathcal{A}} \operatorname{Res}_{z=a} \frac{F'(z)h(z)}{p(z)}. \quad (174)$$

In the independent jet coordinates

$$v(a+w) = \sum_{r \geq 1} u_{a,r} w^r, \quad (u_{a,1}, \dots, u_{a,q_a-1})_{a \in \mathcal{A}}, \quad (175)$$

on K_p , the form $DH_p[h]|_{K_p}$ is the direct sum of the blocks

$$M_{rs}^{(a)}(b) = \begin{cases} -(r+s)b_{a,r+s}, & r+s \leq q_a, \\ 0, & r+s > q_a, \end{cases} \quad 1 \leq r, s \leq q_a - 1. \quad (176)$$

A simple root contributes a block of size zero.

Proof. The rational function h/p is proper. Existence and uniqueness of partial fractions give (173). Conversely, any choice of its n real coefficients yields a polynomial of degree less than n after multiplication by p . This proves the first assertion.

Choose a positively oriented circle enclosing all roots of p and all roots of its sufficiently small coefficient perturbations. The argument principle gives

$$L_F(p) = \frac{1}{2\pi i} \int F(z) \frac{p'(z)}{p(z)} dz.$$

Differentiation in the coefficient direction h , followed by integration by parts on the closed contour, gives

$$DL_F(p)[h] = \frac{1}{2\pi i} \int F(z) \left(\frac{h(z)}{p(z)} \right)' dz = -\frac{1}{2\pi i} \int F'(z) \frac{h(z)}{p(z)} dz.$$

Taking residues proves (174). This calculation does not require differentiable labels for repeated roots.

The map assigning to $v \in V_{n-1}$ all Taylor coefficients of orders $0, \dots, q_a - 1$ at each a is an isomorphism. Indeed, a polynomial in its kernel is divisible by p and has degree less than n , so it is zero; the source and target both have dimension n . By (171), restriction to K_p sets precisely the zeroth coefficient at each root to zero. This proves the independence in (175).

Apply (174) to $F = v^2$. Since $v(a) = 0$,

$$(v^2)'(a+w) = \sum_{r,s \geq 1} (r+s) u_{a,r} u_{a,s} w^{r+s-1}.$$

The residue against (173) is consequently

$$v^\top DH_p[h]v = - \sum_{a \in \mathcal{A}} \sum_{\substack{r,s \geq 1 \\ r+s \leq q_a}} (r+s) b_{a,r+s} u_{a,r} u_{a,s}.$$

These are exactly the blocks (176). $\square$

Lemma A.4 (Positive semidefinite block elimination). *For a root of multiplicity $q \geq 2$, the block in (176) is positive semidefinite if and only if*

$$b_\ell = 0 \quad (3 \leq \ell \leq q), \quad b_2 \leq 0. \quad (177)$$

There is no condition on b_1 .

Proof. Suppose the block is positive semidefinite and some b_ℓ with $\ell \geq 3$ is nonzero. Choose the largest such index ℓ . The diagonal entry in row $r = \ell - 1$ is zero: its coefficient has index $2\ell - 2 > \ell$ and is either outside the range of the block or zero by maximality. For a positive semidefinite matrix, a zero diagonal entry forces its row and column to vanish. For example, positivity of each 2-by-2 principal minor

gives $|M_{rs}|^2 \leq M_{rr}M_{ss} = 0$. But the entry in column 1 of this row is $-\ell b_\ell \neq 0$, a contradiction. Once all b_ℓ with $\ell \geq 3$ vanish, the only potentially nonzero entry is $M_{11} = -2b_2$. This proves both implications, including $q = 2$. $\square$

Proof of Theorem A.2. Suppose $p_j = p + t_j h + o(t_j) \in \mathcal{R}_n$, with $t_j \downarrow 0$. For any $v \in K_p$, Lemma A.1 and polynomial differentiability of H give

$$0 \leq v^\top H(p_j)v = t_j v^\top DH_p[h]v + o(t_j).$$

Division by t_j proves necessity.

For sufficiency, assume that the compression is positive semidefinite. Lemmas A.3 and A.4 give (177) independently at every repeated root. For a root a of multiplicity $q = q_a \geq 2$, put $w = z - a$ and set

$$P_{a,t}(z) = \left[\left(w + \frac{b_{a,1}}{q}t \right)^2 + b_{a,2}t \right] \left(w + \frac{b_{a,1}}{q}t \right)^{q-2}. \quad (178)$$

For a simple root set $P_{a,t}(z) = z - a + b_{a,1}t$. Each factor is monic and has only real roots for every $t \geq 0$. In (178), these roots are

$$a - \frac{b_{a,1}}{q}t \pm \sqrt{-b_{a,2}t}, \quad a - \frac{b_{a,1}}{q}t \quad \text{with multiplicity } q-2.$$

When $q = 2$, the final factor in (178) is 1. Direct differentiation of its polynomial coefficients gives

$$\left. \frac{d}{dt} P_{a,t}(z) \right|_{t=0} = b_{a,1}(z-a)^{q-1} + b_{a,2}(z-a)^{q-2} \quad (q \geq 2).$$

Now define $p_t = \prod_{a \in \mathcal{A}} P_{a,t}$. This is a monic, real-rooted degree- n polynomial for every $t \geq 0$, and its coefficients are polynomials in t . By the product rule,

$$p'_0 = p \left(\sum_{a \in \mathcal{A}} \frac{b_{a,1}}{z-a} + \sum_{\substack{a \in \mathcal{A} \\ q_a \geq 2}} \frac{b_{a,2}}{(z-a)^2} \right) = h,$$

where the last equality is (173) together with (177). This supplies the claimed curve and proves sufficiency. The case $n = 1$ has no nonempty jet block; the simple-root construction proves the assertion in that case as well. $\square$

Corollary A.5 (Root-contact form). *For a root a of p with multiplicity $q_a \geq 2$, write $p = (z-a)^{q_a} R_a$ with $R_a(a) \neq 0$. A direction $h \in V_{n-1}$ belongs to $T_{\mathcal{R}_n}(p)$ if and only if, at every such root,*

$$h^{(j)}(a) = 0 \quad (0 \leq j \leq q_a - 3), \quad \frac{h^{(q_a-2)}(a)}{(q_a-2)!R_a(a)} \leq 0. \quad (179)$$

Simple roots impose no condition, and the list of equalities is empty when $q_a = 2$.

Proof. The coefficient $b_{a,\ell}$ is the Taylor coefficient of order $q_a - \ell$ in h/R_a at a . Vanishing of all the coefficients with $\ell \geq 3$ is equivalent to divisibility of h by $(z-a)^{q_a-2}$. Under that divisibility, the next coefficient is $b_{a,2} = h^{(q_a-2)}(a)/((q_a-2)!R_a(a))$. Apply (177). $\square$

A.2. Normal multipliers and their closedness. Use the coefficient inner product on V_{n-1} and the trace inner product on real symmetric matrices to define the adjoint DH_p^* . The regular normal cone is

$$\hat{N}_{\mathcal{R}_n}(p) = \left\{ \eta \in V_{n-1} : \limsup_{\substack{q \rightarrow p, q \in \mathcal{R}_n \\ q \neq p}} \frac{\langle \eta, q - p \rangle}{\|q - p\|} \leq 0 \right\}.$$

For a cone C , write $C^\circ = \{h : \langle h, c \rangle \leq 0 \text{ for all } c \in C\}$. In finite dimensions the definition of the contingent cone implies $\hat{N}_{\mathcal{R}_n}(p) = T_{\mathcal{R}_n}(p)^\circ$. Indeed, normality gives the inequality on every contingent limit; conversely, a sequence violating normality has a convergent subsequence of normalized displacement vectors, producing a contingent direction with strictly positive pairing.

Theorem A.6 (Exact normal formula and closed multiplier images). *For $p \in \mathcal{R}_n$, define*

$$C_p = \{-DH_p^*(Y) : Y \succeq 0, \text{ range } Y \subseteq K_p\}. \quad (180)$$

Then

$$\hat{N}_{\mathcal{R}_n}(p) = \overline{C_p}. \quad (181)$$

The following conditions are equivalent:

- (1) C_p is closed;
- (2) every root of p has multiplicity at most two;
- (3) there exists $h \in V_{n-1}$ such that $DH_p[h]|_{K_p}$ is positive definite on K_p .

Positive definiteness on the zero-dimensional space is interpreted vacuously. In particular, the closure in (181) cannot be omitted at a polynomial having a root of multiplicity at least three.

Proof. For any $h \in V_{n-1}$,

$$\begin{aligned} h \in C_p^\circ &\iff -\text{Tr}(YDH_p[h]) \leq 0 \text{ for all } Y \succeq 0 \text{ with } \text{range } Y \subseteq K_p \\ &\iff DH_p[h]|_{K_p} \succeq 0. \end{aligned}$$

The last equivalence follows by testing rank-one matrices $vv^\top$, $v \in K_p$, and then using the spectral decomposition of Y . Theorem A.2 gives $C_p^\circ = T_{\mathcal{R}_n}(p)$. The set C_p is a convex cone, so the finite-dimensional bipolar theorem gives

$$\hat{N}_{\mathcal{R}_n}(p) = T_{\mathcal{R}_n}(p)^\circ = (C_p^\circ)^\circ = \overline{C_p}.$$

This proves (181) without a strict-feasibility assumption.

We next describe the multiplier coordinates precisely. Let $E = \bigoplus_{a \in \mathcal{A}} \mathbb{R}^{q_a-1}$ be the jet space in (175), and let $L : E \rightarrow V_{n-1}$ be the inverse jet interpolation map with image K_p . Lemma A.3 says

$$L^\top DH_p[h]L = M(b) := \bigoplus_{a \in \mathcal{A}} M^{(a)}(b). \quad (182)$$

Every positive semidefinite Y with range in K_p can be written uniquely as $Y = LZL^\top$ with $Z \succeq 0$ on E . To see this, use a left inverse L^+ of L and set $Z = L^+Y(L^+)^\top$; conversely congruence preserves positive semidefiniteness. Consequently the normal functional associated with Y is, in the partial-fraction coordinates,

$$h \longmapsto -\text{Tr}(ZM(b)) = -\sum_{a \in \mathcal{A}} \text{Tr}(Z^{(a)}M^{(a)}(b)), \quad (183)$$

where $Z^{(a)}$ is the corresponding diagonal principal block of Z . Each $Z^{(a)}$ is positive semidefinite. Conversely, any tuple of such positive semidefinite blocks is realized by their block-diagonal direct sum. Thus no orthogonality of the jet coordinates has been assumed, and off-diagonal blocks between different roots do not affect the multiplier image.

Suppose first that every $q_a \leq 2$. Every nonempty block $M^{(a)}$ is the scalar $-2b_{a,2}$. Formula (183) then identifies C_p with the cone of functionals

$$h \mapsto \sum_{\substack{a \in \mathcal{A} \\ q_a = 2}} 2u_a b_{a,2}, \quad u_a \geq 0.$$

The b 's are independent linear coordinates by Lemma A.3. This cone is therefore a product of nonnegative rays in a coordinate subspace, carried back by an invertible linear map. It is closed. The same independence allows one to choose $b_{a,2} = -1$ at every double root, producing a positive definite compression. If there are no double roots, $K_p = \{0\}$ and $C_p = \{0\}$.

Now suppose that some root a has $q_a \geq 3$. The functional $\ell(h) = b_{a,3}$ vanishes on the tangent cone, by (177); hence $\ell \in \overline{C_p}$ by (181). If it belonged to C_p , represent it by (183). Its coefficient at $b_{a,2}$ would be $2Z_{11}^{(a)} = 0$. Positive semidefiniteness would then force the first row of $Z^{(a)}$ to vanish. But its coefficient at $b_{a,3}$ would be $6Z_{12}^{(a)} = 0$, instead of 1. Independence of the partial-fraction coordinates prevents contributions from other roots from changing either coefficient. Thus $\ell \notin C_p$, and C_p is not closed. Finally, in this root block the diagonal entry with index $q_a - 1$ is identically zero because $2q_a - 2 > q_a$. The compression can therefore never be positive definite. All three conditions are equivalent. $\square$

Example A.7 (Diverging multipliers at a triple root). Take $p = z^3$ and write $h = c_2 z^2 + c_1 z + c_0$. In the kernel basis z, z^2 , the compressed differential is

$$\begin{pmatrix} -2c_1 & -3c_0 \\ -3c_0 & 0 \end{pmatrix}.$$

Thus the tangent conditions are $c_1 \leq 0$ and $c_0 = 0$, with c_2 free. A multiplier $Z = \begin{pmatrix} u & v \\ v & w \end{pmatrix} \succeq 0$ produces the functional $h \mapsto 2uc_1 + 6vc_0$. Its image consists of all $Ac_1 + Bc_0$ with $A > 0$, together with the zero functional, whereas its closure allows $A = 0$ and arbitrary B . For fixed $B \neq 0$,

$$Z_\varepsilon = \begin{pmatrix} \varepsilon & B/6 \\ B/6 & B^2/(36\varepsilon) \end{pmatrix} \succeq 0$$

has image $2\varepsilon c_1 + Bc_0$, which tends to Bc_0 as $\varepsilon \downarrow 0$. The multiplier norms diverge. In fact, every multiplier sequence representing this limiting functional must be unbounded: a bounded sequence would have a convergent positive semidefinite subsequence representing Bc_0 , contrary to the exact image just computed. Up to coordinate scaling, this is the elementary nonclosed positive semidefinite projection in [33, Example 1].

Remark A.8 (Scope of the coefficient chart). All tangent and normal formulas above use fixed degree and monic normalization. If f_t has actual degree n near $t = 0$, has positive leading coefficient $\lambda(t)$, and $f_0 = \lambda(0)p$, its normalized direction is

$$\left. \frac{d}{dt} \frac{f_t}{\lambda(t)} \right|_{t=0} = \frac{f'_0}{\lambda(0)} - \frac{\lambda'(0)}{\lambda(0)} p.$$

This direction has degree less than n and is the argument to use in DH_p . Degree drops and additional coefficient inequalities require separate arguments. In particular, the exact compression formula is not a general rule asserting that tangents commute with arbitrary intersections of polynomial or matrix constraints.

A.3. A finite-dimensional positive-flow criterion. The monic formula also gives a criterion on a closed cone containing polynomials of every actual degree. The passage to lower degree uses density rather than a change in the size of the Hermite matrix. Write $V_m = \mathbb{R}[z]_{\leq m}$ and define

$$\mathcal{K}_m = \{0\} \cup \{f \in V_m \setminus \{0\} : f \text{ has nonnegative coefficients and only real roots}\}.$$

A real linear map on V_m is called Metzler when its matrix in the ordinary coefficient basis has nonnegative off-diagonal entries.

Theorem A.9 (Hermite criterion for positive polynomial flows). *Let $m \geq 1$, and let $A : V_m \rightarrow V_m$ be Metzler. Then*

$$e^{tA}\mathcal{K}_m \subseteq \mathcal{K}_m \quad (t \geq 0)$$

if and only if every monic degree- m polynomial $p \in \mathcal{R}_m$ whose coefficients are all strictly positive satisfies

$$DH_p[Ap - ([z^m]Ap)p] \big|_{\ker H(p)} \succeq 0. \quad (184)$$

The matrix in this test has size m ; the bracketed direction has degree less than m .

Proof. Let $\mathcal{O} \subset V_m$ be the open cone of strictly positive coefficient vectors. Metzler positivity implies, for $f \in \mathcal{O}$,

$$(e^{tA}f)_i \geq e^{tA_{ii}}f_i > 0 \quad (t \geq 0). \quad (185)$$

For example, choose c so that $A + c\text{Id}$ is entrywise nonnegative and expand its exponential. Keeping the paths that remain at index i gives the stated diagonal lower bound. Thus $\mathcal{O}$ is forward invariant, and the leading coefficient of $e^{tA}p$ remains positive for every polynomial tested in (184).

If the semigroup preserves $\mathcal{K}_m$, normalize this orbit by its leading coefficient. Its derivative at zero is $Ap - ([z^m]Ap)p$, and Theorem A.2 proves necessity.

Conversely, assume (184). For $f \in \mathcal{K}_m \cap \mathcal{O}$, write $f = sp$ with $s = [z^m]f > 0$ and p monic, and put $a = [z^m]Ap$. Theorem A.2 supplies a monic real-rooted curve P_t with $P_0 = p$ and $P'_0 = Ap - ap$. The curve

$$f_t = s(1 + ta)P_t$$

has derivative Af at zero and lies in $\mathcal{K}_m \cap \mathcal{O}$ for all sufficiently small $t \geq 0$. Indeed its scalar factor and all its coefficients are positive by continuity. Hence

$$Af \in T_{\mathcal{K}_m}(f) \quad (f \in \mathcal{K}_m \cap \mathcal{O}). \quad (186)$$

We use the finite-dimensional contingent-cone form of Nagumo's theorem: if a locally Lipschitz vector field takes its values in the contingent tangent cone of a closed set at every point of that set, its unique forward trajectories remain in the set throughout their existence; see [42, 28]. To apply it using only (186), fix $f \in \mathcal{K}_m \cap \mathcal{O}$ and a finite time $T > 0$. The known ambient linear orbit

$$\mathcal{C} = \{e^{tA}f : 0 \leq t \leq T\}$$

is compact and lies in $\mathcal{O}$ by (185). Choose a nonnegative smooth function χ , compactly supported in $\mathcal{O}$, equal to one on a neighborhood of $\mathcal{C}$. The vector field

$B(x) = \chi(x)Ax$ is smooth with compact support, hence globally Lipschitz. At points of $\mathcal{K}_m \cap \mathcal{O}$ it is tangent by (186); at all other points of $\mathcal{K}_m$ it vanishes. The cone $\mathcal{K}_m$ is closed, as verified below, so Nagumo's theorem makes it invariant under B . The curve $e^{tA}f$ solves the same equation on $[0, T]$, since $\chi = 1$ along that curve. Uniqueness therefore proves $e^{tA}f \in \mathcal{K}_m$ throughout $[0, T]$. This argument uses only forward invariance of $\mathcal{O}$ and does not assume real-rootedness of the ambient orbit when choosing χ .

For completeness, nonnegative coefficients persist under coefficient limits. If a nonzero coefficient limit of real-rooted polynomials had a nonreal zero, a small disk around that zero disjoint from the real axis would, by local zero continuity, contain a zero of every sufficiently close polynomial. This is impossible. Including the zero polynomial proves that $\mathcal{K}_m$ is closed, including across degree drops.

Finally, $\mathcal{K}_m \cap \mathcal{O}$ is dense in $\mathcal{K}_m$. For a nonzero $f \in \mathcal{K}_m$ of actual degree d , all roots are nonpositive, so

$$f(z) = c \prod_{j=1}^d (z + r_j), \quad c > 0, \quad r_j \geq 0.$$

For $\varepsilon > 0$, the polynomials

$$f_\varepsilon(z) = c \prod_{j=1}^d (z + r_j + \varepsilon)(1 + \varepsilon z)^{m-d}$$

have degree m , strictly positive coefficients, and only negative roots, and tend coefficientwise to f . Approximate zero by $\varepsilon(1 + z)^m$. Continuity of e^{tA} and closedness of $\mathcal{K}_m$ now extend preservation to the entire cone. $\square$

For $m = 0$, the cone consists of nonnegative constants and every real scalar generator preserves it. No Hermite test is needed in that case.

A.4. Exact localization at products of binary linear forms. Fix integers $m \geq 2$ and $r \geq 1$, and write the coordinates on $\mathbb{R}^{r+2}$ as (T, X, Y) , where $Y = (Y_1, \dots, Y_r)$. Let $\mathcal{H}_{m,r}$ be the set of homogeneous real polynomials of degree m that are monic in T and hyperbolic with respect to $e = (1, 0, 0)$. Equivalently, $f \in \mathcal{H}_{m,r}$ if $T \mapsto f(T, x, y)$ has only real roots for every $(x, y) \in \mathbb{R}^{r+1}$. All tangent cones in this subsection are coefficient-space Bouligand tangent cones. Thus $h \in T_{\mathcal{H}_{m,r}}(p)$ means that there are $\varepsilon_j \downarrow 0$ and $p_j \in \mathcal{H}_{m,r}$ with $p_j = p + \varepsilon_j h + o(\varepsilon_j)$ coefficientwise. In particular, h is homogeneous of degree m and has zero T^m coefficient.

Theorem A.10 (Binary-factor localization). *Let $\lambda_1 < \dots < \lambda_m$ be real numbers, and set*

$$\ell_i = T - \lambda_i X, \quad p = \prod_{i=1}^m \ell_i, \quad q_i = \frac{p}{\ell_i \ell_{i+1}} \quad (1 \leq i < m).$$

Then

$$T_{\mathcal{H}_{m,r}}(p) = \left\{ - \sum_{i=1}^m d_i(X, Y) \frac{p}{\ell_i} - \sum_{i=1}^{m-1} Q_i(Y) q_i(T, X) : \right. \\ \left. d_i \in \mathbb{R}[X, Y]_1, \quad Q_i \in \mathbb{R}[Y]_2, \quad Q_i \geq 0 \right\}. \quad (187)$$

Here the subscripts specify homogeneous degree, and $Q_i \geq 0$ means positive semidefiniteness of the quadratic form on $\mathbb{R}^r$. The forms d_i and Q_i in this representation

are unique. Consequently the tangent cone, in its linear span, is linearly isomorphic to

$$\mathbb{R}^{m(r+1)} \times (\mathbb{S}_+^r)^{m-1},$$

where $\mathbb{S}_+^r$ denotes the cone of real symmetric positive semidefinite $r \times r$ matrices. Every tangent direction has a realizing curve $\varepsilon \mapsto p_\varepsilon \in \mathcal{H}_{m,r}$ whose coefficients are polynomials in $\varepsilon \geq 0$ and such that $p_\varepsilon = p + \varepsilon h + O(\varepsilon^2)$.

The external representation theorem used in the necessity argument is the following precise ternary form of the Helton–Vinnikov theorem: if $f(T, X, Y)$ is homogeneous of degree m , $f(1, 0, 0) = 1$, and f is hyperbolic with respect to $(1, 0, 0)$, then there are real symmetric $m \times m$ matrices A, B with

$$f(T, X, Y) = \det(TI - XA - YB). \quad (188)$$

This statement does not require irreducibility or smoothness. The normalization and the matrix size, including the case in which $f(1, X, Y)$ has degree less than m , are established in [38, Eq. (5) and pp. 2497–2498].

Proof of Theorem A.10. We first prove necessity when $r = 1$. Choose a tangent sequence $p_j = p + \varepsilon_j h + o(\varepsilon_j)$. By (188), write

$$p_j(T, X, Y) = \det(TI - XA_j - YB_j),$$

with A_j, B_j real symmetric. Orthogonally diagonalize A_j , applying the same conjugation to B_j , and arrange its eigenvalues increasingly:

$$A_j = \text{diag}(a_{1,j}, \dots, a_{m,j}).$$

These eigenvalues are the roots of $p_j(T, 1, 0)$. Since the roots of $p(T, 1, 0)$ are distinct, the implicit function theorem for a simple root and the first-order coefficient expansion give real numbers α_i such that

$$a_{i,j} = \lambda_i + \varepsilon_j \alpha_i + o(\varepsilon_j). \quad (189)$$

On setting $X = 0$, the coefficients of $T^{m-1}Y$ and $T^{m-2}Y^2$ in $\det(TI - YB_j)$ show that

$$\text{Tr } B_j = O(\varepsilon_j), \quad e_2(B_j) = O(\varepsilon_j),$$

where $e_2(B_j)$ is the second elementary symmetric function of the eigenvalues. Hence

$$\|B_j\|_F^2 = \text{Tr}(B_j^2) = (\text{Tr } B_j)^2 - 2e_2(B_j) = O(\varepsilon_j).$$

Here symmetry is essential: it identifies $\text{Tr}(B_j^2)$ with a sum of squares. Thus every entry of B_j is $O(\sqrt{\varepsilon_j})$.

The coefficient of Y in the determinant is exactly

$$-\sum_{i=1}^m (B_j)_{ii} \prod_{k \neq i} (T - a_{k,j} X). \quad (190)$$

The limiting forms p/ℓ_i are a basis of the binary forms of degree $m - 1$: evaluation at $(T, X) = (\lambda_i, 1)$ annihilates all but the i th form, whose value is nonzero. Accordingly, the basis matrices in (190) converge to an invertible matrix. Since the coefficient of Y in p_j divided by ε_j converges coefficientwise, there are uniquely determined real numbers β_i such that

$$(B_j)_{ii} = \varepsilon_j \beta_i + o(\varepsilon_j). \quad (191)$$

After passing to a subsequence, the bounds on B_j also give limits

$$\frac{(B_j)_{ik}}{\sqrt{\varepsilon_j}} \longrightarrow c_{ik} \quad (1 \leq i < k \leq m).$$

No continuous choice of representations or diagonalizing matrices is needed for these subsequence limits.

Expand the determinant by permutations. The identity permutation, together with (189) and (191), contributes

$$p - \varepsilon_j \sum_i (\alpha_i X + \beta_i Y) \frac{p}{\ell_i} + o(\varepsilon_j).$$

A permutation moving exactly two indices i, k is a transposition; its contribution to first order is $-\varepsilon_j Y^2 c_{ik}^2 p / (\ell_i \ell_k)$. Every other nonidentity permutation moves at least three indices. Its off-diagonal factors are each $O(\sqrt{\varepsilon_j})$, so its contribution is $O(\varepsilon_j^{3/2})$. Replacing a remaining diagonal factor in a transposition term by its perturbation changes that term by $O(\varepsilon_j^2) + o(\varepsilon_j^2)$. These estimates are coefficientwise, since there are finitely many permutations and all remaining linear factors have bounded coefficients. We obtain

$$h = - \sum_{i=1}^m (\alpha_i X + \beta_i Y) \frac{p}{\ell_i} - Y^2 \sum_{i < k} c_{ik}^2 \frac{p}{\ell_i \ell_k}. \quad (192)$$

In particular, no term of Y -degree at least three occurs in h .

To reduce the pair terms to adjacent pairs, use the polynomial identity

$$\frac{p}{\ell_i \ell_k} = \sum_{j=i}^{k-1} \frac{\lambda_{j+1} - \lambda_j}{\lambda_k - \lambda_i} q_j \quad (i < k). \quad (193)$$

For $X \neq 0$ away from the finitely many vanishing factors, it follows by telescoping

$$\frac{1}{\ell_{j+1}} - \frac{1}{\ell_j} = \frac{(\lambda_{j+1} - \lambda_j)X}{\ell_j \ell_{j+1}};$$

multiplying by p then proves the identity everywhere. All coefficients in (193) are positive. Moreover, $q_1, \dots, q_{m-1}$ are linearly independent: in a relation $\sum_j b_j q_j = 0$, evaluation at $(\lambda_1, 1)$ forces $b_1 = 0$; evaluation successively at $(\lambda_2, 1), \dots, (\lambda_{m-1}, 1)$ forces each remaining coefficient to vanish. They therefore form a basis of the binary forms of degree $m-2$. Equation (192) now proves the required necessity for one spectator variable, with nonnegative coefficients of $Y^2 q_j$.

For arbitrary r , restrict the tangent sequence to $Y = sy$, where $y \in \mathbb{R}^r$ is fixed and s is a single real variable. Restriction preserves monicity in T and hyperbolicity in e , so the preceding ternary conclusion applies to $h(T, X, sy)$ for every y . Decompose h into its components homogeneous in Y . Every component of Y -degree at least three vanishes after all such restrictions, and is therefore identically zero. The Y -degree zero component is a binary form of degree m with zero T^m coefficient; it is divisible by X and consequently has a unique expression $-\sum_i \alpha_i X p / \ell_i$. Each coefficient of Y_a has a unique expansion in the basis $p / \ell_1, \dots, p / \ell_m$. Together these expressions give the unique linear forms $d_i(X, Y)$. Finally, since the q_i are a basis, the quadratic component has a unique expansion

$$- \sum_{i=1}^{m-1} Q_i(Y) q_i(T, X)$$

with real homogeneous quadratic forms Q_i . For every y , the ternary conclusion and uniqueness in the adjacent basis give $Q_i(y) \geq 0$ for every i . Thus each Q_i is positive semidefinite. This proves necessity and uniqueness in all dimensions without a determinantal representation in more than three variables.

For sufficiency, start with any linear forms d_i and positive semidefinite forms Q_i appearing in (187). For $\varepsilon \geq 0$, put

$$L_i = T - \lambda_i X - \varepsilon d_i(X, Y)$$

and define polynomials recursively by

$$P_0 = 1, \quad P_1 = L_1, \quad P_k = L_k P_{k-1} - \varepsilon Q_{k-1}(Y) P_{k-2} \quad (2 \leq k \leq m). \quad (194)$$

The recurrence shows that P_k is homogeneous of degree k , monic in T , and has coefficients polynomial in ε . For each fixed real (X, Y) , it is the characteristic-polynomial recurrence for the real symmetric tridiagonal matrix with diagonal entries $\lambda_i X + \varepsilon d_i(X, Y)$ and adjacent entries $\sqrt{\varepsilon Q_i(Y)}$. All these square roots are real, so all roots of P_m in T are real. Thus $P_m \in \mathcal{H}_{m,r}$ for every $\varepsilon \geq 0$. The matrix entries need not be linear functions of Y ; the polynomial itself is defined by (194) and contains no square roots.

For completeness, expanding the recurrence over matchings M of the path with vertices $1, \dots, m$ gives

$$P_m = \sum_M (-\varepsilon)^{|M|} \prod_{\{i, i+1\} \in M} Q_i(Y) \prod_{j \notin V(M)} L_j.$$

This identity follows inductively by separating matchings according to whether they contain the final edge. The empty matching contributes the first-order changes of the L_j ; the single-edge matchings contribute the quadratic terms; every other matching has a factor ε^2 . Consequently

$$P_m = p - \varepsilon \sum_i d_i \frac{p}{\ell_i} - \varepsilon \sum_i Q_i \frac{p}{\ell_i \ell_{i+1}} + O(\varepsilon^2) = p + \varepsilon h + O(\varepsilon^2).$$

This proves sufficiency and polynomial realization. The linear isomorphism asserted in the theorem follows from the uniqueness of the displayed coordinates. $\square$

Corollary A.11 (Positive stability in the binary-factor stratum). *Let $N \geq 3$, and let*

$$F(z) = \prod_{i=1}^m L_i(z), \quad z \in \mathbb{R}^N,$$

where each L_i has strictly positive coefficients, the L_i are pairwise nonproportional, and their linear span has dimension two. Fix $v \in \mathbb{R}_{>0}^N$. There is a real linear coordinate isomorphism $\Phi : z \mapsto (T, X, Y)$, with $Y \in \mathbb{R}^{N-2}$ and $\Phi(v) = e$, such that

$$\frac{F}{F(v)} = p \circ \Phi, \quad p(T, X, Y) = \prod_i (T - \lambda_i X), \quad \lambda_1 < \dots < \lambda_m.$$

Let $\mathcal{S}_{m,N}^+$ be the homogeneous degree- m real stable polynomials with nonnegative coefficients. Then

$$T_{\mathcal{S}_{m,N}^+}(F) = \{aF + F(v)(h \circ \Phi) : a \in \mathbb{R}, h \in T_{\mathcal{H}_{m,N-2}}(p)\}. \quad (195)$$

Every direction on the right has a realizing curve with coefficients polynomial in time, lying in $\mathcal{S}_{m,N}^+$ for all sufficiently small nonnegative times.

Proof. Normalize the factors by $\tilde{L}_i = L_i/L_i(v)$. Their span is two-dimensional, and each has value one at v . Choose a linear form T in this span with $T(v) = 1$, and a nonzero linear form X in this span with $X(v) = 0$. Then $\tilde{L}_i = T - \lambda_i X$, with distinct λ_i ; reorder the factors increasingly. Complete X to a basis $X, Y_1, \dots, Y_{N-2}$ of the linear forms vanishing at v . Together with T these forms define the asserted isomorphism Φ .

We use the following standard consequence of Gårding's component theorem [31, Corollary 2.7 and Theorem 2.9]: a homogeneous real polynomial hyperbolic in v is hyperbolic in every direction in the connected component of $\{f \neq 0\}$ containing v . In particular, if $v > 0$, f is hyperbolic in v , and f has nonnegative coefficients and is nonzero, then it is hyperbolic in every positive direction. Indeed, f is strictly positive throughout the positive orthant, which is connected and contains v . Such an f is real stable: if $z = x + iy$ with $y > 0$, then $t \mapsto f(x + ty)$ is a nonzero real-rooted polynomial, and so does not vanish at $t = i$. Conversely, a homogeneous real stable polynomial with $f(v) > 0$ is hyperbolic in v , since the real polynomial $t \mapsto f(x + tv)$ has no root in the upper half-plane, hence by conjugation none in the lower half-plane.

Suppose first that $F_j = F + \varepsilon_j g + o(\varepsilon_j) \in \mathcal{S}_{m,N}^+$. For large j , $F_j(v) > 0$. Set $a = g(v)/F(v)$ and normalize:

$$\frac{F_j}{F_j(v)} = \frac{F}{F(v)} + \varepsilon_j \frac{g - aF}{F(v)} + o(\varepsilon_j).$$

Every normalized polynomial is hyperbolic in v by the preceding observation. Pulling back by Φ^{-1} gives a tangent sequence in $\mathcal{H}_{m,N-2}$ at p , proving necessity in (195).

Conversely, let $a \in \mathbb{R}$ and let h be as in the right side of (195). Use the polynomial curve P_ε from Theorem A.10 and put

$$F_\varepsilon(z) = F(v)(1 + a\varepsilon)P_\varepsilon(\Phi(z)).$$

This is homogeneous, its coefficients are polynomials in ε , and

$$F_\varepsilon = F + \varepsilon(aF + F(v)(h \circ \Phi)) + O(\varepsilon^2).$$

Every degree- m monomial has a strictly positive coefficient in F , because every coefficient of every L_i is strictly positive. There are finitely many such coefficients, so all remain strictly positive in F_ε for sufficiently small $\varepsilon \geq 0$. Also $1 + a\varepsilon > 0$ for such times, and F_ε is hyperbolic in v by construction. The component argument above therefore makes F_ε real stable, proving sufficiency. $\square$

Remark A.12. The restriction to a two-dimensional span of pairwise nonproportional factors is part of the theorem. Neither repeated factors nor products whose factors span three or more dimensions are covered by (187). When $r = 0$, the simple-root binary form lies in the open set of monic real-rooted binary forms, and its tangent consists of all homogeneous degree- m perturbations with zero T^m coefficient. When $m = 1$, the monic homogeneous forms are all hyperbolic and their tangent is the full tangent space of the monic affine slice.

A.5. Why fixed univariate restrictions do not determine the tangent. The exactness of Theorem A.2 does not allow one to interchange taking tangents with the family of affine-line restrictions that characterizes multivariate stability. We

give two explicit examples. Both have strictly positive homogeneous coefficients after an invertible real change of coordinates.

We shall use the following standard form of the hyperbolicity component theorem. If a real homogeneous polynomial is hyperbolic in a direction e with $p(e) > 0$, it is hyperbolic in every direction in the connected component of $\{p > 0\}$ containing e ; see [31, Corollary 2.7 and Theorem 2.9]. Here hyperbolicity means that $p(e) \neq 0$ and every polynomial $s \mapsto p(x + se)$, x real, has only real roots. For a homogeneous real polynomial, hyperbolicity in every strictly positive direction is equivalent to real stability: a zero $p(x + iy) = 0$ with $y > 0$ violates the corresponding line test, and a nonreal root of a real line restriction has a conjugate in the upper half-plane. We will repeatedly use the following elementary consequence of the component theorem. If $p_j \rightarrow p$ coefficientwise, each p_j is hyperbolic in d , and a fixed path from d to e has compact image contained in $\{p > 0\}$, then p_j is hyperbolic in e for all sufficiently large j . Indeed, the positive minimum of p on that path persists under coefficient convergence.

A.5.1. *An exact cubic comparison.* In three variables set

$$T = z_1 + z_2 + z_3, \quad X = \frac{z_1 + 2z_2 + 3z_3}{4}, \quad Y = z_2, \quad (196)$$

and define

$$f = T(T - X)(T + X), \quad g_{a,b} = -(aT + bX)Y^2, \quad a, b \in \mathbb{R}. \quad (197)$$

The coordinate change is invertible. Each of $T, T - X, T + X$ has strictly positive coefficients in z , so f is real stable and all its degree-three coefficients are strictly positive. Let $\mathcal{S}_3$ be the set of homogeneous real stable ternary cubics with nonnegative coefficients, together with zero.

For an affine line $z = u + sv$ with $u \in \mathbb{R}^3$ and $v > 0$, put $p(s) = f(u + sv)$ and $h(s) = g_{a,b}(u + sv)$. Its leading coefficient is $c = f(v) > 0$. The normalized base and direction are

$$\bar{p} = \frac{p}{c}, \quad \bar{h} = \frac{h}{c} - \frac{[s^3]h}{c} \bar{p}. \quad (198)$$

Saying that this line passes the Hermite tangent test means $DH_{\bar{p}}[\bar{h}]|_{\ker H(\bar{p})} \succeq 0$.

Theorem A.13 (Positive-direction slice tests are insufficient). *For the family (197), every affine restriction in every strictly positive direction passes the Hermite tangent test if and only if*

$$a + \frac{b}{4} \geq 0, \quad a + \frac{3b}{4} \geq 0. \quad (199)$$

In contrast,

$$g_{a,b} \in T_{\mathcal{S}_3}(f) \iff a \geq |b|. \quad (200)$$

When $a \geq |b|$, the straight perturbation $f + \varepsilon g_{a,b}$ already belongs to $\mathcal{S}_3$ for every sufficiently small $\varepsilon \geq 0$.

Proof. All three affine factors have strictly positive slopes along $v > 0$. If two restricted roots coincide, then $T = X = 0$ there and the third root coincides as well. Thus a restriction has either three simple roots or one triple root. At a simple-root base the monic real-rooted set is open and its tangent is unrestricted. At a triple

root $s = r$, Corollary A.5 gives precisely $\bar{h}(r) = 0$ and $\bar{h}'(r) \leq 0$. The leading-normalization term in (198) vanishes to order three, whereas direct differentiation gives

$$h(r) = 0, \quad h'(r) = -(aT(v) + bX(v))Y(u + rv)^2.$$

For every fixed $v > 0$, a line in direction v can pass through a point with $T = X = 0$ and any specified value of Y . Hence all line tests are equivalent to $aT(v) + bX(v) \geq 0$ for every $v > 0$. The ratio $X(v)/T(v)$ ranges over $(1/4, 3/4)$, so this is equivalent to (199).

We prove the necessity in (200) directly for arbitrary contingent sequences. Suppose $f_j = f + \varepsilon_j g_{a,b} + o(\varepsilon_j) \in \mathcal{S}_3$, where $\varepsilon_j \downarrow 0$. In the coordinates (T, X, Y) the base is

$$p = T^3 - X^2T,$$

which is hyperbolic in $e = (1, 0, 0)$, with positive component $\{T > |X|\}$. The original positive direction $(1, 1, 1)$ becomes $d = (3, 3/2, 1)$. The transformed f_j are hyperbolic in d , and the segment from d to e stays in $\{p > 0\}$. The component argument above therefore makes them hyperbolic in e for all sufficiently large j .

Normalize the coefficient of T^3 to one. This coefficient was $1 + o(\varepsilon_j)$, so normalization does not change the prescribed first variation. Write the normalized polynomials as

$$p_j = T^3 + A_j(X, Y)T^2 + B_j(X, Y)T + C_j(X, Y),$$

where coefficientwise

$$A_j = o(\varepsilon_j), \quad B_j = -X^2 - \varepsilon_j aY^2 + o(\varepsilon_j), \quad C_j = -\varepsilon_j bXY^2 + o(\varepsilon_j).$$

The real translation eliminating the quadratic term produces $T^3 - Q_jT + R_j$, with

$$\begin{aligned} Q_j &= A_j^2/3 - B_j = X^2 + \varepsilon_j aY^2 + o(\varepsilon_j), \\ R_j &= C_j - A_jB_j/3 + 2A_j^3/27 = -\varepsilon_j bXY^2 + o(\varepsilon_j). \end{aligned} \quad (201)$$

All these estimates take place in fixed finite-dimensional spaces of homogeneous coefficient forms. The real-root criterion for a depressed cubic is

$$Q_j(X, Y) \geq 0, \quad |R_j(X, Y)| \leq \frac{2}{3\sqrt{3}}Q_j(X, Y)^{3/2} \quad (X, Y \in \mathbb{R}). \quad (202)$$

For completeness, its discriminant is $4Q_j^3 - 27R_j^2$; equivalently, evaluation at the two critical points $\pm\sqrt{Q_j/3}$ gives the same necessary and sufficient inequalities. At $(X, Y) = (0, 1)$, the first inequality implies $a \geq 0$.

Fix $s > 0$ and evaluate at $(X, Y) = (\pm\sqrt{\varepsilon_j}s, 1)$. Then

$$Q_j(\pm\sqrt{\varepsilon_j}s, 1) = \varepsilon_j(s^2 + a) + o(\varepsilon_j).$$

The difference of the corresponding R_j values has the sharper expansion

$$R_j(\sqrt{\varepsilon_j}s, 1) - R_j(-\sqrt{\varepsilon_j}s, 1) = -2b\varepsilon_j^{3/2}s + o(\varepsilon_j^{3/2}). \quad (203)$$

Indeed, R_j is a homogeneous cubic in X, Y . Its even powers of X cancel in the difference; its X^3 coefficient is $o(\varepsilon_j)$, and its XY^2 coefficient is $-\varepsilon_j b + o(\varepsilon_j)$. This justifies (203) even when the coefficients of the even terms are only $o(\varepsilon_j)$. Apply (202) at both points and use the triangle inequality. After division by $2\varepsilon_j^{3/2}$ and passage to the limit, we obtain

$$|b|s \leq \frac{2}{3\sqrt{3}}(s^2 + a)^{3/2} \quad (s > 0). \quad (204)$$

If $a = 0$, let $s \downarrow 0$ to obtain $b = 0$. If $a > 0$, the minimum of $(s^2 + a)^3/s^2$ is $27a^2/4$, attained when $s^2 = a/2$. Squaring (204) at that value gives $b^2 \leq a^2$. This proves necessity.

For sufficiency, assume $a \geq |b|$. In transformed coordinates the straight perturbation is

$$p_\varepsilon = T^3 - (X^2 + \varepsilon a Y^2)T - \varepsilon b X Y^2.$$

Its depressed coefficient $Q = X^2 + \varepsilon a Y^2$ is nonnegative. Its discriminant is also nonnegative because

$$4(X^2 + \varepsilon a Y^2)^3 - 27\varepsilon^2 b^2 X^2 Y^4 \geq 0.$$

When $\varepsilon Y^2 > 0$, divide this expression by $\varepsilon^3 Y^6$ and put $u = X^2/(\varepsilon Y^2)$; the inequality follows from

$$4(u + a)^3 - 27b^2 u \geq 4(u + a)^3 - 27a^2 u = 4(u + 4a)(u - a/2)^2 \geq 0.$$

The case $\varepsilon Y^2 = 0$ is immediate. Thus p_ε is hyperbolic in e for every $\varepsilon \geq 0$. For sufficiently small ε , all its coefficients in the original z coordinates remain strictly positive, and the fixed segment from e to d remains in its positive set. The component theorem then gives hyperbolicity in d . Positivity of the coefficients makes the entire original positive orthant a connected subset of the same positive component. Hyperbolicity in every positive direction follows, and hence the original polynomial is real stable. This proves sufficiency and the straight-path assertion. $\square$

For example, $a = 0, b = 1$ satisfies every fixed positive-direction Hermite test but is not a coefficient tangent. In this cubic family, enlarging the directions to the full base hyperbolicity cone $T > |X|$ would recover the condition $a \geq |b|$, since $X(v)/T(v)$ then ranges over $(-1, 1)$. The next example shows that this enlargement does not give a general tangent criterion either.

A.5.2. Failure even for all directions in the base cone. Let $\mathcal{H}_5$ denote the monic-in- T real homogeneous quintics hyperbolic in $e = (1, 0, 0)$. Set

$$p = T(T^2 - X^2)(T^2 - 4X^2) = T^5 - 5X^2 T^3 + 4X^4 T, \quad h = -TX^2 Y^2. \quad (205)$$

The positive hyperbolicity component of p containing e is $\Gamma = \{T > 2|X|\}$.

Theorem A.14 (Full-base-cone slice obstruction). *Every affine restriction of (205) in every direction $v \in \Gamma$ satisfies the exact univariate tangent criterion after monic normalization. Nevertheless*

$$h \notin T_{\mathcal{H}_5}(p).$$

The same failure occurs for the tangent to homogeneous real stability at a base with strictly positive coefficients, after the change

$$T = z_1 + z_2 + z_3, \quad X = \frac{z_1 + 2z_2 + 3z_3}{8}, \quad Y = z_2. \quad (206)$$

Proof. The five factors $T - \lambda X$, $\lambda \in \{-2, -1, 0, 1, 2\}$, all have positive slopes in direction $v \in \Gamma$. If any two restricted roots coincide, then $T = X = 0$ at that point, so all five coincide. Simple restrictions impose no tangent condition. At a quintuple root $s = r$, the restricted variation vanishes to order at least three, and its coefficient of $(s - r)^3$ is

$$-T(v)X(v)^2 Y(u + rv)^2 \leq 0.$$

The base leading coefficient is positive, and the monic-normalization term vanishes to order five. Corollary A.5 therefore proves every asserted slice test.

Suppose, to the contrary, that $p_j = p + \varepsilon_j h + o(\varepsilon_j) \in \mathcal{H}_5$, where $\varepsilon_j \downarrow 0$. Write

$$p_j = T^5 + a_{1,j}(X, Y)T^4 + a_{2,j}(X, Y)T^3 + \cdots + a_{5,j}(X, Y).$$

For fixed real (X, Y) , let $M_{2,j}$ be the sum of squares of the five real roots in T , and let $E_{4,j}$ be their fourth elementary symmetric function. Newton's identity gives, coefficientwise,

$$M_{2,j} = a_{1,j}^2 - 2a_{2,j} = 10X^2 + o(\varepsilon_j), \quad E_{4,j} = a_{4,j} = 4X^4 - \varepsilon_j X^2 Y^2 + o(\varepsilon_j). \quad (207)$$

For any five real numbers,

$$|E_4| \leq 5M_2^2. \quad (208)$$

There are five products in E_4 , and each absolute product is bounded by M_2^2 , since each individual absolute root is at most $\sqrt{M_2}$.

At $(X, Y) = (0, 1)$, the first equation in (207) gives $M_{2,j} = o(\varepsilon_j)$. Inequality (208) therefore improves the coefficient of Y^4 in $E_{4,j}$ to $o(\varepsilon_j^2)$. For fixed $s > 0$, average the values at $(X, Y) = (\pm\sqrt{\varepsilon_j}s, 1)$. The odd powers of X cancel; the improved Y^4 coefficient and the remaining even coefficients in (207) give

$$\frac{E_{4,j}(\sqrt{\varepsilon_j}s, 1) + E_{4,j}(-\sqrt{\varepsilon_j}s, 1)}{2} = \varepsilon_j^2(4s^4 - s^2) + o(\varepsilon_j^2).$$

At both evaluation points, $M_{2,j} = 10\varepsilon_j s^2 + o(\varepsilon_j)$. Apply (208) at each point, take the average of the absolute bounds, divide by ε_j^2 , and pass to the limit. It follows that

$$|4s^4 - s^2| \leq 500s^4 \quad (s > 0).$$

This is false whenever $0 < s^2 < 1/504$. Thus no contingent sequence exists.

Finally, (206) is invertible, and all five linear factors $T - \lambda X$ have strictly positive coefficients in z . Their product consequently has every degree-five coefficient strictly positive. The original direction $(1, 1, 1)$ becomes $d = (3, 3/4, 1)$, and the segment from d to e lies in Γ . A hypothetical stable tangent sequence in the original variables would be hyperbolic in d . The component argument would make it hyperbolic in e for large j ; normalization of its T^5 coefficient, which is $1 + o(\varepsilon_j)$, would produce the impossible tangent sequence just excluded. This proves the positive-coefficient version. $\square$

Both failures concern coefficient tangency at a fixed base polynomial. Their proofs use lines whose locations vary on the scale $X = \sqrt{\varepsilon}s$. They do not give a counterexample to a global generator criterion tested at every stable input; such a criterion has additional compatibility across base polynomials.

Remark A.15 (Recovery from the adjacent quotient cone). The localization formula gives a second explanation of these examples. For the cubic roots $-1, 0, 1$, the adjacent quotients are $T - X, T + X$, and

$$aT + bX = \frac{a-b}{2}(T - X) + \frac{a+b}{2}(T + X).$$

Their nonnegative cone gives exactly $a \geq |b|$. For the quintic roots $-2, -1, 0, 1, 2$, let $q_i = p/(\ell_i \ell_{i+1})$ in that order. Direct expansion gives

$$TX^2 = \frac{1}{12}(q_1 - q_2 - q_3 + q_4).$$

Uniqueness of the adjacent quotient coordinates excludes this form from their non-negative cone, although $TX^2 \geq 0$ on Γ . The cubic discriminant proof and quintic moment proof above establish the same conclusions independently of the determinantal representation used in Theorem A.10.

APPENDIX B. DOMAINS, GENERATION, AND POLYNOMIAL-VALUED DYNAMICS

The algebraic restrictions on a stability-preserving generator do not by themselves specify a semigroup on an infinite-dimensional space. We first resolve the domain and approximation questions for summable positive coefficients. We then treat polynomial-valued semigroups, for which the topology forces a different and particularly rigid conclusion.

B.1. An automatic domain theorem. Let E be countable and let $r : E \rightarrow \mathbb{N}$ have finite fibres $E_k = r^{-1}(k)$. Write e_x for the standard basis of $\ell^1(E)$. A nonnegative sequence has interval support if its positive entries form an interval of integers, possibly a singleton or an unbounded interval.

Theorem B.1 (Automatic finite-support domain). *Let $T(t)$ be a positive strongly continuous semigroup on $\ell^1(E)$, with generator A . Suppose that, for each x , and all sufficiently small $t > 0$, the sequence*

$$b_k(t) = \sum_{y \in E_k} (T(t)e_x)_y$$

is log-concave and has interval support. Then $e_x \in D(A)$, and

$$\text{supp}(Ae_x) \subseteq E_{r(x)-1} \cup E_{r(x)} \cup E_{r(x)+1}, \quad E_{-1} = \emptyset.$$

In particular, every finitely supported vector belongs to $D(A)$, and A maps finitely supported vectors to finitely supported vectors.

We give the transition-function argument needed in the proof, to distinguish existence of coordinate derivatives from an a priori finite-rate assumption. The classical result is due to Kolmogorov; see [22, 35].

Lemma B.2. *Let $S(t)$ be a standard countable substochastic transition kernel. Then every off-diagonal limit $S_{yx}(t)/t$, as $t \downarrow 0$, exists and is finite. The limit $(1 - S_{xx}(t))/t$ exists in $[0, \infty]$.*

Proof. Transpose to the row convention and add an absorbing cemetery state. We obtain a stochastic kernel $P(t)$, with $P_{ij}(t) \rightarrow \delta_{ij}$ at zero. Its entries are continuous: the semigroup identity gives

$$|P_{ij}(t+h) - P_{ij}(t)| \leq 1 - P_{ii}(h),$$

and the corresponding left estimate follows by writing $t = (t-h) + h$. Fix $i \neq j$ and $0 < \varepsilon < 1/3$. Choose $\delta_0 > 0$ so that $P_{ii}(s), P_{jj}(s) \geq 1 - \varepsilon$ for $0 \leq s \leq \delta_0$. Consider the discrete chain with transition matrix $P(h)$, started at i . Let τ be its first visit to j , let $f_l = \Pr(\tau = l)$, and put $a_k = \Pr(X_k = i, \tau > k)$. For $kh \leq \delta_0$, first-visit decomposition and $P_{ji}(s) \leq \varepsilon$ give

$$P_{ii}(kh) = a_k + \sum_{l=1}^k f_l P_{ji}((k-l)h) \leq a_k + \varepsilon.$$

Consequently $a_k \geq 1 - 2\varepsilon$, and $f_{k+1} \geq a_k P_{ij}(h)$. If $nh \leq \delta_0$, then

$$P_{ij}(nh) = \sum_{l=1}^n f_l P_{jj}((n-l)h) \geq n(1-\varepsilon)(1-2\varepsilon)P_{ij}(h) \geq n(1-3\varepsilon)P_{ij}(h).$$

Fix $0 < \delta \leq \delta_0$ and take $n = \lfloor \delta/h \rfloor$. Continuity implies

$$\limsup_{h \downarrow 0} \frac{P_{ij}(h)}{h} \leq \frac{P_{ij}(\delta)}{\delta(1-3\varepsilon)} < \infty.$$

First let $\delta \downarrow 0$ along a sequence attaining the lower limit, and then let $\varepsilon \downarrow 0$. The off-diagonal assertion follows.

Subdivision of time, retaining only the path which stays at i , gives $P_{ii}(t) > 0$ for every t . Thus $\phi(t) = -\log P_{ii}(t)$ is continuous, nonnegative, and subadditive. Writing $\delta = nt + s$, $0 \leq s < t$, yields $\phi(\delta) \leq n\phi(t) + \phi(s)$. Hence $\liminf_{t \downarrow 0} \phi(t)/t \geq \phi(\delta)/\delta$. Taking the supremum over $\delta > 0$ proves existence of the limit, possibly infinite. Since $(1 - e^{-u})/u \rightarrow 1$ at zero, this is also the asserted diagonal limit. $\square$

Proof of Theorem B.1. Write $p_{yx}(t) = (T(t)e_x)_y$. Choose M, ω with $\|T(t)\| \leq Me^{\omega t}$, and choose $\lambda > \max(\omega, 0)$. Define the strictly positive bounded weights

$$w_x = \int_0^\infty e^{-\lambda s} \|T(s)e_x\|_1 ds.$$

No uniform positive lower bound on w_x is required. Positivity, Tonelli's theorem, and the semigroup law give

$$\sum_y w_y p_{yx}(t) = e^{\lambda t} \left(w_x - \int_0^t e^{-\lambda s} \|T(s)e_x\|_1 ds \right).$$

Therefore

$$S_{yx}(t) = e^{-\lambda t} \frac{w_y}{w_x} p_{yx}(t)$$

is a standard substochastic kernel, with deficit

$$1 - \sum_y S_{yx}(t) = w_x^{-1} \int_0^t e^{-\lambda s} \|T(s)e_x\|_1 ds = t/w_x + o(t).$$

Lemma B.2 supplies finite off-diagonal derivatives $q_{yx} = \lim_{t \downarrow 0} p_{yx}(t)/t$.

Fix x , put $N = r(x)$, and use the b_k of the theorem. Strong continuity gives $b_N(t) \rightarrow 1$. Finiteness of adjacent fibres and the coordinate derivatives imply $b_{N-1}(t), b_{N+1}(t) = O(t)$. Log-concavity and interval support give the geometric estimates

$$b_{N+j}(t) \leq b_N(t) \left(\frac{b_{N+1}(t)}{b_N(t)} \right)^j, \quad b_{N-j}(t) \leq b_N(t) \left(\frac{b_{N-1}(t)}{b_N(t)} \right)^j.$$

These remain valid when an adjacent coefficient vanishes, since the corresponding tail then vanishes. It follows that

$$\sum_{|k-N| \geq 2} b_k(t) = O(t^2), \quad \sum_{y \neq x} p_{yx}(t) = O(t).$$

The second estimate also uses finiteness of the three central fibres. Since w is bounded and $w_x > 0$, the off-diagonal column sum of $S(t)$ is $O(t)$. Its displayed

deficit is $O(t)$; hence $1 - S_{xx}(t) = O(t)$. The diagonal limit in the preceding lemma is thus finite, and so is

$$q_{xx} = \lim_{t \downarrow 0} \frac{p_{xx}(t) - 1}{t}.$$

On the finite union of the three central fibres, the difference quotient converges coordinatewise and therefore in norm. Outside that union its ℓ^1 norm is $O(t^2)/t = O(t)$. This is precisely strong differentiability of $T(t)e_x$ at zero, with the stated support. $\square$

Corollary B.3. *On any mixed-capacity space with finitely many coordinates, a positive C_0 -semigroup preserving $\mathcal{LP}_{+, \kappa}$ automatically has all polynomials in its generator domain. Its action on each monomial has finite support and changes total degree by at most one.*

Proof. Use the proper grading $r(x) = |x|$. The diagonal specialization of $T(t)z^x$ belongs to the univariate positive Laguerre–Pólya class. Its coefficient sequence is log-concave with interval support: this follows from negative-real-root factorization for its polynomial approximants and passage to coefficient limits. Theorem B.1 applies. $\square$

B.2. Consistent cutoffs and weighted spaces. Let $\mathcal{F}$ denote the finitely supported vectors. Suppose L is a Metzler operator on $\mathcal{F}$, meaning that its off-diagonal matrix entries are nonnegative, and preserves

$$W_N = \text{span}\{e_x : r(x) \leq N\}.$$

Suppose R is a bounded nonnegative operator whose entries raise the grading by exactly one. Write $\rho = \|R\|_1$.

Theorem B.4 (Cutoff generation criterion). *The operator $L + R$ on $\mathcal{F}$ has a positive C_0 -semigroup realization on $\ell^1(E)$, with this action on $\mathcal{F}$, if and only if there are $\tau > 0$ and $B < \infty$ such that*

$$\sup_N \sup_{0 \leq t \leq \tau} \|\exp(tL|_{W_N})\|_1 \leq B.$$

When it exists, the realization is unique, its generator is the closure of $L + R$, and $\mathcal{F}$ is a graph core.

For $s \geq 1$, put

$$X_s = \left\{ f : \|f\|_s = \sum_x |f_x| s^{r(x)} < \infty \right\}.$$

If $\|U(t)\|_1 \leq Me^{\omega t}$ for the semigroup generated by the closure of L , with $M \geq 1$ and $\omega \geq 0$, then compatible semigroups satisfy

$$\|U(t)\|_{X_s} \leq Me^{\omega t}, \quad \|R\|_{X_s} = s\rho, \quad \|T^{(s)}(t)\|_{X_s} \leq M \exp((\omega + Ms\rho)t).$$

Proof. The finite flows are consistent by invariance of the W_N . The asserted bound, followed by subdivision of time, bounds these flows on every compact time interval. They extend from $\mathcal{F}$ to bounded operators $U(t)$ on $\ell^1(E)$. Density proves both the semigroup law and strong continuity. Finite-dimensional positivity proves positivity of the extension. The bounded positive perturbation R can be added by the Dyson series, giving a positive semigroup $T(t)$.

Conversely, subtract R from any proposed generator by bounded perturbation. Each W_N is contained in its domain and is invariant under its action. Uniqueness

of classical solutions identifies the resulting semigroup on W_N with $\exp(tL|_{W_N})$. In particular, it is positive on $\mathcal{F}$, hence positive everywhere, and its local boundedness gives the necessary cutoff estimate. It also proves uniqueness.

Let G denote the generator of the constructed semigroup $U(t)$. For a sufficiently large resolvent parameter λ , its resolvent preserves each W_N , by the Laplace-transform formula. Given $f \in D(G)$, approximate $(\lambda - G)f$ in ℓ^1 by vectors in $\mathcal{F}$ and apply that resolvent. The resulting vectors lie in $\mathcal{F}$ and converge to f in the graph norm of G . Since $G = L$ on $\mathcal{F}$, this proves $G = \bar{L}$. A bounded perturbation has the same domain and an equivalent graph norm, proving the core assertion.

All entries of $U(t)$ vanish unless the output grading is no larger than the input grading. Its weighted column sums are therefore bounded by the unweighted column sums after dividing by the input weight. This proves the bound for $U(t)$ on X_s . Density proves strong continuity there. The identity $\|R\|_{X_s} = s\rho$ follows directly from its one-step grading. The k -th Dyson term has norm at most

$$M^{k+1} e^{\omega t} (s\rho)^k t^k / k!,$$

and summation gives the stated bound. These constructions are compatible under $X_s \hookrightarrow X_1$, term by term. $\square$

Corollary B.5. *If the column sums of $L + R$ are bounded above, the realization in Theorem B.4 exists.*

Proof. The column sums of each finite restriction of L are no larger than those of $L + R$. The positive matrix exponential has norm at most e^{ct} when its column sums are at most c . Apply the cutoff criterion. $\square$

B.3. Growth reduction and the legitimate scaling limit. For mixed capacities, write F and U for the finite and unrestricted coordinates and use $r(x) = |x_U|$. All spaces W_N below are finite dimensional. The next proposition supplies the analytic bridge between local normal-order tests and tests with arbitrary real spectrators.

Proposition B.6. *Let $T(t)$ be a positive C_0 -semigroup on $X_\kappa = \ell^1(\Omega_\kappa)$ preserving $\mathcal{LP}_{+, \kappa}$. Suppose its polynomial action has already been reduced to canonical differential order at most two. Then every canonical coefficient has unrestricted-coordinate degree at most two. The part that increases $|x_U|$ is the bounded operator*

$$R = \sum_{j \in U} \lambda_j z_j + \sum_{\substack{i \in F \\ j \in U}} r_{ij} z_j \partial_i, \quad \lambda_j, r_{ij} \geq 0.$$

For $L = A - R$, the conclusions of Theorem B.4 hold. In particular, for every polynomial f , the expansion $T(t)f = f + tAf + O(t^2)$ holds locally uniformly on every bounded polydisc.

Proof. Corollary B.3 shows that all canonical coefficients are polynomials: the triangular normal-order recursion involves only images of finitely many monomials. Since there are finitely many coefficients after order reduction, and finite-coordinate exponents are bounded, there is a constant C such that

$$\|Az^{(m, x_U)}\|_1 \leq C(1 + |x_U|)^2.$$

Thus the Taylor truncations of $z_F^m e^{a \cdot z_U}$, $a \geq 0$, converge in graph norm. Closedness of A puts this exponential in its domain and identifies its derivative by the differential expression.

Specialize finite variables to $b > 0$; this specialization is bounded because their degrees remain bounded. Dividing the first variation by the initial exponential gives

$$\sum_{|\alpha| \leq 2} (m)_{\alpha_F} b^{-\alpha_F} a^{\alpha_U} P_{\alpha}(b, z_U).$$

Lemma 8.4 makes this polynomial quadratic in z_U . Comparing powers of a , and then the linearly independent falling factorials on the finite m -grid, shows $\deg_{z_U} P_{\alpha} \leq 2$.

Represent the operator as a matrix on the finite-coordinate state space:

$$A_{nm} = \sum_{\substack{|p| \leq 2 \\ |q| \leq 2}} C_{nm;pq} z_U^q \partial_U^p.$$

Terms increasing $|x_U|$ have $(|p|, |q|) = (0, 2), (1, 2)$, or $(0, 1)$. The coefficients with $(0, 2)$ are nonnegative, as coefficients of the image of the input with no unrestricted particles. Poisson tangency at $a = 0$ makes their quadratic part nonpositive on the positive orthant; therefore they vanish.

For $(1, 2)$, fix $p = e_l$. If $q_l > 0$, its coefficient is a slope in the nonnegative affine rate for adding one unrestricted particle. Letting $x_l \rightarrow \infty$ proves that the slope is nonnegative. If $q_l = 0$, the shift removes a particle at l and adds two elsewhere, and its rate is precisely that coefficient times x_l ; the coefficient is again nonnegative. After the $(0, 2)$ elimination, apply Poisson tangency at $a = \varepsilon v$, $v > 0$, divide its quadratic inequality by ε , and let $\varepsilon \downarrow 0$. Every remaining linear-in- a , quadratic-in- z_U coefficient is nonnegative and their sum is nonpositive. They all vanish. The increasing part is consequently

$$R = \sum_{j \in U} z_j K_j,$$

with K_j a nonnegative matrix on the finite state space. This proves boundedness before any scaling is performed.

Subtracting this bounded operator gives $L = A - R$. Its finite restrictions are Metzler and preserve W_N ; the necessity part of Theorem B.4 supplies the positive semigroup $U(t)$ and its compatible weighted realizations. Put

$$S_s : X_s \longrightarrow X_1, \quad (S_s f)(z_F, z_U) = f(z_F, s z_U).$$

This is an isometric isomorphism between the indicated different spaces. For

$$\tilde{U}_s(t) = S_s U^{(s)}(t/s) S_s^{-1}$$

we have $\|\tilde{U}_s(t)\| \leq M e^{\omega t/s}$. On a fixed W_N , every generator entry is

$$s^{-1} s^{|y_U| - |x_U|} L_{yx} = O(s^{-1}).$$

Thus $\tilde{U}_s(t) \rightarrow I$ strongly, uniformly on compact time intervals, first on $\mathcal{F}$, then by density on X_1 . The perturbation of this rescaled semigroup is exactly R , since $S_s R S_s^{-1} = s R$. Its k -th Dyson term is bounded by

$$M^{k+1} e^{\omega t/s} \rho^k t^k / k!,$$

and converges strongly to $t^k R^k / k!$. Dominated summation proves

$$S_s T^{(s)}(t/s) S_s^{-1} \longrightarrow e^{tR}$$

strongly and locally uniformly in time. Positive scaling and coefficient closure of $\mathcal{LP}_+$, Lemma 8.1, show that e^{tR} preserves that class.

Set all unrestricted variables except z_j to zero. We find that $e^{tz_j K_j}$ preserves positive stability. Since this one-step-raising operator is bounded on every X_s , its exponential has a first variation on every bounded polydisc. Finite-site double contacts with the real variable z_j as a spectator force every second-order normal coefficient of K_j to vanish: its product with z_j has to be nonpositive for both signs of z_j . The first-order specialization of Lemma 3.4 then expresses K_j by independent deaths, capacity births, and an affine diagonal. Corollary B.3 applied to 1 excludes the births: a birth in $K_j 1$ would create a degree-two term in $R1$. Positivity gives

$$K_j = \lambda_j I + \sum_{i \in F} (r_{ij} \partial_i + m_{ij} z_i \partial_i), \quad r_{ij} \geq 0.$$

Its remaining diagonal slopes need not yet have a sign.

If $m = m_{ij} \neq 0$, use the input $c + u$ in the corresponding finite variable u , with $c > \max(0, r_{ij}/m)$. Up to a nonvanishing exponential factor, the output is

$$c + e^{mtz} u + \frac{r_{ij}}{m} (e^{mtz} - 1).$$

For $z = iv$, choose $v > 0$ with $\sin(mtv) > 0$. Its root

$$u = -r_{ij}/m - (c - r_{ij}/m)e^{-mtz}$$

then has positive imaginary part, a contradiction. Hence every $m_{ij} = 0$, proving the displayed form of R , including $\lambda_j \geq 0$.

Finally, the generator on every X_s agrees with A on polynomials. If $f \in W_N$, then $A^2 f \in W_{N+2}$. Twice integrating the semigroup equation and using the weighted estimate gives, for $0 \leq t \leq \tau$,

$$\|T(t)f - f - tAf\|_s \leq \frac{t^2}{2} M s^{N+2} e^{(\omega + Ms\rho)\tau} \|A^2 f\|_1.$$

Finite-coordinate radii add only a fixed factor. This proves the final assertion on arbitrary bounded polydiscs. $\square$

Remark B.7. The order reduction used in the preceding proposition needs only the first variation on a small polydisc, which follows directly from the ℓ^1 generator derivative. Negative collision points can be chosen inside that polydisc; polynomial identity then gives the global normal-order conclusion. Arbitrary-radius first variations are proved only after the bounded growth decomposition. Thus no weighted regularity is presupposed in deriving that decomposition.

B.4. Finite approximation with unchanged diagonals.

Theorem B.8. *Let a mixed-capacity operator have the rate form and quadratic-contact inequality of the positive classification, and suppose it has the realization in Theorem B.4. Impose artificial capacities $N_j \geq 2$ at unrestricted sites. Keep its diagonal entries unchanged and replace the rates for immigration and transfer into such a site by*

$$\lambda_j(1 - x_j/N_j), \quad r_{ij}x_i(1 - x_j/N_j),$$

respectively. Denote the finite operator by A_N , and the coordinate projection onto its box by P_N . Then every A_N satisfies the finite positive stability criterion. As all

$N_j \rightarrow \infty$, the following convergence is strong on X_κ , uniformly on compact time intervals:

$$e^{tA_N} P_N \longrightarrow T(t).$$

Proof. In the convention that the second-order expression is $\sum_{i,j} Q_{ij} \partial_i \partial_j$ with Q symmetric, the corrected principal matrix is

$$Q_N(y) = Q(y) - \frac{1}{2} \sum_{\substack{j \in U \\ i \neq j}} \frac{r_{ij}}{N_j} y_j^2 (E_{ij} + E_{ji}).$$

For each admissible contact matrix H , whose entries are nonnegative,

$$\text{Tr}(Q_N(y)H) = \text{Tr}(Q(y)H) - \sum_{\substack{j \in U \\ i \neq j}} \frac{r_{ij}}{N_j} y_j^2 H_{ij} \leq 0.$$

The immigration correction is first order, and the binary sites are unchanged because $N_j \geq 2$. The finite contact theorem applies.

Split $A_N = L_N + R_N$ by unrestricted-degree increase. The increasing parts satisfy $\|R_N\| \leq \rho$ and converge strongly to R . For a nonnegative input in the finite box, variation of constants gives

$$U(t)f - e^{tL_N}f = \int_0^t U(t-s)(L - L_N)e^{sL_N}f ds \geq 0.$$

The finite trajectory is polynomial-valued and belongs to $D(L)$. The integrand is nonnegative because only off-diagonal entries were reduced; their diagonals agree exactly. Therefore $\|e^{tL_N}P_N\| \leq Me^{\omega t}$.

On every fixed W_m , the matrices L_N converge to L once $N_j \geq m$. Finite-dimensional exponential convergence and density, with the preceding uniform bound, imply $e^{tL_N}P_N \rightarrow U(t)$ strongly and uniformly on compact intervals. The Dyson expansions adding R_N have uniformly summable majorants

$$M^{k+1}e^{\omega t}\rho^k t^k / k!.$$

Each fixed term converges to the corresponding term for U, R ; summation proves the claim. The projection P_N , including at time zero, is essential in this full-space statement. $\square$

Corollary B.9. *The rate and contact conditions in the positive mixed-capacity classification, together with the cutoff generation criterion, imply preservation of $\mathcal{LP}_{+, \kappa}$.*

Proof. For stable polynomial input, sufficiently large P_N act as the identity. The finite outputs are stable and nonnegative. Their ℓ^1 limit belongs to $\mathcal{LP}_{+, \kappa}$ by Lemma 8.1, normalizing at $\mathbf{1}$ if the probability-measure form of that theorem is used. A nonzero positive input has nonzero output: the diagonal entry $T(t)_{xx}$ is positive by subdivision of time and strong continuity. Positive Jensen approximants converge in ℓ^1 to every member of $\mathcal{LP}_{+, \kappa}$. Boundedness of $T(t)$ and the same closure theorem complete the proof. $\square$

B.5. Further generation criteria and topology. For comparison with other analytic topologies, let

$$Y_R = \left\{ f = \sum_x f_x z^x : \sum_x |f_x| R^{|x|} < \infty \right\}, \quad \mathcal{E}_{\text{ent}} = \bigcap_{R \geq 1} Y_R.$$

These coefficient seminorms give the usual compact-open topology on entire functions in finitely many variables. The map $f(z) \mapsto f(Rz)$ is an isometry $Y_R \rightarrow Y_1$, rather than a bounded similarity of Y_1 when $R > 1$. In the unrestricted positive canonical expression below, this conjugation replaces

$$\lambda \mapsto R\lambda, \quad b \mapsto b/R, \quad \nu \mapsto \nu/R,$$

and leaves c, M, β, η unchanged. With coordinate weights R_i , the off-diagonal coefficient m_{ij} becomes $m_{ij}R_j/R_i$. These assertions follow by conjugating multiplication and differentiation on each monomial.

Proposition B.10. *Let L be a degree-nonincreasing polynomial operator, and let its consistent finite-degree flows be $U_N(t)$. They extend to a locally equicontinuous C_0 -semigroup on $\mathcal{E}_{\text{ent}}$ if and only if for every $R \geq 1$ and $\tau > 0$ there are $S \geq 1$ and $C < \infty$ such that*

$$\|U_N(t)p\|_{Y_R} \leq C\|p\|_{Y_S} \quad (N \geq 0, 0 \leq t \leq \tau, p \in \mathcal{P}_N).$$

Such an extension is unique.

Proof. Necessity is the definition of local equicontinuity, since finitely many coefficient seminorms can be dominated by their largest radius. For sufficiency, Taylor polynomials are dense in every seminorm. The estimate makes their evolved sequences Cauchy, uniformly in $0 \leq t \leq \tau$, in every target seminorm. Their limits define continuous operators satisfying the same estimate. Density transfers the semigroup law, and uniform approximation by a polynomial orbit proves strong continuity. The same density proves uniqueness. $\square$

Example B.11. The Euler flow $e^{tE}f(z) = f(e^t z)$ acts continuously on all entire functions, but for every fixed R and $t > 0$ its norm on Y_R is infinite, as the degree- n monomial has norm ratio e^{tn} . Conversely, the positive stability-preserving polynomial flow generated by $z\partial_z^2$ cannot extend globally to all positive entire inputs. Indeed,

$$[z]e^{tz\partial_z^2}z^n = n!t^{n-1} \quad (n \geq 1).$$

This follows by applying the lowering operator $n-1$ times and dividing by $(n-1)!$ in its exponential. The coefficient of z in the evolved Taylor truncations of e^{az} , with $a > 0$, is therefore $\sum_{n=1}^N a^n t^{n-1}$, which diverges when $at \geq 1$. Polynomial stability of this flow can be seen on any fixed degree bound N : it is the dilute limit of the one-site generators $(z - z^2/K)\partial_z^2$, $K \geq N$, which satisfy the finite positive criterion. Their restrictions to degree at most N converge as $K \rightarrow \infty$. This example does not satisfy coefficient- ℓ^1 generation and does not contradict Theorem B.13.

For the unrestricted positive canonical operator, write

$$A = cI + \lambda \cdot z + \sum_i \left(b_i + \sum_j m_{ij} z_j \right) \partial_i + ED_\nu - \beta E(E-1) - \sum_i \eta_i z_i^2 \partial_i^2,$$

where $E = \sum_i z_i \partial_i$, $D_\nu = \sum_i \nu_i \partial_i$, and the signs are those in the positive classification. On the probability simplex define

$$F(p) = \nu \cdot p - \beta - \sum_i \eta_i p_i^2.$$

Proposition B.12. *Existence of a positive coefficient- ℓ^1 realization forces $F(p) \leq 0$ throughout the simplex. The strict inequality $\max F < 0$ is sufficient for such a realization.*

Proof. For a finite set of basis states S , let B be the principal matrix of the generator on S . Positivity gives

$$P_S T(t)|_S \geq e^{tB}$$

entrywise. Indeed, the left side, denoted $G(t)$, satisfies $G'(t) = G(t)B + R(t)$, where $R(t) \geq 0$, by differentiating on the finite set of generator-domain columns. Variation of constants proves the comparison.

Choose $x^{(N)}/N \rightarrow p$, with zero coordinates when $p_i = 0$, and retain the finite states $x^{(N)} - \alpha$, where $0 \leq \alpha \leq x^{(N)}$ and $|\alpha| \leq K$. For sufficiently large N , this is the fixed index set of such α supported on $\{i : p_i > 0\}$. Divide their principal matrix by N^2 . It converges to the matrix with diagonal

$$-D(p), \quad D(p) = \beta + \sum_i \eta_i p_i^2,$$

and rates $\nu_i p_i$ from α to $\alpha + e_i$ when $|\alpha| < K$. All linear terms vanish in this scaling. Its exponential has column mass at $\alpha = 0$ equal to

$$e^{-sD(p)} \sum_{j=0}^K \frac{(s\nu \cdot p)^j}{j!}.$$

If $T(t)$ is locally bounded by M , apply the comparison at $t = s/N^2$ and let $N \rightarrow \infty$. Then let $K \rightarrow \infty$. We obtain $M \geq e^{sF(p)}$ for every $s > 0$, forcing $F(p) \leq 0$.

Conversely, the column sum at x is

$$(Az^x)(\mathbf{1}) = |x|^2 F(x/|x|) + O(|x|) + O(1),$$

with a uniform remainder. If $F \leq -\varepsilon$, these column sums are bounded above. Corollary B.5 applies. $\square$

Proposition B.13 (Exact unary generation criterion). *For*

$$A = c + \lambda z + (b + mz)\partial_z + \nu z\partial_z^2 - \chi z^2\partial_z^2, \quad \lambda, b, \nu, \chi \geq 0, \quad c, m \in \mathbb{R},$$

the positive coefficient- ℓ^1 realization exists if and only if

$$\chi > \nu \quad \text{or} \quad \chi = \nu \text{ and } m + b \leq 0.$$

Proof. The terms $cI + \lambda z$ are bounded. Removing them from an existing generator preserves generation; positivity of the resulting semigroup follows from its Metzler restrictions to finite degree cutoffs and density, rather than from an assertion that subtracting a positive operator always preserves positivity. Conversely, add λz by positive bounded perturbation and cI by multiplying the semigroup by e^{ct} . We may set $c = \lambda = 0$. The column sum is

$$(\nu - \chi)n(n-1) + (m+b)n.$$

This proves sufficiency, while Proposition B.12 excludes $\chi < \nu$.

Suppose $\chi = \nu$ and $a = m + b > 0$. Then $A = Q + aE$, where Q is pure death with rates $d_k = bk + \nu k(k-1)$. On the finite cutoff, the elementary finite-state Feynman–Kac formula (obtained by conditioning on the first holding time) gives

$$\|e^{tA} z^n\|_1 = \mathbb{E}_n \exp \left(a \int_0^t N_s ds \right).$$

If $\nu > 0$, fix $t > 0$ and choose K with $\sum_{k>K} d_k^{-1} < t/4$. The independent exponential holding times τ_k give

$$H_n = \sum_{k=K+1}^n \tau_k, \quad Z_n = \sum_{k=K+1}^n k \tau_k.$$

Markov's inequality yields $\Pr(H_n > t) \leq 1/4$, while

$$\mathbb{E} Z_n = \nu^{-1} \log n + O(1), \quad \text{Var } Z_n = \sum_{k=K+1}^n k^2 / d_k^2 = O(1).$$

For large n , Chebyshev's inequality gives $\Pr(Z_n \geq (\log n)/(2\nu)) \geq 3/4$. On an event of probability at least $1/2$, both estimates hold, and the integrated population up to time t is at least Z_n . Thus $\|e^{tA} z^n\|_1 \geq \frac{1}{2} n^{a/(2\nu)}$, contradicting boundedness. If $\nu = 0$, direct solution instead gives

$$\|e^{tA} z^n\|_1 = \left(e^{mt} + b \int_0^t e^{ms} ds \right)^n.$$

The parenthesis is $1 + at + O(t^2) > 1$ for small positive t , giving the same contradiction. $\square$

B.6. Automatic filtration of polynomial-valued semigroups. Let $\mathcal{P} = \mathbb{C}[z_1, \dots, z_d]$, let $\mathcal{P}_N = \{f : \deg f \leq N\}$, and give $\mathcal{P}$ the strict locally convex direct-limit topology of these finite-dimensional spaces. This is the finest locally convex topology on $\mathcal{P}$.

Lemma B.14 (A degree bound from the constant polynomial). *An injective complex-linear stability preserver $T : \mathcal{P} \rightarrow \mathcal{P}$ satisfies*

$$\deg Tp \leq \deg p + \deg T1 \quad (p \neq 0).$$

Proof. Fix a monomial z^α , a coordinate i , and put $G = Tz^\alpha$, $F = T(z_i z^\alpha)$. For every $w \in \mathbb{H}$, the input $z^\alpha(z_i + w)$ is stable, and its nonzero image is $F + wG$. Thus F/G is holomorphic on $\mathbb{H}^d$ and has nonnegative imaginary part there: a value with negative imaginary part would permit the choice $w = -F/G \in \mathbb{H}$.

Choose $v > 0$ where the leading homogeneous parts of both F, G are nonzero. The rational function $F(sv)/G(sv)$ maps $\mathbb{H}$ into its closure. Its numerator degree minus denominator degree is at most one. Indeed, if that difference were $m \geq 2$, its asymptotic form on $s = Re^{i\theta}$ would be $aR^m e^{im\theta}(1 + o(1))$, with $a \neq 0$. As $0 < \theta < \pi$, some fixed angle makes the leading imaginary part negative, contradicting the mapping property for large R . Cancellation does not change the degree difference. Consequently $\deg F \leq \deg G + 1$. Iteration along a monomial multiplication chain, followed by linearity, proves the assertion. $\square$

Theorem B.15 (Automatic polynomial filtration). *Every strongly continuous complex-linear semigroup on $\mathcal{P}$ preserving upper-half-plane stability has an*

everywhere-defined locally finite generator A . Its operators are linear automorphisms of the vector space $\mathcal{P}$, and

$$A\mathcal{P}_N \subseteq \mathcal{P}_N, \quad T(t)\mathcal{P}_N = \mathcal{P}_N \quad (N \geq 0, t \geq 0).$$

The same conclusion holds algebraically if A is assumed locally finite and its finite-orbit exponentials preserve stability.

Proof. A bounded subset of this direct limit lies in some $\mathcal{P}_N$. For the needed implication, otherwise choose polynomials with selected nonzero monomials of strictly increasing degrees. Assign successively positive monomial weights making their weighted coefficient seminorms at least $1, 2, \dots$. Every such weighted seminorm is continuous in the finest locally convex topology, contradicting boundedness.

For fixed f , its orbit over $0 \leq t \leq 1$ is compact, hence contained in a finite-dimensional space. Put

$$W_r = \text{span}\{T(t)f : 0 \leq t \leq r\}.$$

The finite integer dimensions have a minimum for $r > 0$. Choose $\varepsilon > 0$ sufficiently small that $W_\varepsilon = W_{2\varepsilon}$. Then $T(s)W_\varepsilon \subseteq W_\varepsilon$ for $0 \leq s \leq \varepsilon$, and iteration proves invariance for all $s \geq 0$. Finite-dimensional continuous semigroup theory gives an exponential on this space. Its generator agrees on overlapping invariant spaces because it is the derivative of the same orbit. These derivatives define a locally finite linear A on all of $\mathcal{P}$, and the finite-space inverses e^{-tA} agree as well. Thus every $T(t)$ is injective.

Recursive evaluation on monomials gives the canonical locally finite normal expansion $A = \sum_\alpha P_\alpha(z)\partial^\alpha$, with polynomial coefficients but no assumed order bound. Apply the flow to $\prod_i (z_i - y_i)^{\alpha_i}$ and restrict to $z = y + sv$, $v > 0$. This one orbit lies in a finite-dimensional polynomial space, so the restricted degrees are uniformly bounded. At the contact y , the first variation is exactly $\alpha!P_\alpha(y)$. Lemma 3.5 makes it zero when $|\alpha| \geq 3$. Polynomial identity proves $P_\alpha = 0$ for those α . Thus A is a finite-order polynomial differential operator.

The entire orbit of 1 lies in one finite invariant space; choose D with $\deg T(t)1 \leq D$ for all $t \geq 0$. Lemma B.14 gives

$$T(t)\mathcal{P}_N \subseteq \mathcal{P}_{N+D}.$$

Analyticity on finite orbit spaces allows differentiation to every order:

$$A^k\mathcal{P}_N \subseteq \mathcal{P}_{N+D} \quad \text{for every } k, N.$$

Suppose the maximal degree shift of a normal-ordered term in A is $s > 0$, and let $B \neq 0$ be the homogeneous component of shift s . The component of shift ks in A^k is B^k . For $ks > D$, the preceding inclusions force B^k to vanish on every polynomial. This is impossible: if B has differential order r , the principal symbol of B^k is the k -th power of its nonzero symbol in the integral domain $\mathbb{C}[z, \xi]$. Consequently all shifts are nonpositive. This proves invariance of each $\mathcal{P}_N$; its exponential is invertible on that space. The algebraic locally finite case uses the same argument after the first paragraphs, since finite invariant orbit spaces are already available. $\square$

Remark B.16. The topological necessity remains true for any Hausdorff vector-space topology on $\mathcal{P}$ for which the given maps are continuous and the semigroup is strongly continuous. Shkarin's orbit theorem [52, preprint Lemma 5.4] states that an orbit span has either finite dimension or continuum dimension. Countable Hamel dimension forces the first alternative, after which the algebraic proof applies. The

converse classification below is asserted for the specified strict direct-limit topology; continuity in another topology is a separate requirement.

Theorem B.17 (All-polynomial classification). *The semigroups in Theorem B.15 are precisely the polynomial exponentials of*

$$A = cI + \sum_i \left(b_i + i\sigma_i + \sum_j m_{ij}z_j \right) \partial_i - \sum_i q_i(z_i)\partial_i^2 - \sum_{i<j} c_{ij}\partial_i\partial_j,$$

where $c \in \mathbb{C}$, $b_i \in \mathbb{R}$, $\sigma_i \geq 0$, $M = (m_{ij})$ is real Metzler, each q_i is a real nonnegative polynomial of degree at most two, and $c_{ij} \geq 0$. Their flows preserve stability after adjoining any finite collection of inert polynomial variables.

Proof. Filtration and canonical recursion give $\deg P_\alpha \leq |\alpha|$. The preceding contact argument gives order at most two. Double contacts make every second-order coefficient real and nonpositive on $\mathbb{R}^d$. A simple-root contact makes $\Im P_{e_i}(y) \geq 0$ everywhere. Since P_{e_i} is affine, its imaginary part is a constant $\sigma_i \geq 0$; $P_0 = c$ is constant.

We require the following homogeneous sign fact. A homogeneous complex stable polynomial has all coefficients in one common phase, and after removing that phase they are nonnegative. Indeed, homogeneity also gives nonvanishing on the lower half-plane. For $u, v > 0$, $s \mapsto f(u + sv)$ has only real roots and no nonnegative root; its leading coefficient is $f(v) \neq 0$. Thus $f(u)/f(v) > 0$. Fixing v gives a common phase on the positive orthant. The negative real roots give nonnegative directional derivatives there. Letting directions tend to coordinate vectors, and repeating this argument for stable homogeneous derivatives [10], proves nonnegativity of all coefficients.

After removal of the scalar phase $i\Im c$, the highest homogeneous block of the generator is real. Its exponential applied to a monomial has a highest homogeneous part which is stable or zero by scaling and Hurwitz. Near zero time its original monomial coefficient is positive. The homogeneous sign fact makes all other coefficients of that block nonnegative. Its generator matrix is therefore Metzler in every degree.

Put $U_i = P_{2e_i}$. Its homogeneous quadratic part is negative semidefinite. For $j \neq i$, the shift $2e_j - 2e_i$ can only be produced by the z_j^2 coefficient of U_i . Homogeneous Metzler positivity forces that coefficient to be nonnegative; nonpositivity of U_i forces it to vanish. A semidefinite quadratic form with zero diagonal entry has its corresponding row zero. Hence the quadratic part of U_i depends only on z_i , and global nonpositivity also eliminates linear coefficients in other variables. Thus $U_i = -q_i(z_i)$.

For $U_{ij} = P_{e_i+e_j}$, the same unique-shift argument at every coordinate outside $\{i, j\}$ eliminates its outside square coefficients and their rows. Its quadratic part depends only on z_i, z_j . Take an input with $x_i = N, x_j = 1$, all other exponents zero, and inspect the shift $e_i - e_j$. Its rate is

$$N[z_i^2]U_{ij} + m_{ji}.$$

Here all other possible second-order contributions have already been eliminated by the preceding variable-dependence conclusions. Positivity for every N gives $[z_i^2]U_{ij} \geq 0$, hence equality. Interchange i, j . The negative semidefinite quadratic part now has all diagonal entries zero, so vanishes. An affine polynomial nonpositive

on all of $\mathbb{R}^d$ is constant; write it as $-c_{ij}$, $c_{ij} \geq 0$. The degree-one block finally yields $m_{ij} \geq 0$ when $i \neq j$.

Conversely, the affine first-order flow is substitution by

$$z \mapsto e^{tM} z + \int_0^t e^{sM} (b + i\sigma) ds.$$

The matrix e^{tM} is nonnegative with strictly positive diagonal, so this maps $\mathbb{H}^d$ into itself. Each unary term $-q_i(z_i)\partial_i^2$ and each binary term $-c_{ij}\partial_i\partial_j$ preserves every finite coordinate box and satisfies Theorem 3.1. Their flows therefore preserve complex stability, with arbitrary inert variables. On every $\mathcal{P}_N$, Lie products of these flows and the scalar exponential converge to e^{tA} . Hurwitz and invertibility on $\mathcal{P}_N$ give nonzero stable outputs. The consistent finite-dimensional flows are strongly continuous in the strict direct-limit topology, which proves sufficiency. $\square$

B.7. A common space of entire functions. Polynomial-valued dynamics must be distinguished both from coefficient- ℓ^1 generation and from continuous dynamics on all entire functions. A common space for the preceding full complex cone is

$$\mathcal{E}_{2,0} = \left\{ f(z) = \sum_{\alpha} f_{\alpha} z^{\alpha} : \|f\|_R = \sum_{\alpha} |f_{\alpha}| \sqrt{\alpha!} R^{|\alpha|} < \infty \text{ for every } R > 0 \right\}.$$

Positive integer radii suffice to define its Fréchet topology. In particular

$$|f(z)| \leq \|f\|_R \exp \left(\frac{\sum_i |z_i|^2}{2R^2} \right),$$

and Taylor polynomials converge in every seminorm.

Theorem B.18. *Every generator in Theorem B.17 has a unique locally equicontinuous C_0 -semigroup extension to $\mathcal{E}_{2,0}$. Its generator is the same continuous differential operator, defined on all of $\mathcal{E}_{2,0}$. For every $R > 0$ there are $C_R, K_R \geq 0$ such that*

$$\|T(t)f\|_R \leq e^{C_R t} \|f\|_{Re^{K_R t}} \quad (t \geq 0).$$

It preserves the complex Laguerre–Pólya class intersected with $\mathcal{E}_{2,0}$, with zero output allowed.

Proof. Write $q_i = a_i z_i^2 + d_i z_i + e_i$. Work first on any finite degree cutoff, with weight $w_R(\alpha) = \sqrt{\alpha!} R^{|\alpha|}$. For a complex matrix flow, the upper right derivative of the weighted absolute coefficient sum is bounded by the weighted column sums formed from the real diagonal and absolute off-diagonal entries. Their contributions are bounded by

$$\begin{array}{ll} (b_i + i\sigma_i)\partial_i : & |b_i + i\sigma_i| \sqrt{\alpha_i}/R, \\ m_{ij} z_j \partial_i \quad (i \neq j) : & m_{ij} \sqrt{\alpha_i(\alpha_j + 1)}, \\ -d_i z_i \partial_i^2 : & (|d_i|/R) \sqrt{\alpha_i(\alpha_i - 1)}, \\ -e_i \partial_i^2 : & (e_i/R^2) \sqrt{\alpha_i(\alpha_i - 1)}, \\ -c_{ij} \partial_i \partial_j : & (c_{ij}/R^2) \sqrt{\alpha_i \alpha_j}. \end{array}$$

The diagonal contains $-a_i \alpha_i (\alpha_i - 1)$. When $a_i > 0$,

$$\sup_{n \geq 0} \{-a_i n(n-1) + (|d_i|/R) \sqrt{n(n-1)}\} < \infty.$$

When $a_i = 0$, nonnegativity of q_i forces $d_i = 0$. All remaining contributions, including the scalar and linear diagonal, are bounded by $C_R + K_R |\alpha|$, using elementary

arithmetic-geometric mean inequalities. The constants may be chosen to work at every radius $r \geq R$, since the radius-dependent terms decrease.

For the finite polynomial flow put $H(t, r) = \sum_{\alpha} |f_{\alpha}(t)| \sqrt{\alpha!} r^{|\alpha|}$. The upper derivative inequality is

$$D_t^+ H(t, r) \leq C_R H(t, r) + K_R r \partial_r H(t, r), \quad r \geq R.$$

Along $r(s) = R e^{K_R(t-s)}$, the radius derivative cancels the last term. Gronwall proves the asserted estimate, uniformly in every cutoff.

Taylor approximation now constructs $T(t)f$ in each seminorm, uniformly on compact time intervals. The same estimate transfers the semigroup law and strong continuity from polynomials. Every differential monomial in A is continuous between a larger-radius seminorm and a smaller one, since an exponential ratio absorbs all polynomial factors in α . Polynomial approximation in

$$T(t)p - p = \int_0^t T(s) A p \, ds$$

therefore proves the identity for all f , identifies the full generator domain, and proves uniqueness of the extension.

For preservation use the multivariate Jensen polynomials

$$J_n f(z) = \sum_{\alpha_i \leq n} f_{\alpha} \prod_i \frac{(n)_{\alpha_i}}{n^{\alpha_i}} z^{\alpha}.$$

They are stable or zero for complex Laguerre–Pólya inputs by [10, Theorem 5.1 and Lemma 5.2]. Their multipliers lie in $[0, 1]$ and converge pointwise to one, so dominated convergence gives $J_n f \rightarrow f$ in every displayed seminorm. The flow estimate and Hurwitz now prove preservation. $\square$

Proposition B.19 (A sharp full-class obstruction). *For $a, t > 0$ with $4at \geq 1$, the polynomial operator $e^{-t\partial_z^2}$ has no compact-open continuous extension even to the full real Laguerre–Pólya class that agrees with it on real-rooted polynomials.*

Proof. The real-rooted polynomials $g_N(z) = (1 - az^2/N)^N$ converge locally uniformly to e^{-az^2} . Direct differentiation gives

$$(e^{-t\partial_z^2} g_N)(0) = \sum_{k=0}^N \frac{(N)_k}{N^k} \binom{2k}{k} (at)^k.$$

All summands are nonnegative and each fixed multiplier tends to one. The infinite series without those multipliers diverges for $4at \geq 1$, including equality since $\binom{2k}{k} \sim 4^k / \sqrt{\pi k}$. Keeping any finite initial segment and then increasing its length proves divergence of the displayed values. This contradicts continuity of evaluation at zero for the proposed extension. $\square$

B.8. Exactly when a unary flow acts on all entire functions.

Theorem B.20. *Let*

$$A = \gamma I + (u + mz)\partial_z - (az^2 + bz + c)\partial_z^2,$$

where $\gamma \in \mathbb{C}$, $\Im u \geq 0$, $m \in \mathbb{R}$, and $q(z) = az^2 + bz + c$ is real and nonnegative on $\mathbb{R}$. *Its polynomial semigroup extends to a continuous C_0 -semigroup on the compact-open space $\mathcal{H}(\mathbb{C})$ of all entire functions if and only if $q = 0$ or $a > 0$. In these cases the extension is unique, its generator is A on all entire functions, and it preserves*

the compact-open closure of stable polynomials. If q is a nonzero constant, no individual positive-time operator has a continuous extension to $\mathcal{H}(\mathbb{C})$.

Proof. The seminorms $\|f\|_R = \sum_n |f_n| R^n$ define the compact-open topology: they dominate the supremum on the radius- R disk and are at most twice the supremum on the radius- $2R$ disk. If $a > 0$, choose R so large that $\delta_R = a - |b|/R - c/R^2 > 0$. On every finite degree cutoff, absolute weighted column sums with the real diagonal give logarithmic norm at most

$$M_R = \sup_{n \geq 0} \{ \Re \gamma + (m + |u|/R)n - \delta_R n(n-1) \} < \infty.$$

Thus $\|e^{tA}p\|_R \leq e^{M_R t} \|p\|_R$, uniformly in the degree. Every target radius is dominated by a sufficiently large such R . Taylor approximation constructs the unique locally equicontinuous extension and transfers its semigroup law and strong continuity. The differential expression is continuous on $\mathcal{H}(\mathbb{C})$; passing to the limit in the integrated polynomial equation identifies its generator on all of that space. Hurwitz gives preservation of the polynomial-closure class. If $q = 0$, all assertions follow directly from

$$T(t)f(z) = e^{\gamma t} f \left(e^{mt} z + u \int_0^t e^{ms} ds \right).$$

The only remaining case is $a = b = 0, c > 0$. If $T(t)$ extended continuously for a fixed $t > 0$, evaluation at zero after $T(t)$ would be bounded by C times the supremum on some disk of radius R . Polynomial moments give

$$(T(t)e^{wz})(0) = \exp(\gamma t + uB_t w - cC_t w^2),$$

where

$$B_t = \int_0^t e^{ms} ds, \quad C_t = \int_0^t e^{2ms} ds > 0.$$

This identity does not presuppose existence on exponentials: expanding the right side in w gives the unique polynomial solutions for all monomial data, as is verified with the ansatz $e^{v(t)z+d(t)}$, $v' = mv$, $d' = \gamma + uv - cv^2$. Continuity of the proposed extension then permits passage from the Taylor polynomials of e^{wz} . At $w = iY$ the displayed modulus grows like $\exp(cC_t Y^2 + O(|Y|))$, whereas continuity bounds it by $Ce^{R|Y|}$. This contradiction proves necessity. $\square$

Proposition B.21. *For every $t > 0$, the continuous entire-function operator $e^{-tz^2\partial_z^2}$ fails to preserve the class of all entire functions nonvanishing on $\mathbb{H}$, although it preserves the compact-open closure of stable polynomials.*

Proof. Put

$$G_t(z) = \sum_{n \geq 0} e^{-tn(n-1)} \frac{z^n}{n!}.$$

This is $T(t)e^z$. Its order is zero, since

$$\limsup_{n \rightarrow \infty} \frac{n \log n}{tn(n-1) + \log n!} = 0.$$

It is nonconstant, as $G'_t(0) = 1$. A zero-free entire function of order zero is constant, by the zero-order case of Hadamard factorization [9]. Here is a direct proof of the fact used. If such a function is e^h with h entire, its growth gives $\Re h(z) \leq R^{1/2}$ on $|z| \leq R$ for sufficiently large R . For a fixed center a , the nonnegative function

$(R + |a|)^{1/2} - \Re h(a + Re^{i\theta})$ has average $(R + |a|)^{1/2} - \Re h(a)$. Bounding its first Fourier coefficient by its average gives

$$|h'(a)| \leq \frac{2((R + |a|)^{1/2} - \Re h(a))}{R} \rightarrow 0.$$

Thus h is constant. Consequently G_t has a zero. The polynomials $T(t)(1 + z/N)^N$ are real-rooted with nonnegative coefficients, and Theorem B.20 makes them converge locally uniformly to G_t . Hurwitz off the nonpositive real axis, and $G_t(0) = 1$, show that all zeros of G_t are strictly negative real. Choose one, $-r$. The entire function e^{iz} is globally zero-free, but its image $G_t(iz)$ vanishes at $z = ir \in \mathbb{H}$. $\square$

APPENDIX C. STATIC CHANNELS AND STABLE ENCODINGS

This section classifies invertible Hermiticity-preserving, trace-preserving maps with Lee–Yang full Choi tensors, and gives structural results for singular channels. The variables associated with different physical qubits are always distinct. In particular, permutation of physical wires is allowed, whereas an arbitrary change of the computational basis is not.

C.1. Coefficient tensors and complete preservation. For a matrix $M : (\mathbb{C}^2)^{\otimes m} \rightarrow (\mathbb{C}^2)^{\otimes n}$, put

$$G_M(z, w) = \sum_{a \in \{0,1\}^n} \sum_{b \in \{0,1\}^m} M_{ab} z^a w^b. \quad (209)$$

We call M *tensor-LY* if this polynomial is nonzero whenever all its variables belong to the open unit disk $\mathbb{D}$. Zero is excluded from this definition. For a complex-linear map $\Phi : M_{2^m}(\mathbb{C}) \rightarrow M_{2^n}(\mathbb{C})$, our Choi convention is

$$J(\Phi) = \sum_{c,d} \Phi(|c\rangle\langle d|) \otimes |c\rangle\langle d|. \quad (210)$$

Thus full Choi–LY means stability in all $2(m+n)$ variables of $G_{J(\Phi)}$. Hermiticity-preserving and trace-preserving maps are abbreviated HP and TP, respectively.

We use the disk tensor-contraction theorem in the following form: contracting two binary indices of a tensor-LY polynomial with unit radii produces a tensor-LY polynomial or zero. Tensor products, permutation of indices, and coefficientwise conjugation preserve tensor-LY. These facts follow from the classical Asano contraction theorem; the tensor and channel formulation is described in [61, Section 2]. The possible zero result at unit radius will be excluded explicitly where necessary.

Lemma C.1 (The Choi bridge). *Let $\Phi : M_{2^m}(\mathbb{C}) \rightarrow M_{2^n}(\mathbb{C})$ be complex linear. If $J(\Phi)$ is tensor-LY, then Φ , and each of its extensions by identity maps, sends tensor-LY matrices to tensor-LY matrices or zero. Its Hilbert–Schmidt adjoint $\Phi^\dagger$ is also full Choi–LY. For a CPTP map, full Choi–LY is equivalent to preservation of tensor-LY density matrices with every finite qubit reference system left inert.*

Proof. The superoperator and Choi arrays satisfy

$$S(\Phi)_{(a,b),(c,d)} = \Phi(|c\rangle\langle d|)_{ab}, \quad J(\Phi)_{(a,c),(b,d)} = S(\Phi)_{(a,b),(c,d)}.$$

They therefore differ only by a permutation of tensor indices. Applying Φ is contraction of this array with the input coefficient array. Input transposes occurring in the matrix convention also only permute indices. The contraction theorem proves the first assertion, including inert indices. The superoperator of $\Phi^\dagger$ is obtained

by conjugating entries and interchanging the input and output index pairs, which proves the adjoint assertion.

For a CPTP map the output of a density matrix has trace one, so cannot be the zero alternative. Conversely, take the normalized maximally entangled vector between the input and an equally large reference:

$$|\Omega_m\rangle = 2^{-m/2} \sum_c |c\rangle|c\rangle.$$

Its amplitude polynomial is $2^{-m/2} \prod_i (1 + x_i u_i)$, and its density-matrix polynomial is the product of this polynomial and its coefficientwise conjugate in independent variables. Both are stable. Applying Φ to one side produces $2^{-m} J(\Phi)$, proving the converse. $\square$

Ordinary multiplication of tensor-LY matrices is likewise a tensor contraction, with the same zero alternative. In particular, multiplication on either side by an invertible tensor-LY matrix whose inverse is tensor-LY preserves the class of tensor-LY matrices.

C.2. An elementary matrix edge and hyperbolic pencils. Write

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Subscripts denote physical sites, with identity factors suppressed. The real vector spaces needed below are

$$\mathfrak{g} = \text{span}_{\mathbb{R}}\{X, Y, iZ\}, \quad \mathcal{E} = \mathbb{R}I + \mathfrak{g}, \quad \mathfrak{e}_n = \mathbb{C}I + \bigoplus_{j=1}^n \mathfrak{g}_j. \quad (211)$$

Lemma C.2 (Two-sided tangency). *Suppose $M(t)$ is a matrix-valued curve defined for real t near zero, differentiable at zero, with $M(0) = I$ and $M'(0) = B$. If $M(t)$ is tensor-LY for both signs of all sufficiently small t , then $B \in \mathfrak{e}_n$. Conversely, $B \in \mathfrak{e}_n$ implies that e^{tB} is tensor-LY for every real t .*

Proof. Fix a site j and $|\zeta| = 1$. Specializing $z_j = \zeta$ is justified by first using $r\zeta$, $r < 1$, and then applying Hurwitz's theorem. The specialization is stable or zero. At $t = 0$ its value with all remaining variables zero is one, so it is not the zero polynomial for sufficiently small $|t|$. Fix interior values $u = (z_k, w_k)_{k \neq j}$, and put $D(u) = \prod_{k \neq j} (1 + z_k w_k)$. The resulting affine polynomial in w_j is

$$D(u)(1 + \zeta w_j) + tG_B(\zeta, w_j, u) + o(t).$$

Its root $r(t)$ is differentiable at zero, with $r(0) = -\zeta^{-1}$, and stability implies $|r(t)| \geq 1$ for both signs of small t . Differentiating the root equation gives

$$\frac{r'(0)}{r(0)} = \frac{G_B(\zeta, -\zeta^{-1}, u)}{D(u)}.$$

The derivative of $|r(t)|^2$ at its two-sided minimum is zero. Consequently the holomorphic function on the remaining polydisk

$$u \mapsto \frac{G_B(\zeta, -\zeta^{-1}, u)}{D(u)} \quad (212)$$

has purely imaginary values, and is therefore a purely imaginary constant.

Expand B in the Pauli tensor basis, separating site j : $B = \sum_{\sigma,\tau} b_{\sigma,\tau} \sigma_j \otimes \tau$. At $(z_j, w_j) = (\zeta, -\zeta^{-1})$, the one-site polynomials are

$$G_I = 0, \quad G_X = \zeta - \zeta^{-1}, \quad G_Y = i(\zeta + \zeta^{-1}), \quad G_Z = 2.$$

Since the Pauli polynomials on the other sites are linearly independent, constancy of (212) gives, for every nonidentity word τ ,

$$b_{X,\tau}(\zeta - \zeta^{-1}) + ib_{Y,\tau}(\zeta + \zeta^{-1}) + 2b_{Z,\tau} = 0.$$

Taking $\zeta = 1, -1, i$ forces all three coefficients to vanish. As j was arbitrary, every Pauli coefficient of weight at least two is zero. Thus $B = cI + \sum_j (x_j X_j + y_j Y_j + d_j Z_j)$. The remaining scalar contact expression is

$$x_j(\zeta - \zeta^{-1}) + iy_j(\zeta + \zeta^{-1}) + 2d_j.$$

Its real part is zero. The same three values of ζ imply $x_j, y_j \in \mathbb{R}$ and $d_j \in i\mathbb{R}$, as asserted. The scalar c is unrestricted.

For the converse, if $N = aI + xX + yY + izZ \in \mathcal{E}$, direct multiplication gives

$$NZN^* = \delta Z, \quad \delta = a^2 + z^2 - x^2 - y^2 = \det N. \quad (213)$$

When $\delta > 0$, write $G_N(s, w) = A(s) + C(s)w$. Equation (213) implies

$$|A(s)|^2 - |C(s)|^2 = \delta(1 - |s|^2) > 0 \quad (|s| < 1),$$

so N is tensor-LY. Its inverse has the same property. For $b = xX + yY + izZ$, one has $b^2 = (x^2 + y^2 - z^2)I$. The exponential e^{tb} belongs to $\mathcal{E}$ for real t and has determinant one. Therefore it is tensor-LY. Finally, for $B = cI + \sum_j b_j$, $e^{tB} = e^{tc} \otimes e^{tb_j}$, which proves the assertion. $\square$

The space $\mathcal{E}$ is closed under multiplication. Its members sufficiently close to I , and their inverses, are tensor-LY by (213). We call these matrices local stability units. A traceless element $b = xX + yY + izZ \in \mathfrak{g}$ is *hyperbolic* if $x^2 + y^2 - z^2 > 0$. A diagonal phase conjugation followed by conjugation with e^{rY} , where $\tanh(2r) = z/\sqrt{x^2 + y^2}$, carries it to $\sqrt{x^2 + y^2 - z^2}X$. The conjugating matrices and their inverses are local stability units.

Lemma C.3 (Hyperbolic affine pencil). *Let $B = cI + \sum_j b_j \in \mathfrak{e}_n$, and suppose one b_k is hyperbolic. If $I + tB$ is tensor-LY for some nonzero real t , then $b_j = 0$ for all $j \neq k$ and $c \in \mathbb{R}$.*

Proof. Conjugating by a local stability unit, assume $b_k = aX_k$, $a > 0$. For a one-site matrix b define $q_b(z, w) = G_b(z, w)/(1 + zw)$. The function

$$q_X(z, z) = \frac{2z}{1 + z^2}$$

attains every nonreal complex number for $|z| < 1$. Indeed, the inverse equation has two reciprocal roots; neither is on the unit circle for a nonreal target, because the displayed quotient is real there wherever finite. Exactly one inverse root is therefore in the disk.

If $b_j \neq 0$ for another site, q_{b_j} is a nonconstant holomorphic function, since b_j is nonzero and traceless. Freeze all other variable pairs at zero, and divide the polynomial of $I + tB$ by $\prod_i (1 + z_i w_i)$. The result has the form $C + taq_X(z_k, w_k) + tq_{b_j}(z_j, w_j)$. The open image of q_{b_j} allows its variables to be chosen so that the target needed from q_X is nonreal. The preceding inverse construction then gives

an interior zero, a contradiction. Once the other components vanish, an imaginary part of c would make $-(1+tc)/(ta)$ nonreal and give the same contradiction. $\square$

C.3. Routing and factorization of visible outputs. Let $\Phi : M_{2^m}(\mathbb{C}) \rightarrow M_{2^n}(\mathbb{C})$ be HP, TP and full Choi–LY, and put $\phi = \Phi^\dagger$. An output site i is called *transversely visible* if at least one of $\phi(X_i), \phi(Y_i)$ is nonscalar. Denote the set of such sites by A .

Lemma C.4 (Local routing). *For each $i \in A$ there is a unique input site $\pi(i)$ such that, writing $k = \pi(i)$,*

$$\begin{aligned}\phi(X_i) &= b_{ix}I + D_{i,11}X_k + D_{i,21}Y_k, \\ \phi(Y_i) &= b_{iy}I + D_{i,12}X_k + D_{i,22}Y_k, \\ \phi(Z_i) &= \lambda_i Z_k,\end{aligned}\tag{214}$$

where all coefficients are real and $D_i \neq 0$. No injectivity of Φ or D_i is required.

Proof. For a local $U \in \mathfrak{g}_i$, the identity $U^2 = \delta I$, $\delta \in \mathbb{R}$, gives $e^{tU} = c_\delta(t)I + s_\delta(t)U$, with real analytic scalar coefficients satisfying $c_\delta(0) = 1$, $s_\delta(0) = 0$, and $s'_\delta(0) = 1$. These exponentials are stability units. Since ϕ is unital, their images, divided by $c_\delta(t)$, give a two-sided stable pencil $I + r\phi(U)$ near zero. A zero image is excluded near I . Lemma C.2 puts $\phi(U)$ in $\mathfrak{e}_m$. HP then implies

$$\phi(X_i), \phi(Y_i) \in \mathbb{R}I + \sum_k \text{span}_{\mathbb{R}}\{X_k, Y_k\}, \quad \phi(Z_i) \in \mathbb{R}I + \sum_k \mathbb{R}Z_k.$$

Every nonscalar transverse image has a hyperbolic component, so Lemma C.3 confines it to one input site. If two transverse directions used different sites, their sum would have two nonzero hyperbolic components. Thus the whole transverse plane uses the same site whenever its nonscalar image is nonzero.

Choose a transverse U_i with nonzero traceless image on this site, and apply the argument to $U_i + i\varepsilon Z_i$ for a small nonzero real ε . The routed component remains hyperbolic. Lemma C.3 eliminates all other longitudinal components, and its real-scalar conclusion eliminates the scalar coefficient of $\phi(Z_i)$. This proves (214). $\square$

Theorem C.5 (Visible-output decomposition). *Let $\Phi : M_{2^m}(\mathbb{C}) \rightarrow M_{2^n}(\mathbb{C})$ be HP, TP and full Choi–LY. With the preceding notation, put $K = \pi(A)$ and $G_k = \pi^{-1}(k)$. There are HP, TP, full Choi–LY maps $\Xi_k : M_2(\mathbb{C}) \rightarrow M_{2|G_k|}(\mathbb{C})$ such that, after ordering the physical output groups,*

$$\text{Tr}_{A^c} \circ \Phi = \left(\bigotimes_{k \in K} \Xi_k \right) \circ \text{Tr}_{K^c}.\tag{215}$$

The routing function is unique. If Φ is CP, every Ξ_k is CP. The trace on the right is the ordinary, unnormalized partial trace.

Proof. For each $i \in A$, choose a small real open set $\mathcal{O}_i \subset \mathfrak{g}_i$ about a transverse direction whose image is nonscalar. Shrink it so that every $U_i \in \mathcal{O}_i$, and the traceless part of $\phi(U_i)$, is hyperbolic. By (214), $\phi(U_i) \in \mathcal{E}_{\pi(i)}$. These open sets have complex span equal to the entire local traceless matrix space, even when D_i has rank one.

Fix choices $U_i \in \mathcal{O}_i$. For $S \subseteq A$ set

$$F_S(t_S) = \phi \left(\prod_{i \in S} (I + t_i U_i) \right).$$

We prove by induction that, for all sufficiently small real parameters,

$$F_S(t_S) = \prod_{k \in \pi(S)} F_{k,S}(t_{S \cap G_k}), \quad F_{k,S} \in \mathcal{E}_k, \quad F_{k,S}(0) = I, \quad (216)$$

where each factor is multiaffine in its own parameters. Distinct factors act on distinct sites. They and their inverses are stability units near zero. The empty and singleton cases follow from unitality and Lemma C.4.

Suppose (216) holds, and add $i \notin S$, with route k . Put

$$\Delta(t_S) = \phi \left(U_i \prod_{j \in S} (I + t_j U_j) \right), \quad B(t_S) = F_S(t_S)^{-1} \Delta(t_S).$$

The new input product is tensor-LY. Its image is $F_S + t_i \Delta$, which is nonzero near the identity. Left multiplication by the stable inverse gives the two-sided stable pencil $I + t_i B(t_S)$. Lemma C.2 applies to this pencil. At $t_S = 0$ its coefficient is $\phi(U_i)$, whose component on site k is hyperbolic. Continuity and Lemma C.3 therefore imply

$$B(t_S) \in \mathcal{E}_k \quad (217)$$

on a sufficiently small parameter neighborhood.

Fix a parameter t_j belonging to a different group G_ℓ , and fix all the other parameters. Write $F_{\ell,S} = C_0 + t_j C_1$. The matrices C_0, C_1 are linearly independent for sufficiently small fixed values of the other parameters: at zero they are $I, \phi(U_j)$, with a nonzero hyperbolic traceless part in the second matrix. Remove the fixed other-site factors from $\Delta = F_S B$. An affine function of t_j is thereby written as

$$(C_0 + t_j C_1) \otimes R(t_j), \quad R(t_j) = F_{k,S} B(t_S).$$

Choose linear functionals α, β with

$$\alpha(C_0) = 1, \quad \alpha(C_1) = 0, \quad \beta(C_0) = 0, \quad \beta(C_1) = 1.$$

Applying them shows that both $R(t_j)$ and $t_j R(t_j)$ are affine. Hence R , and therefore B , is constant in t_j . Repeating this for every parameter outside G_k gives the update

$$F_{S \cup \{i\}} = \prod_{\ell \neq k} F_{\ell,S} (F_{k,S} + t_i F_{k,S} B(t_{S \cap G_k})).$$

The last factor lies in the real algebra $\mathcal{E}_k$, which is closed under multiplication. It is multiaffine, as is seen by setting every other group's parameters to zero in the original definition of $F_{S \cup \{i\}}$. This completes the induction; only finitely many neighborhood shrinkings are needed.

Setting the other groups' parameters to zero identifies each $F_{k,S}$ as the image of its own group product. Extracting coefficients in (216) consequently gives

$$\phi \left(\prod_{i \in S} U_i \right) = \prod_{k \in \pi(S)} \phi \left(\prod_{i \in S \cap G_k} U_i \right).$$

This is a complex multilinear identity. Its validity on a product of real open sets extends first to their real spans and then to their complex spans. Adjoining identity factors spans the entire matrix algebra on A . Thus the restriction of ϕ to that algebra is the tensor product of maps on the groups, each with image on its routed input site.

For an observable O on G_k , extract the one-site image of $\phi(O \otimes I)$ by normalized partial trace over the other input sites. This gives a unital HP map ψ_k , CP if ϕ is CP. Its adjoint Ξ_k is TP and has the same positivity properties. Equivalently, Ξ_k is obtained by preparing every other input maximally mixed, applying Φ , and tracing the other outputs. These operations have tensor-LY Choi arrays. Their contraction is nonzero because Ξ_k is TP, so Ξ_k is full Choi-LY. Taking adjoints proves (215). The adjoint of the partial trace in that formula tensors with identity, which also verifies its normalization. $\square$

Theorem C.6 (Invertible static classification). *An invertible complex-linear HP, TP map $\Phi : M_{2^n}(\mathbb{C}) \rightarrow M_{2^n}(\mathbb{C})$ is full Choi-LY if and only if*

$$\Phi = (\Phi_1 \otimes \cdots \otimes \Phi_n) \circ \mathcal{P}, \quad (218)$$

where $\mathcal{P}$ permutes the physical input wires and the Φ_i are invertible one-qubit HP, TP, full Choi-LY maps. The map Φ is CP if and only if all its local factors are CP.

Proof. Apply the preceding results to $\phi = \Phi^\dagger$. The induced map on the real space of transverse fields modulo scalars is injective. Indeed, if $\phi(U) = bI$, unitality gives $\phi(U - bI) = 0$; injectivity forces $U = bI$, and a traceless transverse field is then zero. In particular every output is visible, each of its two-dimensional transverse planes maps injectively to its routed two-dimensional input plane, and two such planes cannot route to the same input plane. With n planes on each side, the routes form a permutation and all groups have one site.

Theorem C.5 now gives a tensor product of local factors. Each factor is invertible: a nonzero kernel vector of one factor, tensored with suitable nonzero vectors for the other factors, would give a kernel vector of the full map. The same theorem proves inheritance of CP. Conversely, tensor products and physical wire permutations preserve full Choi-LY and all the stated properties. $\square$

Remark C.7. The proof of Theorem C.6 does not require a semigroup representation, complete positivity, or injectivity of an unproved local restriction. For completeness, its adjoint local longitudinal coefficient in (214) is positive: stability of $I + s\lambda_i Z$, obtained from $I + sZ$ for small $s > 0$, gives $|1 + s\lambda_i| \geq |1 - s\lambda_i|$, hence $\lambda_i \geq 0$; injectivity excludes zero.

Example C.8 (Ordinary static preservation is weaker). Let $n \geq 2$, $d = 2^n$, $\tau = \text{diag}(3/4, 1/4)$, and $\sigma = \tau^{\otimes n}$. For

$$0 < a < \frac{1}{4^n + 1}, \quad \Phi_a(M) = aM + (1 - a) \text{Tr}(M)\sigma,$$

the map Φ_a is invertible and CPTP, and sends every density matrix to a matrix with a polynomial nonzero on the closed unit polydisk. Nevertheless it is neither a product of local channels up to a physical permutation nor full Choi-LY.

Indeed, it fixes σ and multiplies the trace-zero subspace by a , proving invertibility. On the closed polydisk,

$$|G_\sigma| \geq 2^{-n} = d^{-1}, \quad |G_\rho| \leq d$$

for every density matrix ρ . The second inequality follows by bounding the two monomial-vector norms by $\sqrt{d}$ and $\|\rho\|_{\text{op}}$ by one. Hence

$$|G_{\Phi_a(\rho)}| \geq (1 - a)d^{-1} - ad > 0.$$

On input $|0^n\rangle\langle 0^n|$, the occupation indicators at any two distinct output sites have covariance $a(1-a)/16 > 0$. The output is not product, whereas a product of local channels, with or without a wire permutation, sends product inputs to products.

One may also exhibit the Choi zero directly. Its polynomial is

$$a \prod_i (1 + z_i x_i)(1 + w_i y_i) + (1-a)G_\sigma(z, w) \prod_i (1 + x_i y_i).$$

Set all output variables except z_1, w_1 to zero, and set $z_1 = w_1 = t$, $x_1 = y_1 = v$. Define

$$p(t) = (3/4)^{n-1}(3/4 + t^2/4), \quad C(t) = \frac{1-a}{a}p(t), \quad C_0 = C(1).$$

Choose $v_0 = e^{i\theta}$ with $\cos \theta = -1/(1+C_0)$, $\pi/2 < \theta < \pi$. Then $(1+v_0)^2/(1+v_0^2) = -C_0$. For small $\varepsilon > 0$, put

$$t = 1 - \varepsilon, \quad v = (1 - \varepsilon)v_0, \quad q = -\frac{(1+tv)^2}{C(t)(1+v^2)} - 1.$$

As $\varepsilon \downarrow 0$, $q \rightarrow 0$, so all of $|t|, |v|, |q|$ are smaller than one for sufficiently small positive ε . Set $x_2 = y_2 = \sqrt{q}$ and all remaining reference variables to zero. The Choi polynomial becomes $a(1+tv)^2 + (1-a)p(t)(1+v^2)(1+q) = 0$, at an interior point. Lemma C.1 turns this into an ancillary physical witness. The same example is the late-time channel $e^{t(\mathcal{R}_\sigma - \text{Id})} = e^{-t} \text{Id} + (1 - e^{-t})\mathcal{R}_\sigma$, where $\mathcal{R}_\sigma(M) = \text{Tr}(M)\sigma$. Its ordinary strict-preservation property for $t > \log(4^n + 1)$ therefore does not imply preservation at times approaching zero.

C.4. Saturated columns and reflected polynomial factors. For a polynomial p , let $\|p\|_2$ be the Euclidean norm of its coefficient vector. If p is multiaffine in a specified finite set S , define its full reflection by

$$p^*(z) = z^S \overline{p(1/\bar{z})}. \quad (219)$$

The set S is part of the definition, and may include variables absent from p . In contrast, $\bar{p}(w) = \overline{p(\bar{w})}$ below denotes coefficientwise conjugation without reflection.

Lemma C.9 (Saturation and disjoint factors). *Let $F(z, x) = \sum_{b \in \{0,1\}^m} A_b(z)x^b$ be a stable multiaffine polynomial. Suppose $\|A_{e_i}\|_2 = \|A_0\|_2$ for every input site i . Then, over the rational function field in z ,*

$$F(z, x) = A_0(z) \prod_{i=1}^m \left(1 + x_i \frac{A_{e_i}(z)}{A_0(z)} \right). \quad (220)$$

Writing the quotients in coprime form q_i/p_i , there is a polynomial factorization

$$F(z, x) = C(z) \prod_{i=1}^m (p_i(z) + x_i q_i(z)) \quad (221)$$

in which the output-variable supports of the displayed factors are pairwise disjoint. All these factors are stable, and $\|p_i\|_2 = \|q_i\|_2$.

Proof. Setting all input variables except x_i to zero gives a stable affine polynomial $A_0 + x_i A_{e_i}$, whence $|A_{e_i}(z)| \leq |A_0(z)|$ in the output polydisk. The inequality extends continuously to the output torus. Parseval and norm equality imply equality of these moduli everywhere on that torus. At a torus point where $A_0 \neq 0$, boundary Hurwitz makes the input polynomial stable; its nonzero constant coefficient

excludes the zero alternative. Write this input polynomial as $P_i(x_{\neq i}) + x_i Q_i(x_{\neq i})$. The holomorphic ratio Q_i/P_i is contractive on the remaining input polydisk, and has modulus one at its origin. It is therefore constant by the maximum modulus principle. Applying this to every input gives (220) at these torus points. The zero set of the nonzero polynomial A_0 has torus measure zero. Clearing denominators and using uniqueness of Fourier coefficients proves the rational identity everywhere.

Each $p_i + x_i q_i$ is primitive over $\mathbb{C}[z]$ and irreducible over the rational function field, where it is linear in its own input variable. Gauss's lemma applied to the primitive part of F , considered as a polynomial in x , gives (221) with C polynomial. Degrees in any one output variable add under multiplication in this integral domain. Since every such degree of F is at most one, each output variable belongs to at most one displayed factor. Stability of F implies stability of every factor and, by setting $x_i = 0$, of p_i . Finally coefficient norms multiply on disjoint supports. Comparing A_0 with A_{e_i} gives the claimed local norm equality. $\square$

Lemma C.10 (Reduced reflected pairs). *Let $p + xq$ be stable and multiaffine, where p, q are coprime, have equal coefficient norms, and their union of output supports is S . Then*

$$q = e^{i\theta} p^* \quad (222)$$

for some real θ . Conversely, for any stable multiaffine p on a specified set S , the polynomial $p + e^{i\theta} x p^*$ is stable.

Proof. Stable affine slices and equality of norms imply $|p| = |q|$ on the output torus. Multiplication by z^S gives $pp^* = qq^*$. Coprimality implies $p \mid q^*$; write $q^* = ph$, so $p^* = qh$. The constant coefficient of the stable polynomial p is nonzero. Thus p^* has degree one in every variable of S . If h depends on z_j , degree additivity makes q independent of z_j , and the support-union property makes p depend on z_j . Independence of q implies $z_j \mid q^* = ph$. Since p has nonzero constant term, $z_j \mid h$, and then $z_j \mid p^* = qh$. The latter says p is independent of z_j , a contradiction. Hence h is constant. Reflecting twice shows $|h| = 1$, proving (222).

For the converse, p^*/p is holomorphic on the polydisk and has modulus at most one. To justify this assertion even in the presence of boundary zeros, use $p_r(z) = p(rz)$, $r < 1$. The polynomial p_r has no closed-polydisk zero, and p_r^*/p_r has unit modulus on the torus. Successive maximum modulus principles make it contractive inside. Letting $r \uparrow 1$ gives the asserted bound. Hence $1 + e^{i\theta} x p^*/p \neq 0$ for $x \in \mathbb{D}$. $\square$

Reflection descriptions of rational inner functions are classical; see [36]. The elementary proof above specifies the full support set and the coprime multiaffine reduction used here.

C.5. All stable isometries and their coding obstruction.

Theorem C.11 (Stable isometries). *Let $V : (\mathbb{C}^2)^{\otimes m} \rightarrow (\mathbb{C}^2)^{\otimes n}$ be an isometry. Its coefficient polynomial is stable if and only if, after a physical output permutation,*

$$V = |\psi\rangle \otimes V_1 \otimes \cdots \otimes V_m, \quad (223)$$

where ψ is a normalized stable pure state on an optional prepared block S_0 . The remaining nonempty, disjoint blocks S_i carry one-input isometries whose polynomials are

$$G_{V_i}(z, x_i) = \frac{p_i(z) + e^{i\theta_i} x_i p_i^*(z)}{\|p_i\|_2}, \quad \gcd(p_i, p_i^*) = 1, \quad [z^{S_i}] p_i(z)^2 = 0. \quad (224)$$

Each p_i is stable and multiaffine, and reflection uses the whole block S_i . The sets $S_0, S_1, \dots, S_m$ partition the physical outputs. In particular, an isometric channel $M \mapsto VMV^*$ is full Choi-LY exactly for these isometries.

Proof. Write $G_V = \sum_b A_b(z)x^b$. Isometry gives $\langle A_b, A_c \rangle = \delta_{bc}$, so Lemma C.9 applies. In its factorization, disjointness makes both norms and inner products multiplicative. Comparing columns 0 and e_i yields $\|p_i\|_2 = \|q_i\|_2 > 0$ and $\langle p_i, q_i \rangle = 0$. A factor with no output variable could not satisfy the latter equality, so every logical block is nonempty. Normalize the common prepared factor and each pair. There is no residual normalization, because the all-zero column has norm one. Variables absent from the polynomial are fixed-zero prepared outputs.

Lemma C.10 gives the reflected form. Furthermore,

$$\langle p_i, p_i^* \rangle = \overline{\sum_{T \subseteq S_i} (p_i)_T (p_i)_{S_i \setminus T}} = \overline{[z^{S_i}] p_i(z)^2},$$

so orthogonality is exactly the final condition in (224). Conversely that condition gives orthogonal columns, reflection preserves their norms, and Lemma C.10 gives stability. Tensor products prove the converse for V . The isometric channel's Choi polynomial is $G_V(z, x)\overline{G_V(w, y)}$ in independent variables, proving the final equivalence. $\square$

Lemma C.12 (Strict longitudinal bias). *If p is stable and multiaffine on a nonempty specified set S , and $\gcd(p, p^*) = 1$, then for every $j \in S$,*

$$\|p|_{z_j=0}\|_2^2 > \|\partial_j p\|_2^2. \quad (225)$$

Proof. Write $p = a + z_j b$. Stable affine slices imply $|b| \leq |a|$ on the remaining polydisk, hence $\|b\|_2 \leq \|a\|_2$. If $b = 0$, strictness follows from $a \neq 0$. Otherwise suppose equality holds. Torus saturation gives $aa^* = bb^*$, reflecting both polynomials over $R = S \setminus \{j\}$, while the full reflection is $p^* = b^* + z_j a^*$. Let $g = \gcd(a, b)$ and $h = a/g + z_j b/g$. This polynomial is primitive over $\mathbb{C}[z_R]$ and has positive degree in z_j . Over $\mathbb{C}(z_R)[z_j]$,

$$p^* = \frac{a^*}{b} (a + z_j b).$$

Thus h divides both p and p^* over the fraction field. Gauss's lemma gives the same divisibility in the polynomial ring, contradicting coprimality. This proof also covers empty R and variables absent from either slice. $\square$

Corollary C.13 (Single-output erasure). *For a reduced logical block in Theorem C.11, every physical output $j \in S_i$ satisfies*

$$V_i^* Z_j V_i = \delta_j Z_i, \quad \delta_j = \frac{\|p_i|_{z_j=0}\|_2^2 - \|\partial_j p_i\|_2^2}{\|p_i\|_2^2} > 0. \quad (226)$$

Erasure of any such output is not exactly correctable on the logical space. Every nontrivial stable isometric quantum code therefore has distance one.

Proof. Reflection complements occupied subsets, so the two diagonal expectations of Z_j have opposite signs. The cross term is zero because complementary subsets cancel:

$$\langle p_i, Z_j p_i^* \rangle = \overline{\sum_{T \subseteq S_i} (-1)^{\mathbf{1}_{j \in T}} (p_i)_T (p_i)_{S_i \setminus T}} = 0.$$

Lemma C.12 now proves (226). For erasure of output j , the two Kraus operators may be taken as $E_a = \langle a|_j \otimes I$, $a = 0, 1$. The Knill–Laflamme criterion [37] requires $V^* E_a^* E_b V$ to be scalar for all a, b . Subtracting the diagonal cases requires $V^* Z_j V$ to be scalar, contrary to (226). Equivalently, this weight-one phase observable is not detectable by the code. $\square$

Remark C.14. Outputs in the prepared block are exempt from the erasure conclusion. Nor is there a uniform positive lower bound on the individual bias: for $p(z_1, z_2) = 1 + rz_1$, $0 \leq r < 1$, reflected over $\{1, 2\}$, one has $p^* = z_2(r + z_1)$, coprime orthogonal columns, and $\delta_1 = (1 - r^2)/(1 + r^2) \rightarrow 0$. The conclusion concerns exact correction only.

Example C.15 (Coherent information invisible in one-site marginals). The repetition isometry $V|0\rangle = |00\rangle$, $V|1\rangle = |11\rangle$ has polynomial $1 + xz_1z_2$. The square channel $\Phi(M) = V \text{Tr}_2(M) V^*$ is full Choi–LY, with polynomial

$$(1 + x_1z_1z_2)(1 + y_1w_1w_2)(1 + x_2y_2).$$

All its single-output transverse adjoints vanish, although $\Phi^\dagger(X_1X_2) = X_1 \otimes I_2$. Thus Theorem C.5 cannot identify its omitted outputs as a classical or input-independent sector.

Proposition C.16 (No stable Stinespring representation in general). *There is a two-qubit CPTP full Choi–LY map with no Stinespring isometry whose coefficient tensor is LY, for any finite qubit environment.*

Proof. Let Δ be complete computational-basis dephasing on two qubits, let P swap the qubits, and set $\Phi = \frac{1}{2}\Delta + \frac{1}{2}\text{Ad}_P \circ \Delta$. This is CPTP. With $r_j = z_j w_j$ and $s_j = x_j y_j$, its Choi polynomial is

$$\begin{aligned} K(r, s) &= \frac{1}{2}(1 + r_1s_1)(1 + r_2s_2) + \frac{1}{2}(1 + r_1s_2)(1 + r_2s_1) \\ &= 1 + \frac{1}{2}(r_1 + r_2)(s_1 + s_2) + r_1r_2s_1s_2. \end{aligned} \quad (227)$$

For fixed $s_1, s_2 \in \mathbb{D}$, this is the symmetric multiaffine polarization in r_1, r_2 of $(1 + rs_1)(1 + rs_2)$. The Grace–Walsh–Szegő coincidence theorem on the convex circular domain $\mathbb{D}$ implies its nonvanishing for $r_1, r_2 \in \mathbb{D}$; see [11]. Thus the full Choi tensor is stable.

However, $\Phi^\dagger(Z_1) = \Phi^\dagger(Z_2) = (Z_1 + Z_2)/2$. If a stable Stinespring isometry existed, apply Theorem C.11 to its physical and environmental output qubits together. Every one-site observable on a physical output would pull back either to a scalar or to an operator on one logical input, because its output belongs to one disjoint encoding block or to the prepared block. Tracing the environment merely inserts identities at its sites. The displayed two-input longitudinal observable is incompatible with this conclusion. $\square$

C.6. All Schur channels.

Lemma C.17 (A scalar equal-diagonal criterion). *For a nonzero PSD matrix $H = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$, its tensor polynomial is stable if and only if $a > 0$ and $a \geq d$. For arbitrary complex β, γ , the polynomial $1 + \beta x + \gamma y + xy$ is stable if and only if $\gamma = \bar{\beta}$ and $|\beta| \leq 1$.*

Proof. A stable polynomial must have nonzero constant coefficient, so $a > 0$. The necessary boundary root-modulus inequality is

$$|a + \bar{b}\zeta|^2 - |b + d\zeta|^2 = (a - d)(a + d + 2\Re(\bar{b}\zeta)) \geq 0, \quad |\zeta| = 1.$$

Its average gives $a \geq d$. Conversely, PSD gives $|b|^2 \leq ad$, and $a \geq d$ gives $|b| \leq a$. If $|b| < a$, the affine constant factor $a + \bar{b}z$ is nonzero on the closed disk, and the displayed inequality and the maximum modulus principle bound the other coefficient divided by this one by one. This proves stability. If $|b| = a$, PSD and $a \geq d$ force $d = a$, and the polynomial factors as $a(1 + (\bar{b}/a)z)(1 + (b/a)w)$, also stable.

For the second assertion, boundary root-modulus inequalities give $|1 + \beta\zeta| \geq |\gamma + \zeta|$, and the analogous inequality with β, γ exchanged. Squaring and averaging gives $|\beta| = |\gamma|$. The remaining nonnegative linear trigonometric polynomial has mean zero, so vanishes identically, yielding $\beta = \bar{\gamma}$. Setting one variable to zero gives $|\beta| \leq 1$. The first assertion applied to the resulting equal-diagonal PSD matrix proves the converse. $\square$

Theorem C.18 (Stable correlation matrices and Schur channels). *Let $C \succeq 0$ be a 2^m -by- 2^m matrix with $C_{aa} = 1$. Its tensor polynomial is stable if and only if*

$$C = \bigotimes_{i=1}^m \begin{pmatrix} 1 & \eta_i \\ \bar{\eta}_i & 1 \end{pmatrix}, \quad |\eta_i| \leq 1. \quad (228)$$

Equivalently, a CPTP map fixing every computational-basis state is full Choi-LY if and only if it is a tensor product of one-qubit phase/dephasing channels. Parameters $\eta_i = 0$ are allowed.

Proof. Separate one physical site and write $C = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix}$. For arbitrary $z \in \mathbb{D}^{m-1}$, compress the other sites against the product vector $v(z) = \sum_a z^a |a\rangle$. The two-by-two compression is PSD. Its tensor polynomial is an interior specialization of G_C , hence stable, and its constant coefficient is strictly positive. By Lemma C.17,

$$\langle v(z), (A - D)v(z) \rangle \geq 0.$$

At any fixed positive interior coordinate moduli, the angular average of this expression is $\sum_a (A_{aa} - D_{aa})|z|^{2a} = 0$. Continuity and nonnegativity force pointwise equality on every such torus. Varying the moduli proves the polynomial identity $\langle v(z), (A - D)v(z) \rangle = 0$. Independence of the monomials $\bar{z}^a z^b$ gives $A = D$ entry-wise.

Writing u for the remaining independent row and column variables, the full polynomial has the form

$$a(u)(1 + xy) + b(u)x + c(u)y.$$

The specialization $a(u)$ is nonzero throughout the polydisk. Lemma C.17 gives $c(u)/a(u) = \bar{b(u)}/a(u)$ and $|b(u)/a(u)| \leq 1$ for every complex u there. Both ratios are holomorphic on this connected domain, so they are constant. This factors off one matrix in (228). The remaining factor A is PSD, has diagonal one, and is tensor-LY. Induction proves the assertion. The converse follows from the one-site criterion and multiplication on disjoint variables.

For a Schur multiplier $\mathcal{S}_C(M) = C \circ M$, CP and TP are equivalent to $C \succeq 0$ and diagonal one. Its full Choi polynomial is $G_C(r, s)$, where $r_i = z_i u_i$ and $s_i = w_i v_i$. Each disk value is attainable by these products, so full Choi stability is exactly

tensor stability of C . Finally, any CPTP map fixing all computational-basis states has diagonal Kraus operators: each Kraus image of a basis vector must lie in the one-dimensional range of the corresponding pure output. It is therefore a Schur map, proving the equivalent formulation. $\square$

C.7. Channels with pure computational-basis outputs.

Theorem C.19 (Pure-basis-output classification). *Let $\Phi : M_{2^m}(\mathbb{C}) \rightarrow M_{2^n}(\mathbb{C})$ be CPTP, and suppose every computational-basis output $\Phi(|a\rangle\langle a|)$ is pure. The outputs need not be orthogonal or distinct. Then Φ is full Choi-LY if and only if it has the following form.*

Partition the inputs as $E \sqcup K = [m]$, and the outputs into an optional prepared block S_0 and nonempty disjoint blocks S_i , $i \in K$. Prepare a normalized stable pure state ψ on S_0 . On each S_i take a reduced stable two-column matrix V_i with normalized, linearly independent columns v_{i0}, v_{i1} . Reduced means that their amplitude polynomials are coprime. Equivalently,

$$G_{V_i}(z, x_i) = \frac{p_i(z) + e^{i\theta_i} x_i p_i^*(z)}{\|p_i\|_2}, \quad \gcd(p_i, p_i^*) = 1, \quad (229)$$

where p_i is stable and multiaffine and reflection uses the full block. Put $\mu_i = \langle v_{i0}, v_{i1} \rangle$, and choose $|\eta_i| \leq 1$ subject to

$$\mu_i \eta_i = 0. \quad (230)$$

With

$$C_i = \begin{pmatrix} 1 & \eta_i \\ \bar{\eta}_i & 1 \end{pmatrix}, \quad \mathcal{B}_i(M) = V_i(C_i \circ M)V_i^*,$$

the channel, after output permutation, is exactly

$$\Phi(\rho) = |\psi\rangle\langle\psi| \otimes \left(\bigotimes_{i \in K} \mathcal{B}_i \right) (\text{Tr}_E \rho). \quad (231)$$

Empty K is allowed. If $\mu_i = 0$, the block is a stable isometric encoding preceded by arbitrary local phase/dephasing. If $\mu_i \neq 0$, it measures the input in the computational basis and prepares the corresponding nonorthogonal reflected pure state.

Proof. Assume first that Φ is full Choi-LY. Choose normalized output vectors ψ_a , with amplitude polynomials p_a . Complete input dephasing is full Choi-LY; its Choi polynomial is a product of factors $1 + x_i y_i u_i v_i$. After composing with this map and grouping each input row/column pair into its product, we obtain the stable polynomial

$$F(z, w, s) = \sum_a A_a(z, w) s^a, \quad A_a(z, w) = p_a(z) \bar{p}_a(w). \quad (232)$$

Grouping is an equivalence of disk stability because disk-variable products cover the disk. Each A_a has coefficient norm one, being the polynomial of a normalized rank-one projection. Lemma C.9 gives, for every nonempty subset $T \subseteq [m]$,

$$A_T A_0^{|T|-1} = \prod_{i \in T} A_{e_i}. \quad (233)$$

Substitute (232) into this identity. The rational function

$$R_T(z) = \frac{p_T(z) p_0(z)^{|T|-1}}{\prod_{i \in T} p_{e_i}(z)}$$

satisfies $R_T(z)\overline{R_T(\bar{w})} = 1$ in independent variables. Fixing a generic w shows that it is a constant ξ_T of modulus one. We may therefore align the output column phases so that the polynomial

$$G(z, x) = p_0(z) \prod_i \left(1 + x_i \frac{p_{e_i}(z)}{p_0(z)} \right) \quad (234)$$

has exactly those phase-aligned columns. More explicitly, its coefficient at x^T is $\xi_T^{-1} p_T$; hence it is a multiaffine polynomial, not merely a rational expression. The stable constant slice $A_0 = p_0 \bar{p}_0$ makes p_0 zero-free in the output polydisk. The singleton modulus domination for (232), specialized to $w = \bar{z}$, gives $|p_{e_i}/p_0| \leq 1$. Thus (234) is stable. It defines a matrix V with normalized columns; no orthogonality is being asserted.

Apply Lemma C.9 to G , whose column norms are all one, and then Lemma C.10 to its primitive factors. We obtain a stable prepared factor and disjoint reduced reflected pairs with equal local norms. A dependent primitive pair $q_i = c_i p_i$ must have both polynomials constant, so has empty output support; conversely every nonempty primitive pair has independent columns. Assign the empty factors to E and the others to K . Their equal norms imply $|c_i| = 1$. Replacing each erased input variable by $c_i^{-1} x_i$ is a disk-preserving phase change of columns, and makes the two erased-input columns identical. Normalize the prepared factor and all retained pairs. The total normalization is one because the all-zero full column has norm one. All aligned output vectors now have the form

$$v_a = \psi \otimes \bigotimes_{i \in K} v_{i, a_i}, \quad (235)$$

independent of a_E . Common prepared factors and variables absent from G belong to S_0 . This proves all the polynomial and support assertions about the pairs in the theorem.

Choose Kraus operators K_ℓ of Φ . Since the positive sum of their output projections on input a has rank one, every $K_\ell|a\rangle$ is proportional to v_a . Writing $K_\ell|a\rangle = c_{\ell a} v_a$ gives

$$\Phi(|a\rangle\langle b|) = C_{ab} |v_a\rangle\langle v_b|, \quad C_{ab} = \sum_\ell c_{\ell a} \bar{c}_{\ell b}, \quad (236)$$

where $C \succeq 0$ and $C_{aa} = 1$. Original trace preservation is exactly

$$C_{ab} \langle v_b, v_a \rangle = \delta_{ab}. \quad (237)$$

In particular, if $a_K = b_K$, the aligned vectors in (235) are identical, so

$$C_{ab} = \delta_{ab} \quad \text{whenever } a_K = b_K. \quad (238)$$

For each retained block, its independent normalized columns have positive definite Gram matrix

$$H_i = V_i^* V_i = \begin{pmatrix} 1 & \mu_i \\ \bar{\mu}_i & 1 \end{pmatrix}, \quad |\mu_i| < 1.$$

Both V_i^* and

$$H_i^{-1} = \frac{1}{1 - |\mu_i|^2} \begin{pmatrix} 1 & -\mu_i \\ -\bar{\mu}_i & 1 \end{pmatrix}$$

are tensor-LY, the latter by Lemma C.17. Consequently $L_i = H_i^{-1}V_i^*$ is tensor-LY and satisfies $L_i V_i = I$. Contraction cannot give zero because of this identity. The rectangular matrix $L = \langle \psi | \otimes \bigotimes_{i \in K} L_i$ is stable and obeys $L v_a = |a_K\rangle$.

The CP map $\Psi(M) = L\Phi(M)L^*$ is full Choi-LY or zero, and satisfies

$$\Psi(|a\rangle\langle b|) = C_{ab}|a_K\rangle\langle b_K|.$$

Its trace is $C_{ab}\delta_{a_K, b_K} = \delta_{ab}$ by (238). Thus Ψ is TP and nonzero. The compression itself need not be TP on arbitrary output matrices; this argument proves TP only for its composition with Φ , which is all that is required.

The Choi polynomial of Ψ is

$$\sum_{a,b} C_{ab} z^{a_K} w^{b_K} u^a v^b.$$

For $i \in K$, replace $z_i u_i, w_i v_i$ by r_i, s_i ; for $i \in E$, put $r_i = u_i, s_i = v_i$. The variables r_i, s_i range independently over the disk, so stability of this Choi polynomial is precisely tensor stability of the entire correlation matrix C . Theorem C.18 gives $C = \bigotimes_i \begin{pmatrix} 1 & \eta_i \\ \bar{\eta}_i & 1 \end{pmatrix}$, with $|\eta_i| \leq 1$. Taking strings differing only at an erased input in (238) forces $\eta_i = 0$ there. Taking strings differing at a retained input in (237) gives $\eta_i \bar{\mu}_i = 0$, equivalently (230). Combining these tensor-product coefficients with (235) proves (231); the erased coefficient is exactly δ_{a_E, b_E} , hence ordinary partial trace.

Conversely, start from the displayed data. Each C_i is a PSD correlation matrix, so its Schur map is CP. Following it by $M \mapsto V_i M V_i^*$ preserves CP. Normalization of the columns gives the correct diagonal traces, while (230) makes the off-diagonal traces zero. Thus each $\mathcal{B}_i$ is TP and has pure basis outputs. Its two component Choi tensors are stable by Lemma C.10 and Theorem C.18; their composition is nonzero because it is TP. Stable pure preparation, partial trace, tensor products and physical permutations prove the converse. The descriptions of the two cases follow directly from $\mu_i \eta_i = 0$. $\square$

Theorems C.5, C.18, and C.19 concern different singular strata. In particular, none asserts a classification of general channels with mixed computational-basis outputs. Proposition C.16 prevents extending the isometry theorem to that class by an assumption of a stable Stinespring representation.

C.8. Pauli-diagonal maps without invertibility.

Lemma C.20 (Factorization on slit domains). *Let $\Omega_1, \dots, \Omega_n$ be nonempty connected open subsets of $\mathbb{C}$ with $\mathbb{C} \setminus \Omega_j \subseteq \mathbb{R}$. A multiaffine polynomial nonvanishing on $\prod_j \Omega_j$ is a nonzero scalar times a product of one-variable affine polynomials, with constant factors permitted.*

Proof. Write $p(r_1, r') = a(r') + r_1 b(r')$. If $b = 0$, omit that coordinate. Otherwise, on a nonempty connected open neighborhood where $b \neq 0$, the holomorphic function $-a/b$ avoids Ω_1 and is therefore real-valued. It is constant, say $c \notin \Omega_1$. The polynomial identity principle gives $a + cb = 0$ everywhere, so $p = (r_1 - c)b$. The remaining factor is nonvanishing on the remaining domain. Induction proves the assertion. $\square$

Theorem C.21 (All-rank Pauli-diagonal rigidity). *Suppose a complex-linear HP, TP map on n qubits is diagonal in the Pauli-string basis: $\Phi(\sigma_\alpha) = \lambda_\alpha \sigma_\alpha$, where*

$\sigma_\alpha = \bigotimes_j \sigma_{\alpha_j}$ and $\sigma_0 = I$. If Φ is full Choi-LY, then

$$\lambda_\alpha = \prod_j \lambda_{\alpha_j^{(j)}}, \quad \Phi = \bigotimes_j \Phi_j, \quad (239)$$

where $\alpha_j^{(j)}$ denotes the string with the stated Pauli at site j and identities elsewhere. Each local factor is HP, TP and full Choi-LY. If Φ is CP, all factors are CP, and the converse holds. Zero eigenvalues are allowed.

Proof. HP makes the λ_α real, and TP gives $\lambda_0 = 1$. Pauli orthogonality yields

$$J(\Phi) = 2^{-n} \sum_\alpha \lambda_\alpha \sigma_\alpha \otimes \sigma_\alpha^T.$$

The four one-site polynomials are $f_I = 1 + zw$, $f_X = z + w$, $f_Y = i(z - w)$, and $f_Z = 1 - zw$; the transpose contributes a minus sign only for Y .

At every site select a desired axis, independently of the others. For axis X , set the output variables to $z = w = 1$. Only the identity and X terms remain, and their normalized ratio is

$$r_X = \frac{x + y}{1 + xy}.$$

Its image on the bidisk is $\Omega_\perp = \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty))$. For a real $|r| \geq 1$, the inverse equation would require $y = (r - x)/(1 - rx)$, whereas

$$|r - x|^2 - |1 - rx|^2 = (r^2 - 1)(1 - |x|^2) \geq 0.$$

The endpoint equations also require a boundary variable. Conversely, setting $x = y = u$ gives an inverse quadratic with reciprocal roots. A unit-circle root would place r on the omitted rays. For every nonzero $r \in \Omega_\perp$, exactly one inverse root is therefore in the disk; zero is attained at $u = 0$.

For axis Y , set $z = i, w = -i$. The output factor is $f_Y = -2$, and the input transpose changes the sign again. The surviving ratio is $r_Y = i(x - y)/(1 + xy)$, with the same image $\Omega_\perp$, by replacing (x, y) with $(ix, -iy)$. For axis Z , set $z = s, w = -s, x = t, y = t$, all interior. The output specialization kills X and the input specialization kills Y . The ratio is

$$r_Z = \frac{1 + s^2}{1 - s^2} \frac{1 - t^2}{1 + t^2}.$$

Each factor ranges over the right half-plane, so their product ranges exactly over $\Omega_Z = \mathbb{C} \setminus (-\infty, 0]$. Indeed products have arguments strictly between $-\pi$ and π , and the principal square root realizes every point of this slit plane as such a product.

For axes $a_1, \dots, a_n$, divide out the nonzero identity factors after these substitutions. The resulting polynomial is

$$p_a(r) = \sum_{S \subseteq [n]} \lambda_{a_S} \prod_{j \in S} r_j, \quad p_a(0) = 1, \quad (240)$$

where a_S has the chosen axes on S . Boundary specialization at the X, Y sites gives stable or zero by Hurwitz. The zero alternative is impossible: the ratio maps have the nonempty open product image just computed, and the nonzero polynomial (240) cannot vanish identically there. The Z substitutions are interior. Thus p_a is nonvanishing on the corresponding product of slit domains.

Lemma C.20 and $p_a(0) = 1$ give $p_a(r) = \prod_j (1 + \lambda_{a_j^{(j)}} r_j)$. The normalization at zero is algebraic and does not require zero to belong to Ω_Z . Comparing coefficients,

and varying the axes, proves (239) for every Pauli string. Local factors inherit the Choi and positivity properties by the marginal construction in the proof of Theorem C.5. Tensor products prove the converse. $\square$

For CPTP local Pauli factors the exact one-site condition is

$$\lambda_z \geq \max\{|\lambda_x|, |\lambda_y|\},$$

or equivalently $\min(p_0, p_z) \geq \max(p_x, p_y)$ in the Pauli-error probabilities. This is the Pauli specialization of the local classification in Theorem D.1; its sufficiency is also [61, Theorem 3]. Because the Pauli transform between error probabilities and eigenvalues is an invertible tensor-product transform, (239) says that the entire error distribution is a product distribution.

C.9. Classical channels and the purity boundary. Consider the channel that measures m input bits in the computational basis and prepares a diagonal distribution $P(b | a)$ on n output bits. Its full Choi polynomial reduces to

$$K(r, s) = \sum_{a,b} P(b | a) r^b s^a = \sum_a A_a(r) s^a, \quad K(1, s) = \prod_i (1 + s_i). \quad (241)$$

The coefficients are nonnegative. As above, reducing each row/column pair to its product is an equivalence of disk stability. Define the collision purity of column a by $\mathcal{C}(a) = \sum_b P(b | a)^2 = \|A_a\|_2^2$.

Theorem C.22 (The constant-purity boundary). *Every stable stochastic kernel (241) satisfies $\mathcal{C}(a) \leq \mathcal{C}(0)$ for every input. Equality at every singleton input e_i holds if and only if, for a partition $S_0, S_1, \dots, S_m$ of the output variables, the kernel is*

$$K(r, s) = C(r_{S_0}) \prod_{i=1}^m (p_i(r_{S_i}) + s_i p_i^*(r_{S_i})). \quad (242)$$

Here C and the p_i are stable, multiaffine, have nonnegative coefficients, satisfy $C(1) = p_i(1) = 1$, and $\gcd(p_i, p_i^*) = 1$; reflection is over the full indicated block. Empty blocks S_i are allowed and contribute $1 + s_i$. Every column of such a kernel has the same collision purity.

Proof. Fix the output variables in the polydisk. For any input subset T , set its variables equal to t and the other input variables to zero. The resulting univariate polynomial is nonvanishing for $|t| < 1$, has constant coefficient A_0 , and top coefficient A_T in degree $|T|$. If that top coefficient is nonzero, the product of root moduli is at least one, whence $|A_T| \leq |A_0|$. If it is zero the inequality is immediate. Continuity and torus Parseval prove the purity inequality.

Under singleton equality, Lemma C.9 gives disjoint primitive factors $C \prod_i (p_i + s_i q_i)$. The stable constant slice makes the constant coefficients of C, p_i nonzero. Fix their phases to make these constants positive. Each coefficient of each factor can be isolated as a coefficient of K by selecting the constant monomial in all the other factors; disjoint supports therefore make all coefficients of C, p_i, q_i nonnegative. Lemma C.10 gives $q_i = e^{i\theta_i} p_i^*$, and nonnegativity forces the phase to be one. Stochasticity at output variables one gives $p_i(1) = q_i(1) > 0$. Normalizing these common values and the prepared factor gives (242), with no scalar remainder because every column has mass one. Variables absent from K are fixed-zero prepared outputs.

Conversely, Lemma C.10 makes each factor stable. The displayed normalization makes the nonnegative coefficient kernel stochastic. Reflection preserves coefficient norms and disjoint supports make them multiply, so all columns have equal collision purity. $\square$

Corollary C.23 (Deterministic classical channels). *A deterministic stochastic kernel is full Choi–LY if and only if every output bit is identically zero or copies one input bit. An input may be copied to any disjoint block of outputs or discarded.*

Proof. Write $P(b \mid a) = \mathbf{1}_{b=f(a)}$. All column purities equal one. The rational product identity in Lemma C.9 therefore gives

$$r^{f(a)} = r^{f(0) + \sum_{i:a_i=1} (f(e_i) - f(0))}.$$

Stable evaluation at the full origin requires $f(0) = 0$. Each output coordinate is consequently a sum of its singleton Boolean values. Since it must still be a bit when all inputs are one, at most one singleton value can be one. This proves necessity. Conversely the kernel of such a map is $\prod_i (1 + s_i \prod_{j \in S_i} r_j)$, with disjoint copied-output blocks, and is manifestly stable. $\square$

Theorem C.24 (One classical output). *A stochastic kernel from m input bits to one output bit is full Choi–LY if and only if*

$$P(1 \mid a) = \alpha + \sum_i \lambda_i a_i, \quad \alpha \geq 0, \quad \lambda_i \geq 0, \quad 2\alpha + \sum_i \lambda_i \leq 1. \quad (243)$$

Thus it chooses to copy input i with probability λ_i , to output zero with probability $1 - 2\alpha - \sum_i \lambda_i$, or to output a fresh fair bit with probability 2α .

Proof. Write $K(r, s) = A(s) + rB(s)$. Stochasticity gives $A + B = \prod_i (1 + s_i)$. Put $H = (A - B)/(A + B)$, so

$$K = \frac{\prod_i (1 + s_i)}{2} ((1 + r) + (1 - r)H(s)).$$

The quotient $(1 + r)/(1 - r)$ ranges over the open right half-plane; thus stability is exactly $\Re H(s) \geq 0$ on the input polydisk. Set $u_i = (1 - s_i)/(1 + s_i)$. Then H becomes a real multiaffine polynomial in the right-half-plane variables u_i , because every $s^a / \prod_i (1 + s_i)$ is a product of factors $(1 \pm u_i)/2$.

A real polynomial with nonnegative real part on a product of right half-planes has the form $c + \sum_i \lambda_i u_i$, with $c, \lambda_i \geq 0$. To see this, restrict to $u_i = v_i t$, $v_i > 0$, $\Re t > 0$. A real one-variable polynomial of degree at least two has a leading term with negative real part along some interior right-half-plane ray at large radius. Every such restriction therefore has degree at most one. Varying v in its open positive orthant forces all higher homogeneous parts to vanish. Positive real evaluations near zero, or with one variable tending to infinity, show that the remaining coefficients are nonnegative.

Evaluating the transformed polynomial identity at $u_i = (-1)^{a_i}$ gives $1 - 2P(1 \mid a) = c + \sum_i \lambda_i (-1)^{a_i}$. This is polynomial evaluation, including at vertices corresponding to poles of the original change of variables. Consequently (243) holds with $\alpha = (1 - c - \sum_i \lambda_i)/2$. The original probability at $a = 0$ makes $\alpha \geq 0$, and $c \geq 0$ gives the last inequality. Conversely these conditions define the stated stochastic mixture and a positive-real polynomial H , proving stability. $\square$

Every single-output marginal of a multibit classical full Choi–LY channel therefore has the form (243). Marginalization sets the other output variables to one; boundary Hurwitz applies, and stochastic normalization excludes zero. This does not determine the joint output correlations.

Proposition C.25 (Nonconvexity among reset channels). *For $n \geq 3$, the distribution $\nu_\theta = (1 - \theta)\delta_{0^n} + \theta \text{Unif}(\{0, 1\}^n)$, $0 \leq \theta \leq 1$, has a stable polynomial precisely when*

$$0 \leq \theta \leq \frac{1}{1 + \cos^n(\pi/n)} \quad \text{or} \quad \theta = 1. \quad (244)$$

For $n = 1, 2$, all $\theta \in [0, 1]$ are allowed. Consequently the set of full Choi–LY channels is not convex, even among classical reset channels.

Proof. The distribution polynomial is $f_\theta(r) = 1 - \theta + 2^{-n}\theta \prod_j (1 + r_j)$. The endpoints $\theta = 0, 1$ are stable. For $0 < \theta < 1$, an interior zero means that a product of n points in the open disk centered at one with radius one equals $-2^n(1 - \theta)/\theta$. Write these nonzero points as $\rho_j e^{i\alpha_j}$, where $|\alpha_j| < \pi/2$ and $0 < \rho_j < 2 \cos \alpha_j$. A negative real product requires $\sum_j \alpha_j$ to be an odd multiple of π , which is impossible for $n = 1, 2$. For $n \geq 3$, concavity of $\log \cos$ implies

$$\prod_j \rho_j < 2^n \prod_j \cos \alpha_j \leq [2 \cos(\pi/n)]^n.$$

An odd multiple of larger absolute value can only reduce the bound. Every smaller positive product magnitude is attained by setting all arguments to π/n and choosing equal moduli. Therefore an interior zero occurs exactly when $2^n(1 - \theta)/\theta < [2 \cos(\pi/n)]^n$, the complement of (244) away from the separately treated endpoint. Equality requires boundary points and is stable for the open-disk definition. Resetting to this distribution multiplies its polynomial by the stable input trace factor $\prod_i (1 + s_i)$, so the same criterion applies to the channel. $\square$

APPENDIX D. THE COMPLETE ONE-QUBIT STATIC CLASSIFICATION

We give exact parameter criteria for a single qubit, including every singular stratum. The stability classification first requires only Hermiticity preservation and trace preservation. Complete positivity is then imposed by explicit matrix or scalar inequalities.

Write a complex-linear Hermiticity-preserving, trace-preserving map in Bloch coordinates as

$$\Phi(I) = I + t \cdot \sigma, \quad \Phi(\sigma_j) = \sum_i T_{ij} \sigma_i, \quad t \in \mathbb{R}^3, \quad T \in \mathbb{R}^{3 \times 3}, \quad (245)$$

where $\sigma = (X, Y, Z)$. On trace-one Hermitian matrices its action is $r \mapsto t + Tr$. Invertibility means complex-linear invertibility on $M_2(\mathbb{C})$, equivalently $\det T \neq 0$. Full Choi stability always refers to all four variables of $F_{J(\Phi)}$.

Theorem D.1 (All one-qubit HP/TP maps). *The map (245) has a full Lee–Yang Choi kernel if and only if it belongs to exactly one of the following two classes.*

(1) *There is a nonzero real 2×2 matrix D such that*

$$T = \begin{pmatrix} D & 0 \\ 0 & \lambda \end{pmatrix}, \quad \lambda > 0, \quad t = (\tau, 0), \quad \tau \in \mathbb{R}^2, \quad \|\tau\| < 1, \quad (246)$$

and, for every $n \in \mathbb{R}^2$ with $\|n\| = 1$,

$$\|D^\top n\|^2 \leq m_\lambda(\tau \cdot n), \quad m_\lambda(r) = \min_{-1 \leq x \leq 1} ((x+r)^2 + \lambda^2(1-x^2)). \quad (247)$$

(2) The transverse linear action vanishes and

$$T = \text{diag}(0, 0, \lambda), \quad \lambda \geq 0, \quad t_z \geq 0, \quad \|t_\perp\| \leq 1. \quad (248)$$

For $|r| < 1$, the minimum in (247) is

$$m_\lambda(r) = \begin{cases} \lambda^2(1-r^2/(1-\lambda^2)), & 0 < \lambda < 1, |r| \leq 1-\lambda^2, \\ (1-|r|)^2, & 0 < \lambda < 1, |r| > 1-\lambda^2, \\ (1-|r|)^2, & \lambda \geq 1. \end{cases} \quad (249)$$

The complex-linear rank is $2 + \text{rank } D$ in the first class and $1 + \mathbf{1}_{\{\lambda > 0\}}$ in the second. In particular the first class is invertible exactly when D is invertible.

D.1. Boundary kernels and the proof of the classification.

Lemma D.2 (Trace-one scalar kernels). *A Hermitian trace-one matrix with transverse Bloch norm R and longitudinal coordinate h has a stable coefficient polynomial if and only if $R \leq 1$ and $h \geq 0$. For $0 \leq R \leq 1$, the function*

$$K_R(z, w) = \frac{1 + zw + R(z + w)}{1 - zw} \quad (250)$$

has strictly positive real part on $\mathbb{D}^2$, and the infimum of that real part is zero.

Proof. A rotation about Z makes the transverse component $(R, 0)$ and only rotates the disk variables. The coefficient polynomial becomes

$$f(z, w) = \frac{1 + h + R(z + w) + (1 - h)zw}{2} = \frac{1 - zw}{2} (K_R(z, w) + h).$$

For $a = |z| < 1$, $b = |w| < 1$ and $0 \leq R \leq 1$, multiplication by $|1 - zw|^2$ shows that the real part of the numerator of K_R is at least

$$1 - a^2b^2 - R(a(1 - b^2) + b(1 - a^2)) \geq (1 - ab)(1 - a)(1 - b) > 0.$$

This proves sufficiency. Setting $z = w = is$ and letting $s \uparrow 1$ shows that the infimum of $\Re K_R$ is zero.

For necessity, stability gives the boundary root-modulus inequality obtained by treating f as affine in w . At $z = e^{i\theta}$ its squared difference between the constant and linear coefficients is $h(1 + R \cos \theta)$. Its average is h , so $h \geq 0$. If $h > 0$, the same inequality for all θ gives $R \leq 1$. If $h = 0$ and $R > 1$, setting $w = 0$ gives the interior zero $z = -1/R$. $\square$

Lemma D.3 (The local block form). *Every map in Theorem D.1 whose full Choi kernel is stable satisfies*

$$T = D \oplus \lambda, \quad \lambda \geq 0. \quad (251)$$

If $D \neq 0$, then $t_z = 0$ and $\lambda > 0$.

Proof. Let $\phi = \Phi^\dagger$. It is unital and Hermiticity preserving. Lemma C.1 makes it a preserver of complex matrix-LY polynomials, with a possible zero output. Near the identity that zero alternative is excluded by continuity. Apply Lemma C.2 to

the two-sided stable curves $\phi(e^{sX})$, $\phi(e^{sY})$, and $\phi(e^{siZ})$. Hermiticity preservation, and its anti-Hermitian counterpart, give

$$\begin{aligned}\phi(X) &= t_x I + D_{11}X + D_{12}Y, \\ \phi(Y) &= t_y I + D_{21}X + D_{22}Y, \\ \phi(Z) &= t_z I + \lambda Z.\end{aligned}\tag{252}$$

The stable matrix $I + sZ$, $s > 0$ small, maps to $(1 + st_z)I + s\lambda Z$. Its root-modulus inequality is

$$|1 + st_z + s\lambda|^2 - |1 + st_z - s\lambda|^2 = 4s\lambda(1 + st_z) \geq 0,$$

so $\lambda \geq 0$.

If $D \neq 0$, choose a real transverse field A such that the transverse component of $\phi(A)$ is nonzero. For small nonzero real ε , both $A + i\varepsilon Z$ and the traceless part of its image are hyperbolic. The input pencil $I + s(A + i\varepsilon Z)$ is stable for both signs of sufficiently small s : its exponentials are stability units by Lemma C.2, and the identity $(A + i\varepsilon Z)^2 \in \mathbb{R}I$ rewrites these exponentials as scalar multiples of the pencil near the identity. Its image is a stable pencil whose scalar part is $t_A + i\varepsilon t_z$. Lemma C.3 forces this scalar to be real, hence $t_z = 0$.

Finally specialize the two input Choi variables to $u = e^{i\theta}$, $v = -e^{-i\theta}$. The input matrix is $Z + in \cdot \sigma_\perp$, with n ranging over the real unit circle. The output matrix is $\lambda Z + i(Dn) \cdot \sigma_\perp$. Hurwitz's theorem makes its polynomial stable or zero. If $\lambda = 0$, choose n with $Dn \neq 0$. The polynomial is then nonzero but its constant coefficient is zero, contradicting stability. This proves $\lambda > 0$. $\square$

Proof of Theorem D.1. By Lemma D.3, it remains to treat the two possible transverse blocks. Suppose first that $D = 0$, and rotate about Z so that $t_\perp = (R, 0)$. Direct expansion of the Choi kernel gives

$$F_{J(\Phi)} = \frac{(1 + uv)(1 - zw)}{2} \left(K_R(z, w) + t_z + \lambda \frac{1 - uv}{1 + uv} \right).\tag{253}$$

The last quotient ranges over the right half-plane. Thus $R \leq 1$, $t_z \geq 0$, and $\lambda \geq 0$ imply strict nonvanishing by Lemma D.2. Conversely, specialization $u = v = 0$ gives a trace-one Hermitian kernel with transverse norm R and longitudinal coordinate $t_z + \lambda$, so $R \leq 1$. If $\lambda = 0$, that same lemma gives $t_z \geq 0$. If $\lambda > 0$ and $t_z < 0$, choose z, w with $\Re(K_R + t_z) < 0$. A value of $(1 - uv)/(1 + uv)$ in the right half-plane then cancels the parenthesis in (253), a contradiction. This proves the second class completely, without positivity.

Now suppose $D \neq 0$. Lemma D.3 gives $\lambda > 0$ and $t_z = 0$. Set

$$\begin{aligned}A_0 &= (1 + \lambda)/2, & B_0 &= (1 - \lambda)/2, & \zeta &= (t_x - it_y)/2, \\ a &= (D_{11} + D_{22} + i(D_{12} - D_{21}))/2, & b &= (D_{11} - D_{22} + i(D_{12} + D_{21}))/2.\end{aligned}\tag{254}$$

The full polynomial has the reciprocal decomposition

$$F_{J(\Phi)} = p + vp^\sharp, \quad p = g + uh, \quad g = A_0 + \zeta w + \bar{\zeta}z + B_0zw, \quad h = \bar{b}w + \bar{a}z,\tag{255}$$

where $p^\sharp(z, w, u) = zwu \overline{p(1/\bar{z}, 1/\bar{w}, 1/\bar{u})}$. This is an algebraic identity with no positivity assumption.

If the full polynomial is stable, its specializations p and g are stable, since their constant coefficient is $A_0 > 0$. Lemma D.2 applied to g gives $R = \|t_\perp\| \leq 1$.

The root test in u and continuity give $|h| \leq |g|$ on the output two-torus. Write $z = e^{i(\theta+\varphi)}$, $w = e^{i(\theta-\varphi)}$ and $n = (\cos \varphi, -\sin \varphi)$. Direct expansion yields

$$|h|^2 = \|D^\top n\|^2, \quad |g|^2 = (\cos \theta + t_\perp \cdot n)^2 + \lambda^2 \sin^2 \theta. \quad (256)$$

Minimization in $x = \cos \theta$ proves (247). If $R = 1$, rotate so that $t_\perp = e_1$. The endpoint $x = -1$ gives $\|D^\top n\| \leq 1 - n_1$ for every unit n . At $n = e_1$ the first row of D vanishes. Setting $n = (\cos \alpha, \sin \alpha)$ then gives $|\sin \alpha| \|D^\top e_2\| \leq 1 - \cos \alpha$. Division by $|\sin \alpha|$ and passage to $\alpha = 0$ kills the second row. Since $D \neq 0$, necessarily $R < 1$.

Conversely, assume (246) and (247). The polynomial g has no zero on the closed bidisk. Indeed its constant coefficient in w , namely $A_0 + \bar{\zeta}z$, has no zero on the closed disk, and at $|z| = 1$ the squared root-modulus difference is

$$|A_0 + \bar{\zeta}z|^2 - |\zeta + B_0 z|^2 = \lambda(1 + 2\Re(\bar{\zeta}z)) > 0.$$

Here $2|\zeta| = R < 1$. The maximum modulus principle in z proves the asserted closed-bidisk nonvanishing. Equations (247) and (256), followed by the maximum modulus principle in each variable, imply $|h/g| \leq 1$ on the bidisk. Hence $p = g + uh$ is stable.

For any stable multiaffine p , one has $|p^\#|/|p| \leq 1$ in the open polydisk. To include boundary zeros, apply the maximum modulus principle to $p_r(z, w, u) = p(rz, rw, ru)$, $r < 1$, which has no zeros on the closed polydisk. On its distinguished boundary $|p_r^\#| = |p_r|$, so $|p_r^\#|/|p_r| \leq 1$ inside. Let $r \uparrow 1$. This proves the assertion, and hence $p + vp^\# \neq 0$ when $|v| < 1$. Sufficiency follows.

The formula (249) follows by minimizing the quadratic $(1 - \lambda^2)x^2 + 2rx + r^2 + \lambda^2$: for $\lambda < 1$ its critical point is $-r/(1 - \lambda^2)$, and otherwise its minimum on the interval is at an endpoint. Finally the Pauli transfer matrix is $\begin{pmatrix} 1 & 0 \\ t & T \end{pmatrix}$, whose rank is $1 + \text{rank } T$. This gives all rank assertions. $\square$

D.2. Complete positivity and finite inequalities.

Theorem D.4 (The CPTP subfamilies). *In the first class of Theorem D.1, complete positivity is equivalent to*

$$J = \begin{pmatrix} A_0 & 0 & \zeta & a \\ 0 & B_0 & \bar{b} & \zeta \\ \bar{\zeta} & b & B_0 & 0 \\ \bar{a} & \bar{\zeta} & 0 & A_0 \end{pmatrix} \succeq 0, \quad (257)$$

with coefficients (254). For a positive trace-preserving map of this form with $0 < \lambda < 1$, the stability criterion (247) is equivalent to the finite matrix inequality

$$\frac{DD^\top}{\lambda^2} + \frac{\tau\tau^\top}{1 - \lambda^2} \preceq I_2. \quad (258)$$

At $\lambda = 1$, positivity forces $\tau = 0$, and the stability criterion is $\|D\|_{\text{op}} \leq 1$.

For the singular rank-three subclass, input and output rotations about Z put $D = \text{diag}(d, 0)$ with $d > 0$ and $\tau = (p, q)$. The complete CPTP and full-Choi-LY conditions are the two scalar inequalities

$$q^2 + (d + \sqrt{p^2 + \lambda^2})^2 \leq 1, \quad \frac{d^2}{\lambda^2} + \frac{p^2}{1 - \lambda^2 - q^2} \leq 1, \quad d, \lambda > 0. \quad (259)$$

The first inequality makes the denominator in the second strictly positive.

In the transverse-erasure class, the complete conditions are

$$t_z, \lambda \geq 0, \quad \|t_\perp\|^2 + (t_z + \lambda)^2 \leq 1. \quad (260)$$

Proof. The matrix in (257) is the Choi matrix in the order (00,01,10,11), with each pair ordered as (output,input). Its partial trace over the output equals the input identity. Choi's criterion gives the first assertion.

For the finite inequality, let $k = 1 - \lambda^2 > 0$ and fix a transverse unit vector n . Reversing n if necessary, set $r = \tau \cdot n \geq 0$ and $s^2 = \|D^\top n\|^2$. The desired scalar form of (258) is

$$Q = \lambda^2 - s^2 - \lambda^2 r^2 / k \geq 0. \quad (261)$$

If $r \leq k$, this is exactly the first branch of (249). If $r > k$, put $a = k/r \in (0, 1)$. Positivity means that the affine image of the Bloch unit ball lies in the unit ball. Its support in the unit output direction $(an, \sqrt{1 - a^2})$ therefore satisfies

$$ar + \sqrt{a^2 s^2 + \lambda^2 (1 - a^2)} \leq 1.$$

Squaring the nonnegative sides gives

$$0 \leq k - 2ar + a^2(r^2 - s^2 + \lambda^2) = \frac{k^2}{r^2} Q.$$

Thus positivity itself implies (261) in the clipped region. Conversely (261) bounds the unconstrained minimum over all real x , hence implies (247) on $[-1, 1]$. Varying n proves (258). Positivity at the input Bloch points $\pm e_z$ gives $\|\tau\|^2 + \lambda^2 \leq 1$, proving both $\lambda \leq 1$ and the stated endpoint assertion.

Suppose now D has rank one. Real rotations in the transverse input and output planes put it in the form $\text{diag}(d, 0)$, $d > 0$; these are implemented by rotations about Z and preserve both properties under consideration. In these coordinates the Choi matrix satisfies $2J = I + H$, where

$$H = pX \otimes I + qY \otimes I + dX \otimes X + \lambda Z \otimes Z.$$

Pauli multiplication gives

$$H^2 = (p^2 + q^2 + d^2 + \lambda^2)I + 2pdI \otimes X - 2d\lambda Y \otimes Y.$$

The last two Pauli matrices anticommute. The eigenvalues of H^2 are $q^2 + (\sqrt{p^2 + \lambda^2} \pm d)^2$, each twice. Also $\text{Tr } H = \text{Tr } H^3 = 0$, by Pauli orthogonality. Newton's identities make the characteristic polynomial of H even, so its spectrum is symmetric about zero. Its least eigenvalue is consequently $-\sqrt{q^2 + (\sqrt{p^2 + \lambda^2} + d)^2}$. This proves the first inequality in (259) as the exact CP condition. Since $d, \lambda > 0$, it also implies $\lambda < 1$ and $1 - \lambda^2 - q^2 > 0$. Taking the Schur complement of the lower-right entry in (258) now gives precisely the second inequality in (259).

Finally, a transverse-erasure map measures in the Z basis and prepares

$$\rho_\pm = \frac{1}{2}(I + t_x X + t_y Y + (t_z \pm \lambda)Z). \quad (262)$$

Its Choi matrix is block diagonal on the input and is positive exactly when both prepared matrices are positive, equivalently $\|t_\perp\|^2 + (t_z \pm \lambda)^2 \leq 1$. Since $t_z, \lambda \geq 0$, the two inequalities reduce to (260). $\square$

Remark D.5. Full Choi stability does not require both states in (262) to be individually Lee–Yang: the quantity $t_z - \lambda$ may be negative. The choice $t_\perp = 0$, $t_z = \lambda = 1/3$ is a rank-two CPTP example with positive longitudinal drift. A rank-three example retaining transverse information is $t = 0$, $D = \text{diag}(1/5, 0)$, $\lambda = 2/5$.

At rank one, the maps are precisely resets $M \mapsto \text{Tr}(M)\rho$; their CPTP full-Choi-LY condition is $t_z \geq 0$ and $\|t\| \leq 1$. These singular examples explain why the zero longitudinal drift conclusion for an invertible map cannot be extended without a rank hypothesis.

APPENDIX E. COHERENT STABILITY AT HIGHER CAPACITIES

The dynamical classification at arbitrary finite spins is Theorem 7.22. This appendix gives two complementary results. The infinitesimal classification determines all Hermitian Hamiltonians whose coherent Gibbs kernels are stable at every temperature. A separate, static argument classifies all independently rotation-covariant trace-preserving maps, including maps between different spin representations. The covariance assumption in the latter result is explicit throughout.

E.1. Coherent normalization and the Hamiltonian cone. At site i , fix a positive integer κ_i and put $j_i = \kappa_i/2$. Let $\mathcal{H}_{\kappa_i} = \mathbb{C}^{\kappa_i+1}$ have orthonormal basis $|a_i\rangle$, $0 \leq a_i \leq \kappa_i$. For multi-indices a set

$$b_a = \prod_i \binom{\kappa_i}{a_i}, \quad |a\rangle \longleftrightarrow \sqrt{b_a} z^a. \quad (263)$$

Thus the monomial inner product is $\langle z^a, z^b \rangle = \delta_{ab}/b_a$. All adjoints in this section refer to the physical orthonormal basis, or equivalently to this weighted polynomial inner product. The coherent kernel of a matrix M is

$$F_M(z, w) = \sum_{a,b} \sqrt{b_a b_b} M_{ab} z^a w^b. \quad (264)$$

In particular, $F_I(z, w) = \prod_i (1 + z_i w_i)^{\kappa_i}$. We call M coherently Lee–Yang if F_M has no zero when all its variables belong to $\mathbb{D}$. Both weights in (264) are essential.

The spin matrices have the following differential realization:

$$S_i^x = \frac{(1 - z_i^2)\partial_i + \kappa_i z_i}{2}, \quad S_i^y = \frac{(1 + z_i^2)\partial_i - \kappa_i z_i}{2i}, \quad S_i^z = \frac{\kappa_i}{2} - z_i \partial_i. \quad (265)$$

They satisfy $(S_i^x)^2 + (S_i^y)^2 + (S_i^z)^2 = j_i(j_i + 1)I$. Write $S_i^\perp = (S_i^x, S_i^y)^\top$.

Theorem E.1 (Higher-spin Hamiltonians). *For a Hermitian operator H on $\bigotimes_i \mathcal{H}_{\kappa_i}$, the following conditions are equivalent:*

- (1) $e^{-\beta H}$ is coherently Lee–Yang for every $\beta \geq 0$;
- (2) this holds for every $0 \leq \beta < \varepsilon$, for some $\varepsilon > 0$;
- (3) H admits the expression

$$\begin{aligned} H = & h_0 I + \sum_i (h_i^x S_i^x + h_i^y S_i^y + h_i^z S_i^z) + \sum_{i: \kappa_i \geq 2} (S_i^\perp)^\top B_i S_i^\perp \\ & + \sum_{i < j} ((S_i^\perp)^\top J_{ij} S_j^\perp + e_{ij} S_i^z S_j^z), \end{aligned} \quad (266)$$

where all coefficients are real, $B_i = B_i^\top$, and

$$h_i^z \leq 0, \quad B_i \succeq 0, \quad e_{ij} \leq -\|J_{ij}\|_{\text{op}}. \quad (267)$$

The coefficient $(B_i)_{12}$ multiplies $S_i^x S_i^y + S_i^y S_i^x$. At a capacity-one site all single-site quadrupoles reduce to scalars or zero and are omitted from (266).

We give the normalization and coordinate computations in the proof, so that the finite-capacity infinitesimal theorem can be applied without changing the physical adjoint.

Lemma E.2 (Polarization and the coherent Cayley transform). *Define the Dicke isometry*

$$W_i|a\rangle = \binom{\kappa_i}{a}^{-1/2} \sum_{|x|=a} |x\rangle, \quad W = \bigotimes_i W_i. \quad (268)$$

The binary matrix kernel of WMW^* is the normalized polarization of F_M , separately in each row and column block. Moreover, let

$$C_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} i & -i \\ 1 & 1 \end{pmatrix}, \quad C_i = W_i^* C_1^{\otimes \kappa_i} W_i, \quad C = \bigotimes_i C_i. \quad (269)$$

For Hermitian H , coherent stability of e^{-tH} for all sufficiently small $t \geq 0$ is equivalent to the condition that $\exp(-tCHC^*)$ is a complete upper-half-plane stability preserver for every $t \geq 0$.

Proof. At binary words of weights a, b , the matrix entry of WMW^* is $M_{ab}/\sqrt{b_a b_b}$. Identifying the variables in their respective blocks multiplies this coefficient by $b_a b_b$, giving exactly (264). The disk polarization theorem therefore identifies stability of the two kernels; see [11, Theorem 2.1]. Polarization, binary tensor contraction, and diagonal specialization show that a coherently stable matrix sends each stable coherent vector to a stable vector or zero. An invertible matrix cannot produce the latter alternative. This argument does not require WMW^* to be invertible outside the symmetric subspace.

The matrix C_i is unitary. On polynomials it acts by

$$f(z_i) \mapsto \left(\frac{s_i + i}{\sqrt{2}} \right)^{\kappa_i} f\left(\frac{s_i - i}{s_i + i} \right), \quad (270)$$

which identifies disk stability with upper-half-plane stability. Consequently coherent stability of e^{-tH} implies ordinary stability preservation by e^{-tCHC^*} . Preservation for a sufficiently small time interval implies preservation for all times by composition. Theorem 3.2 then gives complete preservation.

For the converse, put $D_i = \text{diag}_{0 \leq a \leq \kappa_i}((-1)^{\kappa_i - a})$ and $D = \bigotimes_i D_i$. Symmetric powers of $C_1 C_1^\top$ give $CC^\top = D$, and

$$F_D(s, t) = \prod_i (s_i t_i - 1)^{\kappa_i} \quad (271)$$

is upper-half-plane stable. Indeed $st = 1$ is impossible for two points of the upper half-plane. A complete preserver $N = e^{-tCHC^*} = C e^{-tH} C^*$ applied in the s variables to this kernel produces the coherent coefficient matrix $ND = C e^{-tH} C^\top$. This is precisely the separate Cayley transform of both groups of variables in $F_{e^{-tH}}$. It is nonzero because N is invertible. Applying the inverse transforms proves the required disk stability. $\square$

Proof of Theorem E.1. Put $A = -CHC^*$. By Lemma E.2, Theorem 3.2 applies to A . We recall the parts of its conclusion used here. The generator belongs to the degree-at-most-two part of the enveloping algebra of the local $\mathfrak{sl}_2$ actions. Its second-order differential coefficients are real and nonpositive on the appropriate

real lines or planes. If $A = R + iS$ in the real monomial basis, then its imaginary part has the form

$$S = cI + \sum_i J_{i,\kappa_i}(q_i), \quad J_{i,\kappa_i}(q) = q\partial_i - \frac{\kappa_i}{2}q', \quad q_i \in \mathbb{R}[z_i]_{\leq 2}, \quad q_i \geq 0 \text{ on } \mathbb{R}. \quad (272)$$

Conversely these conditions, together with preservation of the finite degree box, characterize the generators.

The degree-two operator space of Proposition 3.4 decomposes as the direct sum of the scalar space, the local spin spaces, the local traceless symmetric quadrupoles when $\kappa_i \geq 2$, and the spaces spanned by $S_i^a S_j^b$ for $i < j$. Its dimension is

$$1 + 3n + 5\#\{i : \kappa_i \geq 2\} + 9\binom{n}{2}. \quad (273)$$

For clarity, the local assertion follows from the usual decomposition of the square of the three-dimensional spin representation into its scalar, antisymmetric, and traceless symmetric parts: the antisymmetric part is the spin commutator, the scalar part is the Casimir, and the rank-two part is nonzero precisely for $\kappa_i \geq 2$. Distinct sites give independent tensor factors. This also proves the stated directness. Since the Cayley action is a local spin rotation, it preserves this operator space. Thus an arbitrary Hermitian H satisfying condition (2) already has total spin degree at most two, with the stated capacity-one exceptions.

Hermiticity of A and the real diagonal monomial inner product give $R^* = R$ and $S^* = -S$. For one site write

$$J^- = \partial, \quad J^0 = z\partial - \kappa/2, \quad J^+ = z^2\partial - \kappa z.$$

In the weighted inner product, $(J^-)^* = -J^+$ and $(J^0)^* = J^0$. Consequently skew-adjointness of $aJ^- + bJ^0 + cJ^+$, for real a, b, c , forces $b = 0$ and $a = c$. The scalar in (272) vanishes, and hence

$$q_i(s) = \alpha_i(1 + s^2), \quad \alpha_i \geq 0, \quad iJ_{i,\kappa_i}(q_i) = -2\alpha_i S_i^y. \quad (274)$$

The Cayley rotations are

$$CS_i^x C^* = -S_i^z, \quad CS_i^y C^* = S_i^x, \quad CS_i^z C^* = -S_i^y. \quad (275)$$

Thus the original longitudinal field contributes $h_i^z S_i^y$ to A , and (274) gives $h_i^z = -2\alpha_i \leq 0$. Each mixed longitudinal-transverse quadratic term in H gives an imaginary second-order coefficient in A . The direct decomposition above, or the linearly independent principal polynomials at each site, prevents cancellation between distinct such terms. They all vanish.

It remains to compute the real second-order inequalities. The principal coefficients of (S^x, S^y, S^z) in (265) are $((1 - s^2)/2, -i(1 + s^2)/2, -s)$. Put

$$u(s) = \left(\frac{2s}{1 + s^2}, \frac{1 - s^2}{1 + s^2} \right)^\top;$$

its range is dense in the real unit circle. For the pair term $(S_i^\perp)^\top J_{ij} S_j^\perp + e_{ij} S_i^z S_j^z$, the coefficient q_{ij} of $\partial_i \partial_j$ in A satisfies

$$\frac{4q_{ij}(s, t)}{(1 + s^2)(1 + t^2)} = e_{ij} - u(s)^\top J_{ij} u(t). \quad (276)$$

Its nonpositivity is therefore equivalent to $e_{ij} \leq -\|J_{ij}\|_{\text{op}}$. Before fixing the local Casimir gauge, write a surviving local quadratic term as $(S_i^\perp)^\top K_i S_i^\perp + c_i (S_i^z)^2$, with K_i real symmetric. For $\kappa_i \geq 2$ its coefficient q_i of ∂_i^2 satisfies

$$\frac{4q_i(s)}{(1+s^2)^2} = c_i - u(s)^\top K_i u(s). \quad (277)$$

Thus $c_i \leq \lambda_{\min}(K_i)$. The Casimir identity rewrites this term as $(S_i^\perp)^\top B_i S_i^\perp$ plus a scalar, where $B_i = K_i - c_i I_2 \succeq 0$. This proves necessity.

Conversely, (267) makes (276) and (277) real and nonpositive and makes the imaginary part satisfy (272). The expression preserves the degree box because it is built from the spin operators. Theorem 3.2 and Lemma E.2 give coherent stability at every time. This proves (3) $\Rightarrow$ (1); (1) $\Rightarrow$ (2) is immediate. $\square$

Corollary E.3 (Exact symmetric-qubit realization). *Every Hamiltonian in Theorem E.1 has a Hermitian qubit extension $\hat{H}$, invariant under permutations within the capacity blocks, whose pair coefficients satisfy the qubit Lee–Yang cone and such that*

$$\hat{H}W = WH, \quad e^{-\beta\hat{H}}W = We^{-\beta H} \quad (\beta \in \mathbb{R}). \quad (278)$$

Proof. For each $B_i \succeq 0$, let $\lambda_i = \lambda_{\max}(B_i)$ and use the exact identity

$$(S_i^\perp)^\top B_i S_i^\perp = (S_i^\perp)^\top (B_i - \lambda_i I_2) S_i^\perp - \lambda_i (S_i^z)^2 + \lambda_i j_i(j_i + 1)I. \quad (279)$$

Replace each spin in this expression and in (266) by $\hat{S}_i^a = \frac{1}{2} \sum_{r=1}^{\kappa_i} \sigma_{ir}^a$. Retain the last term of (279) as the indicated scalar. Interblock pairs have transverse matrix $J_{ij}/4$ and longitudinal coefficient $e_{ij}/4$. Intrablock pairs have transverse matrix $(B_i - \lambda_i I_2)/2$ and longitudinal coefficient $-\lambda_i/2$. If $b_i = \lambda_{\min}(B_i)$, then

$$\|B_i - \lambda_i I_2\|_{\text{op}} = \lambda_i - b_i \leq \lambda_i.$$

All pairs therefore satisfy the qubit cone. Identical-slot symmetric off-diagonal products vanish by Pauli anticommutation; the remaining identical-slot terms are scalars. Each collective spin preserves the symmetric sector and intertwines the physical spin through W_i . This proves both identities in (278). $\square$

The reduction of higher spins to spin one-half and the corresponding anisotropic sufficient Lee–Yang theorems are classical; see, in particular, [53, Theorem 2 and equations (2.38)–(2.44)]. The corollary supplies explicit coefficients for every Hamiltonian in the necessary-and-sufficient cone, including single-ion quadrupoles. It is an invariant-sector identity and does not assert a spectral comparison with the other qubit spin sectors.

E.2. Independently covariant maps between different capacities. Let U_k denote the spin- $k/2$ representation on $\mathcal{H}_k = \mathbb{C}^{k+1}$, now allowing $k = 0$. For input capacities $\kappa = (\kappa_1, \dots, \kappa_n)$ and output capacities $\ell = (\ell_1, \dots, \ell_n)$, a complex-linear map $\Phi : \text{End}(\bigotimes_i \mathcal{H}_{\kappa_i}) \rightarrow \text{End}(\bigotimes_i \mathcal{H}_{\ell_i})$ is independently covariant if

$$\Phi(U_\kappa(g) M U_\kappa(g)^*) = U_\ell(g) \Phi(M) U_\ell(g)^*, \quad U_k(g) = \bigotimes_i U_{k_i}(g_i), \quad (280)$$

for every $g = (g_1, \dots, g_n) \in SU(2)^n$. The input and output sites in this definition are matched in advance.

Use the Choi convention $J(\Phi) = \sum_{a,b} \Phi(|a\rangle\langle b|) \otimes |a\rangle\langle b|$. Its coherent polynomial is

$$F_{J(\Phi)}(z, w, u, v) = \sum_{c,e,a,b} \sqrt{b_c^\ell b_e^\ell b_a^\kappa b_b^\kappa} [\Phi(|a\rangle\langle b|)]_{ce} z^c w^e u^a v^b. \quad (281)$$

Full coherent Choi stability means stability in all four groups of variables. For one matched pair of sites put

$$K_o = 1 + zw, \quad K_i = 1 + uv, \quad A = (1 + zu)(1 + wv), \quad B = K_o K_i. \quad (282)$$

Theorem E.4 (Covariant root classification). *Suppose Φ is complex linear, trace preserving, and independently covariant as in (280). Its full coherent Choi kernel is Lee–Yang if and only if $\Phi = \bigotimes_i \Phi_i$, where each local factor is specified by an integer $0 \leq d_i \leq \min(\kappa_i, \ell_i)$ and an unordered list $s_{i1}, \dots, s_{id_i} \geq 0$ as follows. With $q_i(r) = \prod_{a=1}^{d_i} (r + s_{ia})$, its local kernel is*

$$F_{J(\Phi_i)} = \frac{K_{o,i}^{\ell_i - d_i} K_{i,i}^{\kappa_i - d_i} \prod_{a=1}^{d_i} (A_i + s_{ia} B_i)}{(\ell_i + 1) \int_0^1 q_i(x) dx}. \quad (283)$$

Every such map is completely positive, although neither positivity nor Hermiticity preservation was assumed. Its complex-linear rank is

$$\text{rank } \Phi = \prod_i (d_i + 1)^2. \quad (284)$$

It is injective exactly when $d_i = \kappa_i$ at every site, surjective exactly when $d_i = \ell_i$ at every site, and invertible exactly when $d_i = \kappa_i = \ell_i$ at every site.

We prove the theorem together with a factorization through an intermediate spin representation. The representation theory of covariant maps and the general covariant completely positive simplex are established in [1]; coherent descriptions and multipliers are also studied in [7]. We use the multiplicity-free commutant description, and derive the additional root restriction directly from stability.

Lemma E.5 (Range of the coherent ratio). *The image of A/B on $\mathbb{D}^4$ is precisely $\Omega = \mathbb{C} \setminus (-\infty, 0]$.*

Proof. For $a \in \mathbb{D}$, the disk automorphisms

$$f_a(x) = \frac{x - a}{1 - \bar{a}x}, \quad h_a(y) = \frac{y + \bar{a}}{1 + ay}$$

satisfy

$$1 + f_a(x)h_a(y) = \frac{(1 - |a|^2)(1 + xy)}{(1 - \bar{a}x)(1 + ay)}.$$

Apply f_a to z, v and h_a to u, w . The ratio A/B is unchanged. Choosing $a = z$ makes the new z zero, and reduces the ratio to $(1 + w'v')/(1 + u'v')$. It is nonzero. If it equalled $-t$ for some $t > 0$, then $1+t = -v'(w'+tu')$, whereas $|v'|(|w'|+t|u'|) < 1+t$. This proves inclusion in Ω . Conversely, setting $u = z$ and $v = w = -z$ gives

$$\frac{A}{B} = \left(\frac{1 + z^2}{1 - z^2} \right)^2.$$

The successive maps $z \mapsto z^2$, $x \mapsto (1+x)/(1-x)$, and $y \mapsto y^2$ map onto the disk, the right half-plane, and Ω , respectively. This proves surjectivity. $\square$

Lemma E.6 (Polynomial factorization on a product of slit planes). *If a nonzero polynomial $Q \in \mathbb{C}[r_1, \dots, r_n]$ has no zero on Ω^n , then*

$$Q(r) = c \prod_i q_i(r_i), \quad c \neq 0, \quad (285)$$

where each q_i is monic and all its roots belong to $(-\infty, 0]$.

Proof. Write $Q(x, y) = \sum_{k=0}^d a_k(y)x^k$, with $a_d \neq 0$. On any connected open patch of Ω^{n-1} where $a_d \neq 0$, every root of the monic polynomial $Q(x, y)/a_d(y)$ is real and nonpositive. Its coefficients a_k/a_d are therefore real-valued holomorphic functions, and hence constant on the patch. Polynomial identity gives $Q(x, y) = a_d(y)p(x)$ everywhere, for a fixed monic polynomial p with all roots in $(-\infty, 0]$. Since p is nonzero on Ω , the polynomial a_d has no zero on Ω^{n-1} . Induction proves the claim, skipping variables of degree zero. This reasoning includes repeated roots and slices on which the degree might initially drop. $\square$

For $0 \leq r \leq k$, let $R_{k \rightarrow r}$ be partial trace from a Dicke-encoded k -qubit input to r qubits, identified with $\text{End}(\mathcal{H}_r)$. The reduced operator is supported on the symmetric sector: a vector antisymmetric in a retained pair is orthogonal to the fully symmetric input sector. Define the adjoint map

$$C_{r \rightarrow k}(N) = W_k^* (W_r N W_r^* \otimes I_2^{\otimes(k-r)}) W_k. \quad (286)$$

These maps are completely positive and covariant; $R_{k \rightarrow r}$ is trace preserving. At $r = 0$ they are the ordinary trace and scalar multiplication of I_{k+1} , respectively.

Lemma E.7 (Coherent compression kernels). *For $0 \leq r \leq \min(\kappa, \ell)$, the map $T_r^{\kappa, \ell} = C_{r \rightarrow \ell} R_{\kappa \rightarrow r}$ has coherent Choi kernel*

$$A^r K_o^{\ell-r} K_i^{\kappa-r}. \quad (287)$$

These $\min(\kappa, \ell) + 1$ maps form a basis of all complex-linear $SU(2)$ -covariant maps between the two matrix spaces. Their trace effects are

$$(T_r^{\kappa, \ell})^*(I_{\ell+1}) = \frac{\ell+1}{r+1} I_{\kappa+1}. \quad (288)$$

Proof. The Dicke coefficients give, for $c = a + k$ and $e = b + k$,

$$[C_{r \rightarrow \ell}(|a\rangle\langle b|)]_{c,e} = \frac{\sqrt{\binom{r}{a} \binom{r}{b} \binom{\ell-r}{k}}}{\sqrt{\binom{\ell}{c} \binom{\ell}{e}}}, \quad (289)$$

and zero if the two shifts do not agree. In particular,

$$F_{C_{r \rightarrow \ell}(N)}(z, w) = (1 + zw)^{\ell-r} F_N(z, w). \quad (290)$$

Adjunction gives the coefficients of $R_{\kappa \rightarrow r}$, with ℓ replaced by κ . If its input indices are $a + m, b + m$, the two square-root binomial weights from the output and input of the Choi kernel cancel exactly the two denominators in these formulas. What remains is

$$\binom{r}{a} \binom{r}{b} \binom{\ell-r}{k} \binom{\kappa-r}{m},$$

the coefficient in the expansion of (287).

Covariance is equivalent to the Choi matrix commuting with $U_\ell \otimes \overline{U_\kappa}$. The Clebsch–Gordan decomposition is multiplicity free, with spins $|\ell - \kappa|/2, |\ell + \kappa|/2 +$

$1, \dots, (\ell + \kappa)/2$. Its commutant therefore has dimension $\min(\kappa, \ell) + 1$. The displayed covariant kernels are independent, since division by $K_o^\ell K_i^\kappa$ gives the successive powers of A/B , whose range is open by Lemma E.5. They consequently span the entire complex commutant, without a Hermiticity assumption.

Finally $C_{r \rightarrow \ell}(I_{r+1}) = I_{\ell+1}$: the projection onto the symmetric sector of the first r qubits acts identically on the fully symmetric ℓ -qubit sector. Covariance and irreducibility imply that $C_{r \rightarrow \ell}^*(I_{\ell+1})$ is a scalar multiple of I_{r+1} . Taking traces identifies this scalar as $(\ell + 1)/(r + 1)$. Composing with the trace-preserving $R_{\kappa \rightarrow r}$ proves (288). $\square$

Proof of Theorem E.4. For the independent product group, the commutant is the tensor product of the local commutants in Lemma E.7. There is therefore a unique polynomial Q of separate degrees at most $\min(\kappa_i, \ell_i)$ such that

$$F_{J(\Phi)} = \prod_i K_{o,i}^{\ell_i} K_{i,i}^{\kappa_i} Q\left(\frac{A_1}{B_1}, \dots, \frac{A_n}{B_n}\right). \quad (291)$$

The separate degree bounds make this a polynomial. The prefactor never vanishes in the polydisk. The variables of different sites are independent, so Lemma E.5 shows that full coherent stability is equivalent to nonvanishing of Q on Ω^n . Lemma E.6 gives $Q = c \prod_i q_i$, where $q_i(r) = \prod_{a=1}^{d_i} (r + s_{ia})$ and $s_{ia} \geq 0$. The kernel thus factors, up to one nonzero scalar, into the numerators of (283).

For one factor write $q(r) = \sum_{a=0}^d q_a r^a$ and set $\Psi_{\kappa, \ell, q} = \sum_{a=0}^d q_a T_a^{\kappa, \ell}$. Every coefficient q_a is nonnegative, so this map is completely positive. By (288) its trace effect is

$$\Psi_{\kappa, \ell, q}^*(I_{\ell+1}) = c_\ell(q) I_{\kappa+1}, \quad c_\ell(q) = (\ell + 1) \sum_{a=0}^d \frac{q_a}{a+1} = (\ell + 1) \int_0^1 q(x) dx > 0. \quad (292)$$

In the global factorization, trace preservation forces $c \prod_i c_{\ell_i}(q_i) = 1$. Thus the initial scalar is positive real and the map is the tensor product of the individually normalized completely positive maps in (283). Conversely these maps are covariant and trace preserving by construction, and their kernels are stable because q_i has no root in Ω . This proves the classification and automatic complete positivity.

To prove the rank assertion, first consider the square case $d = \kappa = \ell$. The map $\Psi_{d,d,q}$ is also the Dicke compression

$$\Psi_{d,d,q}(M) = W_d^* \left[\left(\bigotimes_{a=1}^d \mathcal{T}_{s_a} \right) (W_d M W_d^*) \right] W_d, \quad \mathcal{T}_s(N) = N + s \operatorname{Tr}(N) I_2, \quad (293)$$

where the tensor product in brackets acts on the d qubit matrix factors. Its kernel is $\prod_a (A + s_a B)$ by diagonalization of the binary variables, which verifies the identity with $\Psi_{d,d,q}$. As an operator on Hilbert–Schmidt space, $\mathcal{T}_s$ is positive definite: its eigenvalues are 1 on traceless matrices and $1 + 2s$ on the scalar space. The embedding $M \mapsto W_d M W_d^*$ is a Hilbert–Schmidt isometry. Its compression of the positive definite tensor product is therefore positive definite and invertible. Define

$$\Lambda_{d,q} = \frac{\Psi_{d,d,q}}{c_d(q)}, \quad \Lambda_{0,1} = \operatorname{Id}_{\mathbb{C}}. \quad (294)$$

The multiplication identity (290) gives $C_{d \rightarrow \ell} C_{a \rightarrow d} = C_{a \rightarrow \ell}$, and adjunction gives $R_{d \rightarrow a} R_{\kappa \rightarrow d} = R_{\kappa \rightarrow a}$. Hence

$$\Psi_{\kappa, \ell, q} = C_{d \rightarrow \ell} \Psi_{d, d, q} R_{\kappa \rightarrow d}. \quad (295)$$

The map $C_{d \rightarrow \ell}$ is injective by (290); its adjoint $R_{\ell \rightarrow d}$ is therefore surjective, and the same holds for $R_{\kappa \rightarrow d}$. The middle map in (295) is invertible on a space of dimension $(d+1)^2$. This proves local rank $(d+1)^2$. Ranks multiply under tensor products, which gives (284) and all assertions about injectivity, surjectivity, and invertibility. $\square$

Corollary E.8 (Trace, invertible core, and symmetric cloning). *Each local map in Theorem E.4 has the exact factorization*

$$\Phi_i = C_{d_i \rightarrow \ell_i} \circ \Lambda_{d_i, q_i} \circ R_{\kappa_i \rightarrow d_i}, \quad C_{d \rightarrow \ell} = \frac{d+1}{\ell+1} C_{d \rightarrow \ell}. \quad (296)$$

All three maps are trace preserving and completely positive, and the middle map is an invertible coherent Lee–Yang map. For equal input and output capacities the local factors are also unital.

Proof. The first two assertions follow from (295), (292), and $c_d(q)/c_\ell(q) = (d+1)/(\ell+1)$. For equal capacities, covariance makes $\Phi(I)$ scalar, and trace preservation makes this scalar one. $\square$

The normalized map $C_{d \rightarrow \ell}$ is the universal symmetric cloner of [57, Eq. (3.3)]; its dimension-ratio normalization and concatenation law are classical. Here stability forces the entire factorization (296). The integer d specifies an intermediate spin dimension and the exact algebraic rank; it is not an assertion of noiseless recovery or an operational quantum communication capacity. If a specified covariance relation permutes the matched output sites, the theorem applies after undoing that permutation; an invertible map then matches equal capacities.

Example E.9 (The spin-one spectral region). For $\kappa = \ell = 2$, let λ_1, λ_2 be the multipliers of a trace-preserving covariant map on the dipole and quadrupole subspaces. The exact coherent Choi condition is

$$0 \leq \lambda_2 \leq \lambda_1, \quad 3\lambda_1 \leq 2 + \lambda_2, \quad 3\lambda_1^2 + \lambda_2^2 \geq 4\lambda_2. \quad (297)$$

Indeed, in the degree-two case write $q(r) = r^2 + ar + b$. Its roots are real and nonpositive exactly when $a, b \geq 0$ and $a^2 \geq 4b$. The maps with kernels A^2 , AB , and B^2 have scalar multipliers 1, 3/2, 3, respectively, dipole multipliers 1, 1, 0, and quadrupole multipliers 1, 0, 0. For the middle map, partial trace annihilates the quadrupole $\text{diag}(1, -2, 1)$, while the difference of the two extreme projectors is sent to S^z ; covariance determines its action on each irreducible operator space. Therefore

$$\lambda_1 = \frac{1+a}{1+3a/2+3b}, \quad \lambda_2 = \frac{1}{1+3a/2+3b}.$$

Solving for a, b gives (297) when $\lambda_2 > 0$. Degree one gives $\lambda_2 = 0$ and $0 < \lambda_1 \leq 2/3$, and degree zero gives the origin, proving all endpoint cases. In particular, the ordinary qutrit depolarizing map with $\lambda_1 = \lambda_2 = t$ fails coherent Choi stability for every $0 < t < 1$, although both endpoints are stable.

Remark E.10. Theorem E.4 assumes independent local covariance, not only covariance under a common rotation. Its proof is static and does not use Theorem E.1. It also does not apply the invertible qubit classification to a polarized higher-spin map: a Dicke lift is generally singular on the full qubit matrix space. No unrestricted noncovariant higher-spin static classification is asserted here.

APPENDIX F. CONNECTED PRESERVERS AND THE OBSTRUCTION TO FLOW GENERATION

The exact infinitesimal cone does not generate every connected static preserver, even after taking limits of products. We prove this on the fixed multiaffine space $V_n = \mathbb{C}[z_1, \dots, z_n]_{\text{ma}} \simeq (\mathbb{C}^2)^{\otimes n}$. Let S_n be the semigroup of invertible complex-linear operators that preserve disk stability. For its ordinary monomial coefficient matrix M , write

$$G_M(z, w) = \sum_{a, b \in \{0, 1\}^n} M_{ab} z^a w^b. \quad (298)$$

For $M \in \text{GL}(V_n)$, the symbol criterion is

$$M \in S_n \iff G_M \text{ is stable on } \mathbb{D}^{2n}. \quad (299)$$

For necessity, fix $w \in \mathbb{D}^n$ and apply M to the stable polynomial $\prod_i (1 + w_i z_i)$. Invertibility excludes the zero output. Conversely, tensor contraction of G_M with a stable input proves preservation, again excluding zero by invertibility. This is the multiaffine disk-symbol convention of the preceding sections. Hurwitz's theorem shows that S_n is closed relative to $\text{GL}(V_n)$: an invertible limit cannot have identically zero symbol. Nonzero scalar multiplication preserves S_n .

Define

$$\begin{aligned} W_n &= \{A : e^{tA} \in S_n \text{ for every } t \geq 0\}, \\ E_n &= \langle e^A : A \in W_n \rangle, \quad T_n = \overline{E_n}^{\text{GL}(V_n)}, \quad H_n = S_n \cap S_n^{-1}. \end{aligned} \quad (300)$$

Here E_n consists of finite products. The closure T_n is a different semigroup. The group H_n is the group of reversible preservers, and H_n^0 denotes its identity component.

Theorem F.1 (Global connectedness and a strict flow obstruction). *For every $n \geq 1$, S_n is path connected. Nevertheless,*

$$E_n \cap H_n = H_n^0, \quad T_n \cap T_n^{-1} = H_n^0. \quad (301)$$

For $n \geq 2$, every nontrivial coordinate permutation π satisfies

$$\pi \in S_n \setminus T_n. \quad (302)$$

Thus a positive-radius matrix neighborhood of π contains no finite product of admissible one-parameter flows, even with unbounded product lengths and generator norms.

Path connectedness. Let $D_r z^a = r^{|a|} z^a$, $0 < r \leq 1$. It is an invertible preserver, and

$$G_{D_r M D_r}(z, w) = G_M(rz, rw). \quad (303)$$

For $M \in S_n$, its constant coefficient M_{00} is nonzero. A path of nonzero scalars normalizes it to one. With this normalization, $D_r M D_r$ converges as $r \downarrow 0$ to the vacuum projector $|0\rangle\langle 0|$.

Consider the convex open set in the complex affine hyperplane $X_{00} = 1$ given by

$$\mathcal{B} = \left\{ X : X_{00} = 1, \sum_{(a,b) \neq (0,0)} |X_{ab}| < 1 \right\}. \quad (304)$$

Every member has a kernel nonvanishing on the closed polydisk, by strict constant-term domination. For small positive r, s , both $D_r M D_r$ and D_s belong to $\mathcal{B}$ and are invertible. Indeed the nonconstant coefficient sum for D_s is $(1+s)^n - 1$.

The invertible part of $\mathcal{B}$ is path connected. To see this, take invertible $X, Y \in \mathcal{B}$ and set $L(\zeta) = (1-\zeta)X + \zeta Y$. Its determinant is a nonzero polynomial in ζ , since it is nonzero at zero. The real segment $L([0, 1])$ is compactly contained in $\mathcal{B}$, so a sufficiently thin complex neighborhood of $[0, 1]$ still maps into $\mathcal{B}$. A path from 0 to 1 in this neighborhood avoids the finitely many determinant zeros. Its image connects X to Y through invertible members of $\mathcal{B}$. Now connect M to $D_r M D_r$ using (303), connect this point to D_s inside $\mathcal{B}$, and connect D_s to I by increasing s to one. Every point is in S_n . The singular vacuum projector is used only as a limit to enter $\mathcal{B}$; it is never a point of the path. $\square$

For reference, there is also an explicit two-variable path. On $\text{span}(z_1, z_2)$ let $P_{\pm}$ be the symmetric and antisymmetric projections, and let U_{θ} act as $P_+ + e^{i\theta} P_-$ there and as the identity on $\text{span}(1, z_1 z_2)$. For $r = 1/4$,

$$\begin{aligned} G_{D_r U_{\theta}} &= 1 + r a_{\theta}(z_1 w_1 + z_2 w_2) + r b_{\theta}(z_1 w_2 + z_2 w_1) + r^2 z_1 z_2 w_1 w_2, \\ a_{\theta} &= (1 + e^{i\theta})/2, \quad b_{\theta} = (1 - e^{i\theta})/2. \end{aligned} \quad (305)$$

Its nonconstant coefficient sum is at most $2\sqrt{2}r + r^2 < 1$, and its determinant is $r^4 e^{i\theta} \neq 0$. The paths $I \rightarrow D_r, D_r U_{\theta}$ for $0 \leq \theta \leq \pi$, and $D_s \pi$ for $r \leq s \leq 1$ connect the identity to the coordinate exchange. These paths are not asserted to have preserving forward increments.

Lemma F.2 (Reversible factors). *If $A, B \in S_n$ and $AB \in H_n$, then $A, B \in H_n$. Moreover every element of H_n^0 is a nonzero scalar times a tensor product of one-site invertible matrices, whereas no nontrivial coordinate permutation has this form.*

Proof. The identities $A^{-1} = B(AB)^{-1}$ and $B^{-1} = (AB)^{-1}A$ prove the first assertion. The group H_n is a closed subgroup of $\text{GL}(V_n)$ and hence a real Lie group. Its Lie algebra is $W_n \cap (-W_n)$. By Corollary 3.11, in half-plane coordinates this algebra is

$$\mathbb{C}I + \bigoplus_i \mathfrak{sl}_2(\mathbb{R})_i. \quad (306)$$

The fixed tensor Cayley transformation preserves products and coordinate permutations. A connected Lie group is generated by the exponentials of its Lie algebra. Exponentiating (306) and taking finite products therefore gives only scalar multiples of local tensor products. A nontrivial coordinate permutation cannot have this form: varying one input factor varies a different output factor, whereas an invertible local tensor product varies only the corresponding output factor. Equivalently, for a two-factor exchange, an identity $Ax \otimes By = cy \otimes x$ with fixed nonzero x and varying independent y would force the fixed vector Ax to be proportional to every y . $\square$

The exact-product assertion in Theorem F.1. If $e^{A_m} \cdots e^{A_1} \in H_n$, with every $A_j \in W_n$, Lemma F.2 makes every exponential factor a unit. For $0 \leq s \leq 1$, factor $e^{A_j} = e^{(1-s)A_j} e^{sA_j}$. Both factors preserve stability, so the same lemma gives $e^{sA_j} \in H_n$. Thus each factor belongs to H_n^0 , and so does their product. Conversely H_n^0 is generated by exponentials of its Lie algebra, which is contained in W_n . This proves $E_n \cap H_n = H_n^0$. $\square$

For the closure assertion we use Neeb's classical connected-unit theorem: if G is a connected Lie group and $T \subseteq G$ is a closed submonoid satisfying

$$T = \overline{\langle \exp L(T) \rangle}^G, \quad L(T) = \{A : \exp(tA) \in T \text{ for every } t \geq 0\},$$

then $T \cap T^{-1}$ is connected. This is [44, Definition II.1(3) and Proposition III.2, pp. 171, 178]; see also [51, Section 6.1].

The closure assertion in Theorem F.1. The ambient real Lie group $G = \mathrm{GL}_{2n}(\mathbb{C})$ is connected. By definition T_n is a closed submonoid and $T_n \subseteq S_n$. Every $A \in W_n$ generates a one-parameter semigroup in E_n , while $L(T_n) \subseteq L(S_n) = W_n$. Hence

$$L(T_n) = W_n, \quad T_n = \overline{\langle \exp L(T_n) \rangle}^G.$$

Neeb's theorem therefore makes $T_n \cap T_n^{-1}$ a connected subgroup of H_n , so it is contained in H_n^0 . The reverse inclusion follows because the exponentials of both signs of (306) belong to T_n and generate H_n^0 .

If a nontrivial coordinate permutation π belonged to T_n , its finite order would imply $\pi^{-1} \in T_n$. It would therefore be a unit of T_n , and hence belong to H_n^0 , contrary to Lemma F.2. This proves (302). Relative closure in the general linear group and ordinary matrix-norm closure agree at the invertible point π , proving the neighborhood statement. $\square$

Corollary F.3 (Exact local divisibility). *Call $M \in S_n$ exactly locally divisible if, for every identity neighborhood U in $\mathrm{GL}(V_n)$, it is a finite product of members of $S_n \cap U$. The exactly locally divisible units of S_n are precisely H_n^0 . In addition, if a continuous path $M : [0, 1] \rightarrow S_n$ starts at I and satisfies $M(t)M(s)^{-1} \in S_n$ for $s \leq t$, then a unit endpoint must belong to H_n^0 .*

Proof. The identity component of a Lie group is open, so choose U with $H_n \cap U \subset H_n^0$. Any exact factorization of a unit into $S_n \cap U$ consists entirely of units by Lemma F.2, hence has endpoint in H_n^0 . Conversely, subdividing a finite exponential representation of a member of H_n^0 puts all its factors in any prescribed identity neighborhood. For the path assertion, write $M(1) = [M(1)M(t)^{-1}]M(t)$. A unit endpoint makes $M(t)$ a unit for every t , so the continuous path stays in H_n and ends in H_n^0 . $\square$

Remark F.4. The identity for the closure in (301) is $T_n \cap T_n^{-1} = H_n^0$. It does not assert $T_n \cap H_n = H_n^0$ for arbitrary units of the larger semigroup; finite order supplies the inverse inclusion needed for permutations. The result concerns flows of the infinitesimal cone. Static apolar maps may lie on other branches: for $F(s, t) = s - t$, $B_F = \pi - I$, so $I + B_F = \pi$, whereas

$$e^{tB_F} = \frac{1 + e^{-2t}}{2}I + \frac{1 - e^{-2t}}{2}\pi.$$

Allowing such a static permutation as an additional primitive changes the factorization question. No description of the remaining noninvertible boundary or the nonunit part of T_n is claimed.

APPENDIX G. QUANTITATIVE RIGIDITY AND CORRELATED TESTS

The finite-test rigidity of Theorem 7.3 is an exact statement. Its quantitative behaviour depends on the allowed inputs. We prove that products of two GHZ states detect precisely the failure of a diagonal Hamiltonian to have additive energies. The approximate-additivity theorem of Kalton and Roberts then gives a dimension-independent comparison with distance from one-site dynamics. In contrast, even all product Lee–Yang inputs admit no such comparison.

Throughout this section, a density matrix acts on $\mathcal{H}_n = (\mathbb{C}^2)^{\otimes n}$, and

$$F_X(z, w) = \sum_{x, y \in \{0, 1\}^n} X_{xy} z^x w^y,$$

$$\mathcal{Y}_n = \{\rho \geq 0 : \text{Tr } \rho = 1, F_\rho \neq 0 \text{ on } \mathbb{D}^{2n}\}.$$

The condition is zero-freeness, not merely that F_ρ is a nonzero polynomial. We use the unhalved trace norm $\|\cdot\|_1$ and diamond norm $\|\cdot\|_\diamond$, and write $\text{dist}_1(X, \mathcal{Y}_n) = \inf_{\rho \in \mathcal{Y}_n} \|X - \rho\|_1$. Let LocHP_n be the real vector space of maps

$$\mathcal{M} = \sum_{i=1}^n \mathcal{M}_i \otimes \text{Id}_{\{1, \dots, n\} \setminus \{i\}},$$

where every $\mathcal{M}_i$ is complex linear and Hermiticity preserving. Let LocGKLS_n be the cone obtained by requiring each $\mathcal{M}_i$ to be a one-qubit GKLS generator. In particular, local Hamiltonian generators belong to both classes.

G.1. The diagonal quotient norm. Identify a subset $S \subseteq [n]$ with the computational basis vector whose occupied sites are exactly S . For a real function h on $2^{[n]}$, put

$$H = \text{diag}(h(S) : S \subseteq [n]), \quad \mathcal{L}_H(X) = -i[H, X],$$

$$\text{osc } h = \max_S h(S) - \min_S h(S).$$

Proposition G.1. *For every diagonal Hermitian H ,*

$$\begin{aligned} d(H) &:= \text{dist}_\diamond(\mathcal{L}_H, \text{LocHP}_n) \\ &= \text{dist}_\diamond(\mathcal{L}_H, \text{LocGKLS}_n) \\ &= \min_{b \in \mathbb{R}^n} \text{osc} \left(h(S) - \sum_{i \in S} b_i \right). \end{aligned} \tag{307}$$

If $h(S) = h([n] \setminus S)$ for all S , then $d(H) = \text{osc } h = \|\mathcal{L}_H\|_\diamond$.

Proof. Fix $\mathcal{M} \in \text{LocHP}_n$ and set $E_{xy} = |x\rangle\langle y|$. The coefficient of E_{xy} in $\mathcal{M}(E_{xy})$ equals

$$\langle x | \mathcal{M}(E_{xy}) | y \rangle = \sum_i \langle x_i | \mathcal{M}_i(E_{x_i y_i}) | y_i \rangle.$$

If $x_i = y_i$, the summand is real. The summands for $(x_i, y_i) = (0, 1)$ and $(1, 0)$ are conjugates. Thus, for a vector $b \in \mathbb{R}^n$ independent of x, y , its imaginary part is

$b \cdot (y - x)$. Since $\|E_{xy}\|_1 = 1$ and every matrix entry is bounded by the trace norm,

$$\|\mathcal{L}_H - \mathcal{M}\|_\diamond \geq \max_{x,y} |(h(x) - b \cdot x) - (h(y) - b \cdot y)| = \text{osc}(h - b \cdot x).$$

Here and below $h(x)$ is the subset notation written in occupation coordinates.

For a Hermitian matrix D one has

$$\| -i[D, \cdot] \|_\diamond = \lambda_{\max}(D) - \lambda_{\min}(D). \quad (308)$$

Indeed, subtract the midpoint of the spectral interval from D . The trace-norm ideal inequality, also after adjoining an identity ancilla, gives the upper bound $2\|D\|_{\text{op}}$ for the resulting commutator. The matrix unit between eigenvectors at the two spectral endpoints gives the reverse bound. Apply this identity to $D = H - H_b$, where $H_b = \sum_i b_i |1\rangle\langle 1|_i$. The local Hamiltonian generator $\mathcal{L}_{H_b}$ attains the corresponding oscillation. Because $\text{LocGKLS}_n \subseteq \text{LocHP}_n$, the two distance infima agree with the infimum of these oscillations. It is a minimum: the objective is continuous and

$$\text{osc}(h - b \cdot x) \geq \text{osc}(b \cdot x) - \text{osc } h = \|b\|_{\ell^1} - \text{osc } h.$$

Suppose now that h is invariant under complementation. Replacing x by $1 - x$ shows that $\text{osc}(h - b \cdot x) = \text{osc}(h + b \cdot x)$. The oscillation is a convex function of its argument, so

$$\text{osc } h \leq \frac{1}{2} \text{osc}(h - b \cdot x) + \frac{1}{2} \text{osc}(h + b \cdot x) = \text{osc}(h - b \cdot x).$$

Taking $b = 0$ proves the last assertion. $\square$

G.2. Two correlated blocks. For disjoint nonempty $A, B \subseteq [n]$, define

$$\begin{aligned} |\psi_{A,B}\rangle &= \frac{|0_A\rangle + |1_A\rangle}{\sqrt{2}} \otimes \frac{|0_B\rangle + |1_B\rangle}{\sqrt{2}} \otimes |0_{[n]\setminus(A \cup B)}\rangle, \\ \rho_{A,B} &= |\psi_{A,B}\rangle\langle\psi_{A,B}|. \end{aligned} \quad (309)$$

The pure-state coefficient polynomial is $\frac{1}{2}(1 + \prod_{i \in A} z_i)(1 + \prod_{i \in B} z_i)$. It is zero-free on $\mathbb{D}^n$, and hence $\rho_{A,B} \in \mathcal{Y}_n$. Set

$$\begin{aligned} \Delta_{A,B}(H) &= h(A \cup B) - h(A) - h(B) + h(\emptyset), \\ \Delta(H) &= \max_{A \cap B = \emptyset} |\Delta_{A,B}(H)|. \end{aligned}$$

Theorem G.2 (Correlated diagonal rigidity). *Let $n \geq 2$ and let H be a diagonal Hermitian Hamiltonian. For every pair of disjoint nonempty subsets A, B the following limit exists:*

$$\lim_{t \downarrow 0} \frac{\text{dist}_1(e^{-itH} \rho_{A,B} e^{itH}, \mathcal{Y}_n)}{t} = \frac{|\Delta_{A,B}(H)|}{2}. \quad (310)$$

If $\delta(H)$ denotes the maximum of these limits, then

$$\delta(H) = \frac{\Delta(H)}{2}, \quad \delta(H) \leq d(H) \leq 4K \delta(H), \quad (311)$$

where K is any universal constant in the real Kalton–Roberts approximate-additivity theorem. In particular, one may take $K = 45$.

We first establish the polynomial and boundary facts used in the proof.

Lemma G.3 (Block compression). *Let $f(u_1, \dots, u_k, v)$ be multiaffine and zero-free on the open unit polydisk. If $a(v)$ and $d(v)$ are its constant and full-degree coefficients in the u variables, then $a(v) + d(v)s$ is zero-free on the open unit polydisk in (s, v) .*

Consequently, if $V : \mathbb{C}^2 \otimes \mathbb{C}^2 \rightarrow \mathcal{H}_n$ is the isometry

$$V|a, b\rangle = |a_A, b_B, 0_{[n] \setminus (A \cup B)}\rangle,$$

then $F_{V^ \sigma V}$ is zero-free on $\mathbb{D}^4$ for every $\sigma \in \mathcal{Y}_n$.*

Proof. Fix the remaining variables v in their polydisk. The polynomial $g(u) = f(u, \dots, u, v)$ has no zero in $\mathbb{D}$, and $a(v) = g(0) \neq 0$. If $\deg g = k$, all its roots have modulus at least one, and the product-of-roots identity gives $|d(v)| \leq |a(v)|$. If $\deg g < k$, then $d(v) = 0$. Thus $a(v) + d(v)s \neq 0$ for $|s| < 1$. This proves the first assertion; it is the unit-disk form of the Asano coefficient contraction [6].

For the second assertion, first set all row and column variables outside $A \cup B$ to zero. Apply the first assertion successively to the row variables in A , the row variables in B , and the corresponding two groups of column variables. The resulting coefficient polynomial is exactly $F_{V^* \sigma V}$. No normalization or zero-polynomial exception occurs in this argument. $\square$

The compression in Lemma G.3 is completely positive and trace nonincreasing, but need not preserve trace. We use its unnormalized output. For every Hermitian X , $\|V^* X V\|_1 \leq \|X\|_1$, as follows either from trace-norm duality or from the positive and negative parts of X .

Lemma G.4 (A two-qubit boundary contact). *Write $\rho_0 = |++\rangle\langle++|$, where $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, and put*

$$\begin{aligned} u &= |00\rangle + |10\rangle + |01\rangle + |11\rangle, \\ v &= |00\rangle - |10\rangle - |01\rangle + |11\rangle, \quad \ell(X) = \Im\langle v|X|u\rangle. \end{aligned}$$

If $t_j \downarrow 0$, X_j is Hermitian, F_{X_j} is zero-free on $\mathbb{D}^4$, and $\|X_j - \rho_0\|_1 = O(t_j)$, then $\ell(X_j) = o(t_j)$. Moreover,

$$|\ell(X)| \leq 2\|X\|_1 \quad \text{for every Hermitian } X. \quad (312)$$

Proof. The Hermitian observable representing ℓ is

$$R = \frac{|u\rangle\langle v| - |v\rangle\langle u|}{2i}.$$

The vectors u, v are orthogonal and both have norm two. On their span R has eigenvalues 2, -2, and it vanishes on the orthogonal complement. Thus $\|R\|_{\text{op}} = 2$, proving (312).

Consider

$$g_j(s) = F_{X_j}(s-1, s-1, 1, 1) = a_{2,j}s^2 + a_{1,j}s + a_{0,j}.$$

For $X_j = \rho_0$ this polynomial is s^2 . Continuity of its coefficients gives $a_{2,j} = 1 + O(t_j)$ and $a_{1,j}, a_{0,j} = O(t_j)$. The column variables have been specialized to boundary points, so we justify their use. For fixed j , the polynomials $F_{X_j}(s-1, s-1, r, r)$, with $0 < r < 1$, are zero-free on $|s-1| < 1$. Hurwitz's theorem shows that their limit as $r \uparrow 1$ is zero-free there or identically zero. For all sufficiently large j , $a_{2,j} \neq 0$, which excludes the latter alternative.

Let $r_{1,j}, r_{2,j}$ be the two roots of g_j , counted with multiplicity. Vieta's formulas give their sum and product as $O(t_j)$; the quadratic formula then gives $|r_{k,j}| = O(\sqrt{t_j})$. Because $|r_{k,j} - 1| \geq 1$,

$$\Re r_{k,j} \leq \frac{1}{2}|r_{k,j}|^2 = O(t_j).$$

Their real parts have sum $O(t_j)$, so the same real parts are bounded below by $-O(t_j)$. Consequently $\Re r_{k,j} = O(t_j)$ and $\Im r_{k,j} = O(\sqrt{t_j})$. It follows that

$$\Im(r_{1,j}r_{2,j}) = O(t_j^{3/2}), \quad \Im a_{0,j} = O(t_j^{3/2}) + O(t_j^2) = o(t_j).$$

Finally $a_{0,j} = F_{X_j}(-1, -1, 1, 1) = \langle v | X_j | u \rangle$. $\square$

Proof of Theorem G.2. Fix A, B , let V be as in Lemma G.3, and abbreviate $\Delta = \Delta_{A,B}(H)$ and $\rho(t) = e^{-itH} \rho_{A,B} e^{itH}$. The image of V is invariant under H , and $V^* \rho(t) V = e^{-itD} \rho_0 e^{itD}$, where the logical diagonal Hamiltonian D has energies $h(\emptyset), h(A), h(B), h(A \cup B)$. Since $\rho_0 = |u\rangle\langle u|/4$ and $\langle v | u \rangle = 0$, differentiation gives

$$\ell(V^* \rho(t) V) = -\Delta t + O(t^2). \quad (313)$$

Let $t_j \downarrow 0$ be arbitrary, and choose $\sigma_j \in \mathcal{Y}_n$ with

$$\|\rho(t_j) - \sigma_j\|_1 \leq \text{dist}_1(\rho(t_j), \mathcal{Y}_n) + t_j^2.$$

Because $\rho_{A,B} \in \mathcal{Y}_n$ and $\rho(t_j) = \rho_{A,B} + O(t_j)$, both the distance on the right and $\|\sigma_j - \rho_{A,B}\|_1$ are $O(t_j)$. Lemma G.3 shows that $X_j = V^* \sigma_j V$ has a zero-free coefficient polynomial. It is Hermitian and $X_j = \rho_0 + O(t_j)$, so Lemma G.4 applies without rescaling its trace. Equations (312) and (313) imply

$$2\|\rho(t_j) - \sigma_j\|_1 \geq |\ell(V^* \rho(t_j) V - X_j)| = |\Delta|t_j + o(t_j).$$

As the sequence was arbitrary, this proves the lower bound $|\Delta|/2$ for the liminf in (310).

For a matching upper bound, subtract $(\Delta/4)Z_1Z_2$ from D , where $Z = \text{diag}(1, -1)$. The remaining logical energy is affine in the two occupation bits: its mixed difference is zero. Its evolution of the logical vector $|++\rangle$, embedded by V , is therefore a product of two GHZ states with unit-modulus relative phases and a vacuum on the remaining sites. This comparison state belongs to $\mathcal{Y}_n$. Since all four logical configurations have equal probability and the two values of Z_1Z_2 occur equally often, the inner product between the actual and comparison pure vectors is $\cos(t\Delta/4)$. Their trace distance is

$$2|\sin(t\Delta/4)| \leq t|\Delta|/2 \quad (t \geq 0).$$

This proves (310). Empty subsets have mixed difference zero, and the family of pairs is finite, so $\delta(H) = \Delta(H)/2$.

We now invoke the precise inherited input. The Kalton–Roberts theorem [34, Theorem 4.1] supplies a universal $K \leq 45$ with the following property: if a real set function f on a set algebra satisfies $f(\emptyset) = 0$ and

$$|f(A \cup B) - f(A) - f(B)| \leq \eta \quad (A \cap B = \emptyset),$$

then there is an additive real set function μ such that $\sup_S |f(S) - \mu(S)| \leq K\eta$. Apply it on $2^{[n]}$ to $f(S) = h(S) - h(\emptyset)$ and $\eta = \Delta(H)$. On this finite algebra, $\mu(S) = \sum_{i \in S} b_i$ for some real b_i , and hence

$$\text{osc}(h - b \cdot x) \leq 2K\Delta(H).$$

If $\Delta(H) = 0$, the same conclusion follows directly from finite additivity. Proposition G.1 now yields $d(H) \leq 2K\Delta(H) = 4K\delta(H)$.

Finally choose a minimizer b in (307). The flow of H_b consists of independent diagonal one-qubit unitaries. It rotates each polynomial variable by a unit complex number and thus preserves $\mathcal{Y}_n$. The Duhamel formula and unitary trace-norm invariance imply, for every density matrix ρ and $t \geq 0$,

$$\|e^{t\mathcal{L}_H}\rho - e^{t\mathcal{L}_{H_b}}\rho\|_1 \leq t\|\mathcal{L}_H - \mathcal{L}_{H_b}\|_\diamond = td(H).$$

Apply this estimate to each $\rho_{A,B}$, divide by t , and use (310). Taking the maximum gives $\delta(H) \leq d(H)$. $\square$

Remark G.5. The universal constant in (311) is inherited from Kalton–Roberts. The bridge to that theorem is the exact root-contact identity (310). The probes may contain arbitrarily large blocks, their family is exponential in n , and the defect uses distance to the full set $\mathcal{Y}_n$. No efficient certification procedure or corresponding statement for non-diagonal Hamiltonians is asserted.

G.3. Product inputs do not give uniform robustness. Let $\mathcal{P}_n \subseteq \mathcal{Y}_n$ be the set of all product density matrices in $\mathcal{Y}_n$, including mixed products, and put

$$\delta_{\text{prod}}(\mathcal{L}) = \sup_{\rho \in \mathcal{P}_n} \limsup_{t \downarrow 0} \frac{\text{dist}_1(e^{t\mathcal{L}}\rho, \mathcal{Y}_n)}{t}.$$

We give two different mechanisms by which this defect can be small. For use with mixed states, recall the elementary estimate

$$\|e^{-itH}\rho e^{itH} - \rho\|_1 \leq 2|t|\sqrt{\text{Tr}(\rho H^2) - (\text{Tr}(\rho H))^2}. \quad (314)$$

To prove it, purify ρ to a unit vector Ψ and evolve the purification by $H \otimes I$. For a pure-state projection $P = |\Psi\rangle\langle\Psi|$, the two possibly nonzero eigenvalues of $-i[H \otimes I, P]$ are $\pm\sqrt{\text{Var}_\rho H}$. Its trace norm is therefore twice this square root. The variance is constant along the unitary orbit, so integration gives the same bound for purified states. Partial trace gives (314).

Proposition G.6 (A nearest-neighbour obstruction). *On a path with $m \geq 1$ edges and $n = m + 1$ vertices, let*

$$H_m = \frac{1}{2m} \sum_{j=1}^m Z_j Z_{j+1}, \quad \mathcal{L}_m = -i[H_m, \cdot].$$

Then

$$\begin{aligned} \|\mathcal{L}_m\|_\diamond &= d(H_m) = 1, \\ \text{dist}_1(e^{t\mathcal{L}_m}\rho, \mathcal{Y}_n) &\leq \sqrt{\frac{3}{m}}|t| \quad (\rho \in \mathcal{P}_n, t \in \mathbb{R}). \end{aligned}$$

For the equatorial pure product $|\psi_\theta\rangle = \bigotimes_{j=1}^n (|0\rangle + e^{i\theta_j}|1\rangle)/\sqrt{2}$ and its density ρ_θ , the sharper formula is

$$\|e^{t\mathcal{L}_m}\rho_\theta - \rho_\theta\|_1 = 2\sqrt{1 - \cos(t/(2m))^{2m}} \leq \frac{|t|}{\sqrt{m}}. \quad (315)$$

Proof. The diagonal energy is invariant under simultaneous complementation. Equal computational spins have energy $1/2$, alternating spins have energy $-1/2$, and these are the extrema. Thus Proposition G.1 gives the two norm identities.

For a product density, the computational spins are independent signs. Write $W_j = Z_j Z_{j+1}$ for their edge products, regarded as random variables in this diagonal distribution. Disjoint edges have zero covariance, each variance is at most one, and each adjacent covariance has absolute value at most one by Cauchy–Schwarz. Hence

$$\text{Var} \left(\sum_{j=1}^m W_j \right) \leq m + 2(m-1) \leq 3m.$$

Equation (314) gives the stated mixed-state bound. The initial state itself belongs to $\mathcal{Y}_n$ and is an admissible comparison state in the distance.

For ψ_θ , the computational distribution is uniform. The root sign and the m edge products determine all vertex signs bijectively, so the edge products are independent uniform signs. It follows that

$$\langle \psi_\theta | e^{-itH_m} | \psi_\theta \rangle = \cos(t/(2m))^m.$$

The trace distance of unit pure states ψ, ϕ is $2\sqrt{1 - |\langle \psi, \phi \rangle|^2}$; this follows by restricting the difference of their projections to their two-dimensional span. It proves the equality in (315). Independence gives $\text{Var}_{\rho_\theta} H_m = 1/(4m)$, so (314) proves its last inequality. Finally, $\rho_\theta \in \mathcal{Y}_n$, since its coefficient polynomial is a product of linear factors with roots on the unit circle. $\square$

Lemma G.7 (One-qubit physical states). *A one-qubit density matrix $\tau = \begin{pmatrix} a & \bar{b} \\ b & d \end{pmatrix}$ belongs to $\mathcal{Y}_1$ if and only if $a \geq d$.*

Proof. Its polynomial is $f(z, w) = a + bz + \bar{b}w + dzw$. Suppose $a \geq d$. Positivity gives $|b|^2 \leq ad \leq a^2$, and $a > 0$ because $a + d = 1$. The denominator of

$$q(z) = \frac{\bar{b} + dz}{a + bz}$$

has no zero in $\mathbb{D}$. For $|z| = 1$,

$$|a + bz|^2 - |\bar{b} + dz|^2 = (a - d)(a + d + 2\Re(bz)) \geq 0.$$

If $|b| < a$, the maximum principle gives $|q(z)| \leq 1$ in $\mathbb{D}$. The remaining case $|b| = a$ forces $a = d = |b|$, and cancellation makes q a constant of modulus one. Thus in either case $f(z, w) = (a + bz)(1 + wq(z))$ is nonzero for $|z|, |w| < 1$.

Conversely, suppose $a < d$. Then $|b| \leq \sqrt{ad} < (a + d)/2$, so the displayed boundary difference is strictly negative on the unit circle. By continuity, some z inside the circle satisfies $|a + bz| < |\bar{b} + dz|$. The denominator in $w = -(a + bz)/(\bar{b} + dz)$ is nonzero and $|w| < 1$; this gives an interior zero of f . $\square$

Proposition G.8 (An exponential product-input obstruction). *For $n \geq 2$, put $P_n = |1^n\rangle\langle 1^n|$ and $\mathcal{K}_n = -i[P_n, \cdot]$. Then*

$$\|\mathcal{K}_n\|_\diamond = 1, \quad d(P_n) = \frac{n-1}{n}. \quad (316)$$

For every $\rho \in \mathcal{P}_n$ and every real t ,

$$\text{dist}_1(e^{t\mathcal{K}_n}\rho, \mathcal{Y}_n) \leq \|e^{t\mathcal{K}_n}\rho - \rho\|_1 \leq 2^{1-n/2}|t|. \quad (317)$$

For equatorial pure products, the middle expression is exactly

$$4\sqrt{2^{-n}(1 - 2^{-n})} |\sin(t/2)|.$$

Proof. The commutator norm is one by (308). To evaluate (307), average b over coordinate permutations. Convexity of oscillation and permutation symmetry of $h(x) = \mathbf{1}_{\{x=1^n\}}$ show that this cannot increase the objective. We may therefore take $b_i = c$ for all i . The residual energies are

$$\{-ck : 0 \leq k \leq n-1\} \cup \{1-cn\}.$$

If $c \leq 1/n$, the last value minus the value at weight $n-1$ is $1-c \geq (n-1)/n$. If $c \geq 1/n$, the difference between weights zero and $n-1$ has absolute value $c(n-1) \geq (n-1)/n$. For $c = 1/n$, all residual energies lie in $[-(n-1)/n, 0]$ and both endpoints occur. This proves (316).

The coefficient polynomial of a product density is the product of the individual coefficient polynomials in disjoint variables. Thus it is zero-free precisely when all factors are zero-free. Lemma G.7 implies that every occupied population is at most $1/2$. Consequently $p = \text{Tr}(P_n \rho) \leq 2^{-n}$. For a purification Ψ of ρ ,

$$\langle \Psi | (e^{-itP_n} \otimes I) | \Psi \rangle = 1 - p + pe^{-it}.$$

The pure-state trace-distance formula and partial-trace contractivity therefore give

$$\|e^{tK_n} \rho - \rho\|_1 \leq 4\sqrt{p(1-p)} |\sin(t/2)| \leq 2\sqrt{p} |t|.$$

This proves (317), using the initial LY state as comparison. For an equatorial pure product $p = 2^{-n}$, and no partial trace is needed, giving the asserted equality. $\square$

Corollary G.9. *There is no function $\omega : [0, \infty) \rightarrow [0, \infty)$ with $\omega(0) = 0$ and $\lim_{s \downarrow 0} \omega(s) = 0$, independent of n , such that*

$$\text{dist}_\diamond(\mathcal{L}, \text{LocGKLS}_n) \leq \omega(\delta_{\text{prod}}(\mathcal{L}))$$

for every Hamiltonian generator $\mathcal{L}$ of diamond norm one. This remains false if the Hamiltonian is required to be a sum of nearest-neighbour two-site terms on a path.

Proof. Proposition G.6 gives a sequence with distance one and $0 \leq \delta_{\text{prod}}(\mathcal{L}_m) \leq \sqrt{3/m} \rightarrow 0$, contradicting the required behaviour of ω . Proposition G.8 separately gives exponentially small product defect and distance tending to one. $\square$

The normalization of the path interactions is by their total strength; no positive lower bound on each edge coefficient is assumed. These examples concern a global diamond norm and do not obstruct estimation of individual interaction coefficients in other access models. Nor do they contradict Theorem G.2: correlated inputs can detect what all product inputs detect only weakly. For instance, the GHZ state on all n sites belongs to $\mathcal{Y}_n$ and has $\text{Tr}(P_n \rho) = 1/2$, so the product-population bound used above does not extend to arbitrary physical Lee–Yang states.

APPENDIX H. AN EXACT COUNTEREXAMPLE TO A STRONGER GROUND-STATE RADIUS

The qualitative strict radius in Theorem 6.6 cannot in general be replaced by the proposed explicit radius for deformed EPR ground states. We address the stronger clause of Conjecture 1 in [61, Section 5.4], with its stated unweighted-graph hypothesis.

Theorem H.1. *Let G have vertices $\{0, 1, 2, 3, 4, 5\}$ and edges*

$$\{01, 02, 03, 14, 15\}.$$

Put

$$H = - \sum_{ij \in E(G)} (|00\rangle + \frac{1}{2}|11\rangle)(\langle 00| + \frac{1}{2}\langle 11|)_{ij}.$$

The ground state ψ is unique up to scalar. Its multiaffine polynomial $f_\psi(z) = \sum_b \psi_b z^b$ has a zero satisfying

$$|z_i| < \frac{499}{500}\sqrt{2} \quad (0 \leq i \leq 5).$$

Thus the proposed radius $s^{-1/2}$ fails at $s = 1/2$ on an unweighted six-vertex tree.

Proof. We give rational spectral and zero certificates. All inequalities below are verified by the finite procedure at the end of this appendix, using only integer arithmetic and exact fractions.

Write $K = -H$. Index the computational basis by integers $b \in \{0, \dots, 63\}$, with bit b_i at vertex i . For each edge ij and each input with $b_i = b_j = a$, changing both bits to $c \in \{0, 1\}$ contributes $2^{-(a+c)}$ to the corresponding entry of K . All other entries from that edge are zero. This specifies the symmetric rational matrix K completely.

Consider $M = 37I - 8K$. The charge

$$Q(b) = b_0 + b_4 + b_5 - b_1 - b_2 - b_3$$

is conserved. In charge order $-3, -2, \dots, 3$, the block dimensions are

$$1, 6, 15, 20, 15, 6, 1.$$

Order each block by increasing integer b . Successive rational symmetric elimination has pivot signs as follows:

Q	-3	-2	-1	0	1	2	3
negative pivots	0	0	0	1	0	0	0
positive pivots	1	6	15	19	15	6	1.

No pivot is zero; the negative pivot is the first in the charge-zero block. For clarity, the recurrence at step i is

$$p_i = M_{ii}^{(i)}, \quad M_{jk}^{(i+1)} = M_{jk}^{(i)} - M_{ji}^{(i)} M_{ik}^{(i)} / p_i \quad (j, k > i).$$

It proves the stated inertia by exact rational comparison, rather than taking the pivot signs as external data. Sylvester's law of inertia shows that K has exactly one eigenvalue above $37/8$. Since $K_{00} = 5$, its largest eigenvalue is at least 5 and is simple. Every other eigenvalue is below $37/8$.

Define a rational vector v as follows. For a bit string use the key

$$(a, b, i, j) = (b_0, b_1, b_2 + b_3, b_4 + b_5).$$

The amplitude for each individual bit string is the following numerator divided by 10^{15} ; a missing key has amplitude zero.

Key	Numerator
(0, 0, 0, 0)	1000000000000000
(0, 0, 1, 1)	8942179087130
(0, 0, 2, 2)	136748346620
(0, 1, 0, 1)	171232824560490
(0, 1, 1, 2)	2452435031309
(1, 0, 1, 0)	171232824560490
(1, 0, 2, 1)	2452435031309
(1, 1, 0, 0)	97255575326164
(1, 1, 1, 1)	37858301388951
(1, 1, 2, 2)	1200949534669

These are computational-basis amplitudes, not normalized orbit-basis coordinates. Put

$$\hat{\lambda} = \frac{539109343678406}{10^{14}}.$$

Direct rational summation gives

$$\|(K - \hat{\lambda}I)v\|_2^2 < 10^{-24}, \quad \hat{\lambda} - 37/8 > 1/2.$$

Let v_g be the orthogonal projection onto the largest eigenspace of K . Expanding in an orthonormal eigenbasis and using the preceding separation gives

$$\|v - v_g\|_2 \leq 2\|(K - \hat{\lambda}I)v\|_2 < 2 \cdot 10^{-12}.$$

Since $v_{000000} = 1$, v_g is nonzero, hence is a scalar multiple of the true ground state. Both v and v_g have charge zero; in particular they have even parity.

For an even-parity vector w put

$$\tilde{f}_w(x) = f_w(\sqrt{2}x) = \sum_b 2^{|b|/2} w_b x^b.$$

The coefficients of $\tilde{f}_v$ are rational. Set

$$c = \frac{175105 + 981180i}{10^6}, \quad \ell = \frac{-379834 + 921467i}{10^6}.$$

Exact squaring gives $|c|, |\ell| < 499/500$. Fix $x_0 = x_1 = c$ and $x_2 = x_3 = x_4 = \ell$. Multiaffinity yields

$$\tilde{f}_v(c, c, \ell, \ell, \ell, x_5) = A_v + B_v x_5, \quad |A_v|^2 < (85/1000)^2, \quad |B_v|^2 > (852/10000)^2.$$

The displayed inequalities are finite sums over the amplitude table. Each coefficient A or B contains at most 32 terms. A fixed-variable monomial has modulus at most one, and its even tilt factor is at most eight. Therefore

$$|A_{v_g} - A_v|, |B_{v_g} - B_v| < 32 \cdot 8 \cdot 2 \cdot 10^{-12} < 10^{-8}.$$

In particular $B_{v_g} \neq 0$, and

$$x_5 = -A_{v_g}/B_{v_g}, \quad |x_5| < \frac{85/1000 + 10^{-8}}{852/10000 - 10^{-8}} < 499/500.$$

The last inequality is rational. This gives a zero of $\tilde{f}_{v_g}$ with all coordinates of modulus below 499/500. Rescaling by $\sqrt{2}$ proves the theorem. $\square$

The certificate locates a zero of the exact ground-state polynomial. It does not treat a rounded eigenvector as an exact eigenvector. The existence of some radius greater than one, proved above by positive-field strictification, is compatible with this example.

Complete finite verification procedure. The following standard-library Python procedure reproduces all rational assertions used in the proof. The operations implementing a Gaussian rational number are written explicitly. Floating-point arithmetic and numerical eigenvalue computations are not used. The same source is supplied as `certificates/epr_six_vertex_certificate.py`; a second, independent rational certificate is included with the source package.

```

"""Exact standard-library certificate: a zero strictly inside radius sqrt(2).

No floating point or numerical eigensolver is used in any assertion.
K is the negative EPR Hamiltonian at s=1/2 on edges 01,02,03,14,15.
"""

from fractions import Fraction as F

EDGES = [(0, 1), (0, 2), (0, 3), (1, 4), (1, 5)]
N = 64
K = [[F(0) for _ in range(N)] for _ in range(N)]
for b in range(N):
    for i, j in EDGES:
        bi, bj = (b >> i) & 1, (b >> j) & 1
        if bi == bj:
            for out in (0, 1):
                c = (b & ~(1 << i) & ~(1 << j)) | (out << i) | (out << j)
                K[c][b] += F(1, 2) ** (bi + out)
assert all(K[i][j] == K[j][i] for i in range(N) for j in range(N))

# Exact inertia proves that all but the largest eigenvalue are below 37/8.
def charge(b):
    return sum((b >> j) & 1 for j in (0, 4, 5)) - sum((b >> j) & 1 for j in (1, 2, 3))

assert all(not K[i][j] or charge(i) == charge(j) for i in range(N) for j in range(N))

negative = 0
pivot_signs = []
for q in range(-3, 4):
    sector = [b for b in range(N) if charge(b) == q]
    A = [[F(37 if i == j else 0) - 8 * K[i][j] for j in sector] for i in sector]
    signs = []
    for i in range(len(sector)):
        pivot = A[i][i]
        assert pivot != 0
        negative += pivot < 0
        signs.append('+' if pivot > 0 else '-')
        for j in range(i + 1, len(sector)):
            for k in range(j, len(sector)):
                A[k][j] -= A[j][i] * A[k][i] / pivot
                A[j][k] = A[k][j]
        pivot_signs.append((q, ''.join(signs)))
assert negative == 1
assert K[0][0] == 5

states = [(0,0,0,0),(0,0,1,1),(0,0,2,2),(0,1,0,1),(0,1,1,2),
           (1,0,1,0),(1,0,2,1),(1,1,0,0),(1,1,1,1),(1,1,2,2)]
numerators = [1000000000000000,8942179087130,136748346620,171232824560490,
               2452435031309,171232824560490,2452435031309,97255575326164,
               37858301388951,1200949534669]
amps = {state:F(num,10**15) for state,num in zip(states,numerators)}
v = []
for b in range(N):
    state = ((b >> 0)&1,(b >> 1)&1,((b >> 2)&1)+((b >> 3)&1),
            ((b >> 4)&1)+((b >> 5)&1))
    v.append(amps.get(state,F(0)))
assert v[0] == 1

```

```

lam = F(539109343678406,10**14)
r = [sum(K[i][j] * v[j] for j in range(N)) - lam*v[i] for i in range(N)]
assert sum(x*x for x in r) < F(1,10**24)
assert lam - F(37,8) > F(1,2)

# Therefore the orthogonal projection v_g onto the true top eigenspace
# obeys ||v-v_g||_2 < 2*10^-12. In particular v_g is nonzero.

def add(x,y): return x[0]+y[0],x[1]+y[1]
def mul(x,y): return x[0]*y[0]-x[1]*y[1],x[0]*y[1]+x[1]*y[0]
def scale(c,x): return c*x[0],c*x[1]
def norm2(x): return x[0]*x[0]+x[1]*x[1]

zc=(F(175105,10**6),F(981180,10**6))
zl=(F(-379834,10**6),F(921467,10**6))
certified_radius=F(499,500)
assert norm2(zc)<certified_radius**2 and norm2(zl)<certified_radius**2
fixed=[zc,zc,zl,zl,zl]
A=(F(0),F(0));B=(F(0),F(0))
for b in range(N):
    if not v[b]: continue
    degree=b.bit_count()
    assert degree%2==0
    term=(v[b]*2**(degree//2),F(0))
    for j in range(5):
        if (b>>j)&1: term=mul(term,fixed[j])
    if (b>>5)&1:B=add(B,term)
    else:A=add(A,term)

# These wide rational bounds leave much more room than the eigenvector error.
assert norm2(A) < F(85,1000)**2
assert norm2(B) > F(852,10000)**2

# At the five fixed coordinates each monomial has modulus<1, and tilting
# by sqrt(2) multiplies each even coefficient by at most8. Each linear
# coefficient A or B involves at most32 terms. Thus replacing v by v_g
# changes A and B by less than 32*8*2e-12 < 1e-8.
err=F(1,10**8)
assert 32*8*F(2,10**12) < err
assert F(85,1000)+err < F(852,10000)-err
assert F(85,1000)+err < certified_radius*(F(852,10000)-err)

print('PASS: exact inertia has one negative pivot:',pivot_signs)
print('PASS: squared eigenvector residual < 10^-24')
print('PASS: |A_v|<85/1000 and |B_v|>852/10000')
print('PASS: true ground polynomial has all |z_i|<499/500 after sqrt(2) scaling')

```

DISCLOSURE OF ARTIFICIAL INTELLIGENCE USE

Most of the mathematical content, proofs, computational certificates, and text in this article were generated with substantial assistance from OpenAI GPT-6 Pro in ChatGPT chat mode and GPT-6 Astra in Codex, both using the ultra reasoning setting. These were the only AI tools used. The author organized and directed the research through repeated interactions, including questions, choices of research directions, requests for extensions and critical review, and selection and organization of the material. AI assistance was also used for literature searches, proof audits, exact computational checks, and manuscript revision. These checks do not constitute complete proof-assistant verification, which is not claimed. The author takes full responsibility for the accuracy, originality, attribution, and conclusions of the article.

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